

DESIGN AND CONTROL OF ARTIFICIAL MUSCLES FOR ROBOTIC APPLICATIONS

by

Thilina Hemaka Weerakkody

A thesis submitted in partial fulfillment
of the requirements for the
Doctor of Philosophy
degree in Mechanical Engineering in the
Graduate College of
The University of Iowa

December 2024

Thesis Committee: Caterina Lamuta, Thesis Supervisor
Venanzio Cichella
Hiroyuki Sugiyama
Deema Totah

Copyright by
Thilina Hemaka Weerakkody
© 2024
All Rights Reserved

“Everyone wants the prize – but only a few are willing to pay the price.”

This is dedicated to my best friend and love of my life, Niki, our joy of bundles, Thivane and Dilane, my parents, sister, and family, who believed in me and encouraged me through every hardship.

“Soft robotics, with its seamless integration of biology and engineering, breathes life into machines, while control and dynamics orchestrate their movements, transforming them into adaptive and resilient entities.”

ACKNOWLEDGEMENTS

My Ph.D. journey has been nothing short of a thrilling roller coaster—a blend of relentless challenges, rewarding achievements, and countless moments of self-discovery. Moving to a new country and embracing a different culture was daunting and exhilarating. Though demanding and tiresome, the work ethic in the United States constantly drives innovation and creativity, making every struggle worthwhile.

I am deeply grateful to my advisor, Dr. Caterina Lamuta, whose unwavering support and exceptional guidance have been the cornerstone of this journey. Your unique approach to integrating theoretical foundations with experimental validation was precisely the kind of research I aspired to pursue. Your dedication, vision, and mentorship have inspired me to aim higher and explore the depths of my potential. I could not have asked for a better supervisor.

I am equally grateful to Dr. Venanzio Cichella, who introduced me to the practical world of Control theory. Your encouragement, guidance, and confidence in my abilities have been pivotal throughout this endeavor. Your accomplishments and commitment to excellence in research and academia have inspired me.

I extend my heartfelt thanks to my committee members, Prof. Hiroyuki Sugiyama and Dr. Deema Totah. Your insightful feedback and support during my examination processes were invaluable. Studying advanced multibody dynamics under Prof. Sugiyama has been one of the most enriching experiences of my academic journey, shaping my approach to problem-solving in mechanical systems.

I would also like to acknowledge my collaborators, Dr. Venanzio Cichella, Dr. Deema Totah, Prof. Leonardo Pagnotta, Dr. Fatima Toor, and Sandeep T. Laroia, MD. Your contributions and shared expertise have significantly enriched this work.

My gratitude extends to my past and present lab mates at SMMS Lab—Parth Kotak, Mahmudul Alam Shakib, Sean Maxson, Rabiul Mamman, and Marissa McFadden—and at CAS Lab, Maxwell Hammond. Your camaraderie made even the most challenging days bearable, making this demanding adventure unforgettable.

Finally, I owe everything to my loving and incredible wife, Dulanji Niki Amaraweera, whose unwavering support, encouragement, and belief in me have been my greatest source of strength. Your patience, understanding, and sacrifices are the foundation for this achievement. To my children, Thivane and Dilane, thank you for filling our lives with immeasurable joy and for being the bright light that kept me going. My heartfelt thanks also go to my Amma, Thaththa, and my sister, Nadeesha, for their unconditional love, faith, and support that has carried me through every step of this journey.

To everyone who has been part of this roller coaster with me, thank you!

ABSTRACT

This dissertation presents the development of modeling and control architectures for innovative robotic devices powered by artificial muscles. It focuses on developing theoretical models to design and describe the behavior of artificial muscles—specifically twisted and coiled artificial muscles (TCAMs), twisted and spiraled artificial muscles (TSAMs), and NiTi shape memory alloy (SMA) muscles—and on implementing robust control algorithms for real-time applications. The research addresses various applications, including rehabilitation robotics, underwater exploration, and robotic surgery, by modeling, manufacturing, and testing a range of devices.

A physics-based theoretical model and an adaptive robust control architecture are proposed for actuating TCAMs, TSAMs, and SMA muscles. These models and controllers are experimentally validated through several practical applications: a soft exoskeleton for wrist rehabilitation, a variable stiffness Ankle-Foot Orthosis, a soft glove for hand rehabilitation, an octopus-inspired muscular hydrostat for underwater tasks, deployable vortex generators to enhance the aerodynamic performance of small unmanned aerial vehicles operating at low Reynolds numbers, and a surgical robot for paracentesis procedures. This work also includes developing simulation models, data acquisition frameworks, control hardware, and circuitry, which have contributed to numerous peer-reviewed journal publications.

The thesis is structured to provide a comprehensive overview of the theoretical models and control frameworks, their application in various robotic devices, and their experimental validation. It begins with developing a generalized physics-based model for TCAM actuation, followed by the design of an $\mathcal{L}_1$ adaptive control algorithm suitable for the highly nonlinear nature of TCAMs and SMA muscles. Subsequent chapters explore the applications of these models and control algorithms in rehabilitation robotics, underwater environments, and surgical robotics. The research demonstrates significant advancements in developing and controlling robotic devices powered by artificial muscles, highlighting their potential for broader adoption in diverse real-time applications.

PUBLIC ABSTRACT

This dissertation explores the development of new robotic devices powered by artificial muscles, which are a novel type of actuator that mimics the function of biological muscles. Unlike traditional motors, these artificial muscles can be lighter, more flexible, and capable of producing movement in a more natural and efficient way. The research focuses on creating both theoretical models and control systems that allow these muscles to function accurately and reliably in real-world applications.

Artificial muscles, such as twisted and coiled artificial muscles (TCAMs), twisted and spiraled artificial muscles (TSAMs), and shape memory alloys (SMAs), have been used in a range of innovative applications. These include wearable exoskeletons for rehabilitation, devices for robotic surgery, and underwater robots designed for exploration and environmental monitoring. The work presented in this dissertation involves designing, building, and testing these devices, as well as developing computer algorithms that control their movements in real-time.

The findings of this research show that artificial muscles have significant potential to improve the design and functionality of robotic devices. By developing more responsive and adaptable robots, this work could impact several fields, from healthcare and medical rehabilitation to environmental science and industrial automation.

TABLE OF CONTENTS

LIST OF TABLES	xiii
LIST OF FIGURES	xiv
CHAPTER 1. INTRODUCTION AND BACKGROUND	1
1.1. Soft Robotics and Artificial Muscles: State of the Art	1
1.2. Objective and Scope	8
1.3. Scientific Contributions	9
1.4. Thesis outline	9
CHAPTER 2. DYNAMIC MODELLING AND CONTROL OF ARTIFICIAL MUSCLES	12
2.1. Physics-based Actuation Model for TCAMs	13
2.1.1 Introduction	13
2.1.2 TCAM Actuation Model	16
2.1.3 Control Law	22
2.1.4 TCAM Sample Manufacturing Process	26
2.1.5 Simulation and Experiment Results	28
2.1.6 Summary	43
2.2. $\mathcal{L}_1$ Adaptive Controller for TCAM Actuation	44
2.2.1 Introduction	44
2.2.2 TCAM Dynamic Model	46
2.2.3 Problem Formulation	48

2.2.4	Control Law	50
2.2.5	Controller Performance Bounds	52
2.2.6	Experimental Results	55
2.2.7	Summary	60
2.3.	Modelling and Control of Shape Memory Alloy (SMA) Actuators	61
2.3.1	Nomenclature	61
2.3.2	Introduction	62
2.3.3	Shape Memory Alloy Theory and Modelling	63
2.3.4	Model for SMA Phase Transformation	66
2.3.5	Model for SMA Temperature Dynamics	67
2.3.6	Model for SMA Constitutive Law	68
2.3.7	SMA Finite State Machine Approach	70
2.3.8	Model Validation	72
2.3.9	Control Law	78
2.3.10	Experimental Validation	81
CHAPTER 3. ARTIFICIAL MUSCLES FOR REHABILITATION ROBOTICS APPLICATIONS		88
3.1.	Lightweight Bioinspired Exoskeleton for Wrist Rehabilitation Powered by TCAMs	89
3.1.1	Introduction	89
3.1.2	TCAMs-Exo Design	96
3.1.3	Theory of TCAM Insulation Model	108
3.1.4	Experimental Setup	112
3.1.5	Results and Discussion	118

3.1.6	Summary	125
3.2.	Modelling and Control of Ankle-Foot-Orthotics powered by TCAMs	127
3.2.1	Introduction	127
3.2.2	PowAFO Design Requirements	129
3.2.3	Mechanical Design and Hardware	131
3.2.4	PowAFO Actuation Circuitry Design	134
3.2.5	PowAFO Experimental Validation	135
3.3.	Design and Control of Rehabilitation Glove powered by Shape Memory Alloys (SMAs)	137
3.3.1	Prototyping	138
3.3.2	Soft Finger Exoskeleton Powered by SMA	141
3.3.3	Experimental Validation	143
CHAPTER 4. ARTIFICIAL MUSCLES FOR UNDERWATER ROBOTICS		145
4.1.	Octopus-Inspired Muscular Hydrostats Powered By Twisted and Coiled Artificial Muscles	146
4.1.1	Introduction	146
4.1.2	Semi-Automated polymer fishing-line TCAM manufacturing system	149
4.1.3	Contraction muscular hydrostat: Actuation modeling and validation	152
4.1.4	Torsional muscular hydrostat: Actuation modeling and validation	154
4.1.5	Bending muscular hydrostat: Actuation modeling and validation	156
4.1.6	Extension muscular hydrostat: Actuation modeling and validation	158
4.1.7	Hardware layout and transfer function for the underwater actuation and control of muscular hydrostats	159

4.1.8	Control and underwater actuation of TCAM-powered hydrostats	163
4.1.9	Underwater Actuation of MHs for Waves and Currents	165
4.1.10	Rube Goldberg obstacle course	170
4.1.11	Summary	173
4.2.	Control of TSAM Powered Vortex Generators	176
4.2.1	Introduction	176
4.2.2	Manufacturing and Actuation Modelling of TSAMs	181
4.2.3	TSAM Model	181
4.2.4	Controller Implementation, Hardware and Circuitry	186
4.2.5	Wind Tunnel actuation and control of deployable VGs powered by TSAMs	189
4.2.6	Underwater actuation and control of deployable VGs powered by TSAMs .	194
CHAPTER 5. ARTIFICIAL MUSCLE POWERED SEMI-AUTONOMOUS SURGICAL ROBOT		200
5.1.	Design of a Surgical Robot	200
5.1.1	Introduction	200
5.2.	Design of an Artificial Muscle Powered Surgical Robot	207
CHAPTER 6. CONCLUSIONS AND FUTURE WORK		209
6.1.	Conclusions	209
6.2.	Original Contributions	211
6.3.	Future Work	212
APPENDIX A: HARDWARE AND CIRCUITRY		214
A.1.	Automated TCAM manufacturing system circuitry	214

A.2. Rube Goldberg setup actuation circuitry	215
A.3. PowAFO actuation circuitry - 32 TCAMs	216
A.4. TSAMs with VG underwater actuation circuitry	217
A.5. Miniature TCAM actuation circuitry	218
APPENDIX B: MATLAB AND SIMULINK CODES	219
B.1. Temperature Tracking Using FLIR Thermal Camera	219
B.2. Realtime Multiple Points Tracking in Air and Underwater	222
B.3. Simulink SIMSCAPE model for kinematic analysis and control	227
B.4. TSAM Image Processing based Height Measuring	228
B.5. IMU Realtime Angle Tracking - MATLAB	230
B.6. IMU Realtime Angle Tracking - ROS/Linux	232
B.7. Arduino ROS1/roserial publisher for System Actuation	236
B.8. TCAM manufacturing machine - Arduino Code	237
B.9. TSAM Scalable manufacturing system - Arduino Code	242
REFERENCES	243

LIST OF TABLES

Table 2.1.	Physical, thermal and electrical properties of CF/PDMS TCAM samples.	32
Table 2.2.	Physical, thermal, and electrical properties of Nylon TCAM samples.	33
Table 2.3.	Material-dependent physical, thermal, and electrical properties for CF & Nylon TCAM samples.	33
Table 2.4.	TCAM Nominal Parameters	56
Table 2.5.	Control strategy and application overview	64
Table 2.6.	Table of state and transition definition	73
Table 3.1.	EXO device for wrist rehabilitation main features. Abbreviations—Grounded: Gnd. , Un-grounded: Ugnd. , Joint: JT	95
Table 3.2.	Length and body weight percentage for both male (M) and female (F) subjects.	105
Table 3.3.	Muscle attachment points as hand and forearm percentage for both extension/flexion and ulnar/radial deviation motions. (Abbreviations are reported in Figure 3.3 above).	105
Table 3.4.	Load applied for different muscle diameters.	108
Table 3.5.	Adopted phalanx length	140
Table 3.6.	Prototype main dimensions for a finger	142
Table 4.1.	Angles of attack Vs. VG configuration deployed at $Re = 100,000$ and $Re = 140,000$	189
Table 4.2.	Airfoil vortex generator configurations.	191

LIST OF FIGURES

Figure 1.1	Classification of the actuators	3
Figure 1.2	Specific power vs. weight of actuators and natural muscles	5
Figure 1.3	Artificial muscles types: (a) Dielectric Elastomer Actuators (DEAs), (b) IonicPolymer Metal Composites (IPMC), (c) Piezoelectric Artificial Muscles, (d) Conducting Polymers (CP), (e) Shape-Memory Alloys (SMA), (f) Shape-Memory Polymers (SMP), (g) Liquid Crystal Elastomers (LCE), (h) Twisted and Spiraled Artificial Muscles (TSAMs), (i) Twisted and Coiled Artificial Muscles (TCAMs), (j) Pneumatic Artificial Muscles (PAM).	6
Figure 1.4	Stress vs. strain for different types of artificial muscles	7
Figure 2.1	Schematic overview of the proposed theoretical model	15
Figure 2.2	(a) straight fiber before coiling; (b) spontaneously coiled fiber (TCAM) after manufacturing; (c) actuated TCAM.	17
Figure 2.3	Schematic representation of the experimental setups for (a) constant stress, (b) constant strain, and (c) dynamic force configuration for TCAMs testing. . .	28
Figure 2.4	(a) Constant stress configuration setup for CF/PDMS TCAM, (b) Constant strain configuration setup for Nylon TCAM, and (c) Dynamic force configuration setup for Nylon TCAM.	31
Figure 2.5	Dynamic model validation for the three configurations: CF/PDMS TCAM-Sample 1.	34
Figure 2.6	Dynamic model validation for the three configurations: CF/PDMS TCAM-Sample 2.	35
Figure 2.7	Dynamic model validation for the three configurations: CF/PDMS TCAM-Sample 3.	36
Figure 2.8	Dynamic model validation for the three configurations: Nylon TCAM-Sample 1. . .	37

Figure 2.9	Dynamic model validation for the three configurations: Nylon TCAM-Sample 2.	38
Figure 2.10	Dynamic model validation for the three configurations: Nylon TCAM-Sample 3.	39
Figure 2.11	Validation of real-time $\mathcal{L}_1$ adaptive controller for Configuration 1—Position control of TCAM lifting mass tracking (a)-(c) step and (d)-(f) sinusoidal reference.	40
Figure 2.12	Validation of real-time $\mathcal{L}_1$ adaptive controller for Configuration 2—Force control of TCAM under constant strain tracking (a)-(c) step and (d)-(f) sinusoidal reference.	41
Figure 2.13	Validation of real-time $\mathcal{L}_1$ adaptive controller for Configuration 3 — Position control of TCAM pulling antagonistic spring tracking (a)-(d) step and (e)-(h) sinusoidal reference.	42
Figure 2.14	Validation of real-time $\mathcal{L}_1$ adaptive controller for Configuration 3 — Force control of TCAM pulling antagonistic spring tracking (a)-(d) step and (e)-(h) sinusoidal reference.	43
Figure 2.15	Experimental results: step reference tracking	58
Figure 2.16	Experimental results: sinusoidal reference tracking.	59
Figure 2.17	Diagram phase and shape memory effect	65
Figure 2.18	Path for a) mechanical load and unload, b) thermal load, and c) thermal unload.	74
Figure 2.19	a) Temperature evolution over time b) Strain evolution over time c) Stress evolution over time.	75
Figure 2.20	a) Stress-strain b) Strain-temperature c) Stress-temperature evolution on the diagram phase.	75
Figure 2.21	a) Temperature evolution over the time b) Strain evolution over time c) Stress evolution over time.	76
Figure 2.22	a) Stress-strain b) Strain-temperature c) Stress-temperature evolution on the diagram phase.	77
Figure 2.23	a) Temperature over the time b) Stress-temperature evolution on the diagram phase c) Stress-strain evolution.	77

Figure 2.24	a) Temperature over the time b) Stress-temperature evolution on the diagram phase c) Stress-Strain evolution.	78
Figure 2.25	a) Temperature over the time b) Stress-temperature evolution on the diagram phase c) Stress-Strain evolution.	79
Figure 2.26	Schematic representation of the experimental setups for a) constant stress, b) constant strain, and c) dynamic force configuration for TCAMs testing.	82
Figure 2.27	SMA position tracking validation using $\mathcal{L}_1$ adaptive control architecture for constant stress configuration	83
Figure 2.28	SMA force tracking validation using $\mathcal{L}_1$ adaptive control architecture for constant strain configuration	85
Figure 2.29	SMA position tracking validation using $\mathcal{L}_1$ adaptive control architecture for dynamic configuration	86
Figure 2.30	SMA force tracking validation using $\mathcal{L}_1$ adaptive control architecture for dynamic configuration	87
Figure 3.1	(a) Schematic representation of end effector, where the connection with the patient is provided by just one point; (b) grounded exoskeleton. The device joints match the human ones with a connection to an external frame; (c) ungrounded exoskeleton. The device has no attachments with an external frame.	90
Figure 3.2	Muscles mainly involved in the wrist rotations: (a) Extensor Carpi Ulnaris; (b). Extensor Carpi Radialis Longus; (c) Extensor Carpi Radialis Brevis; (d) Flexor Carpi Ulnaris; (e) Flexor Carpi Radialis; (f) Palmaris Longus. (Adapted from Kenhub).	97
Figure 3.3	Design and components of the TCAMs-Exo: (a) schematic representation of the device; (b) the artificial muscles; (c) attachment coordinates.	98

Figure 3.4	chematic representation of the kinematics of the actuation principle of the hand. (a) The generic muscle between the points A and B (black not activated, red activated) from the side plane; (b) the generic muscle contracts providing wrist extension; (c) the generic muscle between A and B attachments from the side plane; (d) the muscle contracts, providing ulnar deviation; (e) back view of the muscle in the cold status; (f) back view of the muscle in the activated status, the muscle provides extension and ulnar deviation; (g) three-dimensional view for the activated status.	101
Figure 3.5	Schematic representation of the kinematic of the actuation principle of the hand. The muscles are attached to the bands placed on the hand and close to the elbow: (a) extensional/flexional attachments; (b) ulnar and radial deviations attachments; (c) length considered for the design.	104
Figure 3.6	(a) Worst case for wrist extension motion. The weight is applied in the center of gravity generating the maximum bending moment around the x axis; (b) after an extension of θ , the bending moment decreases because of the relative angle between the weight and its arm OB; (c) worst case for the ulnar deviation; (d) after a rotation of β , the bending moment decreases.	106
Figure 3.7	(a) Longitudinal section for muscle and insulation tube, where L is the tube length; (b) cross-section showing radius and components; (c) thermal circuit for the system considered, where k_{SR} is 0.16 W/m K.	109
Figure 3.8	Contour lines for the temperature T_3 as a function of D_2 and the tube thickness for a steady-state condition. The temperature of the TCAM is 105 °C.	110
Figure 3.9	Step-by-step procedure to simulate the trend temperature T_3 as a function of the inner radius R_2 and the thickness of the silicone rubber tube.	112
Figure 3.10	(a) Three-dimensional view of the 3D-printed hand and forearm. The center of gravity is given by the steel weights; (b) top view of the 2-DOF revolute joint; (c) side view of the 2-DOF revolute joint.	114
Figure 3.11	Three-dimensionally printed mold where liquid urethane is poured and cured. .	115
Figure 3.12	(a) Three-dimensionally printed outer and inner mold for the tube manufacturing (material: polylactic acid (PLA)); (b) silicone cast inside the molds; (c) a manufactured tube and the mold.	116

Figure 3.13	The block diagram of $\mathcal{L}_1$ adaptive control algorithm architecture.	117
Figure 3.14	(a) Temperature trend for a four-bundle TCAM activated with 0.2 kg of load and 5 % of contraction $R_1 = 0.9$ mm. The inset shows the temperature along with the muscle; (b) temperature trend for the tube. The inset is the temperature along the tube surface in steady-state condition; (c) temperature trend for the tube. The inset is the temperature along the tube surface for 30 s of actuation. A FLIR E95 thermal camera is used for temperature measurements.	119
Figure 3.15	(a) Design of the glove composed of TCAM and tube, NDL 1 and 2, IMU 1 and 2, a sleeve, and the EMGs design of the hand–forearm used as a test bench; a revolute joint substitutes the wrist, and several steel weights are used to reach the CG and the weights of a human hand; (b) motion performed with the artificial wrist: extension angle performed 65° ; (c) flexion angle performed 65° ; (d) radial deviation angle performed 23° ; (e) ulnar deviation angle performed 56° ; (f) maximum wrist motion angle on a ROM polar plot.	121
Figure 3.16	Simulations of rehabilitation exercise. The motion considered in the plot (a–c) are for extension. (a) Rehabilitation exercise performed in passive mode; (b) a rehabilitation exercise to emulate the weak hand; (c) a rehabilitation exercise of a weak hand together with TCAMs–Exo (active–assisted mode); (d) velocity for flexion and extension; (e) velocity for ulnar and radial deviation; (f) ulnar–radial deviation cycle performed at high frequency.	122
Figure 3.17	Tests with the EMGs; (a) worn glove, with EMG placed below the sleeve; (b) measuring panel for the voltage recorded by the EMGs; (c) maximum temperature along the tube during the test; (d) resting position of the hand; (e) the exercise results in a 10° rotation of the hand.	124
Figure 3.18	Ankle dorsiflexion and plantarflexion motions.	127
Figure 3.19	a) Passive AFO, b) Active AFO schematic device, and c) Powered active AFO.	128
Figure 3.20	Stiffness Measurement Apparatus (SMAApp) outline	132
Figure 3.21	CAD models of the PowAFO	132
Figure 3.22	Link segments: Kinematic diagrams of the PowAFO. (a) Isometric view, (b) Sagittal plane view	133

Figure 3.23	Controller hardware circuit	134
Figure 3.24	Controller diagram overview	135
Figure 3.25	Tracking plots for (a) Torque tracking using $\mathcal{L}_1$ adaptive controller, and relevant (b) TCAM temperatures and (c) voltage plots	136
Figure 3.26	Prototype Render	138
Figure 3.27	Real prototype	139
Figure 3.28	Rings for the a) proximal b) middle and c) distal phalanx	139
Figure 3.29	Prototype schematics in: (a) straight configuration; (b) bent configuration.	140
Figure 3.30	The implementation of $\mathcal{L}_1$ adaptive controller for SMA finger angle tracking	143
Figure 3.31	SMA actuated finger module angle tracking using $\mathcal{L}_1$ adaptive controller: (a) angle, temperature, and voltage plots, (b) initial and final instances of the fingertip	144
Figure 4.1	Bioinspiration and TCAM Arrangement for Proposed Hydrostats: (a) Octopus limbs exhibit extraordinary degrees of freedom with fine control. (b) Schematic diagram of octopus limb muscular anatomy with oriented fiber muscle layer. Schematic illustration of the proposed: (c) Bending hydrostat (BH), (d) Contraction hydrostat (CH), (e) Extension hydrostat (EH), and (f) Torsion hydrostat (TH).	147
Figure 4.2	Automated manufacturing machine components and steps	150
Figure 4.3	(a) Manufacturing process for self-coiled muscles; (b) Spontaneous coiling TCAM; (c) Completed self-coiled TCAM on annealing rack; (d) Completed self-coiled TCAM	151

Figure 4.4	(a) CH containing 4 TCAMs attached to 3D printed holders. A silicone elastomer sheet is attached to the holders using hose clamps. (b) CH before and (c) after electrothermal actuation of TCAMs using an input power of 62.4 W for 12 s. A maximum shortening of 4.7 cm (40 %) is achieved on TCAM contraction. (d) Working principle of nylon fishing line TCAMs. Validation of the (e) electrothermal and (f) dynamic model for TCAMs employed in our CH. 3D, three-dimensional.	153
Figure 4.5	TH with theoretical model. (a) TH device. (b) Current TH can achieve up to 75° of rotation. (c) Detail of TH showcasing the gel core, TCAM winding, and opened external skin. (d) Expanded view of torsion device schematic highlighting dual contribution of TCAM actuation and the closure of intercoil spacing along the length of the TCAM can be viewed as an arc length along the circumference of the gel core.	154
Figure 4.6	Images, actuation, and FE simulations of BH. (a) BH consisting of a Voronoi tessellated cylinder with a mechanical instability and a single TCAM attached to the cylinder through 3D-printed holders and eyebolts. (b) Larger less dense Voronoi cells. (c) Small dense Voronoi cells reinforce the back of the cylinder to control the direction of bending. (d) Hydrostat undergoing bending due to TCAM contraction using an input power of 10 W for 3 s. A maximum bend of 72° is achieved on TCAM actuation.	157
Figure 4.7	Application of TCAMs- Soft Bending Manipulator. a) Time-lapse diagram of bending manipulator motion, and b) orientation tracking and voltage provided by the $\mathcal{L}_1$ controller.	158
Figure 4.8	Images, actuation, and FE simulations of extension muscular hydrostat. (a) EH prototype comprising a PET braided sleeve with an elastomer coating, a flexible mesh, 4 TCAMs, and TCAM end terminals; (b) EH after TCAM contraction depicting an extension of 18.3% (2.2 cm). TCAMs are electrothermally actuated using an input power of 40 W for 6 s; Unit cell of the red silicone mesh (c) before and (d) after extension; (e) Braided sleeve frozen into an initial compressed state using a thin layer of silicone elastomer.	159

Figure 4.9	Angle/displacement measurement of each muscular hydrostat. Camera detection markers for the (a) CH, (c) TH, (e) BH, and (g) EH, respectively. Real-time experimental measurement system for the (b) CH, (d) TH, (f) BH, and (h) EH, respectively.	161
Figure 4.10	Schematic layout of the onboard hydrostat control and actuation system. (a) The hardware architecture of the onboard actuation system. (b) The $\mathcal{L}_1$ adaptive control architecture implemented on SIMULINK.	162
Figure 4.11	Control and underwater actuation of hydrostats. (a) The hardware configuration of the underwater onboard system. (b) The power electronic circuit for TCAM actuation (c) The CH before and during actuation. (d) The BH during the actuation.	165
Figure 4.12	Displacement/Angle tracking for muscular hydrostats using the $\mathcal{L}_1$ adaptive controller. CH angle tracking in (a) air and (b) underwater. TH angle tracking in (c) air and (d) underwater.	166
Figure 4.13	Displacement/Angle tracking for muscular hydrostats using the $\mathcal{L}_1$ adaptive controller. BH angle tracking in (e) air and (f) underwater. EH angle tracking in (g) air and (h) underwater. MH, muscular hydrostats.	167
Figure 4.14	University of Iowa Fluid’s lab towing tank facility.	168
Figure 4.15	The hardware and the control architecture for the real-time towing tank testing.	168
Figure 4.16	Tracking plots for contraction muscular hydrostats.	169
Figure 4.17	Tracking plots for Bending muscular hydrostats.	170
Figure 4.18	Tracking plots for Extension muscular hydrostats.	171
Figure 4.19	Tracking plots for Torsion muscular hydrostats.	172
Figure 4.20	Insulation process for TCAMs. (a) Tube mold, central rod, and cured unprepared insulation tube. (b) Split tube with TCAM placed within being painted with DragonSkin FX-Pro. (c) Insulated TCAM ready for use.	173

Figure 4.21	Rube Goldberg underwater obstacle course demonstration. (a) Complete obstacle course at the ready, (b) Marble blocked by torsion gate, (c) Actuated torsion gate allows passage, (d) Marble in chest awaiting illumination, (e) Actuated bending cylinder shines the spotlight on the marble, (f) Marble in its initial position atop the extension ramp, (g) Actuated extension cylinder tilts the ramp and sends the marble down the path, (h) Marble is caught in the contraction cylinder basket as indicated by the red circle, (i) Marble is tipped out of the basket and into the treasure chest again highlighted by the red circle.	174
Figure 4.22	(a) Reynolds number operating regime for various aircraft, including UAVs, under sea level standard conditions. (b) Effect of Reynolds numbers on maximum L/D ratio for smooth and rough airfoils. *Adapted with permission from Lissaman, P B S Annual Review of Fluid Mechanics, 31 Dec 1982, Vol. 15, Issue 1, pages 223–239. Copyright 1983 by Annual Reviews Inc.	176
Figure 4.23	(a) Cuttlefish with actuated papillae muscles (inset showing protrusion of a single papillae). (b) Octopus with actuated papillae muscles (inset showing protrusion of a single papillae). (c) Out of plane vertical actuation of a TSAM (left: flat configuration before actuation; right: actuated configuration upon electrothermal input).	178
Figure 4.24	(a) TSAM with a vortex generator attached on the top and (b) image of a TSAM.	182
Figure 4.25	TSAM position tracking using a camera	187
Figure 4.26	Control architecture for TSAM actuation system	188
Figure 4.27	Circuitry diagram for the TSAM actuation system	188
Figure 4.28	Manufactured airfoil with TSAMs-powered VGs. (a) Airfoil section with the deployable VGs (shown in the inset) and electrical setup for actuating TSAMs. (b) Top view of the TSAMs powered VGs mounted on an insert plate that can be slotted into the nose of the airfoil.	190
Figure 4.29	Temperature plots of VG in (a) raw1 = 8 cm and (b) raw2 = 2 cm.	191
Figure 4.30	(a), (d) Displacement, (b), (e) temperature, and (c), (f) voltage plots for two TSAMs actuated to and maintained at different heights using the $\mathcal{L}_1$ adaptive controller.	192

Figure 4.31	Effects of VGs deployed by electrothermally actuated TSAMs on the lift characteristics of the airfoil (a) Reynolds number 100,000 (b) Reynolds number of 140,000 in air.	194
Figure 4.32	Airfoil inside the towing tank	195
Figure 4.33	Manufactured airfoil with TSAMs-powered VGs. (a) Airfoil section with the deployable VGs (shown in the inset) and electrical setup for actuating TSAMs. (b) Graphical view of the top side, (c) the bottom side, and (d) Actual top view of the TSAMs powered VGs mounted on an insert plate that can be slotted into the nose of the airfoil.	196
Figure 4.34	Effects of VGs deployed by electrothermally actuated TSAMs on the lift characteristics of the airfoil (a) Reynolds number 100,000 (b) Reynolds number of 120,000 underwater.	197
Figure 4.35	Effects of VGs deployed by electrothermally actuated TSAMs on the drag characteristics of the airfoil (a) Reynolds number 100,000 (b) Reynolds number of 120,000 underwater.	198
Figure 5.1	Overview of Phase I design powered by DC motors	201
Figure 5.2	Overview of Phase II design powered by TCAMs	201
Figure 5.3	Design overview of bed and the gantry	202
Figure 5.4	Design overview of ultrasound rail	202
Figure 5.5	Design overview of the mechanism	203
Figure 5.6	Design overview of the slider	203
Figure 5.7	Kinematic diagram of the device	204
Figure 5.8	Yantra 1 - Simscape Model	205
Figure 5.9	SIMULINK SIMSCAPE Control Model and plots	206
Figure 5.10	Yantra 2 - Design Overview 1	207
Figure 5.11	Yantra 2 - Design Overview 2	207

Figure 5.12	Yantra 2 - Design Overview 3	208
Figure 5.13	Yantra 2 - Needle device Simulation	208

CHAPTER 1. INTRODUCTION AND BACKGROUND

1.1 Soft Robotics and Artificial Muscles: State of the Art

The field of robotics has undergone profound changes over the past decades, transitioning from research laboratories to factories and homes. Conventional robots, mainly made from conventional electric motors and rigid links, are characterized by high accuracy and precision and have found wide applications across various fields, ranging from industry to healthcare. However, their low compliance makes them unsuitable for applications requiring safe interaction with humans and soft structures. This limitation has led to the emergence of a new field of robotics known as soft robotics [1].

Soft robotics represents an innovative departure from traditional rigid robots, drawing inspiration from biological systems to develop flexible and adaptable compliant robots. This new generation of robots has proliferated in the past years. It enables new robot abilities for a wide range of applications ranging from wearable devices to search and rescue missions in unstructured environments [2–5].

Constructed from compliant materials, soft robotic devices offer several distinct advantages over rigid robots. This includes unique compliant motions, multidirectional functionality, and adaptability. Furthermore, their compliant nature enables them to interact with the environment more naturally and gently, facilitating navigation through complex and delicate environments with greater ease and safety. The compliance of soft robots makes them great candidates for developing assistive and surgical robotic devices, thanks to the safer interaction they provide when in contact with humans.

Soft robotic devices hold promise for minimally invasive surgical procedures. Their compliant nature allows for better adaptation to the body's tissues and organs, resulting in reduced trauma, faster recovery times, and improved patient outcomes compared to traditional rigid robots.

Soft robotics also presents unique advantages when considering underwater applications. Their flexibility enables soft robots to adapt to varying underwater conditions and interact with

delicate marine environments, making them suitable for underwater exploration and maintenance operations.

To develop soft robots, however, conventional multicomponent actuators, such as electric motors, pneumatic, or hydraulic actuators, are not suitable since they limit the portability and compliance of the soft robotic system. For this reason, in the past decade, we have witnessed the development of artificial muscles [6]. Artificial muscles are inspired by biological muscles' large power/weight ratio and can contract, expand, or rotate within one single component when exposed to different input stimuli. Unlike conventional actuators, artificial muscles exhibit inherent compliance and flexibility, making them well-suited for integration into soft robotic systems. By mimicking the biomechanical properties of living organisms, artificial muscles enable soft robots to achieve smooth and fine movements, enhance adaptability to complex environments, and ensure safe human interactions.

A quick overview and classification of mechanical actuators must be provided to further understand the properties and advantages of artificial muscles.

Mechanical actuators are defined as devices that convert a generic input energy into mechanical output. Mechanical actuators can be classified into two main groups: conventional actuators and artificial muscles (Figure 1.1) [7]. Conventional actuators - e.g., electric motors and hydraulic and pneumatic actuators - require multiple components (such as bearings, gears, coils, bushes, and shafts) to generate an output motion. Conventional actuators are actuated by either electromagnetic or pressurized fluid energy and generate rotary or linear output motions, as shown in Figure 1.1.

Conventional actuators can be classified into Electromagnetic actuators and Fluid power actuators. Electromagnetic actuators (or electric motors) can be DC (Direct Current) motors (e.g., the motor of a hand drill) or AC (Alternating Current) motors (e.g., industrial pumps). DC motors have high precision-control positioning [8], high starting torque, and high efficiency [9]. It is also easy to control their speed and torque [10], and they are able to provide smooth operation at low speeds [11]. In contrast, AC motors are more reliable but are difficult to operate at low

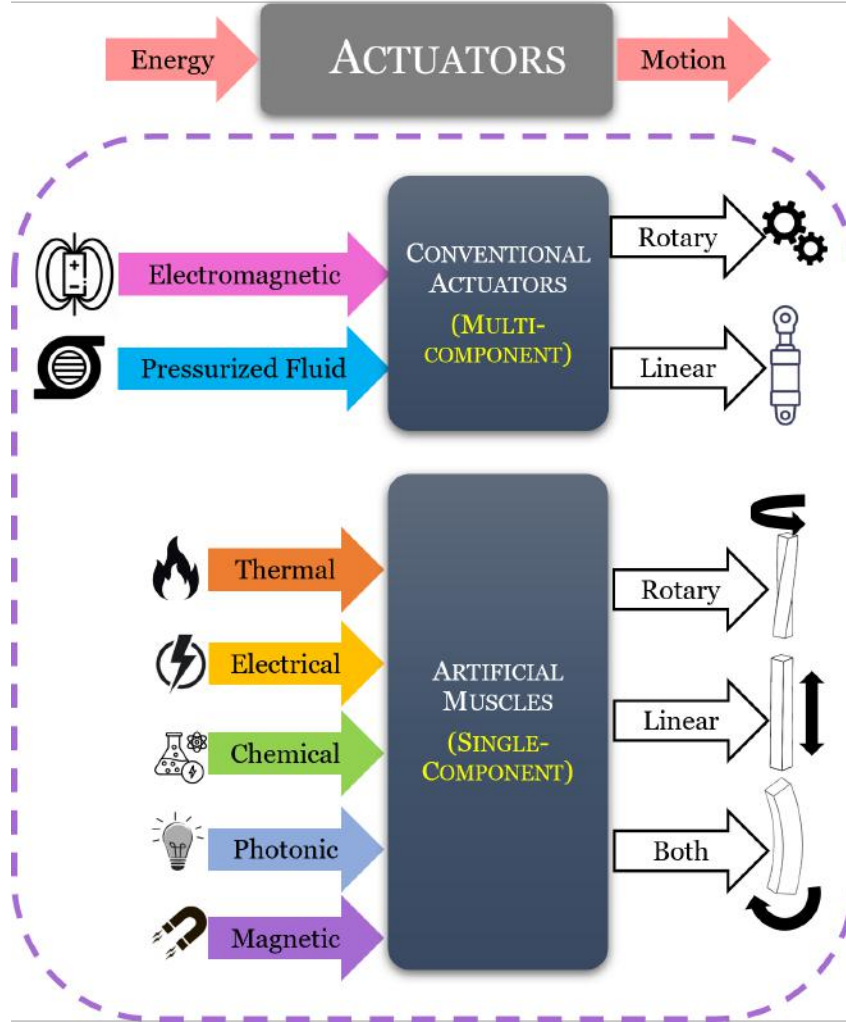


Figure 1.1. Classification of the actuators

speeds [12]. DC motors have high manufacturing costs but low maintenance costs [13]. Since they use DC current, they are inherently bulkier than AC motors [14, 15]. Electric motors find a wide range of applications [16] ranging from the automotive industry to power window actuation [17], and in the aerospace field for aerodynamic surface actuation [18].

Fluid power actuators can be classified into Hydraulic (e.g., hydraulic forklift) and Pneumatic (e.g., pneumatic manipulator employing double-acting cylinder actuators). Hydraulic actuators offer high specific power and force [13], making them suitable for applications requiring substantial strength. They also provide moderate precision control [19, 20]. However, they come with high installation costs [13], and their pumping systems are noisy and prone to leakage [21], posing

environmental risks and potential damage to surrounding areas. On the other hand, pneumatic actuators boast easy installation with low initial costs and moderate maintenance expenses [13]. They are safe, clean, and offer moderate specific power with straightforward handling [22]. Nonetheless, they exhibit moderate efficiency due to pressure losses and air compressibility [13]. Overall, hydraulic actuators excel in providing high force and precision control but come with higher installation costs and maintenance challenges. In contrast, pneumatic actuators offer simplicity, lower initial costs, and ease of handling, albeit with moderate efficiency and precision control.

Differently from conventional actuators, artificial muscles are characterized by a single component and can provide large power/weight ratios [23, 24]. An artificial muscle is defined as any material or device that can reversely contract, expand, or rotate within one single component in response to an external stimulus [25]. The unique actuation behavior mainly relies on the use of smart materials that can be actuated with different input stimuli, such as thermal, electrical, chemical, photonic, and magnetic energy. Their output motions can be rotary (torsion), linear (contraction/elongation), or combined actuation (bending) [26].

Compared to conventional actuators, lightweight artificial muscles have a higher specific power/weight ratio and portability. Specific power is defined as output mechanical work per unit volume provided within one cycle. Figure 1.2 shows a specific power vs. weight comparison of actuators and natural muscles.

As shown in Figure 1.2, DC motors are the lightest among conventional actuators. Even though DC motors are as light as natural muscles, they generate lower power. Hydraulic actuators, on the other hand, can generate more considerable specific power, yet they are heavy, which does not make them a good candidate for soft robotic applications. Artificial muscles are as lightweight as natural muscles and DC motors. However, at the same time, they can generate the most extensive range of specific power compared to natural muscles and conventional actuators. There are numerous types of artificial muscles, including Dielectric Elastomer Actuators (DEAs), Ionic Polymer Metal Composites (IPMC), Conducting Polymers (CP), Piezoelectric Artificial Muscles, Liquid Crystal Elastomers (LCE), Twisted and Coiled Artificial Muscles, Twisted and Spiral Artificial

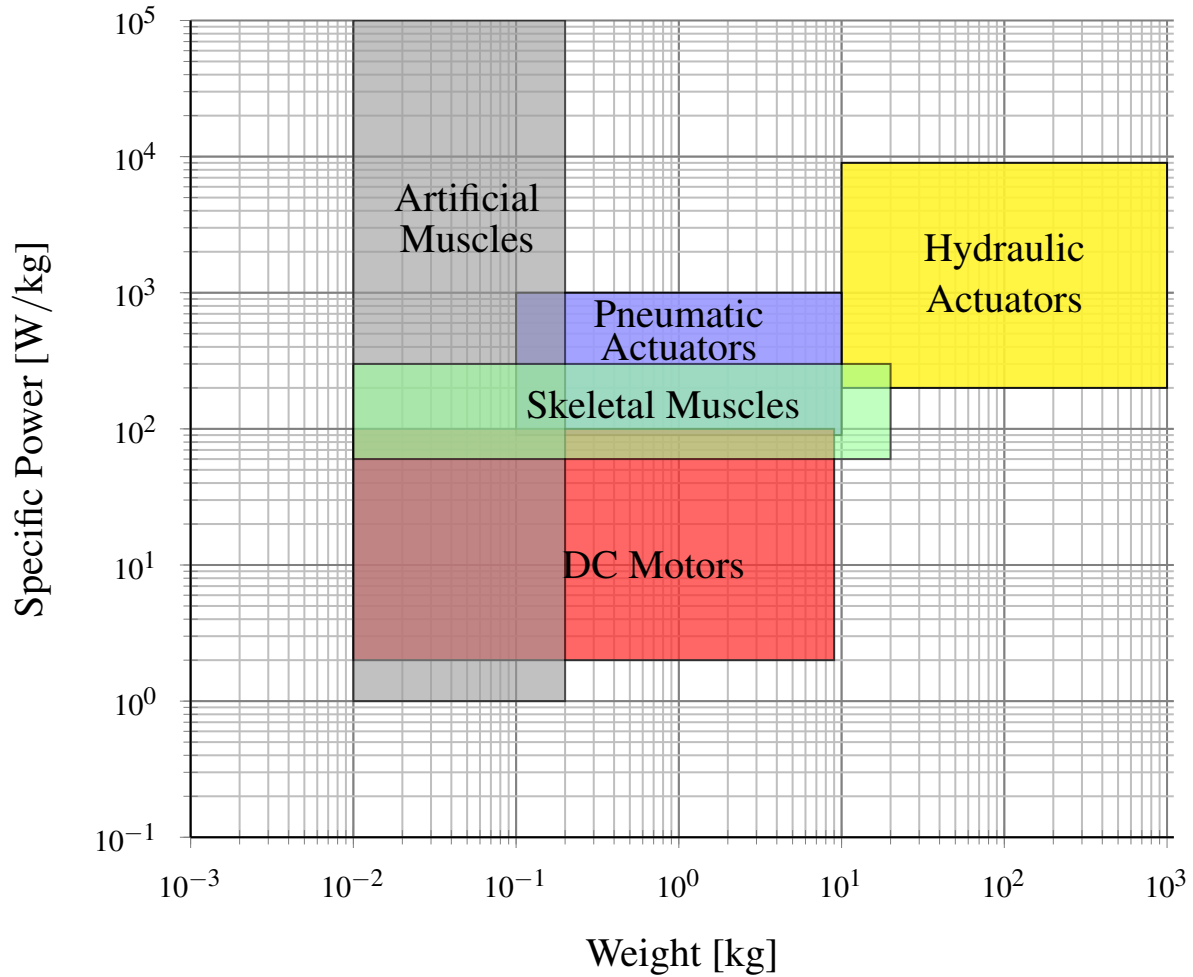


Figure 1.2. Specific power vs. weight of actuators and natural muscles

Muscles, Shape-Memory Alloys (SMA), Shape-Memory Polymers (SMP), Pneumatic Artificial Muscles (PAM), and Hydraulic Artificial Muscles (HAM). Some of these artificial muscles are shown in Figure 1.3. (The figures are acquired from [7]). These actuators are capable of performing both linear and rotary motions as a single unit, making them lighter and more compact than conventional actuators.

Figure 1.2 illustrates the stress vs. strain behavior of different artificial muscles. With the exception of LCE, all other types of artificial muscles can generate considerable stress compared to skeletal muscles across a wide range of strain values. However, only PAMs, HAMs, DEAs, SMPs, and TCAMs exhibit larger strain values compared to skeletal muscles. Among these arti-

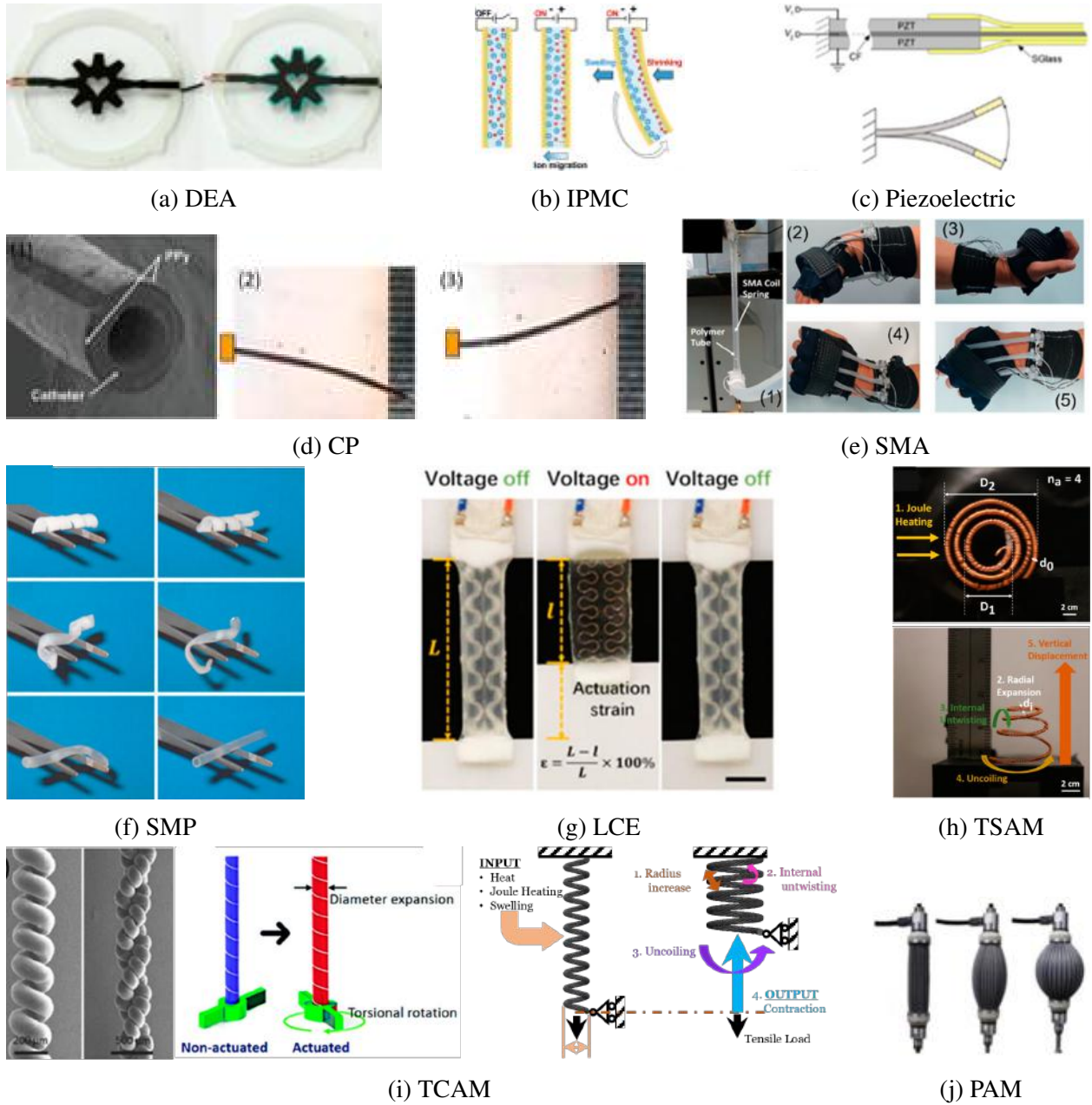


Figure 1.3. Artificial muscles types: **(a)** Dielectric Elastomer Actuators (DEAs), **(b)** Ionic Polymer Metal Composites (IPMC), **(c)** Piezoelectric Artificial Muscles. **(d)** Conducting Polymers (CP), **(e)** Shape-Memory Alloys (SMA), **(f)** Shape-Memory Polymers (SMP), **(g)** Liquid Crystal Elastomers (LCE), **(h)** Twisted and Spiraled Artificial Muscles (TSAMs), **(i)** Twisted and Coiled Artificial Muscles (TCAMs), **(j)** Pneumatic Artificial Muscles (PAM).

ficial muscles, DEAs provide the largest strain for a given stress, while SMAs generate the most significant stress for a given strain. Piezoelectric actuators can generate high stress, but their strain capabilities are limited. HAMs and PAMs can produce more significant strains, but they inherently

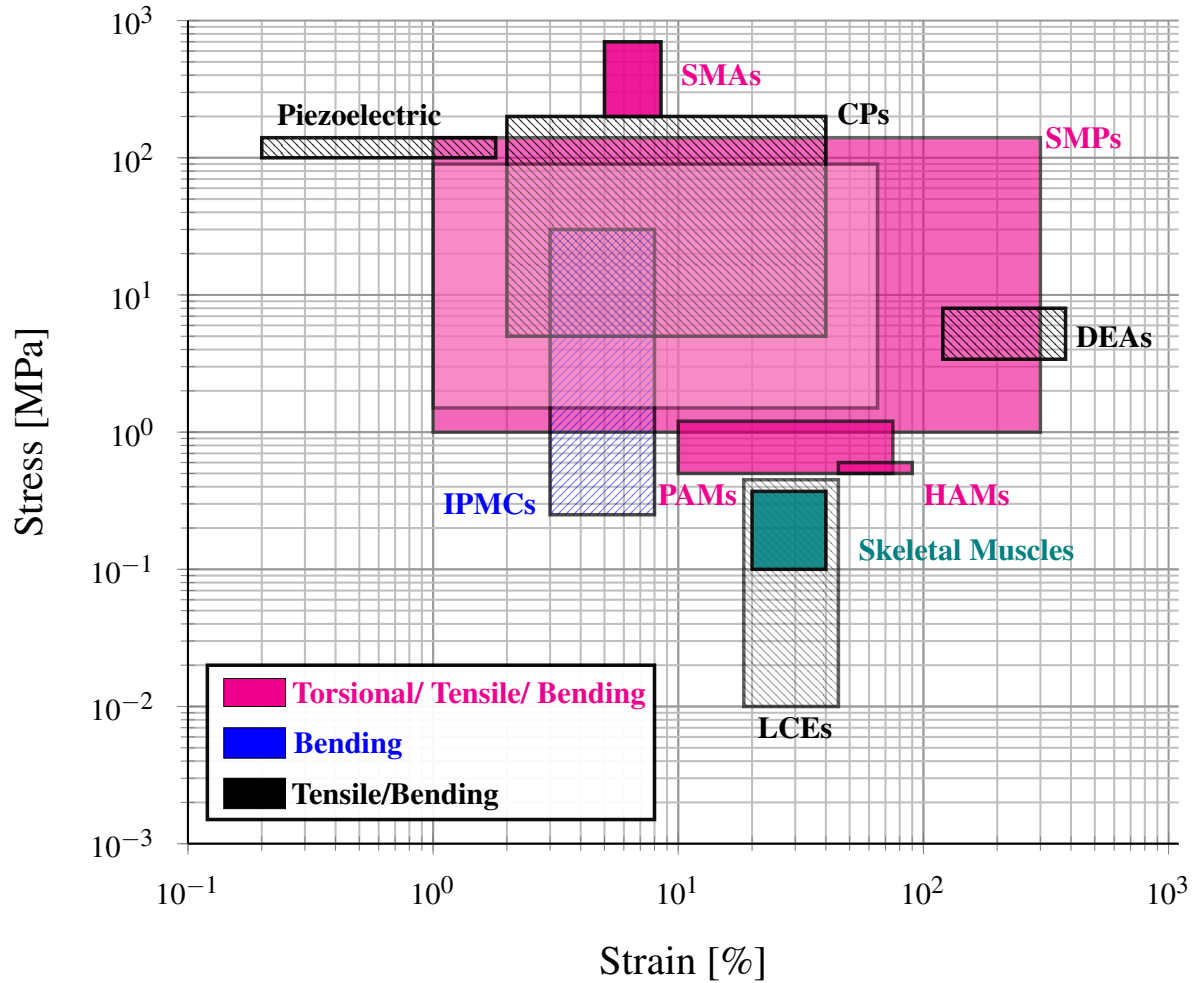


Figure 1.4. Stress vs. strain for different types of artificial muscles

require bulky systems due to external fluid pressure pump systems. IPMCs and LCEs can generate a wide range of stress values, although their strain is limited.

As depicted in Figure 1.4, only TCAMs and SMPs exhibit the most extensive range of strain and stress, making them suitable for a wide range of applications. Compared to SMPs, however, TCAMs do not present the same limitations in terms of costs and hysteretic behavior. IPMCs are limited to bending motion, while Piezoelectric, CP, DEA, and LCE actuators are limited to bending and torsion motions. However, all other actuators can perform torsion, tensile, and bending motions, including TCAMs [7].

In summary, TCAMs demonstrate a wide range of stress and strain capabilities, allowing

them to perform torsional, tensile, and bending motions. Furthermore, they can be actuated electrothermally and through swelling, providing versatility in their applications. With the ability to lift up to 12,600 times their weight with low input voltage (0.1 V/cm) [27], TCAMs hold significant potential as artificial muscle for soft robotic applications.

To fully exploit the potential of soft robotic devices, accurate mathematical modeling and robust control strategies for artificial muscles are necessary [28–34]. These models enable researchers and engineers to understand and predict the behavior of soft robotic systems, thereby optimizing their design and performance. Additionally, robust control strategies are essential for accurately manipulating and controlling the movements of artificial muscles in soft robotic devices, considering their flexible and compliant nature, as well as the specific requirements of the application [4, 35, 36]. In rehabilitation applications, researchers are actively exploring mathematical models and control strategies to enhance the performance and safety of artificial muscles in soft robotic assistive devices [30, 32, 37–43]. Moreover, the focus on accurate mathematical modeling and robust control strategies in surgical robotics leads to significant advancements in soft robotics. The attention is mainly directed towards optimizing the design, performance, and safety of artificial muscles for minimally invasive procedures, ultimately improving patient care [34, 44–46].

In underwater exploration and manipulation, the increasing interest in soft robotics and artificial muscles has led to focused research efforts into the modeling of artificial muscles to enhance their performance and functionality in wet and extreme environments [47, 48].

My research aims to explore the mathematical modeling and control aspects of different types of artificial muscles to develop bioinspired soft robots for a wide range of applications, spanning from healthcare to underwater exploration.

1.2 Objective and Scope

The objective of this dissertation is to develop modeling and control architectures for novel robotic devices powered by artificial muscles. This includes developing theoretical models to design and describe the behavior of both the artificial muscles and the robotic devices and implementing robust

control algorithms for the real-time control of devices powered by artificial muscles. Different devices are modeled, manufactured, and tested for several applications, including rehabilitation robotics, robotic surgery, underwater exploration, and fluid control.

1.3 Scientific Contributions

This PhD research proposes a theoretical physics-based model and a robust control architecture for actuating twisted and coiled artificial muscles (TCAMs), twisted and spiraled artificial muscles (TSAMs), and NiTi shape memory alloy (SMA) artificial muscles. The proposed models and controllers are experimentally validated through practical applications, which include: (1) a Nylon TCAM-powered soft exoskeleton for wrist rehabilitation; (2) a carbon fiber TCAM-powered variable stiffness Ankle-Foot Orthosis device; (3) an SMA-powered soft glove for hand rehabilitation; (4) a TCAM-powered octopus-inspired muscular hydrostat capable of bending, contraction, extension, and twisting underwater; (5) TSAM-powered smart skins for underwater antifouling and flow control; and (6) a TCAM-powered surgical robot for paracentesis procedures. These applications involve hardware development, real-time sensing, data acquisition, and controlled actuation. Additionally, the research encompasses the design and development of simulation models, data acquisition frameworks, control hardware, and circuitry. Throughout the PhD journey, this work has resulted in the publication of nine peer-reviewed journal articles.

1.4 Thesis outline

This doctoral thesis is organized as follows. Chapter 2: presents the theoretical and control frameworks for artificial muscles. Chapter 2.1 presents the generalized physics-based theoretical model for TCAM actuation. The model is experimentally validated for carbon-fiber (CF) TCAMs reinforced with silicone rubber (SR) and nylon-6 fishing lines wrapped with a nichrome wire. The model is presented for three different TCAM attachment configurations: constant-stress, constant-strain, and dynamic stress-strain. The objective of this model is to be used as a simulation model prior to TCAM use in soft robotic applications and to derive the dynamics of the TCAMs for tuning

the real-time impedance controller for applications.

Chapter 2.2 proposes a robust and adaptive control architecture for these highly nonlinear TCAMs in real-time applications. The controller is based on novel $\mathcal{L}_1$ adaptive control theory, which offers significant capability and fast adaptation by decoupling the adaptation loop from the control loop, thus enabling arbitrarily fast adaptation without sacrificing robustness. This is a key advantage for tracking different impedance targets using the same controller parameters in uncertain environments. The $\mathcal{L}_1$ adaptive controller is experimentally validated for TCAM actuation in the three configurations mentioned above (i.e., constant stress, constant strain, and dynamic external force).

Shape Memory Alloy (SMA) actuators are an emerging class that offers high recoverable strain, specific stress, and reliability. However, modeling SMAs as a linear system is challenging due to high hysteresis. Chapter 2.3 presents a mathematical model and an $\mathcal{L}_1$ robust adaptive control architecture for Ni-Ti SMA actuators. The proposed control algorithm compensates for disturbances and uncertainties and is mathematically formulated and experimentally validated for tracking performance. The $\mathcal{L}_1$ adaptive controller's experiments are compared with those of PID tracking.

Chapter 3 extends the aforementioned theoretical framework into application. We propose a modular soft exoskeleton for wrist rehabilitation powered by TCAMs in Chapter 3.1. This section presents the design and development of the device, followed by the theoretical framework presented in the previous chapter. Finally, we present an initial validation of the device actuation using Electromyography (EMG) signals.

Chapter 3.2 describes developing and controlling a variable stiffness actuator for an Ankle-Foot Orthosis (AFO) powered by carbon fiber TCAMs. This chapter covers the device design, control circuitry, and control architecture. The proposed circuitry manages power distribution for the TCAMs, and the controller actuates them individually as required. The variable stiffness actuator's torque generation is experimentally validated for critical static angles, with the results presented in this section.

Chapter 3.3 extends the work from Chapter 2.3 to validate the efficacy of the proposed $\mathcal{L}_1$ adaptive controller for SMA actuators using a finger exoskeleton glove. This chapter presents kinematic calculations and the finger angle tracking results for the developed device.

Chapter 4 presents the control applications of Artificial Muscles in underwater environments. Chapter 4.1 presents the octopus-inspired muscular hydrostats that are developed and actuated by TCAMs. The individual muscular hydrostats can perform bending, contraction, extension, and torsion motions. My contribution to this work was the implementation of the model and control architecture, followed by real-time testing of individual devices in the air and underwater (in still water, wave, and current environments) to evaluate the efficacy of the controller.

Chapter 4.2 discusses the control of Twisted and Spiraled Artificial Muscles (TSAMs) for underwater applications, focusing on the control of deployable Vortex Generators (VG) to enhance the aerodynamic performance of small unmanned aerial vehicles operating at low Reynolds numbers. An array of electrothermally actuated TSAMs with VGs on the aerofoil is experimentally verified by measuring the lift coefficient in a wind tunnel under different wind speeds followed by a towing tank for different flow speeds.

Chapter 5 describes the design and control simulation of a surgical robot powered by (1) conventional actuators in Chapter 5.1 and (2) TCAMs for use inside a Magnetic Resonance Imaging (MRI) machine in Chapter 5.2. This chapter presents the design, kinematic and dynamic analysis, workspace analysis, and preliminary control system.

Finally, Chapter 6 concludes the dissertation.

CHAPTER 2. DYNAMIC MODELLING AND CONTROL OF ARTIFICIAL MUSCLES

Artificial muscles, such as twisted and coiled artificial muscles (TCAMs), twisted and spiraled artificial muscles (TSAMs), and shape memory alloys (SMAs), offer new opportunities for designing more efficient, flexible, and biologically inspired robotic devices. However, the inherent nonlinear behavior and dynamic characteristics of these materials present significant challenges for modeling and control. This chapter addresses these challenges by presenting a comprehensive theoretical modeling approach and robust control frameworks for these novel actuators.

The chapter begins in Section 2.1 with a generalized physics-based model for TCAM actuation, which provides a fundamental understanding of their behavior under different loading and attachment configurations, including constant stress, constant strain, and dynamic stress-strain scenarios. This model serves as a simulation tool to predict the performance of TCAMs in various applications, allowing for optimization and adaptation before implementation in robotic systems.

Building upon this foundation, Section 2.2 introduces a novel $\mathcal{L}_1$ adaptive control architecture designed to handle the high levels of nonlinearity and uncertainty associated with artificial muscle actuation. By decoupling the adaptation loop from the control loop, this control strategy offers significant improvements in response time and robustness, making it well-suited for real-time applications where precise and rapid adaptation is critical.

Additionally, Section 2.3 in this chapter explores the application of these theoretical models and control strategies to SMA actuators, which, despite their high specific stress and strain recovery, present unique modeling challenges due to their hysteretic behavior. A robust $\mathcal{L}_1$ adaptive control framework is proposed to manage these challenges, providing effective disturbance rejection and precise control performance.

Overall, this chapter lays the groundwork for understanding and controlling artificial muscles in various robotic applications, setting the stage for their practical implementation and integration into advanced robotic systems discussed in subsequent chapters.

2.1 Physics-based Actuation Model for TCAMs

2.1.1 Introduction

Many modern engineering innovations are inspired by biological systems. In the case of soft robotics, the compliant nature of the actuation mechanisms used within the musculoskeletal system of the human body provides a benchmark for engineering modern technology. Recently, twisted and coiled artificial muscles (TCAMs) have emerged as lightweight, high-power, soft-robotic actuators able to replicate many of the traits of the system which inspired them [7, 49–51]. In contrast to conventional electromagnetic, pneumatic, and hydraulic actuators, TCAMs are promising for developing complex and compact actuator arrays which can exhibit desirable compliant behavior [52, 53]. TCAMs can be manufactured by highly twisting (until spontaneous coiling is achieved) inexpensive materials such as polymer fishing lines or carbon fibers embedded in a silicone matrix and are capable of lifting more than 12600 times their own weight with less than 0.2 V/cm of electrical input [49, 54–56]. The contraction of TCAMs is due to an anisotropic volume expansion of the highly twisted precursor fibers that can be induced by heating, Joule heating, or chemical solvent absorption [57, 58]. There is a promising future for these lightweight muscles as the next-generation actuators for a wide range of soft robotics applications such as medical devices, prostheses, robotics, biomimetic devices, and self-morphing skins [55, 58–63].

The development of theoretical dynamic models able to describe TCAMs actuation is then essential for the design and control of TCAMs-actuated smart and robotic devices. Existing dynamic models on TCAMs can be classified into fitting-based [64–66] and physics-based [67–76] models. While the first category includes models where the dynamic constants are obtained by fitting experimental data, the models in the second category provide equations to relate the dynamic constants to the physical properties of the materials used. Physics-based models are particularly useful because a deep understanding of the physical phenomena behind TCAMs actuation can lead to an optimization of the TCAMs properties, to the development of novel and efficient control strategies, as well as the development of novel TCAMs types.

Thermal and electro-thermal actuation via Joule heating represents the most popular actuation methods. The modeling of electro-thermally actuated TCAMs can be divided into two steps: 1) thermal model: model the relationship between the input voltage and the TCAM fiber radius; 2) mechanical model: model the relationship between the TCAM radius and the output displacement and/or force. Step 1) is usually developed considering natural convection in the air [68,70,73,74,77]. A few models also consider more complex phenomena such as radiation and the heat generated from damping [75] or electrical resistance change with temperature [67, 78]. Step 2) can be pursued using three main approaches: i) Castigliano’s second theorem; ii) Love’s theory or Knot theory; iii) Cosserat rod theory. In i) the total strain energy of a generic system is modeled under the assumption of small deformations to relate external loads to the displacement field of the system [67, 74, 79]. In ii) the total potential energy of a coiled fiber (obtained by highly twisting) is modeled to relate the untwisting to the vertical displacement [49, 50, 69, 80, 81]. Only the bending and torsional stiffness of the system are considered (shear and tensile strains are neglected). iii) is a more complex and generalized model that considers large deformation and also includes shear and tensile strains. The Cosserat rod theory is usually used to model soft and extremely compliant bodies, usually characterized by hyperelastic properties such as elastomers [68, 82]. Considering that TCAMs are usually obtained from materials with elastic or viscoelastic properties (i.e., carbon fibers, nylon monofilament) which can fall under the assumptions of i) and ii), Castigliano’s and Love’s theory are usually preferred over Cosserat rod theory due to their easier implementation.

In this work we propose a generalized and physics-based dynamic model for TCAMs that includes both a thermal and a mechanical model (see Figure 2.1). The thermal model includes the relationship between the input voltage and the TCAM fiber radius, considering Joule heating, free convection in air, and the anisotropic volume expansion of the precursor materials. Contrary to existing models where the convection heat transfer coefficient is obtained by fitting experimental data, a physics-based approach to relate the convection coefficient to the temperature-changing geometry of TCAMs is proposed. The mechanical model proposed builds upon our previous static

model and simplified dynamic model [49, 81] derived from Love’s theory. Specifically, TCAMs are modeled as a second-order dynamic system with temperature-dependant stiffness and damping coefficients. While in our previous work [81] the damping coefficient was obtained by a fitting procedure, a physics-based approach that considers the Kelvin-Voight viscoelastic model is proposed herein for the first time to obtain a temperature-dependant damping coefficient. As demonstrated via experimental validation, this model is suitable for any type of TCAMs (tests on both carbon fiber and polymer-based TCAMS are performed), and for different working configurations (with different displacement/force boundary conditions).

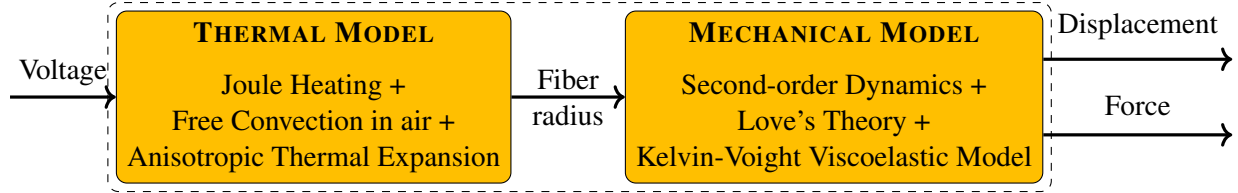


Figure 2.1. Schematic overview of the proposed theoretical model

Moreover, an $\mathcal{L}_1$ adaptive controller for both TCAM displacement and force control is proposed in this Section. This controller is an extension of our previous work [83], which focuses only on position control.

The control of various kinds of artificial muscles has been a well explored area of research spanning a wide range of control methodologies from PID [84] and observer based controllers [85], to iterative learning [86, 87], reinforcement learning [88] and adaptive control methodologies [89]. However, due to their relative novelty, TCAMs have not yet enjoyed the same breadth of controls research and what does exist is primarily limited to feed forward, PID and Lyapunov theory [90–96]. TCAMs provide a challenge to controller design due to the high non-linearity of their respective models and their strong dependence on parameters determined by their material properties and fabrication techniques. The muscle’s actuation by joule heating also lends value to controllers with guaranteed boundedness of unmeasured states like temperature as a preventative measure for system overheating. These challenges lead to the choice of adaptive output feedback control methodologies such as $\mathcal{L}_1$ adaptive control [97, 98] to be optimal for use with TCAMs. The

use of $\mathcal{L}_1$ sampled data control framework further enhances the use of TCAMs in lightweight devices by guaranteeing robustness despite relatively large sampling times and delays, thus allowing for lower cost and weight in system components. The extension of the work from [83] done here to include TCAM force control broadens the work's potential impact by enabling the integration of $\mathcal{L}_1$ adaptive TCAM control in a wider range of potential use cases.

The Section is organized as follows. Subsection 2.1.2 describes the thermal model and the mechanical model for TCAMs actuation. In subsection 2.1.3, we formulate the control law of the proposed $\mathcal{L}_1$ adaptive controller for position and force. Subsection 2.1.4 presents the materials and manufacturing techniques that have been used to fabricate the CF/PDMS and Nylon fishing line TCAMs. Subsection 2.1.5 describes the simulation and experimental results for the theoretical model and $\mathcal{L}_1$ adaptive controller validation. Subsection 2.1.6 concludes this article.

2.1.2 TCAM Actuation Model

TCAMs are usually fabricated using a low-cost manufacturing process. A straight fiber under tensile load is highly twisted until the formation of spontaneous coils (see Figure 2.2 a and b). Two main ingredients are needed for a highly twisted and coiled fiber to be a TCAM: 1) anisotropic volume expansion of the fiber (the fiber expands only along its radius, without any elongation when an external stimulus such as heat or chemical swelling is applied). For some materials such as nylon, [99], a radial expansion is also accompanied by a shortening of the fiber; 2) large volume expansion (the larger the radial expansion, the larger the TCAM contraction). As an example, in composite carbon fibers/silicone rubber TCAMs [49], 1) is given by anisotropic carbon fibers, while 2) is given by silicone rubber, which amplifies the small radial expansion of carbon fibers. In general, the actuation mechanism of TCAMs can be divided into four main steps (see Figure 2 c): (i) radius expansion (and shortening in some cases) due to external stimulus; (ii) untwisting; (iii) uncoiling; (iv) vertical displacement, i.e., contraction.

Building upon our previous static [49] and dynamic [81] models for electro-thermally actuated TCAMs, in this section, we propose: a) a thermal physics-based model to describe the

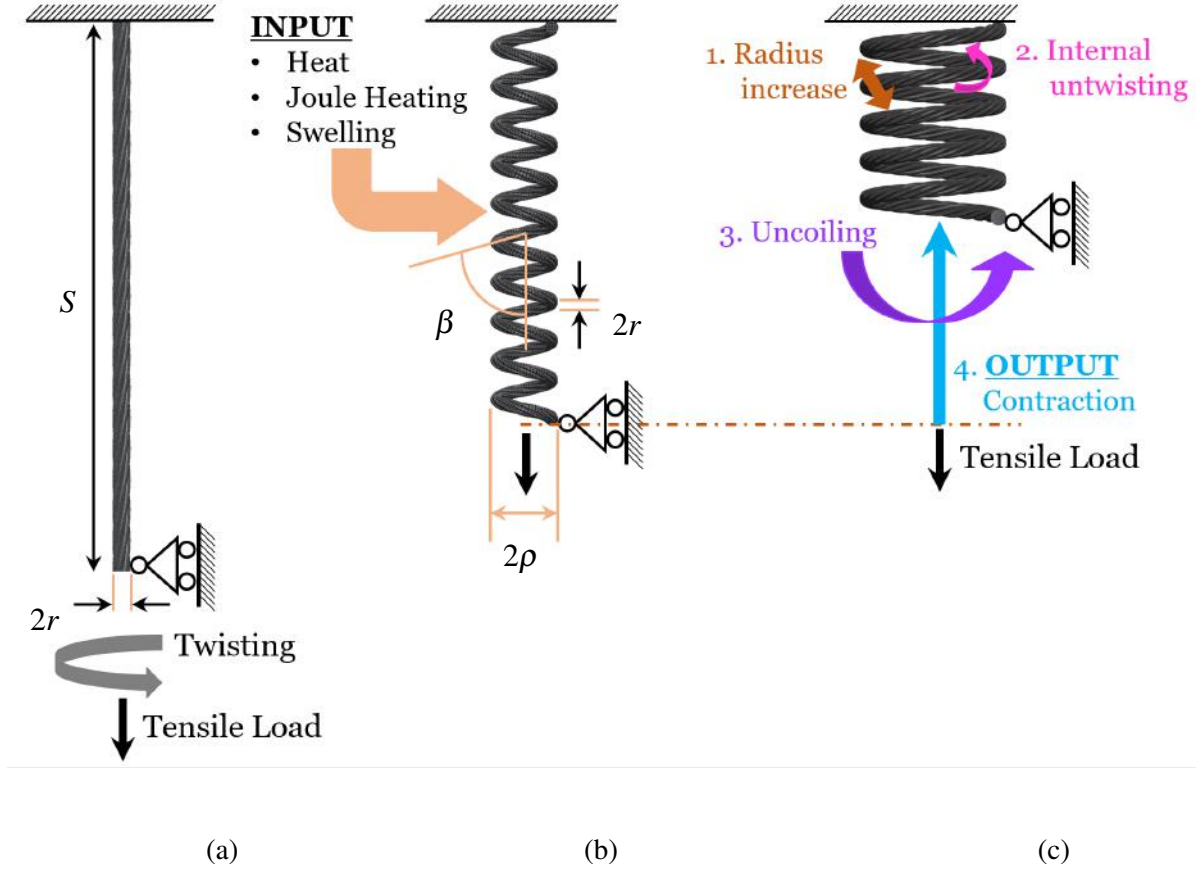


Figure 2.2. (a) straight fiber before coiling; (b) spontaneously coiled fiber (TCAM) after manufacturing; (c) actuated TCAM.

relationship between the input voltage and the TCAM fiber radius; b) a physics-based dynamic model for actuation that describes the generalized relationship between the TCAM fiber radius and the output displacement/force.

2.1.2.1 Thermal Model

The relationship between the input voltage $V(t)$ and the temperature change $T_{\Delta}(t)$ due to Joule heating can be expressed as follows [100]:

$$\dot{T}_{\Delta}(t) = \frac{V^2(t)}{C_t R} - \frac{\lambda(T_{\Delta})}{C_t} T_{\Delta}(t), \quad (2.1)$$

where $T_{\Delta}(t) = T(t) - T_{amb}$, where $T(t)$ is the TCAM temperature and T_{amb} is the ambient temperature, C_t , R , and $\lambda(T_{\Delta})$ are the thermal mass, electrical resistance, and the absolute thermal conductivity of the TCAM, respectively. The thermal mass is given by $C_t = mC_p$, where m and C_p are the TCAM mass and the specific heat capacity, respectively. Moreover, R is measured prior to the actuation and is assumed to be constant throughout the actuation [101]. Finally, $\lambda(T_{\Delta})$ is defined as:

$$\lambda(T_{\Delta}) = h(T_{\Delta})A_s(T_{\Delta}). \quad (2.2)$$

Here, $h(T_{\Delta})$ and $A_s = \pi r^2(T_{\Delta})$ are the convection heat transfer coefficient, and cross-section area of the TCAM, respectively, where $r(T_{\Delta})$ is the fiber radius (see Figure 2 b). To model the heat transfer due to convection, we assume that the TCAM can be modeled as a vertical cylinder in air. Considering that for all TCAM samples $D/L > 35/(Gr)^{0.25}$ (where D and L are the coil diameter and length, respectively, and Gr the Grashof number for air), the equation of a vertical cylinder reduces to the equation of free convection in the air of a vertical plane, according to Churchill and Chu model [18]:

$$h(T_{\Delta}) = \frac{K}{2\rho(T_{\Delta})} \left(0.6 + \frac{0.387Ra_L^{1/6}}{(1 + (0.559/Pr)^{9/16})^{8/27}} \right)^2, \quad (2.3)$$

where Pr , Ra_L , and K are the Prandtl number, Rayleigh number for characteristic length, and thermal conductivity of the TCAM, respectively. The Rayleigh number can be obtained from $Ra_L = Gr \times Pr$. The initial radius of the coil $\rho(T_{\Delta})$ (see Figure 2b) is defined as a function of the temperature as follows:

$$\rho(T_{\Delta}) = \frac{S \sin \beta(T_{\Delta})}{2\pi N}, \quad (2.4)$$

where S is the initial length of the straight fiber/composite yarn (before coiling, as shown in Figure 2a) and $\beta(T_{\Delta})$ and N are the coil angle of the coiled TCAM, and the number of coils formed spontaneously in the TCAM during the manufacturing (see Figure 2b), calculated as:

$$N(T_{\Delta}) = \frac{S}{2\pi} \sqrt{\frac{\bar{m}g}{B(T_{\Delta})}}. \quad (2.5)$$

where B is the bending stiffness of the TCAM (see Equation 7) and $\bar{m}$ is the external mass applied on the TCAM during fabrication (tensile load in Figure 2 a).

The incremental change of the TCAM's temperature (due to Joule heating) causes the radius change of the TCAM fiber. The relationship between the temperature change $T_\Delta(t)$ and the fiber radius $r(T_\Delta)$ can be written as:

$$\frac{r(T_\Delta) - r_0}{r_0} = CTE T_\Delta(t). \quad (2.6)$$

where CTE and r_0 are the linear thermal expansion coefficient in the radial direction, and initial fiber radius, respectively. It is worth noting that only the radius increase is considered as a consequence of anisotropic volume expansion. The initial length of the fiber S is considered constant and any shortening of the fiber is neglected.

2.1.2.2 Mechanical model

The anisotropic volume expansion of the TCAM (i.e., fiber radius increase) causes a stiffening of the TCAM. Specifically, the increase of the bending stiffness (B) and the torsional stiffness (C) due to radial expansion can be written as:

$$B(T_\Delta) = E_x \frac{\pi}{4} r^4(T_\Delta), \quad (2.7)$$

$$C(T_\Delta) = G_{yz} \frac{\pi}{2} r^4(T_\Delta), \quad (2.8)$$

where E_x and G_{yz} are the longitudinal Young's modulus and shear modulus, respectively. The detailed derivation of E_x and G_{yz} is available in the Appendix of [49]. The variation of the TCAM stiffness causes the variation of its coil angle, $\beta(T_\Delta)$, as expressed by Equation (2.9):

$$\cos \beta(T_\Delta) = 1 - \frac{1}{B(T_\Delta) - C(T_\Delta)} \left(\frac{qB(T_\Delta)C(T_\Delta)}{\sqrt{B(T_\Delta)\bar{m}g}} - 2C(T_\Delta) \right), \quad (2.9)$$

where $q = \frac{2\pi n}{S}$ is the total end rotation of the fiber and n is the number of turns applied during the coiling process. Finally, the length of the TCAM after coiling can be calculated as:

$$\bar{\ell} = S \cos \beta(T_\Delta). \quad (2.10)$$

Considering the TCAM as a second-order mass-spring-damper system [81] (see Figure 3a), where, m , $b(t)$ and $k(t)$ are the total mass of the system (the TCAM mass is neglected), the damping coefficient and the spring coefficient of the TCAM, respectively, the dynamic equilibrium can be written as:

$$m\ddot{\ell}(t) + b(T_\Delta)\dot{\ell}(t) + k(T_\Delta)\ell(t) = F_e(t). \quad (2.11)$$

where, $\ell(t)$ is the time-dependent length of the TCAM, $m\ddot{\ell}(t) = F_i$ the inertial force, $b(T_\Delta)\dot{\ell}(t) = F_b$ the damping force, $k(T_\Delta)\ell(t) = F_k$ the elastic force, and $F_e(t)$ the external force applied to the TCAM. Notice that Equation (2.11) emphasizes the dependency of the damping and stiffness coefficients, i.e., $b(T_\Delta)$ and $k(T_\Delta)$, respectively, on the TCAM's temperature, which is explained in details next.

Consider a steady-state configuration where a constant mass $\bar{m}$ is attached to the lower extremity of the TCAM, i.e., $\dot{\ell}(t) = 0$, $\ddot{\ell}(t) = 0$, and $F_e = \bar{m}g$. From Equation (2.10) and (2.11), the spring coefficient $k(T_\Delta)$ can be obtained as a function of temperature and materials properties as:

$$k(T_\Delta) = \frac{\bar{m}g}{S \cos \beta(T_\Delta)}. \quad (2.12)$$

For what concerns the damping coefficient, we consider the Kelvin–Voigt viscoelastic theory [102] to model the viscoelastic behavior of TCAMs. The Kelvin–Voigt theory models a viscoelastic material as a parallel between a spring and a damper, which is the same second-order system we considered to obtain Equation (2.11). Specifically, this theory models the relationship between stress $\sigma(t)$ and strain $\varepsilon(t)$ as:

$$\eta(T_\Delta) \frac{d\varepsilon(t)}{dt} + E\varepsilon(t) = \sigma(t), \quad (2.13)$$

where $\eta(T_\Delta)$ and E are the viscosity and Young's modulus of the TCAM, respectively. The dynamic viscosity $\eta(T_\Delta)$ can be obtained from the kinematic viscosity ν and the density δ at the given temperature of the TCAM as $\eta(T_\Delta) = \delta(T_\Delta) \times \nu$ [103]. Considering a creep test (i.e., a constant external load $\bar{m}g$ is applied for a long time) where $\ddot{\ell}(t) \rightarrow 0$, Equation (2.11) can be written as:

$$b(T_\Delta)\dot{\ell}(t) + k(T_\Delta)\ell(t) = \bar{m}g. \quad (2.14)$$

Strain can be expressed as $\varepsilon(t) = \frac{\ell(t) - \bar{\ell}}{\bar{\ell}} = \frac{\ell(t)}{\bar{\ell}} - 1$. The time derivative of the strain can then be calculated as:

$$\frac{d\varepsilon(t)}{dt} = \frac{\dot{\ell}(t)}{\bar{\ell}}. \quad (2.15)$$

Both Equations (2.13) and (2.14) describe a creep test with the difference that while Equation (2.13) is written in terms of stress, Equation (2.14) is written in terms of force. Considering that Equation (2.13) can be obtained from Equation (2.14) by dividing each term by the cross sectional area of the TCAM, A_s , we can write $\frac{b(T_\Delta)\dot{\ell}(t)}{A_s} = \eta(T_\Delta)\frac{d\varepsilon(t)}{dt}$. Considering Equation (2.15), we can then write the damping coefficient $b(T_\Delta)$ in terms of the temperature and materials properties as follows:

$$b(T_\Delta) = \frac{\eta(T_\Delta)A_s}{\bar{\ell}}. \quad (2.16)$$

Considering the system equilibrium expressed by Equation (2.11), the electro-thermal actuation of the TCAM causes a change of the spring coefficient (Equation (12)) and damping coefficient (Equation (16)), both functions of T_Δ . Equation (11) can then be generalized as:

$$\underbrace{m\ddot{\ell}(t) + \bar{b}\dot{\ell}(t) + \bar{k}\ell(t)}_{F_{init}} + \underbrace{b_\Delta\dot{\ell}(t) + k_\Delta\ell(t)}_{F_{TCAM}} = F_e, \quad (2.17)$$

F_{sys}

where, $\bar{k} = k(0)$ and $\bar{b} = b(0)$ are the stiffness and damping coefficients when $T_\Delta = 0$, contributing to the total dynamic force of the non-actuated system F_{init} . The stiffness and damping coefficients for $T_\Delta(t)$ (i.e., during electro-thermal actuation) are $k_\Delta = k(T_\Delta) - k(0)$ and $b_\Delta = b(T_\Delta) -$

$b(0)$, respectively. The change of these coefficients is then responsible for the generation of a TCAM contraction force, namely F_{TCAM} . The sum of these forces accounts for the force of the system F_{sys} , which balances the external force F_e acting on the muscle.

2.1.3 Control Law

The nonlinearities and importance of temperature in the control of TCAMs motivate the use of $\mathcal{L}_1$ adaptive control methodology to avoid overheating and account for system and environmental uncertainties. In general, this methodology allows us to use a system of equations that is comprised of the sum of its linear and nonlinear components

$$\begin{aligned}\dot{x}(t) &= A_p x(t) + B_p(u(t) + f(t, x(t))), \\ y(t) &= C_p x(t),\end{aligned}\tag{2.18}$$

where (A_p, B_p, C_p) is a known controllable-observable triple with state vector $x(t)$ and output $y(t)$, $u(t)$ is a given input and $f(t, x(t))$ is an unknown nonlinear function accounting for system uncertainties and disturbances, and determine a control law such that the system behaves as a desired system. This desired system takes the form

$$M(s) = C_m(sI - A_m)^{-1}B_m,\tag{2.19}$$

where (A_m, B_m, C_m) are defined as

$$A_m = A_p - B_p K, \quad B_m = B_p, \quad C_m = C_p.\tag{2.20}$$

Here, K is a constant gain such that A_m is Hurwitz and (2.19) has desired dynamics. The response of this desired system is the Laplace transform

$$y_m(s) = M(s)K_g r(s), \quad K_g = -(C_m A_m^{-1} B_m)^{-1},\tag{2.21}$$

where $r(s)$ is the Laplace transform of $r(t)$ given by

$$r(t) = r_d[i], \quad t \in [iT_s, (i+1)T_s), \quad i \in \mathbb{Z}^+, \quad (2.22)$$

and $r_d[i]$ is a given discrete time reference command with sample time period $T_s > 0$.

In order to arrive at a control law to accomplish this, we consider the discrete time output predictor,

$$\begin{aligned} \hat{x}_d[i+1] &= e^{A_m T_s} \hat{x}_d[i] + A_m^{-1} (e^{A_m T_s} - I) (B_m u_d[i] + \hat{\sigma}_d[i]), \\ \hat{y}_d[i] &= C_m \hat{x}_d[i], \quad \hat{x}_d[0] = C_m^\dagger y_0, \end{aligned} \quad (2.23)$$

where T_s is the model's sampling rate, $\hat{x}_d[i]$ is the state, $\hat{y}_d[i]$ is the output, y_0 is the initial output, and $u_d[i]$ is the control input. $\hat{\sigma}_d[i]$ is the estimate of the unknown function $f(t, x(t))$ given by,

$$\hat{\sigma}_d[i] = -\Phi^{-1}(T_s) e^{\Lambda A_m \Lambda^{-1} T_s} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} (\hat{y}_d[i] - y_d[i]), \quad (2.24)$$

where $y_d[i] = y(iT_s)$ is the sampled output,

$$\Phi(T_s) = \int_0^{T_s} e^{\Lambda A_m \Lambda^{-1} (T_s - \tau)} \Lambda d\tau,$$

and

$$\Lambda = \begin{bmatrix} C_m \\ D\sqrt{P} \end{bmatrix},$$

where P is matrix solution to $A_m^\top P + P A_m = -Q$ for a given positive definite matrix Q , $P = \sqrt{P}^\top \sqrt{P}$,

and D is a matrix that satisfies $D \left(C_m (\sqrt{P})^{-1} \right)^\top = 0$. The discrete control input is then defined by

$$\begin{aligned} x_u[i+1] &= e^{A_o T_s} x_u[i] + A_o^{-1} (e^{A_o T_s} - I) (B_o e^{-A_m T_s} \hat{\sigma}_d[i]), \\ u_d[i] &= K_g r_d[i] - C_o x_u[i], \quad x_u[0] = 0, \end{aligned} \quad (2.25)$$

where (A_o, B_o, C_o) is the minimal state variable realization of the transfer function

$$O(s) = C(s)M^{-1}(s)C_m(sI - A_m)^{-1}, \quad (2.26)$$

and $C(s)$ is a strictly proper stable transfer function such that $C(0) = 1$. Finally, the continuous control input is given as

$$u(t) = u_d[i], \quad t \in [iT_s, (i+1)T_s), \quad i \in \mathbb{Z}^+.$$

Further discussion of the conditions and performance bounds of this controller can be found in [83]. The generality of this control architecture allows it to be used in both the displacement and force control of TCAMs with only minor parameter changes in Simulink.

2.1.3.1 Displacement Controller

Displacement control is most useful in the case of a TCAM system under constant-stress or dynamic force, see Figure 2.3. Rewriting Equation (2.11) and (2.17) with $d(t) = \bar{\ell} - \ell(t)$, we can obtain

$$m\ddot{d}(t) + \bar{b}\dot{d}(t) + \bar{k}d(t) = k(T_\Delta)\bar{\ell} - F_e - k_\Delta d(t) - b_\Delta \dot{d}(t). \quad (2.27)$$

Assuming the right-hand side of Equation (2.27) can be approximated by a linear relationship to temperature change $T_\Delta(t)$ as follows

$$cT_\Delta(t) \approx k(T_\Delta)\bar{\ell} - F_e - k_\Delta d(t) - b_\Delta \dot{d}(t),$$

for some $c > 0$, the system can be linearized and put into the form of Equation (2.18). The state vector describing the system is $x(t) = [d(t), \dot{d}(t), T_\Delta(t)]^\top$, and the input is the voltage squared, $u = V^2$, so the matrices (A_p, B_p, C_p) take the form

$$A_p = \begin{bmatrix} 0 & 1 & 0 \\ -\frac{\bar{k}}{m} & -\frac{\bar{b}}{m} & \frac{c}{m} \\ 0 & 0 & -\frac{\bar{\lambda}}{C_t} \end{bmatrix},$$

$$B_p = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{C_t R} \end{bmatrix}, \quad C_p = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}.$$

Here, Equation (2.1) is used to help define the third state, $T_\Delta(t)$, and $\bar{\lambda}$ is given by Equation (2.2) where $\bar{\lambda} = \lambda(0)$. From this triple, the process laid out in the control law section can be used to implement an $\mathcal{L}_1$ adaptive controller for TCAM displacement.

2.1.3.2 Force Controller

Controlling the output force of TCAM within a system allows muscles to be used in more complex situations than displacement controllers are capable of. See for instance Figure 2.3(b), which gives a simple example of constant strain in a system. For this reason, force control is well suited for soft robotic systems and for use with outer loop impedance controllers where nonuniform and nonlinear system displacement may occur as a result of enviromental interactions.

Force control of TCAM systems is predicated on a similar linearizing assumption to the one made in the displacement case. Here,

$$cT_\Delta(t) \approx F_{sys}(t), \tag{2.28}$$

where F_{sys} is the change in the TCAMs applied force from the systems initial state. Defining the state $x(t) = T_\Delta(t)$, the output $y(t) = F_{sys}(t)$, and the input $u(t) = V^2(t)$, we obtain the following

system

$$\begin{aligned}\dot{x}(t) &= -\frac{\bar{\lambda}}{C_t}x(t) + \frac{1}{C_t R}u(t), \\ y(t) &= cx(t).\end{aligned}\tag{2.29}$$

Here, again $\bar{\lambda}$ is given by Equation (2.2) where $\bar{\lambda} = \lambda(0)$. This realization of the system gives $A_p = -\frac{\bar{\lambda}}{C_t}$, $B_p = \frac{1}{C_t R}$, $C_p = c$ and enables the formulation of the $\mathcal{L}_1$ adaptive controller presented above.

2.1.4 TCAM Sample Manufacturing Process

Two types of TCAMs were manufactured using Carbon fiber (CF)/silicone rubber (SR) composite, and polymer fishing lines, respectively. Polydimethylsiloxane (PDMS) is used as the SR, and a 0.8mm monofilament nylon fishing line is used as the polymer fishing line. Even though the actuation principle is the same in both materials, the fabrication process is different yet simple and straightforward, and it can be performed in any environment without sophisticated equipment. Each fabrication process is described below.

2.1.4.1 CF/PDMS TCAM Manufacturing

Commercial 3k carbon fiber bundles containing 3000 fibers with epoxy sizing (from ACP Composites, Inc.) are used. One CF bundle is dipped into a SR mixture of uncured Sylgard 184 PDMS and hexane (2:1 by weight). The hexane reduces the viscosity of the PDMS to ensure a more homogeneous distribution along the CF bundle. Then, in a process known as pre-twisting, the composite yarn is twisted using a drill with a known load attached at the bottom extremity before it is set aside to cure the PDMS for 24 h under a tension load. The pre-twisting is essential to remove the excess PDMS from the CF bundles in order to obtain a yarn with constant radius throughout the entire length. Pre-twisting redistributes the PDMS and fills the gaps between fibers and allows to obtain a PDMS content of $\sim 40\%$ in 1-bundle TCAMs. The number of pre-twisting turns for

1-bundle is 240 turns/m. After the curing of PDMS, the twisted angle of the CF yarn (α) can be measured, which is essential to calculate the Young and shear moduli, E_x and G_{yz} , respectively (see Appendix in [49]). Finally, the straight composite yarn is highly twisted until spontaneous coiling. The external mass used as the load to coil the fiber composite is $\bar{m}$ and is the same as the load used during pre-twisting. The number of turns required to coil the TCAM n is measured for use in the theoretical model.

2.1.4.2 Nylon fishing line TCAM Manufacturing

The nylon TCAMs are fabricated from inexpensive, commercially available fishing lines. (*KastKing Clear Premium Monofilament Fishingline, KastKing Garden City (NY, USA)*). For this work, 0.8 mm diameter nylon fishing lines are used.

There are three main stages of the fabrication process, which are 1) pre-twisting, 2) over-twisting, and 3) annealing and training. In the first step, one end of the polymer fishing line is fixed to a hand drill chuck, and the other end is connected to a 400 g external load to tense the nylon TCAM. A horizontal rod is attached to the bottom end of the TCAM with the tensile load to avoid untwisting during the twisting process. Once the hand drill starts, the polymer fishing line twists. Since the polymer fishing line is a non-conductive material, a 36 AWG stainless steel wire is wrapped around the TCAM during the twisting process to allow actuation via Joule heating. The next step is over-twisting. After twisting for certain turns, the fishing line begins to over-twist and form spontaneous coils. The use of an optimized external load will ensure to produce even coils. Alternatively, a mandrel can be used to form coils, instead of spontaneous coiling, as explained in [99]. The final step is annealing and training. Here, TCAM undergoes two annealing cycles, each at 150 °C for 1 hour. Before each annealing cycle, the TCAM is stretched to separate contacting coils and obtain a desired pitch. This process allows to maintain a specific desired geometry.

It is worth noting that a manual manufacturing method based on spontaneous coiling has been used in this Section. While this method provides very consistent samples when Nylon is

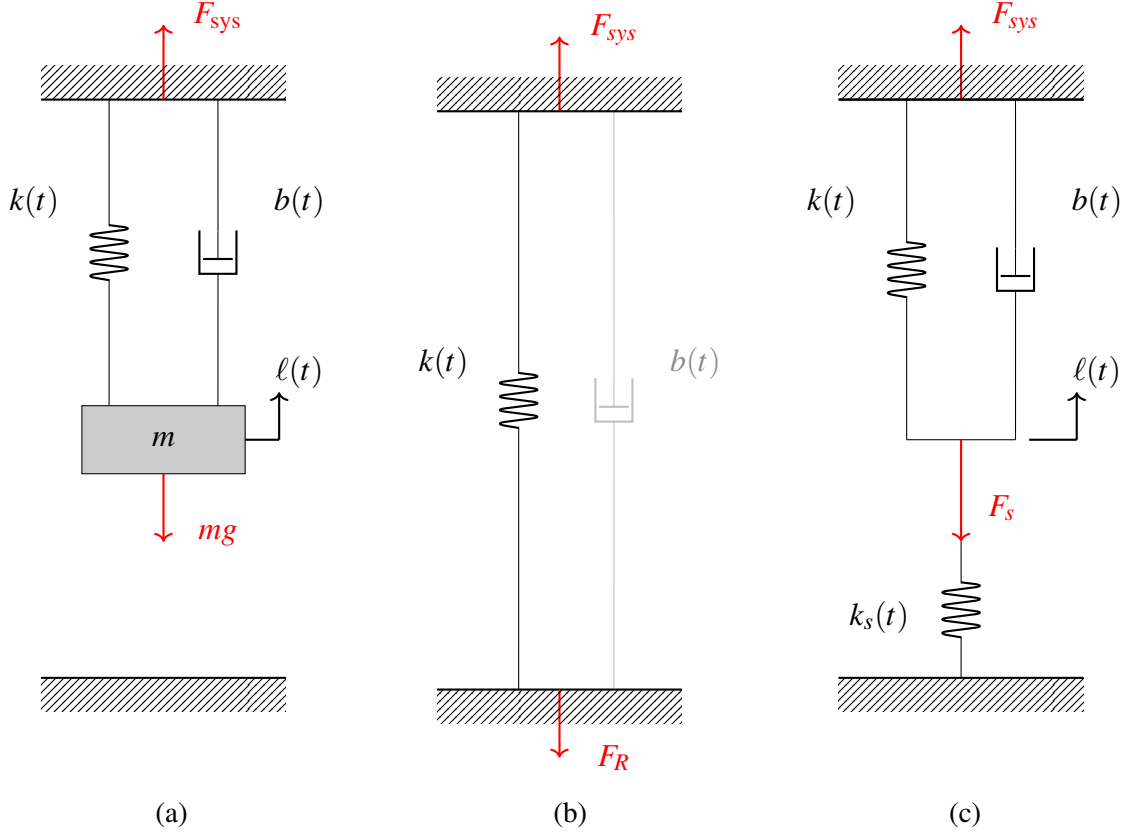


Figure 2.3. Schematic representation of the experimental setups for **(a)** constant stress, **(b)** constant strain, and **(c)** dynamic force configuration for TCAMs testing.

used, as reported elsewhere [50, 56], some inconsistencies in CF/PDMS samples can be observed (mainly due to the presence of PDMS and the compliant nature of the final composite). However, all these issues have been addressed in a recent publication from our group [104].

2.1.5 Simulation and Experiment Results

In this section, we experimentally validate our proposed theoretical model and the $\mathcal{L}_1$ adaptive controller for the CF/PDMS and Nylon fishing-line TCAMs. The TCAMs have been tested in the following three configurations:

1. Constant stress configuration:

a constant load is applied to the TCAM lower extremity and the output displacement (i.e. TCAM contraction calculated as $\ell(t) - \bar{\ell}$, where $\bar{\ell}$ is the initial length of the TCAM) is

measured;

2. Constant strain configuration:

both TCAMs extremities are fixed and the output force F_{sys} is measured;

3. Dynamic force configuration:

a generic variable external force (simulated using a spring in series with the TCAM) is applied to the TCAM lower extremity and the output displacement and force F_{sys} are measured.

The schematic representation of the working configurations is shown in the Figure 2.3.

2.1.5.1 Constant Stress Configuration

Figure 2.3(a) shows the TCAM constant stress configuration schematic representation where the top end is fixed, and the bottom end is connected to an external mass m . During the actuation, only the TCAM length $\ell(t)$ changes and Equation (2.17) can be written as follows:

$$\underbrace{m\ddot{\ell}(t) + \bar{b}\dot{\ell}(t) + \bar{k}\ell(t)}_{F_{init}} + \underbrace{b_{\Delta}\dot{\ell}(t) + k_{\Delta}\ell(t)}_{F_{TCAM}} = mg, \quad (2.30)$$

where $F_e(t) = mg$.

2.1.5.2 Constant Strain Configuration

Figure 2.3(b) shows the constant strain configuration where both extremities of the TCAM are fixed, and the external force is equal to the reaction force from the lower encastre, F_R . During the actuation, TCAMs generate the force F_{TCAM} without any contraction due to the constraints from both ends. Since there is no change in length during the actuation, $\dot{\ell}(t) = 0$ and $\ddot{\ell}(t) = 0$. Equation

(17) can then be written as:

$$\underbrace{\bar{k}\ell(t)}_{F_{init}} + \underbrace{k_{\Delta}\ell(t)}_{F_{TCAM}} = F_R, \quad (2.31)$$

.

2.1.5.3 Dynamic Force Configuration

Figure 2.3(c) shows the schematic representation of the dynamic force configuration. The TCAM top end is fixed, and the bottom end is connected to an external spring. The force exerted by the spring during the TCAM actuation can be expressed as:

$$F_e(t) = F_s = k_s(\bar{\ell} + \ell_s - \ell(t)), \quad (2.32)$$

where k_s is the spring stiffness, ℓ_s is the initial displacement of the spring from its equilibrium length, and $\ell(t)$ is the time-varying TCAM length. The dynamics of the system can be derived from Equation (2.17) as:

$$\underbrace{m\ddot{\ell}(t) + \bar{b}\dot{\ell}(t) + \bar{k}\ell(t)}_{F_{init}} + \underbrace{b_{\Delta}\dot{\ell}(t) + k_{\Delta}\ell(t)}_{F_{TCAM}} = k_s(\bar{\ell} + \ell_s - \ell(t)). \quad (2.33)$$

2.1.5.4 Experimental Setup

The TCAMs were actuated in the three proposed configurations (see Figure 2.4) to validate the effectiveness of the proposed theoretical model. Following the TCAM fabrication process mentioned above, the top end is attached to a fixed point of the setup. The bottom is connected

1. to a horizontal rigid rod (with a specific mass) for *config. 1*),
2. to a fixed point for *config. 2*),
3. to a horizontal rigid rod and a spring that was fixed from the other end for *config. 3*).

As shown in Figure 2.4(a) and 2.4(c), the two ends of the horizontal rod go through rails, preventing the TCAMs from uncoiling during actuation in configuration 1 and 3. The bottom end is fixed in configuration 2 as it shown in Figure 2.4(b).

The input voltage is sent through ROS to the microcontroller (Model: Arduino Mega2560) to actuate the TCAMs. The output displacement, temperature, and force are measured by a laser displacement sensor (LSD) (Keyence LK-G157), a thermal camera (FLIR E95), and a load cell (FUTEK lsb 200), respectively. The data acquisition system (DAQ) consists of a National

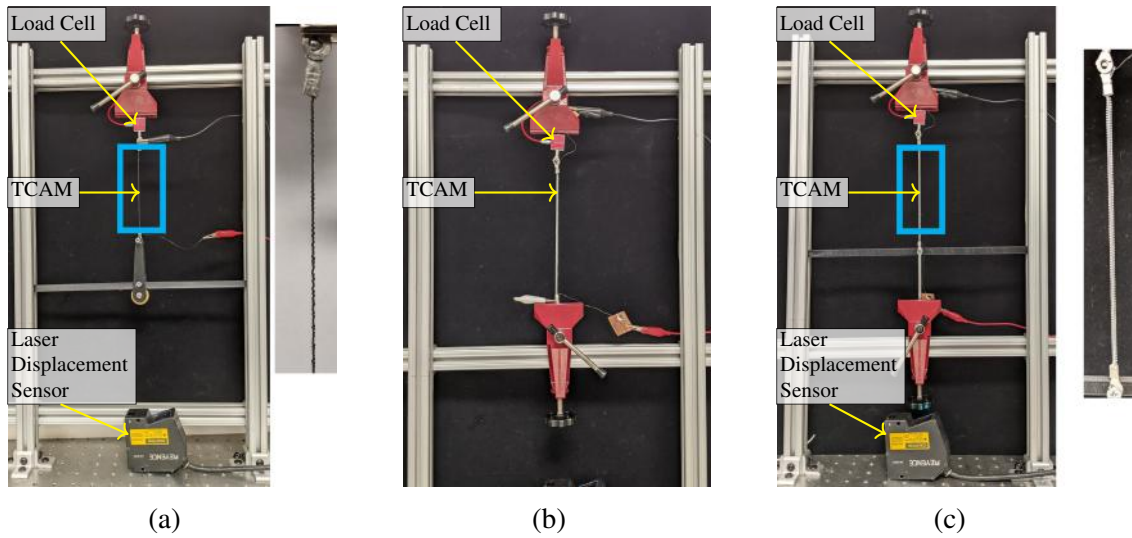


Figure 2.4. (a) Constant stress configuration setup for CF/PDMS TCAM, (b) Constant strain configuration setup for Nylon TCAM, and (c) Dynamic force configuration setup for Nylon TCAM.

Instrument-myDAQ and an Arduino Mega2560. The data acquired by the DAQs is sent through ROS (Robot Operating System) to a Matlab/SIMULINK code in real time.

2.1.5.5 Experimental validation

The physics-based theoretical model described in the previous section has been implemented in MATLAB/SIMULINK, and the theoretical results have been compared with the experimental results obtained for CF/PDMS TCAM samples and Nylon fishing line TCAM samples. Three samples from each type were tested (each sample was tested five times) for all three configurations described in the previous section.

The ambient temperature is $T_{amb} = 24^{\circ}\text{C}$. The thermal mass C_t for the CF/PDMS TCAMs is calculated by

$$C_t = m_{CF} C_{CF} + m_{PDMS} C_{PDMS}. \quad (2.34)$$

Similarly, the thermal mass C_t for the Nylon fishing-line wrapped with stainless steel (SS) wire can be calculated by

$$C_t = m_{nylon} C_{nylon} + m_{SS} C_{SS}. \quad (2.35)$$

During the CF/PDMS TCAM actuation, a constant step input voltage of 12V was provided for 4 seconds, followed by free cooling of 36 seconds. For the Nylon fishing-line TCAMs, a constant step input voltage of 12V was provided for 7 seconds, followed by free cooling of 33 seconds.

Table 2.1. Physical, thermal and electrical properties of CF/PDMS TCAM samples.

CF/PDMS 1-bundle Samples	Sample 1	Sample 2	Sample 3
Number of coiling turns, n ,	120	132	140
Pretwisting twist angle, α , ($^{\circ}$)	5.0	5.8	6.0
CF volumetric fraction, v_f	0.975	0.972	0.975
Mass of CF fiber m_{CF} , (g)	0.15	0.14	0.17
Mass of PDMS m_{PDMS} , (g)	0.06	0.07	0.06
Electrical resistance, R , (Ω)	18	16	18

Table 2.2. Physical, thermal, and electrical properties of Nylon TCAM samples.

Nylon Samples	Sample 1	Sample 2	Sample 3
Number of coiling turns, n ,	240	260	270
Pretwisting twist angle, α , ($^{\circ}$)	2.8	2.8	2.8
Nylon volumetric fraction, v_f	0.96	0.96	0.965
Mass of Nylon fiber m_{nylon} , (g)	1.02	1	1.1
Mass of SS wire m_{SS} , (g)	0.06	0.06	0.05
Electrical resistance, R , (Ω)	20	28	18

E_x and G_{yz} have been calculated according to the model described in the Appendix of [49]. The Prandtl number and the thermal conductivity for air at room temperature are $Pr = 0.71$ and $K = 26.24 \times 10^{-3} \text{ W m}^{-1} \text{ K}^{-1}$, respectively. The stiffness of the spring used for the configuration-3

Table 2.3. Material-dependent physical, thermal, and electrical properties for CF & Nylon TCAM samples.

Property	CF/PDMS Samples	Nylon Samples
Initial fiber radius: r_0 , (mm)	0.29	0.4
Initial length of the straight and untwisted yarn: S , (cm)	25	25
Fabrication mass for TCAMs: $\bar{m}$, (g)	50	400
Specific heat capacity of CF : C_{CF} , ($\text{J g}^{-1} \text{ }^{\circ}\text{C}^{-1}$)	1	–
Specific heat capacity of PDMS : C_{PDMS} , ($\text{J g}^{-1} \text{ }^{\circ}\text{C}^{-1}$)	1.46	–
Specific heat capacity of Nylon : C_{nylon} , ($\text{J g}^{-1} \text{ }^{\circ}\text{C}^{-1}$)	–	1.67
Specific heat capacity of Stainless Steel : C_{SS} , ($\text{J g}^{-1} \text{ }^{\circ}\text{C}^{-1}$)	–	0.4680
Coefficient of thermal expansion of PDMS/Ny : CTE , ($^{\circ}\text{C}^{-1}$)	0.0003	0.0003
Kinematic viscosity of PDMS : ν , ($\text{m}^2 \text{ s}^{-1}$)	0.0050	
Kinematic viscosity of Nylon : ν , ($\text{m}^2 \text{ s}^{-1}$)	–	0.4737
Density at room temperature of PDMS : δ , (kg m^{-3})	965	–
Density at room temperature of Nylon : δ , (kg m^{-3})	–	1140

is 125 N m^{-1} . The physical, thermal and electrical properties of the CF/PDMS TCAM samples are listed in Table 2.1 while the properties of the nylon fishing-line wrapped with stainless steel wire TCAMs are listed in the Table 2.2. The common parameters for all samples of each TCAM types

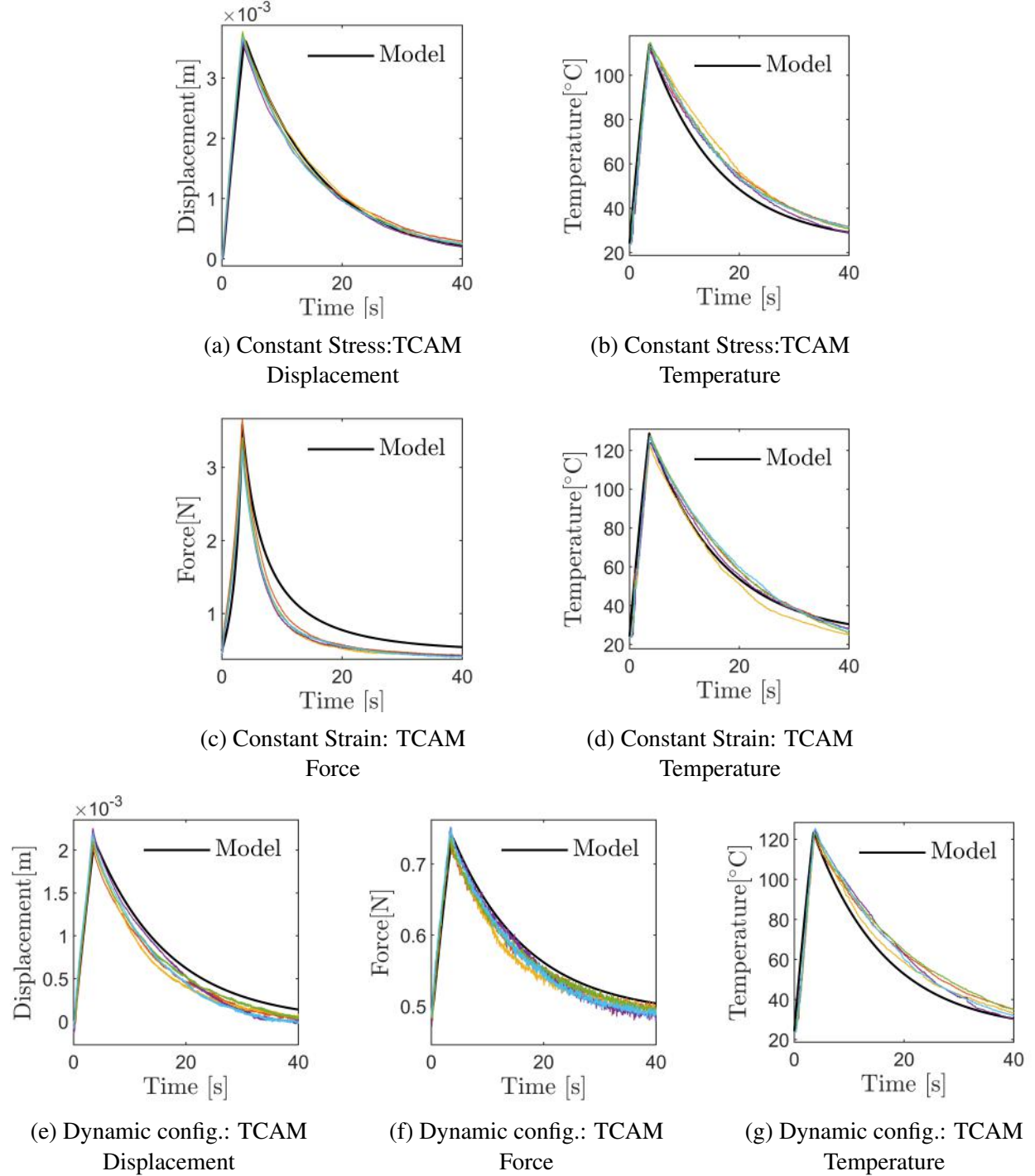


Figure 2.5. Dynamic model validation for the three configurations: CF/PDMS TCAM-Sample 1.

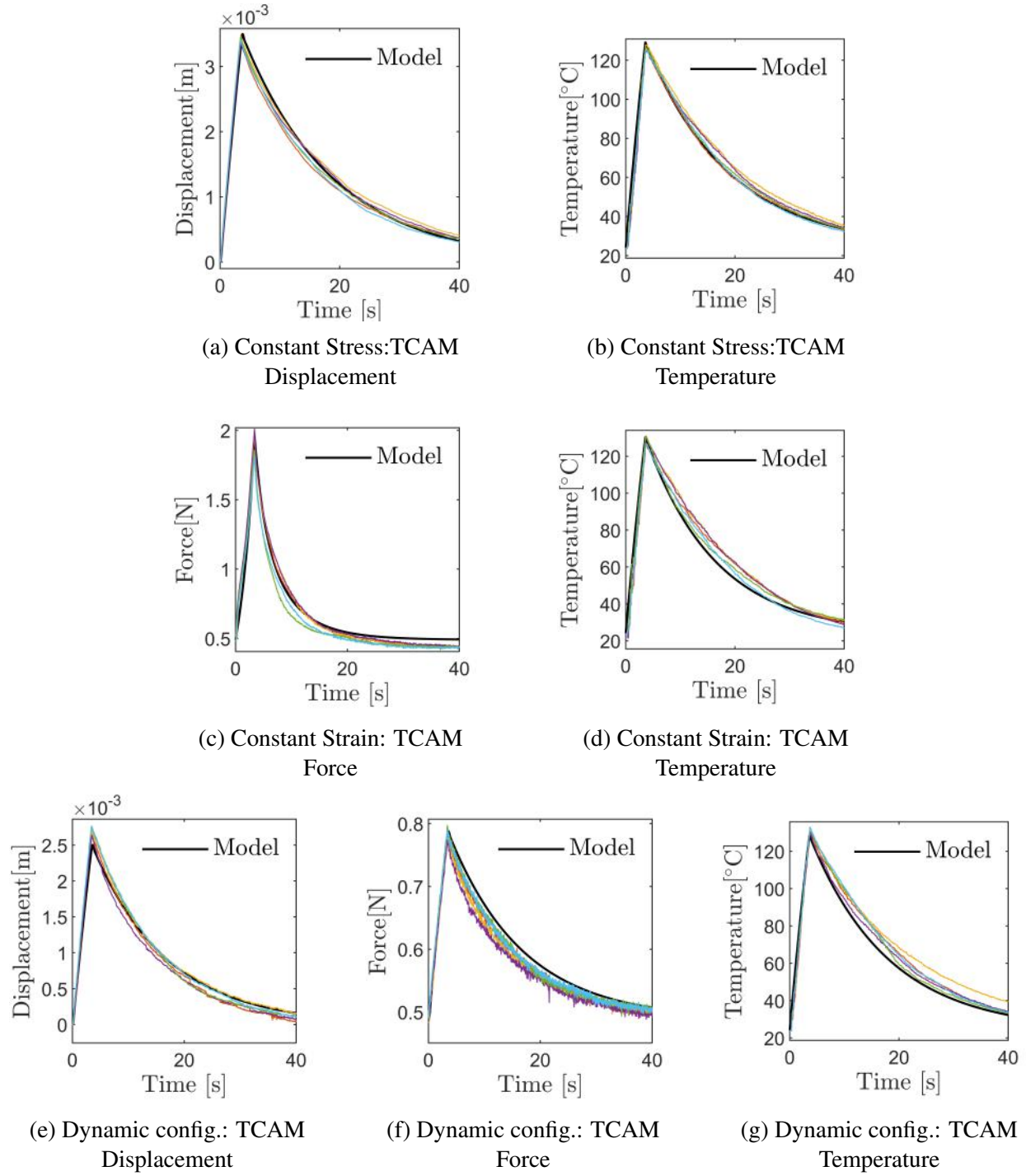


Figure 2.6. Dynamic model validation for the three configurations: CF/PDMS TCAM-Sample 2.

are listed in Table 2.2.

The theoretical and experimental results for the CF/PDMS TCAMs are shown in Figures 2.5, 2.6, and 2.7 while for the nylon fishing line TCAMs are shown in Figures 2.8, 2.9 and 2.10.

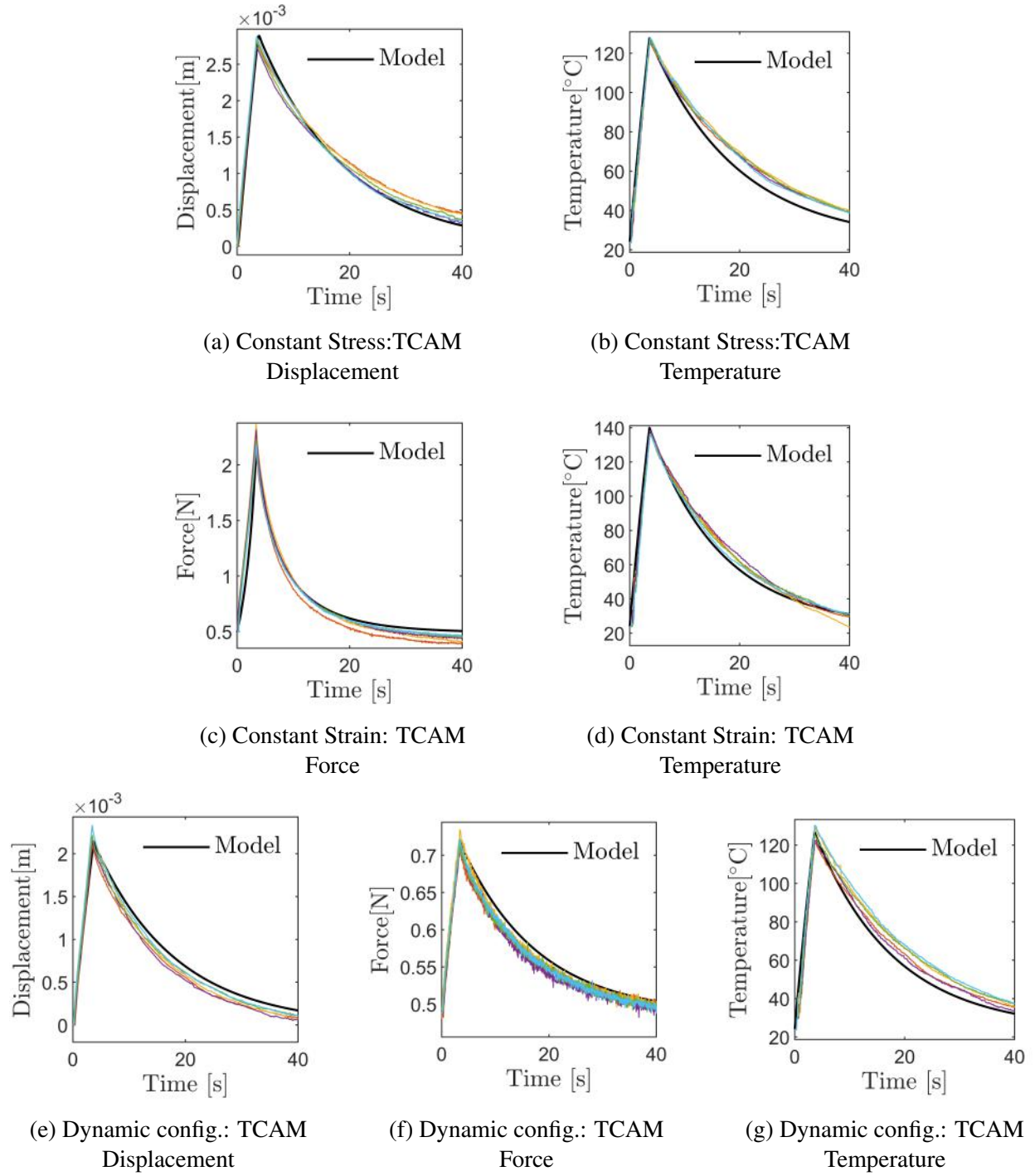


Figure 2.7. Dynamic model validation for the three configurations: CF/PDMS TCAM-Sample 3.

Each figure shows the experimental and theoretical results for the three configurations presented in Figure 2.3 (where different columns represent different configurations). In these figures, the output force plots for configuration-1 and the displacement plots for configuration-2 have been omitted

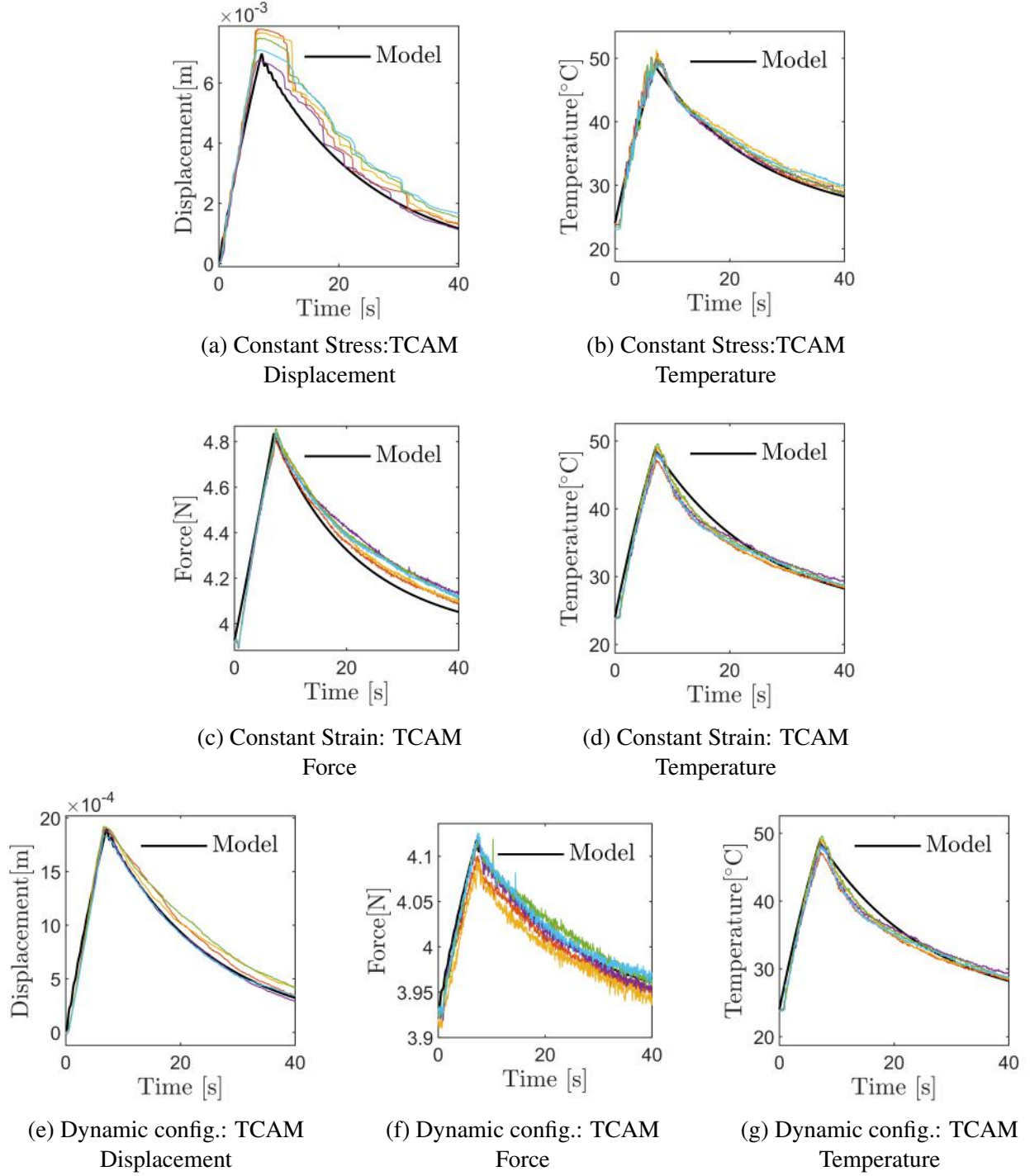


Figure 2.8. Dynamic model validation for the three configurations: Nylon TCAM-Sample 1.

as they display constant values of $\bar{m}g$ and zero, respectively. A good agreement can be observed between experimental data and theoretical predictions (black lines). These results prove that the theoretical model proposed is able to predict the displacement and force generated by different

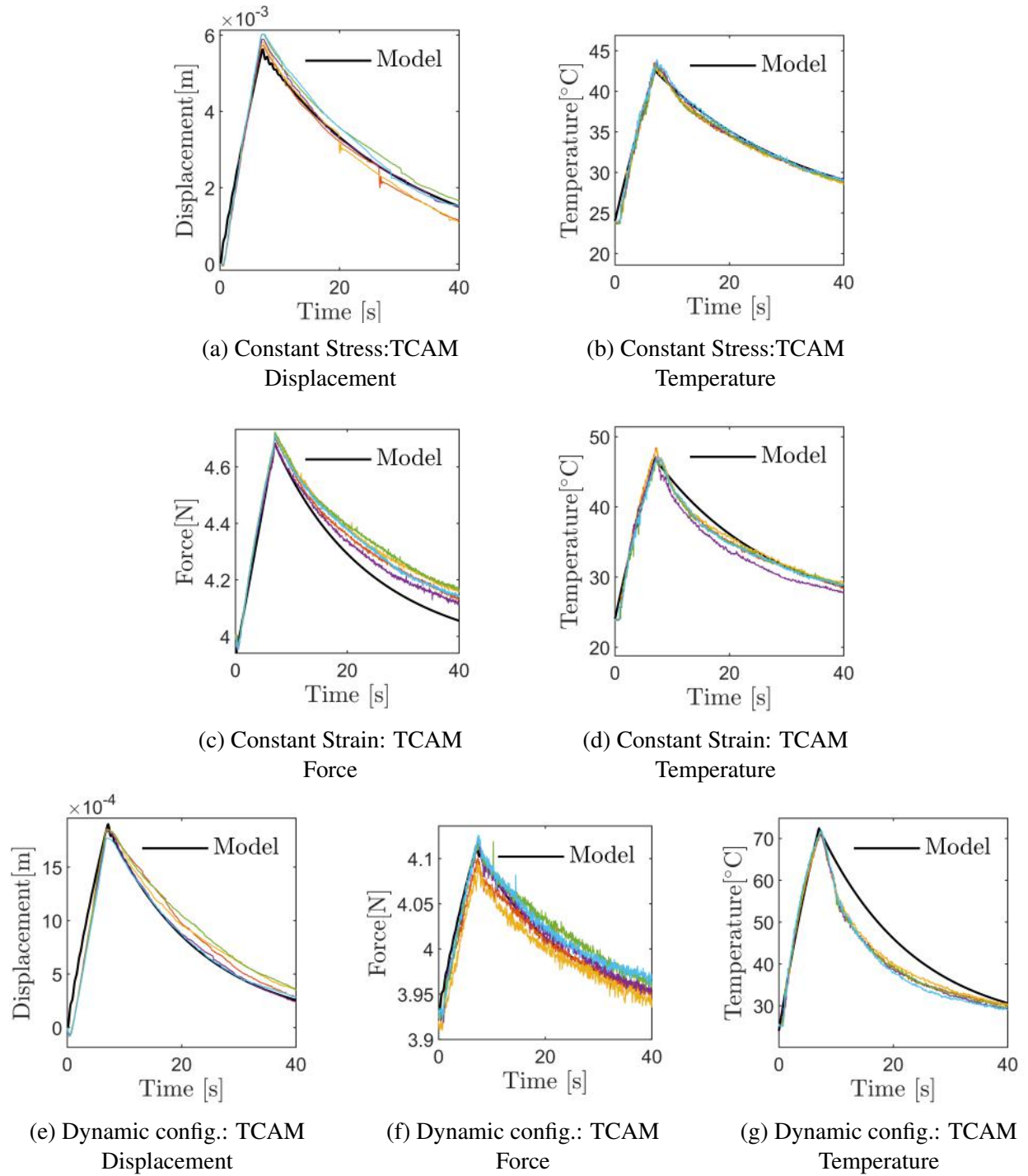


Figure 2.9. Dynamic model validation for the three configurations: Nylon TCAM-Sample 2.

types of TCAMs, in different working configurations. No correlation between the different testing configurations and the errors between theoretical and experimental results can be observed. This led to the conclusion that these errors can be ascribed to the experimental estimation of all the

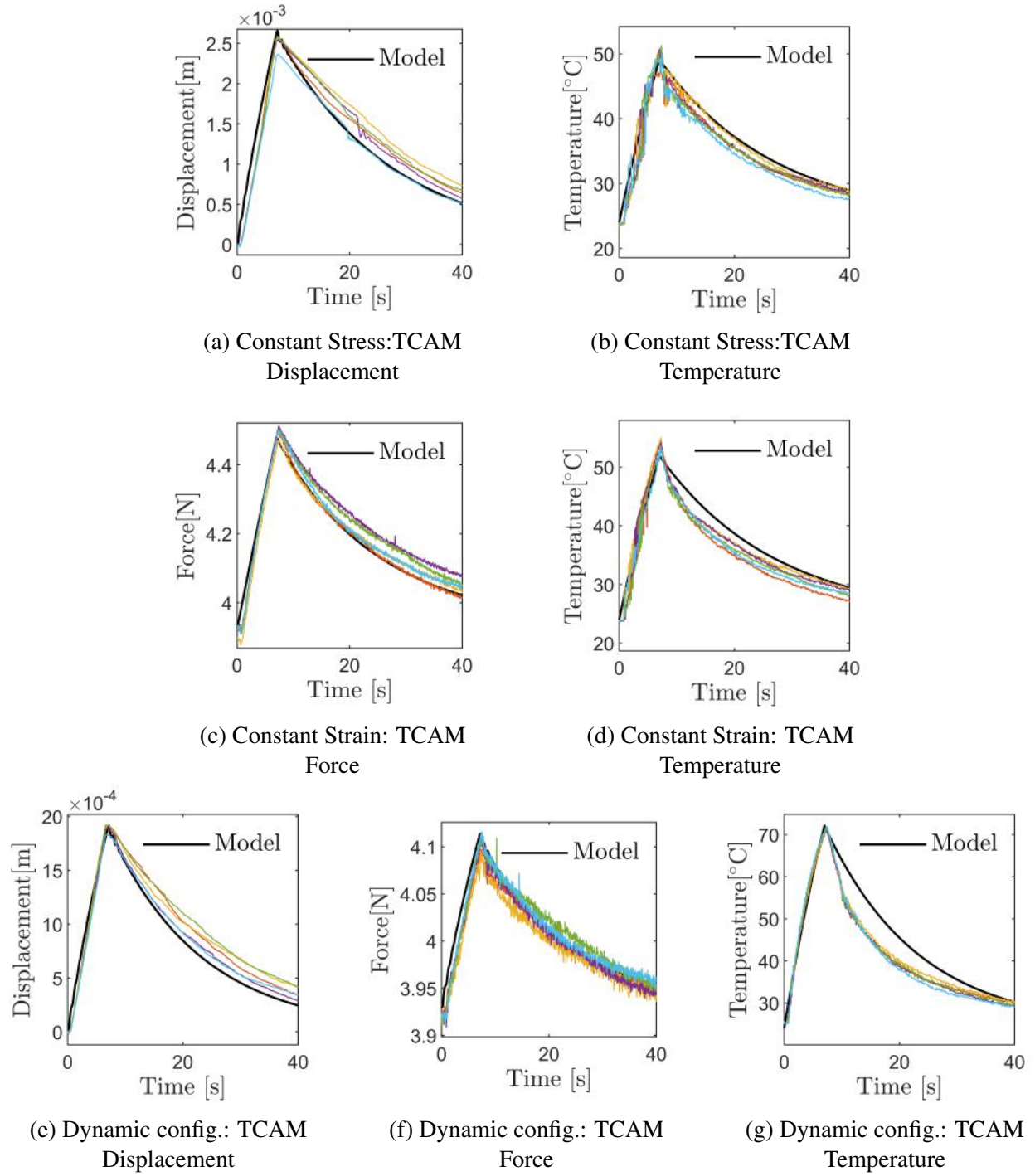


Figure 2.10. Dynamic model validation for the three configurations: Nylon TCAM-Sample 3.

physical, mechanical, thermal, and electrical properties of different samples, inevitably affected by some inaccuracies.

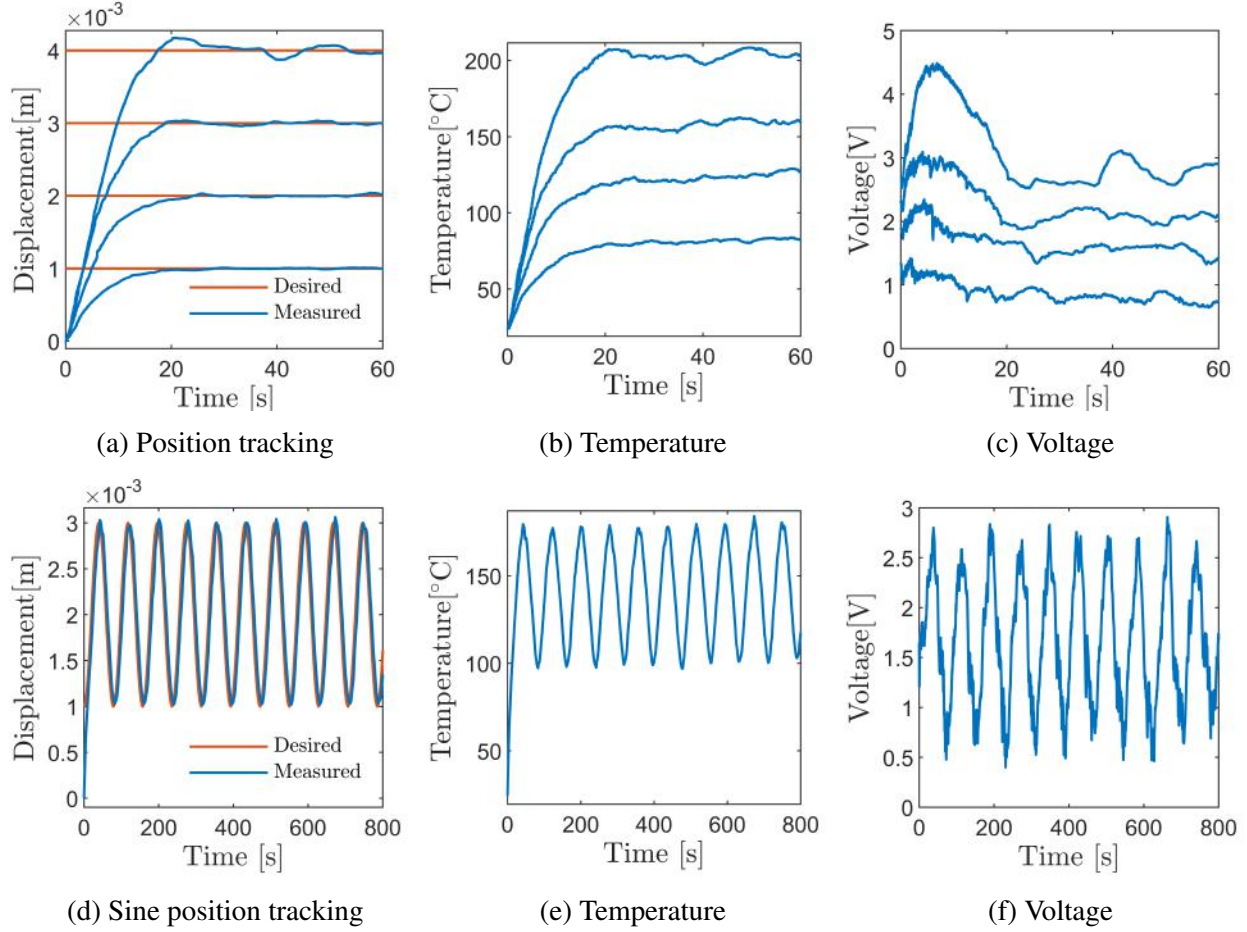


Figure 2.11. Validation of real-time $\mathcal{L}_1$ adaptive controller for Configuration 1—Position control of TCAM lifting mass tracking **(a)-(c)** step and **(d)-(f)** sinusoidal reference.

Figures 2.11-2.14 represent the experimental results for the application of $\mathcal{L}_1$ adaptive position and force control in each of the three configurations discussed in the section above. For the constant-stress configuration, the displacement of the mass was controlled with four-step references, shown in Figure 2.11(a)-(c) and a sinusoidal reference, shown in Figure 2.11(d)-(f). Similarly, the force was controlled for the constant-strain configuration with step (Figure 2.12(a)-(c)) and sinusoidal references (Figure 2.12(d)-(f)). In the final dynamic force configuration, both position (Figure 2.13(a)-(d) and Figure 2.13(e)-(h)) and force (Figure 2.14(a)-(d) and Figure 2.14(e)-(h)) are controlled, again tracking step and sinusoidal references. In each sinusoidal case, the minimum displacement tracked is nonzero in order to avoid slow response resulting from reduced cooling speed of TCAMs nearing ambient temperature. The same position and force controllers are used

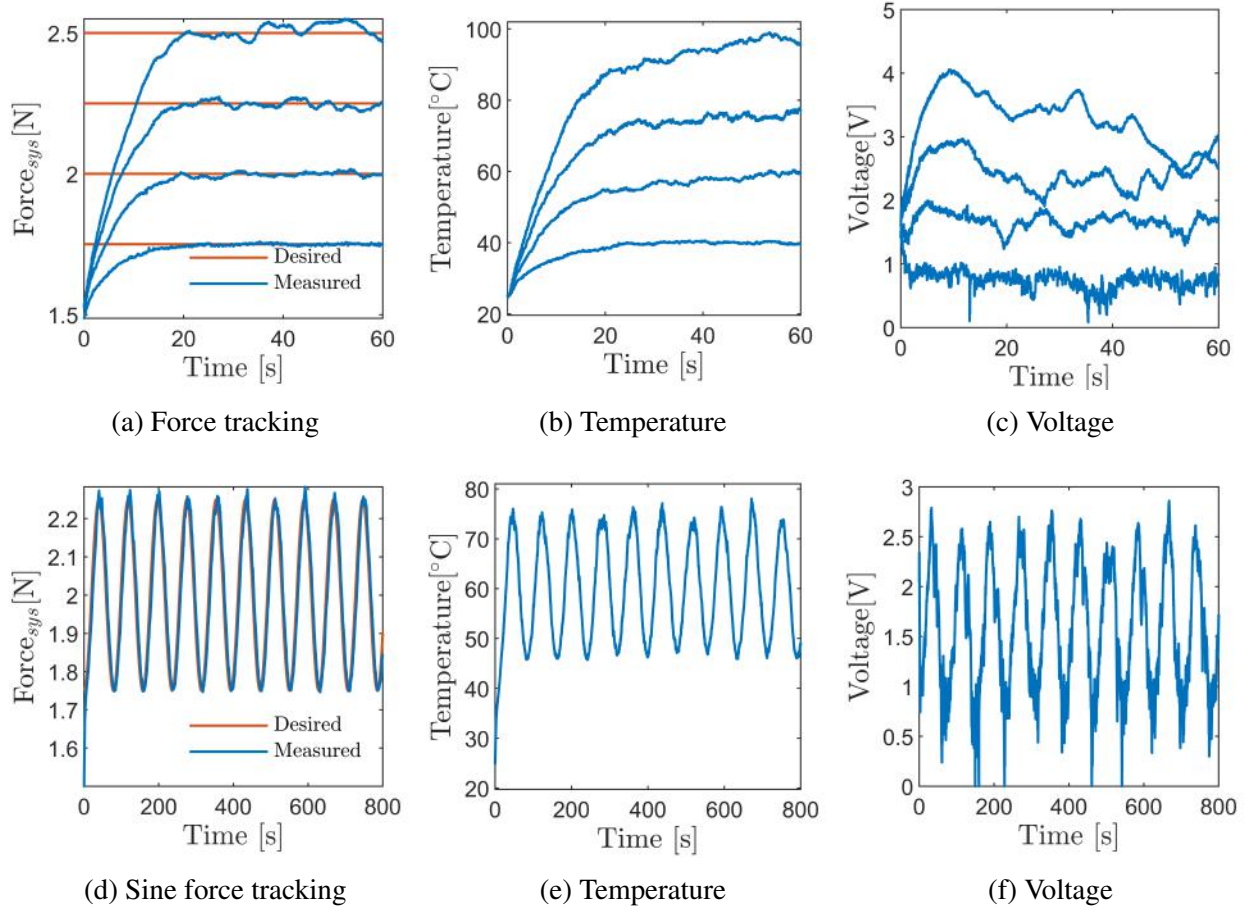


Figure 2.12. Validation of real-time $\mathcal{L}_1$ adaptive controller for Configuration 2—Force control of TCAM under constant strain tracking (a)-(c) step and (d)-(f) sinusoidal reference.

in each test with consistent settling times of twenty and fifteen seconds, respectively, implemented by pole placement of the desired system $M(s)$ within our presented framework. These results demonstrate a key advantage of the $\mathcal{L}_1$ adaptive controllers used in these experiments, consistently achieving desired behavior and convergence to changing position and force reference signals despite model uncertainty without the need for additional tuning, which may be required by other control methods such as proportional integral derivative (PID) controllers [83]. This is shown to be robust in all three cases, with the fast adaptation of the $\mathcal{L}_1$ controller accounting for alternative loading without the need for any changes in the formulation in both the force and position control cases. Lastly, we note that the tests requiring operation at higher temperatures (e.g., the $4 \cdot 10^{-3}\text{m}$ step response in Figure 2.11) present (relatively) diminished performance. This is due to deterio-

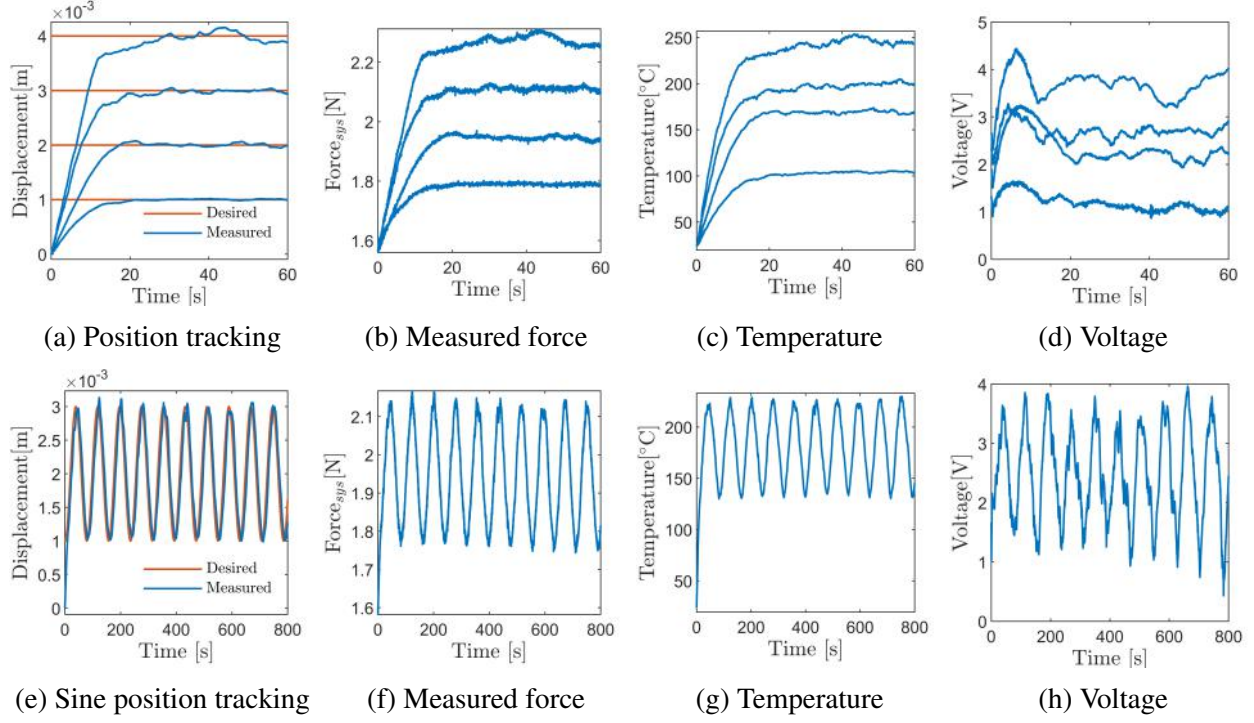


Figure 2.13. Validation of real-time $\mathcal{L}_1$ adaptive controller for Configuration 3 – Position control of TCAM pulling antagonistic spring tracking (a)-(d) step and (e)-(h) sinusoidal reference.

ration of the PDMS in the CF/SR muscle at high temperatures ($> 220^\circ\text{C}$), see [49] for additional details. Nevertheless, we emphasize that despite this significant change in the system dynamics, the transient of the step and frequency responses for all the tests remain consistent by virtue of adaptation.

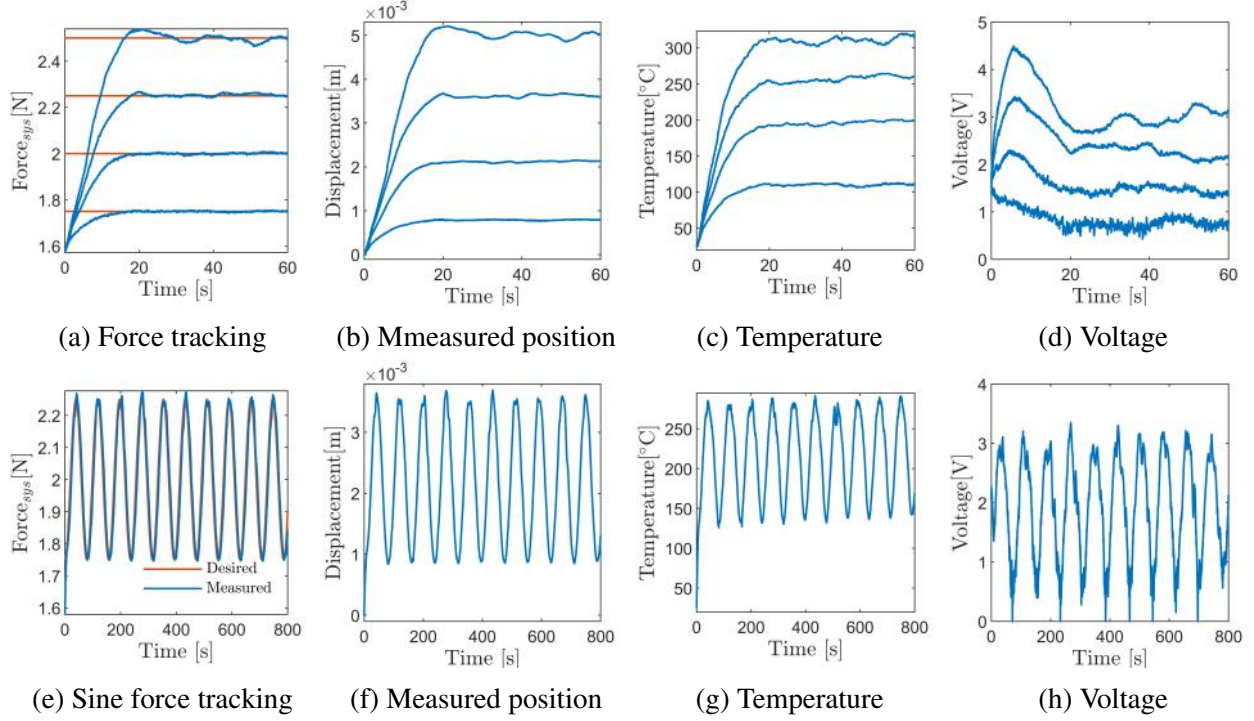


Figure 2.14. Validation of real-time $\mathcal{L}_1$ adaptive controller for Configuration 3 – Force control of TCAM pulling antagonistic spring tracking (a)-(d) step and (e)-(h) sinusoidal reference.

2.1.6 Summary

This section proposes a generalized theoretical model to predict the actuation force and displacement of electro-thermally actuated TCAMs. The model is entirely physics-based and relates the output force and displacement to the physical, thermal, electrical, and mechanical properties of TCAMs. Moreover, we introduce a robust position and force $\mathcal{L}_1$ adaptive controller. This controller ensures stability and performance against disturbances and highly nonlinear TCAM model uncertainties. Finally, the experimental results obtained for different types of TCAMs validate the proposed theoretical model and demonstrate the efficacy of the $\mathcal{L}_1$ adaptive controller for both position and force control in real-time applications.

2.2 $\mathcal{L}_1$ Adaptive Controller for TCAM Actuation

2.2.1 Introduction

Conventional robotic devices such as humanoid robots, manipulators, biomedical and wearable devices are often driven by servo motors and rigid links [42, 105–107], and are able to provide large output force, fast and precise motion, and ease of controllability. However, the high power required by the motors and the presence of rigid components makes these devices heavy and bulky. Soft robotics has been recently proposed as a valid alternative to servo motors and rigid links. Soft robotics devices achieve motions by means of hydraulic or pneumatic actuators consisting of molded elastomeric chambers with fiber reinforcements that induce specific trajectories under fluid pressurization [108–110]. Although this technology increases portability by significantly decreasing a device’s total weight, final products are still bulky due to a need for compressors. These limitations represent a serious obstacle to the development of lightweight robotic devices and have resulted in a growing interest and development of Twisted and Coiled Artificial Muscles (TCAMs) [27, 111, 112]. Inspired by the high power/weight ratio of skeletal muscles [112], TCAMs generate large force and stroke within one single, flexible, and lightweight component. These properties make TCAMs valid substitutes for traditional heavy electromagnetic and pneumatic actuators in applications where low weight, large contractile work, and fine motion are required. Examples of such applications include medical devices [113, 114] prostheses [115], robotics [116, 117] and biomimetic devices [55, 56, 118]. Nevertheless, the operation of novel TCAM-based devices raises several challenges from a control theoretical and engineering perspective, some of which are addressed in this chapter.

In recent years, the topic of control of muscle-actuated systems has been the subject of considerable research and development effort, especially concerning 1 or 2-link robots actuated by artificial muscles. Relevant work includes methods based on traditional PID control [84], observer-based control [85], iterative learning [86, 87], sliding mode control [119–121], fuzzy control [122–124], reinforcement learning [88], and adaptive control [89, 125]. On the other hand, the literature

on control of TCAMs is still relatively young. Recent work can be found in [91, 94–96, 126, 127], where controllers based on PID design or Lyapunov theory are presented.

When designing control algorithms for TCAMs actuating robotics devices, several concerns should be taken into account. For example, as pointed out in [27, 81], the dynamic model that governs the motion of thermally and electro-thermally actuated TCAMs is highly non-linear and strongly dependent on intrinsic material parameters and on the fabrication process. Moreover, since the TCAMs are actuated by (joule) heating, control of the TCAMs must be achieved while preventing the device from overheating. This issue is especially relevant when dealing with robotic devices that can harm humans, e.g., wearable devices. Often, to reduce the weight and cost of the device, sensors that measure the TCAMs' temperature might not be available. Thus, adaptive output feedback controllers must be sought to guarantee boundedness of non-measurable states, such as the temperature, in spite of unmodeled dynamics and partially unknown parameters. Furthermore, in order to lower the requirements for the system's hardware, thus lowering the cost and weight of robots, sampled-data controllers must be designed to guarantee robustness for relatively large sampling periods and time delays.

This Section's main contribution lies in designing a sampled-data robust and adaptive output feedback controller for TCAMs. The controller adaptively adjusts the voltage applied to a TCAM in order to reach a desired TCAM's displacement with desired temporal specifications. The control framework is borrowed from the $\mathcal{L}_1$ adaptive control theory [97, 98] and guarantees stability and performance against disturbance and model uncertainties. The main benefit of the controller is its ability to yield fast and robust adaptation, as demonstrated in [98]. Unlike conventional adaptive control architectures, the $\mathcal{L}_1$ adaptive control methodology provides a systematic framework for control design, potentially easing the implementation process. The experimental results are provided to validate the efficacy of the controller and discuss the implementation and design procedure.

This Section is organized as follows. Subsection 2.2.2 describes the dynamic model governing the motion of the TCAMs and introduces the notation adopted in this section. In Subsec-

tion 2.2.3 formulate the control problem, while in 2.2.4 presents the control strategy that solves such problem. Subsection 2.2.5 states the main robustness and stability results of this section. Finally, Subsection 2.2.6 presents the simulation and experimental results validating the theoretical findings.

2.2.2 TCAM Dynamic Model

In what follows, the dynamic model of the TCAMs under consideration is reported. For the details on the derivation of the model, the reader is referred to [27, 81]. The TCAM's motion is governed by the following dynamic equation:

$$m\ddot{\ell}(t) + b\dot{\ell}(t) + k\ell(t) = F_e(t). \quad (2.36)$$

In Equation (2.36), $F_e(t)$ is an external force applies to the TCAM, m is the total mass of the system, b and k are the damping and spring coefficients of the TCAM, respectively. Here, $\ell(t)$ is the TCAM length before the actuation. Previously, the authors in [81] show that the spring coefficient can be expressed as a function of $T_\Delta = T(t) - T_{amb}$, where $T(t)$ is the temperature of the TCAM at time t , and T_{amb} is ambient temperature. Based on Equations (7) and (16) in [81], $k(T_\Delta)$ can be expressed as follows.

$$k(T_\Delta) = \frac{\bar{m}g}{S \left[1 - \frac{1}{B(T_\Delta) - C(T_\Delta)} \left(\frac{qB(T_\Delta)C(T_\Delta)}{\sqrt{B(T_\Delta)\bar{m}g}} - 2C(T_\Delta) \right) \right]}, \quad (2.37)$$

where

$$B(T_\Delta) = E_x \frac{\pi}{4} (CTE T_\Delta(t) r_o + r_0)^4, \quad (2.38)$$

$$C(T_\Delta) = G_{yz} \frac{\pi}{2} (CTE T_\Delta(t) r_o + r_0)^4, \quad (2.39)$$

S is the initial length of the straight fiber/composite yarn (before coiling), $q = \frac{2\pi n}{S}$ is the total end

rotation of the fiber, and n is the number of turns applied during the coiling process, CTE is the linear thermal expansion coefficient in the radial direction, r_0 is the initial fiber radius, E_x is the longitudinal Young's modulus, and G_{yz} is the shear modulus. $\bar{m}$ is the external mass used and on fabrication [27] (which differs from the mass of the system, m introduced in Equation (2.36)).

In [81] it is shown, using theoretical arguments and experimental observations, that by increasing (decreasing) the temperature of the artificial muscle, the spring coefficient k in Equation (2.37) increases (decreases), thus causing artificial muscle's contraction (expansion). Let us define the muscle contraction as $d(t) = \bar{\ell} - \ell(t)$, where $\ell(t)$ was introduced above and $\bar{\ell} > 0$. For example, the constant $\bar{\ell}$ can be selected to be the TCAM's length when $T_\Delta = 0$ and the external force is the weight applied during fabrication, namely $\bar{\ell} = \frac{\bar{m}g}{k(0)}$.

With the above setup, the dynamic system in Equation (2.36) can be rewritten as

$$m\ddot{d}(t) + b\dot{d}(t) + k(0)d(t) = k(T_\Delta)\bar{\ell} - F_e - k_\Delta d(t) \quad (2.40)$$

where $k_\Delta = k(T_\Delta) - k(0)$. Temperature variation, and therefore muscle contraction, can be induced by applying a voltage difference to the extremities of the TCAM. The dynamic relationship that governs the change of TCAM's temperature for given voltage is as follows:

$$\dot{T}_\Delta(t) = \frac{V^2(t)}{C_t R} - \frac{\lambda}{C_t} T_\Delta(t) \quad (2.41)$$

where R is the TCAM's electrical resistance, C_t is its thermal mass, and λ is its absolute thermal conductivity. Uncertainties in these terms are handled in Equations (2.42) and (2.49).

The control problem addressed in this section is the one of formulating a control law for $V(t)$ that guarantees convergence of the contraction $d(t)$ to some reference command, $r(t)$. Samples of the displacement $d(t)$ are measured by a sensor at some sampling rate T_s and are available for feedback.

2.2.3 Problem Formulation

Throughout this section, $\|x(t)\|_{\mathcal{L}_p}$ denotes the $\mathcal{L}_p$ -norm of the signal $x(t) : [0, \infty] \rightarrow \mathbb{R}^n$ with $1 \leq p \leq \infty$, while $\|\cdot\|_p$ represents the vector or matrix p -norms (see, for example, [[97], Appendix A]).

To mathematically formulate the above problem as a control problem, let the state vector describing the system at hand be defined as $x(t) = [d(t) \dot{d}(t) T_\Delta(t)]^\top$. Let the control input be $u(t) = V^2(t) - \frac{V_{max}^2}{2}$, where V_{max} is the maximum voltage that can be applied to the TCAM. Let the output be $y(t) = d(t)$. With this setup, the system given by Equations (2.40) and (2.41) can be reduced to

$$\begin{aligned}\dot{x}(t) &= A_p x(t) + B_p(u(t) + f(t, x(t))), \\ y(t) &= C_p x(t),\end{aligned}\tag{2.42}$$

where

$$\begin{aligned}A_p &= \begin{bmatrix} 0 & 1 & 0 \\ -\frac{k(0)}{m} & -\frac{b}{m} & \frac{c}{m} \\ 0 & 0 & -\frac{\lambda}{C_t} \end{bmatrix}, \\ B_p &= \begin{bmatrix} 0 \\ 0 \\ \frac{1}{C_t R} \end{bmatrix}, \quad C_p = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix},\end{aligned}$$

and $f(t, x(t))$ represents time-varying model uncertainties and disturbances. Notice that in the above equations the following assumption is used to linearize,

$$cT_\Delta(t) \approx k(T_\Delta)\bar{\ell} - F_e - k_\Delta d(t)$$

see Equation (2.40). The coefficient c can be estimated experimentally by setting the temperature to an arbitrary constant value T_{Δ}^* , then measuring the resulting contraction at steady state, $d^* = \ell_{amb} - mg/k(T_{\Delta}^*)$, with $k(T_{\Delta}^*)$ computed using Equation (2.37), and finally using

$$cT_{\Delta}^* = k_{amb}d^*. \quad (2.43)$$

Next, let's introduce the desired system

$$M(s) = C_m(sI - A_m)^{-1}B_m \quad (2.44)$$

where (A_m, B_m, C_m) are defined as

$$A_m = A_p - B_pK, \quad B_m = B_p, \quad C_m = C_p. \quad (2.45)$$

In the equations above, K is a constant gain that can be selected such that $A_p - B_pK$ is Hurwitz, and such that Equation (2.44) exhibits desired dynamics. Then, the desired response $y_m(t)$ is the Laplace transform

$$y_m(s) = M(s)K_g r(s), \quad K_g = -(C_m A_m^{-1} B_m)^{-1} \quad (2.46)$$

and $r(s)$ is the Laplace transform of $r(t)$ given by

$$r(t) = r_d[i], \quad t \in [iT_s, (i+1)T_s), \quad i \in \mathbb{Z}^+, \quad (2.47)$$

and $r_d[i]$ is a given discrete time reference command with time period $T_s > 0$. With this setup, the problem at hand reduces to the one of designing control laws for $u(t)$ such that the output $y(t)$ of system in Equation (2.42) tracks the desired response $y_m(t)$. The following assumptions are imposed.

Assumption 2.2.1 *The reference command is bounded as follows:*

$$\|r_d[i]\|_\infty \leq M_r \quad \forall i \in \mathbb{Z}^+.$$

Assumption 2.2.2 *For any $\delta > 0$ there exists F_δ and L_0 such that*

$$\|f(t, x_2) - f(t, x_1)\|_\infty \leq F_\delta \|x_2 - x_1\|_\infty,$$

$$\|f(t, 0)\|_\infty \leq L_0$$

hold uniformly for all $\|x_i\| \leq \delta, i \in \{1, 2\}, t \geq 0$.

2.2.4 Control Law

This section presents the sampled-data controller based on the control architecture introduced in [128]. The controller is comprised of three main elements: an output predictor, an adaptation law, and a control law. These three elements are described below.

Output Predictor: Consider the following discrete-time output predictor

$$\begin{aligned} \hat{x}_d[i+1] &= e^{A_m T_s} \hat{x}_d[i] + A_m^{-1} (e^{A_m T_s} - I) (B_m u_d[i] + \hat{\sigma}_d[i]), \\ \hat{y}_d[i] &= C_m \hat{x}_d[i], \quad \hat{x}_d[0] = C_m^\dagger y_0, \end{aligned} \tag{2.48}$$

where T_s is the sampling rate of the model, $\hat{x}_d[i] \in \mathbb{R}^3$ is the state, $\hat{y}_d[i] \in \mathbb{R}$ is the output, y_0 is the known initial output, while $\hat{\sigma}_d[i]$ and $u_d[i] \in \mathbb{R}$ are the estimates and control input, respectively, which will be defined later. The output predictor replicates the closed-loop dynamics, with the estimate $\hat{\sigma}_d[i]$ replacing the unknown function $f(t, x(t))$.

$$\hat{\sigma}_d[i] = -\Phi^{-1}(T_s)e^{\Lambda A_m \Lambda^{-1} T_s} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} (\hat{y}_d[i] - y_d[I]) \quad (2.49)$$

where $y_d[i]$ is the sampled output of (2.42), i.e., $y_d[i] = y(iT_s), i \in \mathbb{Z}^+, \hat{y}_d[i]$ is computed in (2.48),

$$\Lambda = \begin{bmatrix} C_m \\ D\sqrt{P} \end{bmatrix},$$

where $\sqrt{P}$ is such that $P = \sqrt{P}^\top \sqrt{P}$, $P \in \mathbb{R}^{3 \times 3}$ is a positive definite matrix solving $A_m^\top P + PA_m = -Q$ for a given positive definite matrix Q , and $D \in \mathbb{R}^{2 \times 3}$ is a matrix that satisfies $D(C_m(\sqrt{P}^{-1}))^\top = 0$, and

$$\Phi(T_s) = \int_0^{T_s} e^{\Lambda A_m \Lambda^{-1} (T_s - \tau)} \Lambda d\tau.$$

Control Law: The control input $u(t)$ is given by

$$u(t) = u_d[i], \quad t \in [iT_s, (i+1)T_s), \quad i \in \mathbb{Z}^+$$

where $u_d[i]$ is defined by

$$\begin{aligned} x_u[i+1] &= e^{A_o T_s} x_u[i] + A_o^{-1} (e^{A_o T_s} - I) (B_o e^{-A_m T_s} \hat{\sigma}_d[i]), \\ u_d[i] &= K_g r_d[i] - C_o x_u[i], \quad x_u[0] = 0, \end{aligned} \quad (2.50)$$

where K_g and $r_d[i]$ are introduced in Equations (2.46) and (2.47), (A_o, B_o, C_o) is the mini-

mal state-space realization of the following transfer function

$$O(s) = C(s)M^{-1}(s)C_m(sI - A_m)^{-1} \quad (2.51)$$

and $C(s)$ is a strictly proper stable transfer function such that $C(0) = 1$.

Finally, the controller at hand consists of Equations (2.48) to (2.51) subject to the following two conditions:

$$C(s)M^{-1}(s) \text{ is proper,} \quad (2.52)$$

and, for a given ρ_0 , there exists $\rho_r > \rho_0$ such that the following $\mathcal{L}_1$ norm condition holds

$$\|G(s)\|_{\mathcal{L}_1} < \frac{\rho_r - \rho_1 - \rho_2}{L_{\rho_r}\rho_r + L_0}, \quad (2.53)$$

where

$$\begin{aligned} G(s) &= H_0(s)(I - C(s)), \\ H_0(s) &= (sI - A_m)^{-1}B_m \\ \rho_1 &= \|s(sI - A_m)^{-1} - sH_1(s)H_{\text{in}}\|_{\mathcal{L}_1}\rho_0, \\ \rho_2 &= \|H_0(s)K_g\|_{\mathcal{L}_1}M_r, \\ H_{\text{in}}(s) &= C_m(sI - A_m)(I - C_m^\dagger C_m), \\ H_1(s) &= H_0(s)C(s)M^{-1}(s) \\ L_{\rho_r} &= \frac{\bar{\gamma}_0 + \rho_r}{\rho_r}(F_{\bar{\gamma}_0 + \rho_r} + \|K\|_\infty), \end{aligned} \quad (2.54)$$

2.2.5 Controller Performance Bounds

In order to analyze the performance of the controller presented in the previous section, the following reference system is introduced

$$\begin{aligned}
\dot{x}_{\text{ref}}(t) &= A_p x_{\text{ref}}(t) + B_p (u_{\text{ref}}(t) + f(t, x_{\text{ref}}(t))), \\
u_{\text{ref}}(s) &= K_g r(s) - C(s) \sigma_{\text{ref}}(s), \\
y_{\text{ref}}(t) &= C_p x_{\text{ref}}(t), \quad x_{\text{ref}}(0) = x_0,
\end{aligned} \tag{2.55}$$

where

$$\sigma_{\text{ref}}(s) = \omega_{\text{ref}}(s) + H_{\text{in}}(s)x_0,$$

$\omega_{\text{ref}}(s)$ is the Laplace transform of

$$\omega_{\text{ref}}(t) = K x_{\text{ref}}(t) + f(t, x_{\text{ref}}(t)).$$

The output of the reference system can be written as

$$\begin{aligned}
y_{\text{ref}}(s) &= M(s) K_g r(s) + M(s) (I - C(s)) \sigma_{\text{ref}}(s) \\
&\quad + C_m (sI - A_m)^{-1} C_m^\dagger y_0,
\end{aligned}$$

i.e., the reference systems in (2.55) mitigates only the uncertainty $\sigma_{\text{ref}}(t)$ that is within the bandwidth of $C(s)$. Since $C(0) = 1$, the Final Value Theorem implies that the reference system recovers the desired response introduced in (2.44).

The control input of the reference system depends on the uncertainty $f(t, x_{\text{ref}}(t))$ and the unknown state vector $x_{\text{ref}}(t)$, and thus is not implementable. In turn, the reference system in (2.55) is introduced only for the purpose of performance analysis. The following lemma establishes stability results and performance bounds concerning the reference system.

Lemma 2.2.1 *Consider the closed-loop reference system in (2.55) subject to (2.52) and (2.53).*

If $\|x_0\|_\infty \leq \rho_0$, then

$$\|x_{\text{ref}}(t)\|_{\mathcal{L}_\infty} \leq \rho_r \quad (2.56)$$

and

$$\|u_{\text{ref}}(t)\|_{\mathcal{L}_\infty} \leq \rho_{ur} \quad (2.57)$$

where ρ_r was introduced in (2.53) and

$$\begin{aligned} \rho_{ur} = & \|C(s)\|_{\mathcal{L}_1} (L\rho_r\rho_r + L_0) + \|sC(s)M^{-1}(s)H_{\text{in}}(s)\|_{\mathcal{L}_1}\rho_0 \\ & + \|K_g M_r\|_\infty \end{aligned}$$

The proof of Lemma 2.2.1 can be found in [128, Lemma 3.4].

Finally, in the following theorem the stability and performance bounds of the closed-loop system is presented in (2.42) with the controller proposed in Equations (2.48) to (2.51), in relation to the reference system (2.55).

Theorem 2.2.1 *Consider the system given by Equation (2.42) and the control laws proposed in Equations (2.48)–(2.51) subject to conditions (2.52) and (2.53). If $\|x_0\|_\infty \leq \rho_0$, then*

$$\|x_{\text{ref}}(t) - x(t)\|_{\mathcal{L}_\infty} \leq \gamma_x, \quad (2.58)$$

and

$$\|u_{\text{ref}}(t) - u(t)\|_{\mathcal{L}_\infty} \leq \gamma_u, \quad (2.59)$$

with $\lim_{T_s \rightarrow 0} \gamma_x = \lim_{T_s \rightarrow 0} \gamma_u = 0$.

The proof of Theorem 2.2.1 follows steps similar to [128, Th.3.1].

Remark 2.2.1 *Theorem 2.2.1 shows that the output of the closed-loop system, $y(t)$, tracks the desired response, $y_m(t)$, both in transient and steady state, with uniform performance bounds that can be improved by decreasing T_s . The inverse of the sampling time T_s plays the role of an adaptation gain found in typical adaptive control architectures. This parameter can be directly related with the sampling rate of the CPU. The sampling time should always be selected as low as possible, within the limits of the CPU, in order to achieve high controller performance. Further details on the derivation and definition of the performance bounds γ_x and γ_u can be found in [98].*

Also notice that Theorem 2.2.1 together with Lemma 2.2.1 provide uniform bounds for the norm of the state $x(t)$ and the input $u(t)$, which in turn can be used to relate the desired reference $r(t)$ with system's feasibility and safety metrics. These include maximum displacement error allowed, temperature limits for overheating prevention, and maximum voltage allowed.

Finally, in the proposed control architecture, $C(s)$ represents the low-pass filter applied to the input, which attenuates high frequencies in the adaptive control signals. The filter, which must be selected to satisfy conditions (2.52) and (2.53), provides trade off between robustness to time-delays and performance in terms of disturbance attenuation. The reader is referred to [97, 129] for an in-depth discussion on optimization methods for the design of the filter.

2.2.6 Experimental Results

A model for the TCAM dynamics is derived using the parameters reported in Table 2.4 (taken from [81]), and is given by

$$A_p = \begin{bmatrix} 0 & 1 & 0 \\ -44.6652 & -16.6667 & 0.0026 \\ 0 & 0 & -0.0291 \end{bmatrix}$$

$B_p = [0 \ 0 \ 0.1213]^\top$, and $C_p = [1 \ 0 \ 0]$. The system is stable and the eigenvalues are $\lambda_1 = -3.3555$, $\lambda_2 = -13.3112$, and $\lambda_3 = -0.0291$. The eigenvalue λ_3 is the slow eigenvalue and

Table 2.4. TCAM Nominal Parameters

Description	Parameter	TCAM Data
External mass	m	0.106 kg
Fabrication mass	m	0.106 kg
External load	$F_e = mg$	1.04 N
Damping coefficient	b	2.0 Ns
Ambient Temperature	T_{amb}	20 °C
Untwisted fiber yarn length	S	0.06 m
No. of coiled turns per meter	q	2.9322×10^3 turns/m
composite yarn Young's modulus along the axial direction	E_x	4.67×10^8 Pa
composite yarn Shear modulus in the transverse section	G_{yz}	2.2×10^7 Pa
Linear thermal expansion coefficient in the radial direction	CTE	3×10^{-4} °C ⁻¹
Initial fiber radius	r_0	3.6×10^{-4} m
Thermal mass	C_t	0.162 J°C ⁻¹
Electrical Resistance of the TCAM	R	2.5 Ω
Absolute thermal conductivity	λ	0.0086 W°C ⁻¹

it defines the dynamic response. To improve the transient response and decrease the settling time of the closed-loop system, the desired system is defined $M(s)$ through pole placement in order to position the slow eigenvalue, λ_3 , at $\frac{-3}{t_{set}}$. Here, t_{set} is the desired settling time of the response and can be tuned by the user at will. This choice gives $K = [-3.7476 \quad -0.1009 \quad 4.7072]$, which together with Equation (2.45) gives the desired system

$$A_m = \begin{bmatrix} 0 & 1 & 0 \\ -44.6652 & -16.6667 & 0.0026 \\ 0.4545 & 0.0122 & -0.6000 \end{bmatrix}, \quad (2.60)$$

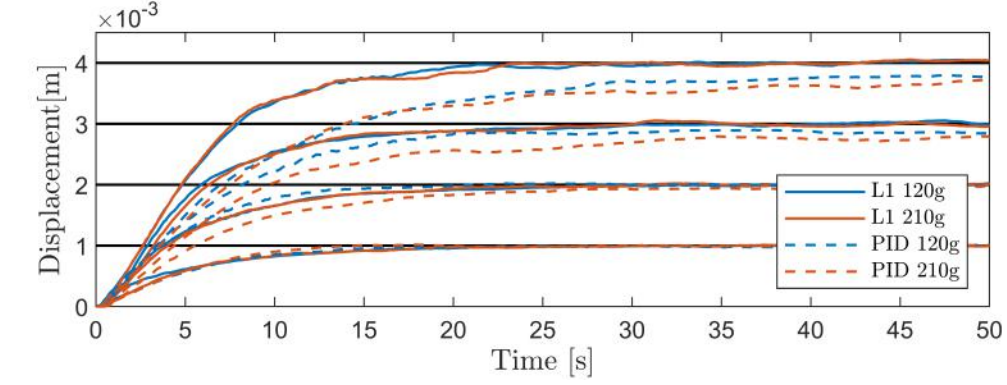
$$B_m = B_p, \quad C_m = C_p.$$

The filter, Q matrix, and sampling time is choosen as follows

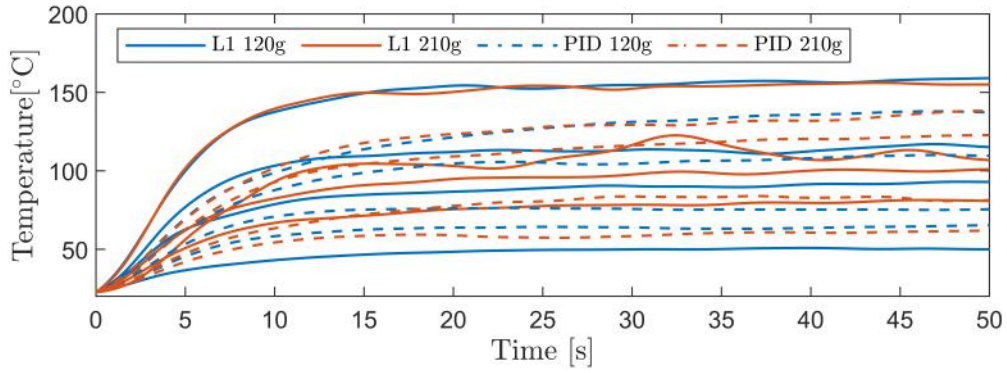
$$C(s) = \frac{25}{(s+1)(s+5)^2}, \quad Q = I_3, \quad T_s = 0.005s. \quad (2.61)$$

Then, with the above setup the controller outlined in Equations (2.48) to (2.51) is implemented.

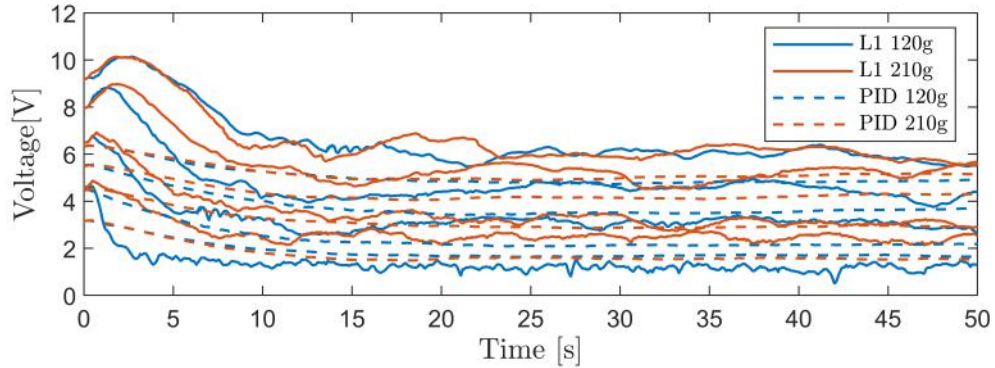
To experimentally validate the efficacy of the proposed controller, TCAM samples are manufactured by following the fabrication process suggested by Lamuta et al. [27] and Giovinco et al. [81]. The controller introduced in Equations (2.60) and (2.61) as well as a PI was implemented on MATLAB/SIMULINK. The controller sends voltage commands through ROS to a microcontroller. The commands are then provided to the TCAM by means of a motor controller. The real-time displacement data are measured using a laser sensor pointed towards the bottom horizontal surface of the mass attached to the TCAM. The displacement measurement is sent to SIMULINK for feedback using a Data Acquisition device. For analysis purpose only, also the temperature of the TCAM is measured using a thermal camera. Figure 1 illustrates the system response for both the $\mathcal{L}_1$ adaptive and PI controllers when the references to be tracked are steps of different amplitudes. The PI controller was tuned such that it would converge well for a one millimeter step with an attached mass of 120 g. To accomplish this the controller's proportional gain was set to 10000 and the integral gain to 500. For fair comparison, the $\mathcal{L}_1$ adaptive controller was tuned by using pole placement to set the settling time of the desired system to 20 s. Each controller tracked steps of one, two, three and four millimeters twice, once with a mass of 120 g, and a second time with a



(a) Displacement tracking



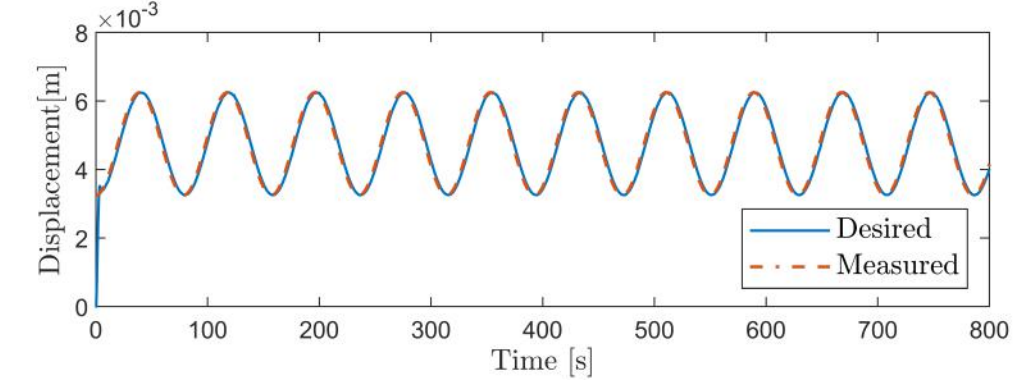
(b) Temperature



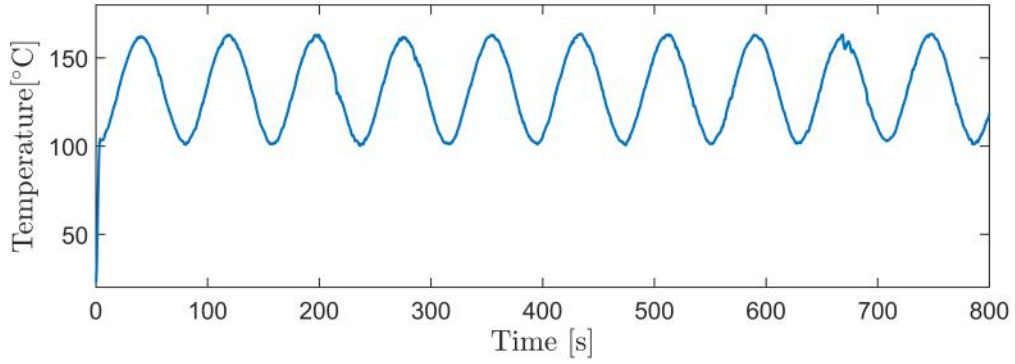
(c) Voltage

Figure 2.15. Experimental results: step reference tracking

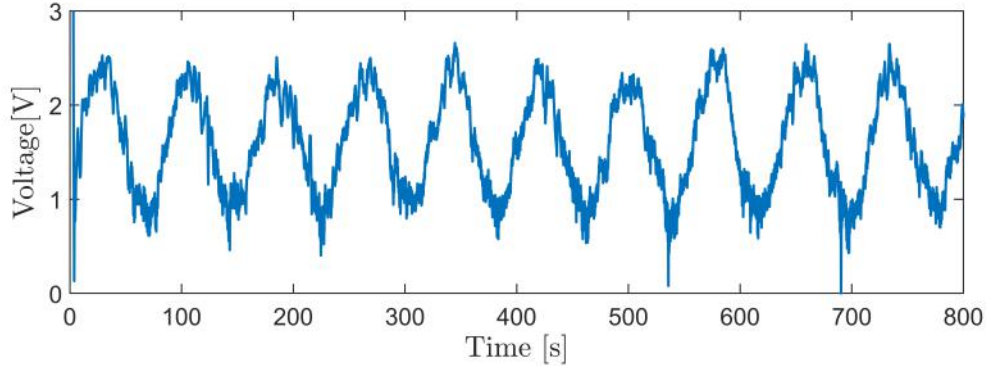
mass of 210 g attached. This comparative test illustrates a key advantage of the $\mathcal{L}_1$ adaptive controller by demonstrating its consistency despite two sources of external change. Where the settling time and behavior of the PI controller changed for both step and mass changes, even to a point when it did not converge within the time of the test, the $\mathcal{L}_1$ adaptive controller exhibited consistent behavior. Figure 1(a) depicts the TCAM's displacement for each controller, which converges to the



(a) Displacement tracking



(b) Temperature



(c) Voltage

Figure 2.16. Experimental results: sinusoidal reference tracking.

desired value. Figures 1(b) and 1(c) illustrate the temperature and voltage, respectively, required to converge. Figure 2 shows the TCAM response to the $\mathcal{L}_1$ adaptive controller when the reference to be tracked is a sampled sinusoidal signal, i.e.,

$$r(t) = 0.00475 + 0.0015 \sin(0.08iT_s + 3\pi/2),$$

$t \in [iT_s, (i+1)T_s), i \in \mathbb{Z}^+$. Figure 2(a) demonstrates agreement of the actual TCAM displacement and the desired value. Finally, Figures 2(b) and 2(c) illustrate the TCAM's temperature and the voltage applied, respectively.

2.2.7 Summary

This section proposed a sample-data adaptive output feedback control strategy for Twisted and Coiled Artificial Muscles (TCAMs). The proposed controller, which is borrowed from the literature on $\mathcal{L}_1$ adaptive control, guarantees robustness of the closed-loop system. The performance analysis takes into account digital implementation of the algorithm that arise from using CPUs and sensors with limited sampling rates. The efficacy of the controller was validated through experimental tests.

2.3 Modelling and Control of Shape Memory Alloy (SMA) Actuators

2.3.1 Nomenclature

Material model

σ	Stress
σ_0	Initial stress
σ_f	Critical stress for martensite finish
σ_s	Critical stress for martensite start
ε	Total strain
ε_0	Initial strain
ε_L	Total recoverable deformation/strain
ε_{s0}	Initial strain in the stress phase
ε_{T0}	Initial transformation-induced strain
ξ	Total martensite fraction
ξ_s	Martensite fraction induced by temperature
ξ_T	Martensite fraction induced by stress
A_f	Austenite finish temperature
A_s	Austenite start temperature
C_a	Slope of Austenite transformation
C_m	Slope of Martensite transformation
D_A	Young's modulus of Austenite
D_{DTW}	Young's modulus of Martensite de-twinned
D_{TW}	Young's modulus of Martensite twinned

M_f Martensite finish temperature [$^{\circ}\text{C}$]

M_s Martensite start temperature [$^{\circ}\text{C}$]

T Temperature [$^{\circ}\text{C}$]

Dynamic model

$\dot{q}_i$	Derivative of i-th generalized coordinate [rad s^{-2}]
ω_i	Angular velocity of i-th phalanx [rad s^{-1}]
τ_i	Generalized force/torque acting on i-th joint (actuator torque) [N m]
g	Gravitational acceleration [m s^{-2}]
k_i	Torsional stiffness of i-th joint (actuator torque) [N m]
l_i	Length of i-th phalanx [m]
l_w	Length of wire - [m]
M_i	Mass of i-th phalanx [kg]
q_i	i-th generalized coordinate (joint angle) [rad]
r_i	Distance from the center of mass of i-th phalanx to its joint [m]
v_w	Contraction velocity of wire [m s^{-1}]
y_i	Vertical distance of the center of mass of i-th phalanx from a reference point [m]

2.3.2 Introduction

Soft robotics has become increasingly popular across various industries, including healthcare, manufacturing, transportation, and surveillance [109, 130–132]. Inspired by biological systems, soft robots have traditionally been limited by conventional actuators, which constrain their ability to mimic natural movements. This limitation has led to the development of artificial muscle actuators in recent years [7, 26, 133]. Among these, Shape Memory Alloys (SMAs) stand out due to their desirable characteristics, such as high recoverable strain, high force generation capability, portability, and low weight, especially when compared to conventional actuators like DC motors or pneumatic/hydraulic systems. Of the various SMAs, the binary nickel-titanium (NiTi) system is particularly noteworthy for its efficient shape recovery, excellent mechanical response, corrosion resistance, and biocompatibility [7, 134, 135]. The impressive properties of SMAs arise from their ability to exist in two distinct phases with different crystal structures. Austenite, the high-temperature phase, features an ordered and compact atomic arrangement akin to a neatly organized stack of blocks. This structure imparts strength and elasticity to the SMA, enabling it to return to its original shape after deformation—a phenomenon known as superelasticity. In contrast, martensite, the low-temperature phase, exhibits a "twinned" structure, where the atomic blocks are shifted and intertwined. This flexible arrangement allows the SMA to undergo significant deformation without sustaining permanent damage. The shape memory effect hinges on the transformation between these two phases. When a NiTi SMA is deformed in its low-temperature martensitic state, its internal structure adjusts to accommodate the stress while retaining the "memory" of its original austenitic form. Upon heating, the SMA transitions back to the austenite phase, during which the reordering of atoms causes the material to snap back to its original shape, even after significant deformation in the martensitic state [136, 137].

These properties make Nitinol SMAs ideal for various soft robotics applications such as grippers [138] due to their ability to conform and exert force, making them perfect for delicate grasping tasks; micro-robots [139], thanks to their miniature size and lightweight nature, are well-suited for minimally invasive surgical procedures. As wearable assistive devices, they can provide

variable and controlled assistance for rehabilitation or human augmentation [140, 141].

Numerous control algorithms have been proposed for SMA actuation, including PID control [142–145], state-space control, model predictive control, observer-based control, fuzzy-logic control, reinforcement control, Lyapunov theory-based control, and adaptive control. Designing a control algorithm for SMAs poses significant challenges due to the high non-linearity of the material and the phase changes it undergoes, as well as changes in the governing equations during the heating and cooling cycles. Additionally, because SMAs are actuated by Joule heating, careful temperature management is essential to prevent overheating [146]. This aspect is particularly critical in rehabilitation/wearable and other healthcare applications to ensure user safety.

Such devices also experience varying loads, which alters the tension acting on the SMAs. Therefore, it is crucial to develop adaptive output feedback controllers that can manage non-measurable states, such as temperature and material properties, as well as other partially unknown kinematic and dynamic parameters. Moreover, to circumvent the need for costly system hardware, a sampled data controller with relatively larger sampling periods and time delays will be implemented, optimizing control efficiency and cost.

In this paper, we proposed an actuation model and a design of a robust and adaptive output feedback controller for SMAs based on sampled data. The model enables the tune and optimize the controller parameters. The controller adaptively adjusts the output current supply to the SMA by varying the voltage to reach the desired strain and stress. The control framework is borrowed from the $\mathcal{L}_1$ adaptive control theory [97, 98] that guarantees stability and performance against disturbance and model uncertainties. The control architecture selection was inspired by the previous applications of $\mathcal{L}_1$ adaptive control theory on twisted and coiled artificial muscles [147–150] and twisted and spiraled artificial muscles [151].

2.3.3 Shape Memory Alloy Theory and Modelling

Shape memory alloys are special materials that can return to their original shape after being exposed to heating or other appropriate temperature-changing stimulus. This is known as the shape

Table 2.5. Control strategy and application overview

Control Method	Control Parameter(s)	Application
Passive control	Heat transfer, constitutive model, Short-time pulse activation, continuous deformable structure, curvature, stroke control, stiffness to force, motion control, periodic current control, modulation of current force coupled with displacement, acceleration and angular velocity, Passive force to length of wires, High-speed thermally-induced transformations	Rigid, soft and bio-inspired robots, coil systems, SMALLbug, Rolling robots, Continuous manipulator, Frog like robot
Sequential control	Kinematic model for motion control to displacement and force	Starfish robot
BB and IL	Iterative learning control, Positional asymmetric excitation	Mesh-worm, Flexible micro-robot
PP	Path planning control	Starfish-like robot
Closed loop	Speed and force	Robotic swimmer
PID	Position, Bending curvature, Positional asymmetric excitation	Caudal fin, Snake robot, Flexible microrobot
PWM	Differential friction, Force to stroke, Peristaltic motion mechanism, peristaltic motion and the orientation	Omegabot, Micro in-pipe, Inchworm robot, Soft robot
ON/OFF control	Liang dynamic model	Flexible SMA actuators

memory effect [152, 153]. Titanium nickel (NiTi) is an alloy of the equiatomic composition of titanium and nickel, about 55wt% Nickel and 45wt% Titanium [154].

The internal rearrangement of atoms during their phase transformation, known as the shape memory effect, leads to measurable changes in physical characteristics, such as electrical resistance. This phenomenon allows SMAs to act as self-sensors, providing information about their internal state and external conditions. However, currently, there is no comprehensive model that can accurately estimate the applied stress, deformation, and temperature based solely on the electrical properties of SMAs. The development of robust and reliable self-sensing models for SMAs is crucial for realizing their full potential in different applications like Structural Health Monitoring, Biomedical Implants, Robotics, and Automation. Only a limited number of multi-physics mathematical models are available to predict the stress, strain, and temperature relationship of Ni-Ti

SMA. Physics-based models are useful over data-driven models in understanding the theory and physical properties of the material [146].

The shape memory alloys have several interesting properties, one of these is the shape memory effect, that allow to use this material as actuator. This effect consists in the recovery of an initial shape after a pre-deformation (pre-strain) and a thermal activation. Thanks the huge level of pre-strain (around 4-6% and also called maximum recoverable strain ϵ_L) the macroscopic effect is like a stroke in a conventional actuator.

To understand the need for a robust and adaptive controller, a clear comprehension of the physics behind this behavior is required. The shape recovery, under thermal load, is explained thanks the diagram phase Figure 2.17. This complex behaviour is characterized from an high hysteresis visible during the increasing or decreasing of the thermal load. In particular the recovery is due to a atomic structure transformation, this means that the material change its shape macroscopically and also its thermo-mechanical properties, such as electrical resistance or Young's module.

When the SMA material (like Ni-Ti alloy) undergoes a transition from austenite to martensite or vice versa, significant changes in its electrical resistance may occur [155]. In general, by

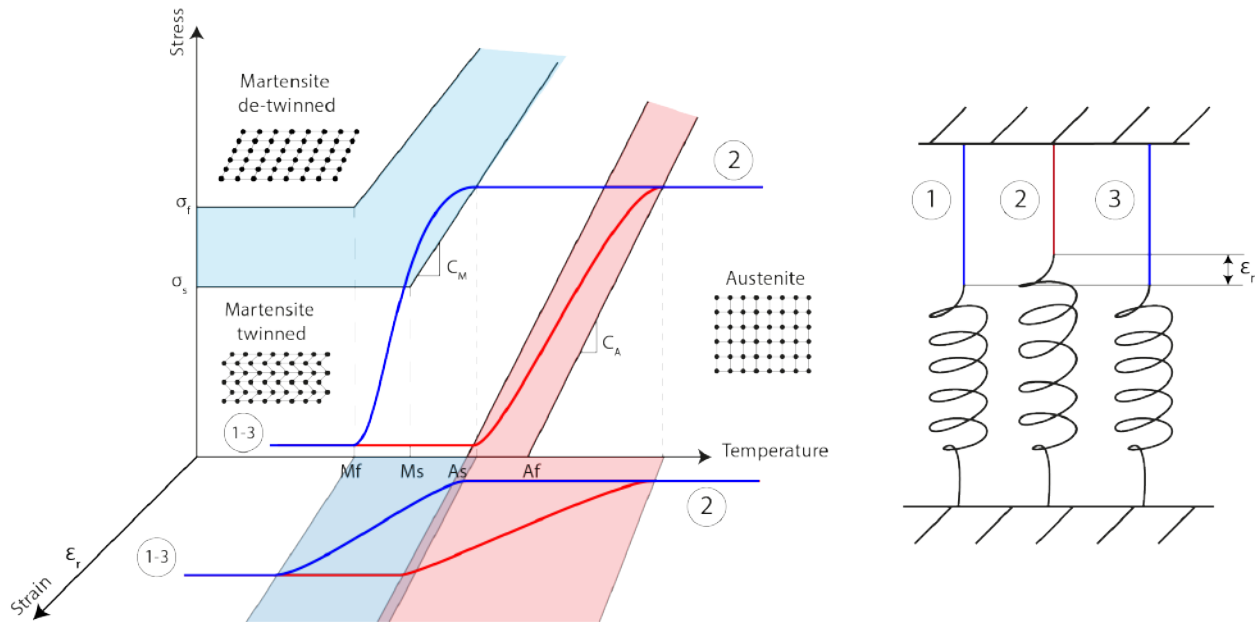


Figure 2.17. Diagram phase and shape memory effect

increasing the temperature, Ni-Ti tends to become more austenitic (point 2 in Figure 2.17) and thus may show a decrease in electrical resistance. On the contrary, by decreasing the temperature, it can switch to a martensitic phase (points 1-3 in Figure 2.17) and show an increase in electrical resistance. This behavior is associated with changes in electronic mobility and electrical conduction through the crystal structure in different stages.

There are different models to describe the resistance change [156, 157], that is affected by several factors like dimensions and boundary condition. One of the most important factor in the behaviour change is the mechanical stress that modify also the resistivity [158]. The cycle of hysteresis widens and could shifts (in the diagram resistance-current or resistance-temperature) when the applied stress increase. Starting from this behaviour, the relation from stress, strain, temperature and electrical properties could be established.

The generalized SMA model consists of three equations: the modeling of phase transformation, temperature dynamics, and constitutive law.

2.3.4 Model for SMA Phase Transformation

The modeling equations depend upon the SMA's heating or cooling stage and current phase, either Martensite or Austenite. Hence, the system can be modeled as a two-state system. The Martensite state is at a lower temperature, while the Austenite state is at a higher temperature. The present phase transformation of the SMA to Austenite by heating can be described using the kinetic laws as [159, 160],

$$\xi(t) = \frac{\xi_0}{2} \left\{ \cos \left[a_A \left(T(t) - A_s - \frac{\sigma(t)}{C_A} \right) \right] + 1 \right\}, \quad (2.62)$$

$$\text{for } C_A(T(t) - A_f) < \sigma(t) < C_A(T(t) - A_s) \quad \text{where} \quad a_A = \frac{\pi}{A_f - A_s}.$$

Similarly, the phase fraction $\xi(t)$ for the cooling of the SMA from its present state to Martensite can be expressed using the kinetic laws as

$$\xi(t) = \frac{1 - \xi_0}{2} \cos \left[a_M \left(T(t) - M_f - \frac{\sigma(t)}{C_M} \right) \right] + \left(\frac{1 + \xi_0}{2} \right), \quad (2.63)$$

$$\text{for } C_M(T(t) - M_s) < \sigma(t) < C_M(T(t) - M_f) \quad \text{where} \quad a_M = \frac{\pi}{M_s - M_f}.$$

where $T(t)$, $\sigma(t)$, and $\xi(0)$ are the temperature of the SMA, the stress of the SMA, and the initial Martensite phase. The material property constants C_A and C_M describe the relationship of temperature and critical stress to induce transformation. A_s, A_f, M_s and M_f are transition temperatures for Austenite start, Austenite finish, Martensite start, and Martensite finish such that $M_f < M_s < A_s < A_f$.

2.3.5 Model for SMA Temperature Dynamics

The thermal characteristics of the SMA can be predicted using the Joule heating equation of

$$\dot{T}_\Delta(t) = \frac{1}{mc_p(\xi)} \left[R(\xi)I^2(t) - hAT_\Delta(t) \right], \quad (2.64)$$

where $T_\Delta = (T(t) - T_0)$, T_0 is the ambient temperature, $I(t)$ is the current drawn through the SMA, h is the heat transfer coefficient, and A is the cross-sectional area of the SMA that impact for the heat loss due to convection. The electrical resistance of the SMA $R(\xi)$ can be expressed as follows:

$$R(\xi) = \frac{4L}{\pi d^2} (\rho_A(1 - \xi) + \rho_M \xi). \quad (2.65)$$

Here, d is the diameter, and L is the length of the SMA wire. The detailed expressions for h can be found in our previous work [146]. The resistivity of the SMA at Austenite phase (ρ_A) and Martensite phase (ρ_M) can be expressed by the following partial differential equations (PDE).

$$\rho_A = \rho_{OA} + (T_{OA} - T) \frac{\partial \rho_A}{\partial T} + \sigma \frac{\partial \rho_A}{\partial \sigma} \quad (2.66)$$

$$\rho_M = \rho_{OM} + (T_{OM} - T) \frac{\partial \rho_M}{\partial T} + \sigma \frac{\partial \rho_M}{\partial \sigma} + \varepsilon \frac{\partial \rho_M}{\partial \varepsilon} \quad (2.67)$$

2.3.6 Model for SMA Constitutive Law

There are several models to predict the SMA behavior [161–164]. The Tanaka and Brinson model has a similar thermodynamic approach that leads to the following constitutive law of

$$d\sigma = \frac{\partial \sigma}{\partial \varepsilon} d\varepsilon + \frac{\partial \sigma}{\partial \xi} d\xi + \frac{\partial \sigma}{\partial T} dT. \quad (2.68)$$

This leads to the following generalized expression of

$$d\sigma = D(\varepsilon, \xi, T) d\varepsilon + \Omega(\varepsilon, \xi, T) d\xi + \Theta(\varepsilon, \xi, T) dT. \quad (2.69)$$

This formulation starts from principles of thermodynamics the energy balance and Clausius-Duhem inequality, so the material functions are defined as

$$\begin{aligned} D(\varepsilon, \xi, T) &= \rho_0 \frac{\partial^2 \Phi}{\partial \varepsilon^2} \\ \Omega(\varepsilon, \xi, T) &= \rho_0 \frac{\partial^2 \Phi}{\partial \varepsilon \partial \xi} \\ \Theta(\varepsilon, \xi, T) &= \rho_0 \frac{\partial^2 \Phi}{\partial \varepsilon \partial T} \end{aligned} \quad (2.70)$$

in which Φ is the Helmholtz free energy, ρ is the density and respectively $D(\varepsilon, \xi, T)$ represent the modulus of the material, $\Omega(\varepsilon, \xi, T)$ can be considered the transformation tensor, and $\Theta(\varepsilon, \xi, T)$ is related to the thermal coefficient of expansion.

Moreover, the internal state variable for the phase fraction (ξ) was considered a simple percentage of material transformed to martensite so far. A separation, based on micro-mechanics behavior, in two contributes could be formulated as

$$\xi = \xi_S + \xi_T \quad (2.71)$$

in which ξ_S is the fraction of single variant martensite due to stress and ξ_T is the multi-variant martensite part induced by the temperature. This separation transforms the Eq. 2.68 and Eq. 2.69 into

$$d\sigma = \frac{\partial \sigma}{\partial \varepsilon} d\varepsilon + \frac{\partial \sigma}{\partial \xi_S} d\xi_S + \frac{\partial \sigma}{\partial \xi_T} d\xi_T + \frac{\partial \sigma}{\partial T} dT \quad (2.72)$$

Assuming the material functions as constant the Eq.2.72 could be written

$$d\sigma = Dd\varepsilon + \Omega_S d\xi_S + \Omega_T d\xi_T + \Theta dT \quad (2.73)$$

solving the differential form

$$\sigma - \sigma_0 = D(\varepsilon - \varepsilon_0) + \Omega_S(\xi_S - \xi_{S0}) + \Omega_T(\xi_T - \xi_{T0}) + \Theta(T - T_0) \quad (2.74)$$

To derive the value of Ω_S and Ω_T two different material restriction cases are applied. A first case of maximum residual strain with the material initially 100% austenite, $\xi_{S0} = 0$ and $\xi_{T0} = 0$ and the remaining conditions $\sigma_0 = \varepsilon_0 = 0$, $\sigma = 0$, $\varepsilon = \varepsilon_L$, $\xi_S = 1$, $\xi_T = 0$. A second case in which there is initially 100% undeformed martensite, $\xi_{S0} = 0$ and $\xi_{T0} = 1$ and the remaining conditions equally to previous conditions. In both cases $T = T_0$.

$$\Omega_S = -\varepsilon_L D \quad (2.75)$$

$$\Omega_T \equiv 0$$

The Young's modulus of each phase (martensite twinned and detwinned and austenite) is different, and the Young's modulus of the material depends by the percentage of martensite fraction. A formulation suggested by Liang and Tanaka define this quantity as

$$D(\varepsilon, \xi, T) = D(\xi) = D_A + \xi(D_M - D_A) \quad (2.76)$$

Accordingly with the simplifications made by [159], the Eq. 2.69 could be rewritten as

$$\sigma - \sigma_0 = D(\xi)\varepsilon - D(\xi_0)\varepsilon_0 + \Omega(\xi)\xi_S - \Omega(\xi_0)\xi_{S0} + \Theta T_\Delta. \quad (2.77)$$

The SMA constitutive model shows the relationship between stress rate ($\dot{\sigma}$), strain rate ($\dot{\varepsilon}$) and temperature rate ($\dot{T}_\Delta$) [160, 165]:

$$\dot{\sigma} = D\dot{\varepsilon} + \Omega\dot{\xi} + \theta\dot{T}_\Delta \quad (2.78)$$

2.3.7 SMA Finite State Machine Approach

In this section, a new approach to defining the Shape Memory Alloy (SMA) model is presented. Building upon the material configuration states outlined in the Brinson model's phase diagram, we have developed a finite state flow chart to simulate material hysteresis and overall behavior. The resulting diagram is shown in Fig. 2.18, where material configurations are denoted by the capital letter "C", while the letter "S" indicates the conditions for transitions between configurations. We applied the Finite State Machine Flowchart methodology to define this complex behavior and track the material's path history. The material's response is heavily influenced by previous load conditions and configurations, with abrupt transitions between states possible. This is due to the transformation between different crystal structures occurring only within transformation bands. When a load condition (mechanical or thermal) reverses, the internal configuration remains unchanged until a new intersection with a transformation band is reached.

To better understand how the model works, the entire algorithm is schematized in Tab. 2.6, which presents the equations for the configurations and the conditions for transitions. Starting from the *pre-strain* path shown in Fig. 2.18(a), the material begins in an unloaded state at ambient temperature. When a mechanical load (stress or strain) is applied, the transformation from *twinned* to *detwinned* martensite occurs within the band between σ_s and σ_f (S1). Complete transformation takes place when the load condition exceeds σ_f (S2). The transformation can be halted and reversed from either the fully *detwinned* martensite state (S3) or midway through the transformation

band (S18). If the load decreases below σ_s (S4), a recoverable strain remains.

After the *pre-strain*, the material is ready for activation. This process is described in Fig. 2.18(b), where multiple paths are possible. Various initial conditions may exist for this new load path, such as residual load from a previous load history. For each starting condition, the temperature load path begins when the temperature exceeds A_s (S5-S13-S15). As the material crosses the transformation band between martensite and austenite (S6), the process concludes with a full austenite structure (S7). The inverse transformation also allows for different paths, as illustrated in Fig. 2.18(c). Starting from either a complete transformation (S8) or an intermediate condition (S9-S17), the transformation begins, and the material may become *twinned* martensite (S12→S16), *detwinned* martensite (S10→S14), or remain in an intermediate state (S10→S11).

The initial conditions, denoted by the subscript 0 , are influenced by the load history. At each load condition change (e.g., from loading to unloading), the initial conditions are updated with the last computed value before the inversion. Consequently, the algorithm employs numerous variables to detect when external conditions change and a new configuration is required. The switch between different cases occurs by taking into account external conditions such as external load, imposed strain, temperature, and their derivatives. When the material crosses the transformation bands (during the transformation phases), the Newton-Raphson method is used to compute the stress or strain (where the trial variable for iteration is denoted by $*$) for imposed strain or stress conditions, respectively.

The reference Figure 2.18 illustrates the primary transition states for complete transformations from martensite to austenite and vice versa. However, it's important to note that the model also includes transition states that allow for the simulation of incomplete transformations. These additional states are based on the inversion of the temperature derivative's sign. This feature enables the model to capture more nuanced material behaviors, such as partial phase transformations that may occur when temperature changes are reversed before a complete transformation is achieved. By incorporating these intermediate states, the model can accurately represent the complex behavior of SMAs under a wider range of thermal conditions, including scenarios where

the material experiences temperature fluctuations that do not result in full phase transitions. This comprehensive approach ensures that the model can simulate both complete and partial transformations, providing a more realistic representation of SMA behavior in various applications and environmental conditions.

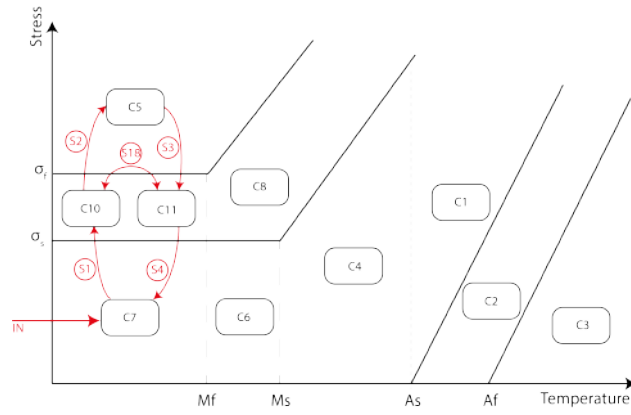
As explained in Experimental Validation section, the different boundary condition should be: imposed stress, in this case the strain is iterative computed to be compatible with the known stress/load; imposed strain, the stress is iterative computed to be compatible with the known strain/displacement.

2.3.8 Model Validation

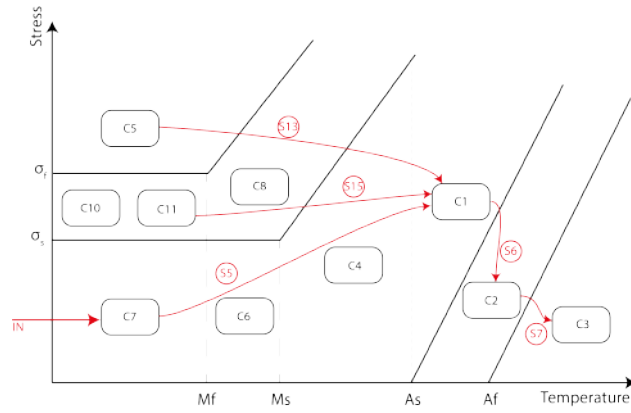
In order to validate the Shape Memory Alloy (SMA) model, a comprehensive series of simulations was conducted using a variety of input conditions. These simulations incorporated both mechanical and thermal stimuli, allowing for thorough testing of the model's response across a wide range of scenarios. The input parameters used for these simulations are clearly illustrated in Figure 2.19, which provides a visual representation of the mechanical loads and temperature variations applied to the SMA material over time. To facilitate a direct comparison between the model's predictions and the expected material behavior across different phases, Figure 2.20 presents a comprehensive visualization of the SMA's behavior overlaid on the phase diagram. This overlay effectively demonstrates the material's transitions between austenite and martensite phases in response to the applied mechanical and thermal loads. The initial version of the model, based on the concept of equilibrium between the imposed external load and the load generated within the material, is represented in Figure 2.19 and Figure 2.20. These figures showcase the model's robustness, particularly in handling complex loading scenarios. Figure 2.19 illustrates the evolution of temperature, strain, and stress over time, with each parameter presented in separate subfigures (2.19(a), 2.19(b), and 2.19(c), respectively). The case analyzed here involves a kinematic mechanism that develops a variable antagonistic load, further demonstrating the model's ability to handle dynamic loading conditions. 2.20 provides additional insights into the material's behavior by presenting the stress-

Table 2.6. Table of state and transition definition

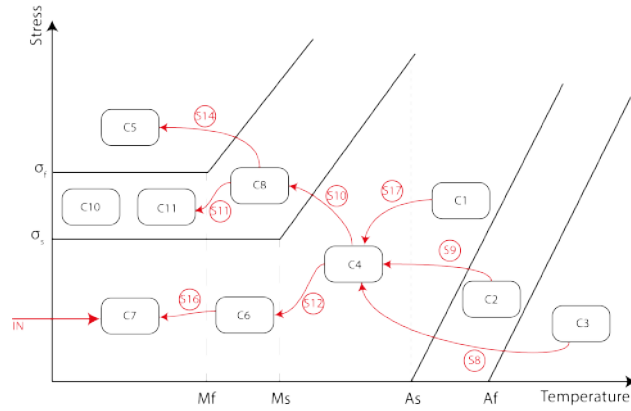
Configuration	Nomenclature	Equations	Transition (IN)
Martensite	C10	$\begin{cases} \xi_S = \frac{1 - \xi_{S0}}{2} \cos \left[\frac{\pi}{\sigma_s - \sigma_f} (\sigma^* - \sigma_f) \right] + \frac{1 + \xi_{S0}}{2}; \\ \xi_T = \xi_{T0} - \frac{\xi_{T0}}{1 - \xi_{S0}} (\xi_S - \xi_{S0}) + \Delta \xi_T; \end{cases}$	S1-S18
	C5	$\xi_S = 1; \quad \xi_T = 0;$	S2-S14
	C7	$\xi_S = \xi_{S0}; \quad \xi_T = \xi_{T0};$	IN-S4-S16
	C11	$\xi_S = \xi_{S0}; \quad \xi_T = \xi_{T0};$	S3-S11-S18
Austenite	C1	$\xi_S = \xi_{S0}; \quad \xi_T = \xi_{T0};$	S5-S15-S13
	C2	$\begin{cases} \xi = \frac{\xi_{S0}}{2} \cos \left[\alpha_A \left(T - A_s - \frac{\sigma^*}{Ca} \right) \right] + 1; \\ \xi_S = \xi_{S0} - \frac{\xi_{S0}}{\xi_0} (\xi_0 - \xi); \\ \xi_T = \xi_{T0} - \frac{\xi_{T0}}{\xi_0} (\xi_0 - \xi); \end{cases}$	S6
	C3	$\xi_S = \xi_{S0}; \quad \xi_T = \xi_{T0};$	S7
	C4	$\xi_S = \xi_{S0}; \quad \xi_T = \xi_{T0};$	S8-S9-S17
	C6	$\xi_S = \xi_{S0}; \quad \xi_T = \frac{1 - \xi_{T0} - \xi_{S0}}{2} \left[\cos (\alpha_M (T - M_f)) + 1 \right] + \xi_{T0}.$	S12
	C8	$\begin{cases} \xi_S = \frac{1 - \xi_{S0}}{2} \cos \left[\frac{\pi}{\sigma_s - \sigma_f} (\sigma^* - \sigma_f - Cm(T - M_s)) \right] + \frac{1 + \xi_{S0}}{2} \\ \xi_T = \xi_{T0} - \frac{\xi_{T0}}{1 - \xi_{S0}} (\xi_S - \xi_{S0}) \end{cases}$	S10
Transitions	S1 S2 S3 S4 S5 S6 S7 S8 S9 S10 S11 S12 S13 S14 S15 S16 S17 S18	$\begin{aligned} &\sigma > \sigma_s \\ &\sigma > \sigma_f \\ &\sigma < \sigma_f \\ &T < M_f, \sigma < \sigma_s \\ &T > A_s, \sigma > Ca(T - A_s) \\ &T > A_s, Ca(T - A_s) > \sigma > Ca(T - A_f) \\ &T > A_f, \sigma < Ca(T - A_f) \\ &\sigma_s + Cm(T - M_s) \geq \sigma \geq Ca(T - A_s) \\ &T < A_s, \sigma < \sigma_s + Cm(T - M_s) \\ &\sigma_f + Cm(t - M_s) < \sigma < \sigma_s + Cm(T - M_s) \\ &T < M_s, \sigma_s < \sigma < \sigma_f \\ &M_f < T < M_s, \sigma < \sigma_s \\ &T > A_s, \sigma > Ca(T - A_s) \\ &\sigma \geq \sigma_f + Cm(T - M_s) \\ &T > A_s, \sigma > Ca(T - A_s) \\ &T < M_s, \sigma < \sigma_s \\ &\frac{dT}{dt} < 0, T < A_s, \sigma < \sigma_s + Cm(T - M_s) \\ &\frac{d\sigma}{dt} < 0 \vee \frac{d\sigma}{dt} > 0 \end{aligned}$	
General Equations		$\begin{cases} \xi = \xi_S + \xi_T; \\ D = D_A + \xi(D_{TW} + \xi_s(D_{DTW} - D_{TW}) - D_A); \\ \Omega = -\varepsilon_L D; \\ \varepsilon = (\sigma - \sigma_0 + D_0 \varepsilon_0 - \Omega \xi_s + \Omega \xi_s)/D; \\ \sigma = \sigma_0 + D\varepsilon - D_0 \varepsilon_0 + \Omega \xi_s - \Omega_0 \xi_{S0}; \end{cases}$	



(a)



(b)



(c)

Figure 2.18. Path for a) mechanical load and unload, b) thermal load, and c) thermal unload.

strain relationship, strain evolution as a function of temperature, and stress evolution as a function of temperature in subfigures 2.20(a), 2.20(b), and 2.20(a), respectively. These graphs offer a comprehensive view of the SMA's behavior under the simulated conditions, allowing for a thorough

validation of the model's predictive capabilities. Through these simulations, it was possible to validate the model's ability to accurately predict the stress-strain relationship under various loading conditions, phase transformations in response to temperature changes, the hysteresis behavior characteristic of SMAs, and the material's response to complex loading scenarios involving both mechanical and thermal inputs. The close alignment between the simulated behavior and the theoretical phase diagram, as well as the consistent results across different representations of the data, provides strong evidence for the model's validity and its capability to capture the unique properties of Shape Memory Alloys. This comprehensive validation process ensures that the model can reliably simulate SMA behavior across a wide range of conditions, making it a valuable tool for the design process.

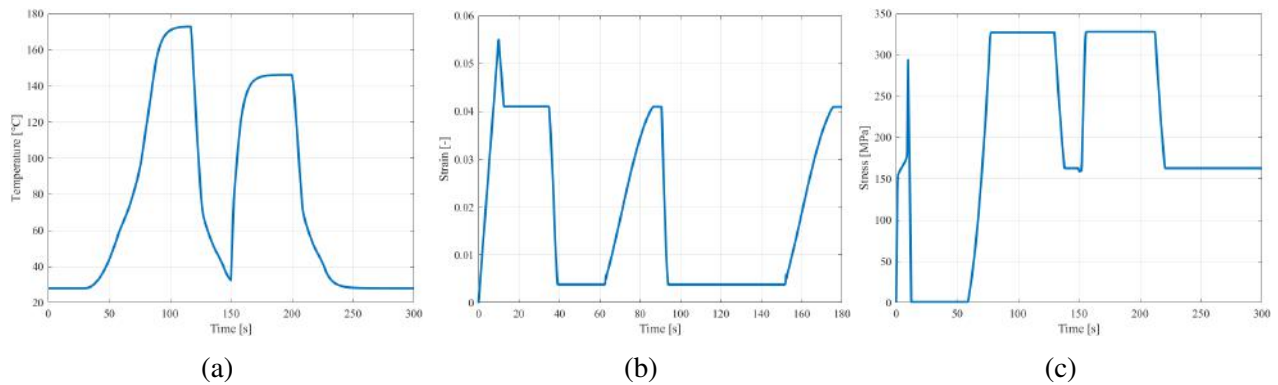


Figure 2.19. a) Temperature evolution over time b) Strain evolution over time c) Stress evolution over time.

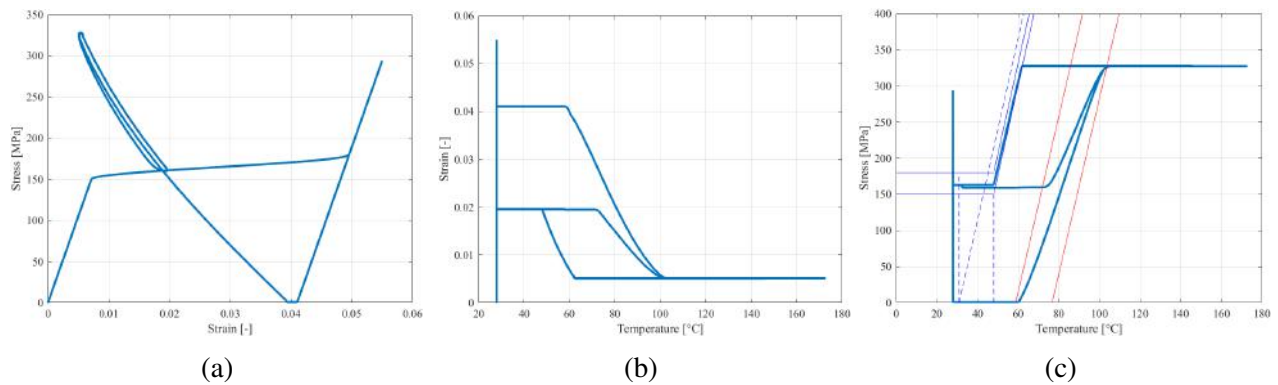


Figure 2.20. a) Stress-strain b) Strain-temperature c) Stress-temperature evolution on the diagram phase.

Following the initial validation, the model underwent a significant refinement to better align with experimental data. This second version of the model was specifically adjusted to account for the complete recovery observed when the material retransforms into martensite, a phenomenon clearly demonstrated in the experimental results. Figure 2.21 and Figure 2.22 illustrate this behavior, showcasing the model's improved accuracy in predicting the material's full recovery cycle. It is important to note that all simulations, including those for this refined model, were conducted with an antagonistic element in place. This element introduced a variable load that changed non-proportionally with displacement, thus replicating operating conditions more accurately. This approach not only tested the model's robustness but also its ability to handle complex, non-linear loading scenarios. The incorporation of such a dynamic loading condition further validates the model's applicability to a wide range of practical situations where SMAs might be employed. The refined model's ability to accurately predict behavior under these challenging conditions underscores its potential as a powerful tool for both research and practical applications in the field of Shape Memory Alloys.

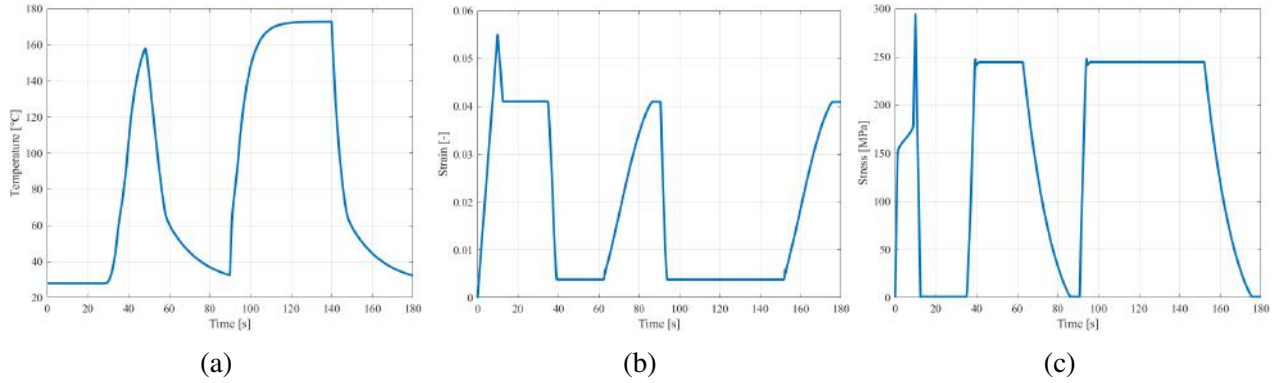


Figure 2.21. a) Temperature evolution over the time b) Strain evolution over time c) Stress evolution over time.

Figure 2.23 demonstrates the model's robust performance in simulating cyclic transformation with partial activation levels. This simulation showcases the model's ability to accurately predict SMA behavior under complex, real-world conditions. It's important to note that this simulation was conducted by coupling the material model with a multibody model simulating a mass suspended by the SMA wire. Consequently, the system's dynamics significantly influence the

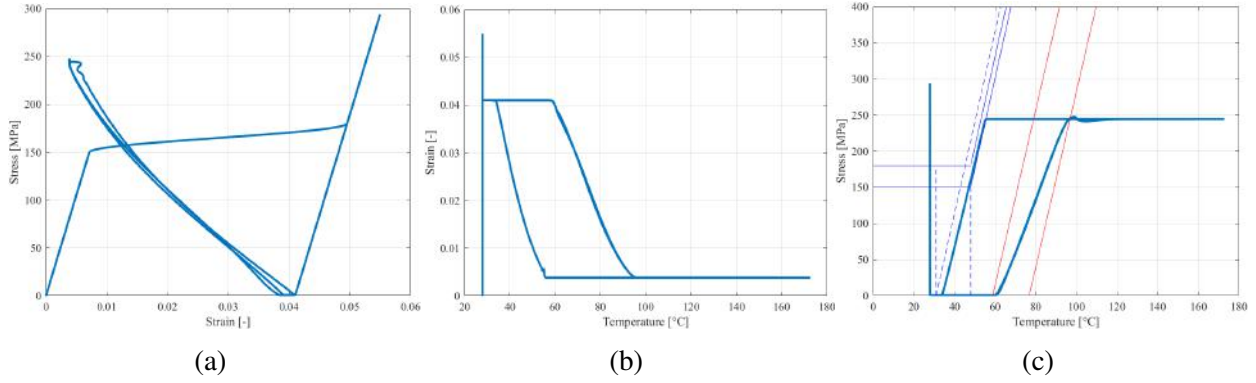


Figure 2.22. a) Stress-strain b) Strain-temperature c) Stress-temperature evolution on the diagram phase.

graphs presented. This coupled approach provides a more comprehensive representation of the SMA's behavior in practical applications, where the material's properties interact with the mechanical system it's integrated into. The figure clearly illustrates how the model effectively captures the SMA's response to repeated partial activations, accounting for both the material's intrinsic properties and the external dynamic forces acting upon it. This level of detail in the simulation underscores the model's versatility and its potential.

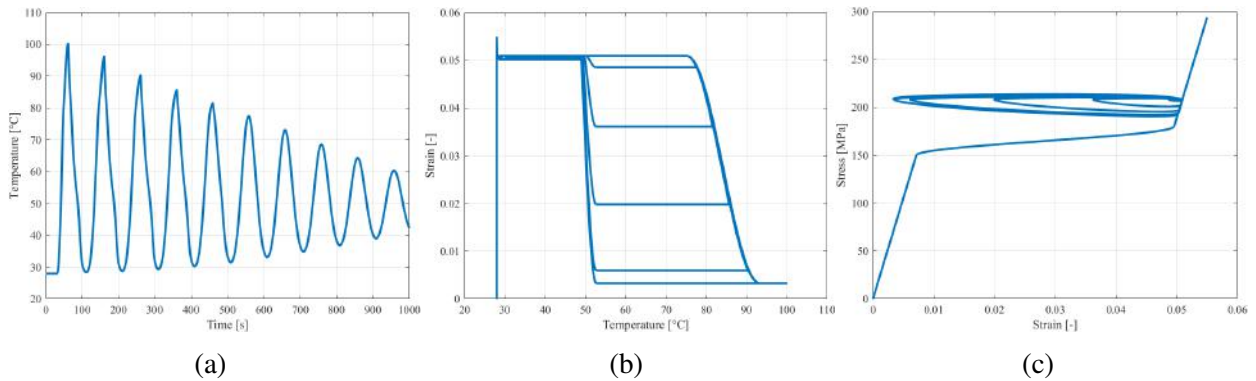


Figure 2.23. a) Temperature over the time b) Stress-temperature evolution on the diagram phase c) Stress-strain evolution.

Figures 2.24 and 2.25 present additional cases of complex inputs, highlighting the model's capability to simulate partial transformations during both cooling and heating processes. These examples further demonstrate the model's ability to handle a wide range of scenarios, confirming its flexibility in addressing various loading and temperature conditions. The accurate simulation

of partial transformations in both thermal directions is particularly significant, as it represents a common challenge in SMA modeling and reflects situations frequently encountered in practical applications of these materials.

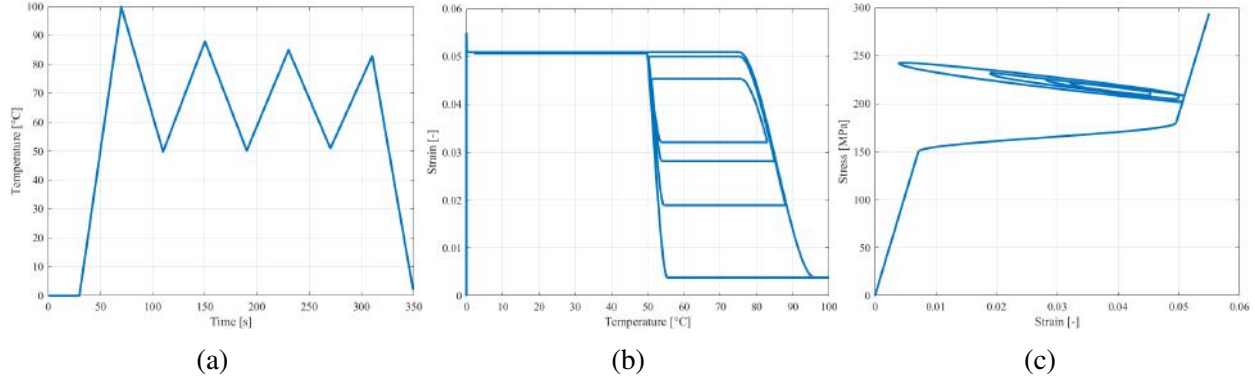


Figure 2.24. a) Temperature over the time b) Stress-temperature evolution on the diagram phase c) Stress-Strain evolution.

It's worth noting that the tests in both figures were conducted using the same thermal input, but with different loading conditions. In Figure 2.24, the external load was generated by coupling a multibody model of a suspended mass with a spring in series. This setup clearly reveals all the dynamic aspects of the system, including oscillations and transient responses. In contrast, Figure 2.25 shows the results when a constant resistive load was applied. This comparison allows for a comprehensive evaluation of the model's performance under both dynamic and static loading conditions, further validating its robustness and versatility in simulating SMA behavior across various real-world scenarios.

2.3.9 Control Law

The high hysteresis, actuation non-linearities and significance in temperature control motivated to use $\mathcal{L}_1$ adaptive control theory. Consider a nonlinear time-invariant affine-in-control system in the form of:

$$\dot{x}(t) = \mathbf{f}(x(t)) + \mathbf{g}(x(t), u(t)), \quad (2.79)$$

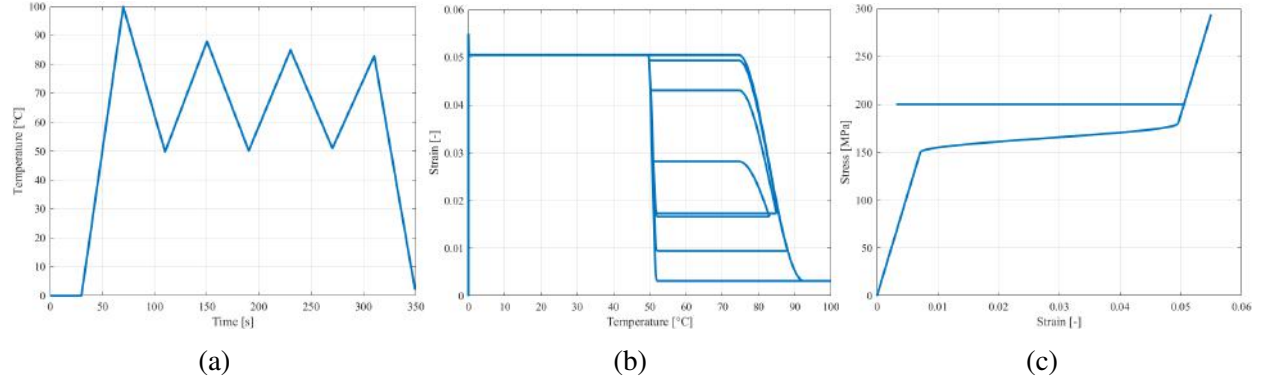


Figure 2.25. a) Temperature over the time b) Stress-temperature evolution on the diagram phase c) Stress-Strain evolution.

where $x(t) \in \mathbb{R}^n$ is the state vector of the sum, $u(t) \in \mathbb{R}^m$ is the input vector. $\mathbf{f}(x(t)) : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $\mathbf{g}(x(t), u(t)) : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$ are vector-valued smooth piece wise-continuous functions that satisfy the Lipschitz condition.

The nonlinear system represented in Equation (2.79) can be transformed into a state-space system of

$$\begin{aligned}\dot{x}(t) &= \mathbf{A}_p x(t) + \mathbf{B}_p (u(t) + \mathbf{f}(t, x(t))), \\ \mathbf{y}(t) &= \mathbf{C}_p x(t) + \mathbf{D}_p u(t),\end{aligned}\tag{2.80}$$

The dynamic characteristics of SMA are defined by Equation (2.62) or (2.63) together with (2.78) and (2.64). Let's define d_e as the integral of errors, i.e.,

$$\dot{d}_e(t) = \varepsilon(t) - \varepsilon_{ref}\tag{2.81}$$

where $\varepsilon(t)$ is the SMA strain and ε_{ref} is the reference trajectory.

Considering the discrete time, the above mathematical problem can be converted into a control problem. The state vector $x(t)$, input vector $u(t)$, and the output vector $y(t)$ can be defined as follows.

$$x(t) = \begin{bmatrix} \varepsilon(t) \\ T_{\Delta}(t) \\ \xi(t) \\ d_e(t) \end{bmatrix}, \quad u(t) = I^2(t), \quad y(t) = \varepsilon(t). \quad (2.82)$$

2.3.9.1 Linearize the Heating Dynamics

The key nonlinear dynamics are described by the phase fraction $\xi(t)$ during the heating and cooling in Equations (2.62) and (2.63). Let's linearize these equations around the operating point (T_0, σ_0) . The perturbations around the operating point $\Delta T(t) = T(t) - T_0$, $\Delta \sigma(t) = \sigma(t) - \sigma_0$. The first-order Taylor expansion of the cosine term gives the heating dynamics as

$$\cos \left[a_A \left(T(t) - A_s - \frac{\sigma(t)}{C_A} \right) \right] \approx \cos(\theta_A) \cdot a_A \left(\Delta T(t) - \frac{\Delta \sigma(t)}{C_A} \right),$$

where

$$\theta_A = a_A \left(T_0 - \frac{\sigma_0}{C_A} \right)$$

Thus, the linearized phase fraction during heating becomes:

$$\xi(t) \approx \frac{\xi_0}{2} (\cos \theta_A + 1) - \frac{\xi_0}{2} \sin \theta_A \cdot \alpha_A \left(\Delta T(t) - \frac{\Delta \sigma(t)}{C_A} \right)$$

Taking the derivatives with respect to time:

$$\dot{\xi}(t) \approx -\frac{\xi_0}{2} \sin \theta_A \cdot \alpha_A \left(\dot{T}(t) - \frac{\dot{\sigma}(t)}{C_A} \right). \quad (2.83)$$

2.3.9.2 Linearize the Cooling Dynamics

Let the cooling Equation 2.63 linearizing around (T_0, σ_0) :

$$\xi(t) \approx \frac{1-\xi_0}{2} \cos \theta_M + \frac{1+\xi_0}{2} - \frac{1-\xi_0}{2} \sin \theta_M \cdot \alpha_M \left(\Delta T(t) - \frac{\Delta \sigma(t)}{C_M} \right)$$

where

$$\theta_M = \alpha_M \left(T_0 - M_f \frac{\sigma_0}{C_M} \right)$$

Taking the derivatives with respect to time:

$$\dot{\xi}(t) \approx -\frac{1-\xi_0}{2} \sin \theta_M \cdot \alpha_M \left(\dot{T}(t) - \frac{\dot{\sigma}(t)}{C_M} \right). \quad (2.84)$$

The linearized state-space model around the operating point (T_0, σ_0) is:

$$A_p = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & -\frac{hA}{mc_p(\xi_0)} & 0 & 0 \\ 0 & -\frac{\xi_0}{2} \sin(\theta_A) \cdot a_A \left(\frac{1}{mc_p(\xi_0)} \cdot R(\xi_0) - \frac{hA}{mc_p(\xi_0)} \right) & -\frac{\xi_0}{2} \sin(\theta_A) \cdot a_A & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

$$B_p = \begin{bmatrix} 0 \\ \frac{R(\xi_0)}{mc_p(\xi_0)} \\ -\frac{\xi_0}{2} \sin(\theta_A) \cdot a_A \cdot \frac{R(\xi_0)}{mc_p(\xi_0)} \\ 0 \end{bmatrix},$$

$$C_p = [1 \quad 0 \quad 0 \quad 0],$$

$$D_p = [0].$$

2.3.10 Experimental Validation

In this section, the proposed $\mathcal{L}_1$ robust adaptive control algorithm is experimentally validated for the Ni-Ti SMAs. The schematic representation is presented to provide the experimental setup and SMA arrangement and to denote the notations. The tracking results are presented afterward.

2.3.10.1 Experimental setup for the Stress-strain Feedback Controller

One end of the SMA (the top end) was fixed to the experimental setup, while the other end functioned either as a sliding joint or a fixed joint, depending on the specific configuration described later. The top fixed end was connected to a load cell (Model: FUTEK LSB200) to measure the force generated by the SMA during actuation. The displacement of the bottom end (when configured as a sliding joint) was measured using a laser displacement sensor (Model: Keyence LK-G157). The SMA temperature was monitored using a handheld thermal camera (Model: FLIR E95).

Actuation of the SMA was achieved using a DC power supply (limited to 1.1 A) and a voltage control circuit incorporating a Darlington pair transistor (Model: TIP122). The base of the Darlington pair was driven by a low-voltage PWM signal from a microcontroller (Model: Arduino Mega 2560). Sensor data were acquired through a data acquisition module (Model: National Instruments myDAQ) and transmitted to a SIMULINK/MATLAB program, where the $\mathcal{L}_1$ adaptive controller was implemented. The controller's output, corresponding to the desired actuation voltage, was sent back to the microcontroller to regulate the SMA input. The SIMULINK/MATLAB controller was operated in real-time based on the data acquisition module's time stamp.

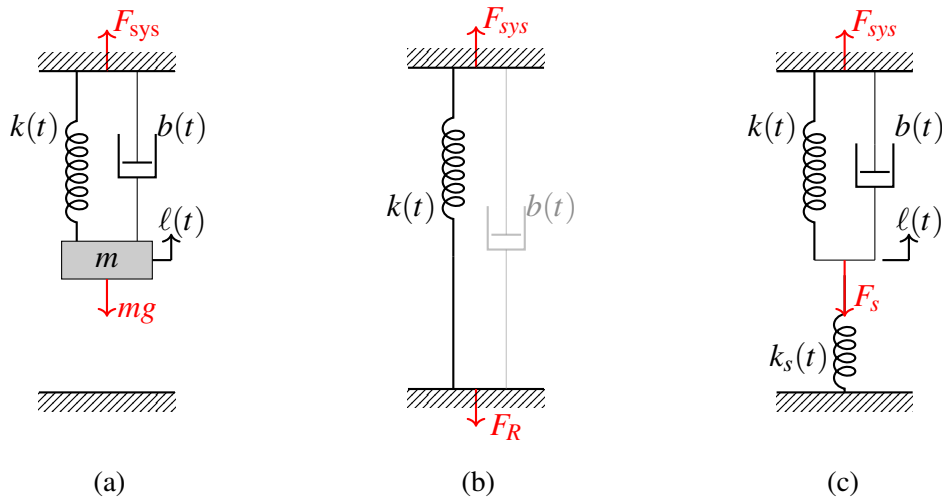


Figure 2.26. Schematic representation of the experimental setups for a) constant stress, b) constant strain, and c) dynamic force configuration for TCAMs testing.

The SMA is actuated for three generalized configurations to evaluate the tracking performance. The three configurations are 1) Constant stress configuration, 2) Constant strain configuration, and 3) Dynamic force configuration, as in Figure 2.26.

2.3.10.2 Constant stress configuration

An external load ($F_e = mg$) is attached to the SMA, and the length $\ell(t)$ is measured using the displacement sensor. Initial length of the SMA is indicated by $\bar{\ell}$. Figure 2.26(a) shows the schematic representation of configuration 1. Since only the displacement changes, the generalized dynamic equation can be rewritten as,

$$m\ddot{\ell}(t) + b\dot{\ell}(t) + k\ell(t) = F_e \quad (2.85)$$

The $\mathcal{L}_1$ displacement controller was implemented using Equation (2.27), as derived in Section 2. Tracking performance for different SMA displacements over time (ranging from 1 mm to 4 mm) is plotted in Figure 2.27. The controller parameters were initially tuned for 1 mm displacement tracking and then applied unchanged to track the other displacement setpoints. Corresponding temperature measurements and voltage inputs to the SMA are also shown in the same figure. The $\mathcal{L}_1$ tracking results demonstrate accurate performance and rapid adaptation across all cases.

For comparison, a PID controller was also implemented and tuned for 1 mm displacement.

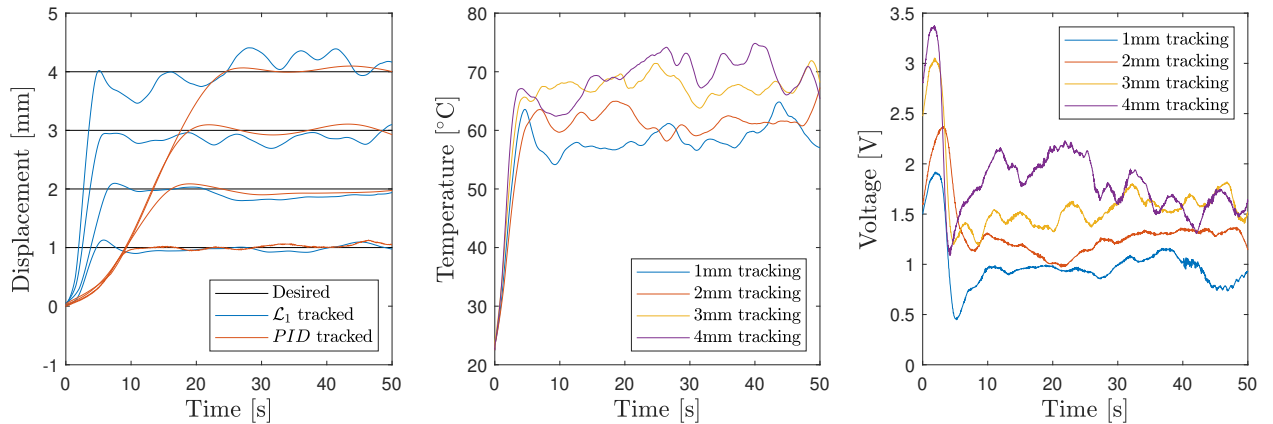


Figure 2.27. SMA position tracking validation using $\mathcal{L}_1$ adaptive control architecture for constant stress configuration

The same PID parameters were used to track the remaining displacements. While the PID controller achieved relatively smoother responses, its tracking was slower and less responsive.

It is important to emphasize that fast adaptation is crucial for controlling highly nonlinear materials like SMAs, which must operate within strict temperature limits to prevent permanent deformation. These results clearly indicate that the $\mathcal{L}_1$ adaptive controller outperforms the PID controller in terms of both tracking accuracy and responsiveness.

2.3.10.3 Constant strain configuration

This configuration represents a scenario in which the SMA acts purely as a force-generating actuator without experiencing any deformation. Consequently, the displacement remains zero while internal stress varies within the material. Both ends of the SMA were rigidly fixed, and the output force, denoted as F_{sys} , was measured using a load cell. In this setup, $F_{\text{sys}} = F_R$, where F_R represents the reaction force generated during SMA actuation. A schematic of Configuration 2 is shown in Figure 2.26(b), and the system dynamics are governed by:

$$k\bar{\ell}(t) = F_R \quad (2.86)$$

The $\mathcal{L}_1$ force controller was implemented based on Equation (2.28) in Section 2. The force tracking performance for various desired force levels is illustrated in Figure 2.28. Consistent experimental procedures and parameter tuning were used for both the $\mathcal{L}_1$ and PID controllers to ensure a fair comparison. An initial prestress of 4.9 N was applied to standardize initial conditions across trials.

As observed in Configuration 1, the $\mathcal{L}_1$ robust adaptive controller demonstrated better tracking performance and rapid adaptation. Although it exhibited slightly greater fluctuations than the PID controller, this trade-off was offset by its faster response—an essential feature when dealing with highly nonlinear materials such as SMAs.

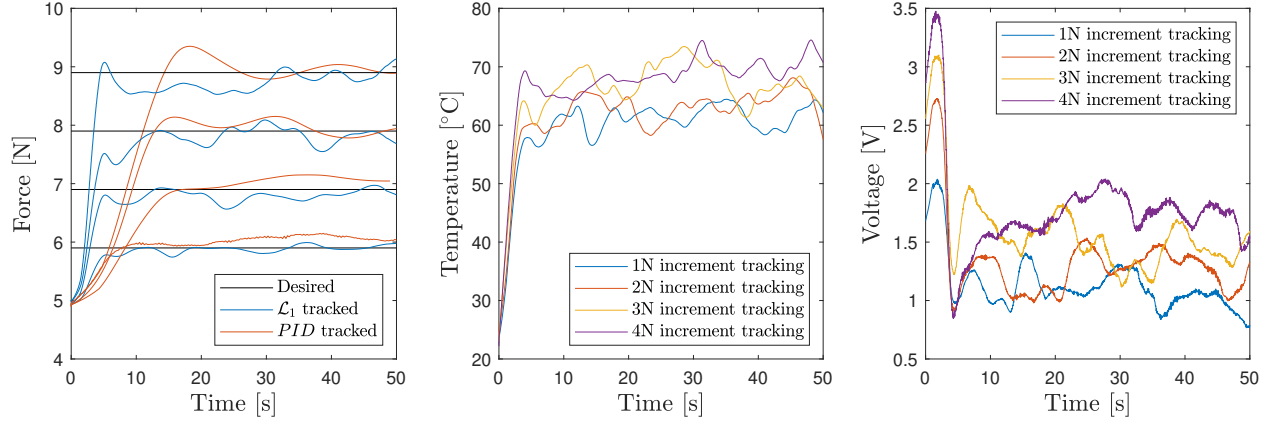


Figure 2.28. SMA force tracking validation using $\mathcal{L}_1$ adaptive control architecture for constant strain configuration

2.3.10.4 Dynamic force configuration

In Configuration 3, a variable external force is applied to the SMA via a series-connected Hookean spring. During SMA actuation, both the system force and the displacement of the SMA's free end are measured using a load cell and a displacement sensor, respectively. The force exerted by the spring, denoted as F_s , is given by:

$$F_{sys} = k_s(\bar{\ell} + \ell_s - \ell(t)), \quad (2.87)$$

where k_s is the spring stiffness, ℓ_s is the initial displacement of the spring from its equilibrium position, and $\bar{\ell}$ is the initial length of the SMA wire. Based on this setup, the second-order dynamics of the system can be described as:

$$m\ddot{\ell}(t) + b\dot{\ell}(t) + k\ell(t) = k_s(\bar{\ell} + \ell_s - \ell(t)). \quad (2.88)$$

A schematic of Configuration 3: Dynamic Force Configuration is shown in Figure 2.26(c). In this arrangement, the SMA wire is connected in series with the external spring and tensions the system until it reaches the desired initial force condition (4.9 N).

The validation for this configuration is twofold. In **Step 1**, the SMA underwent displacement tracking using Equation (2.27), as derived in Section 2. The experimental protocol was the

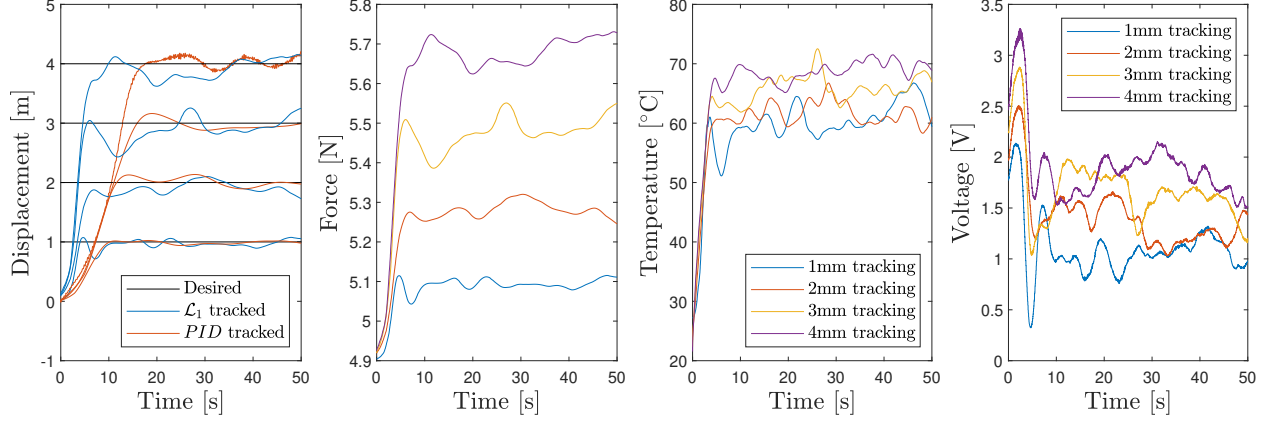


Figure 2.29. SMA position tracking validation using $\mathcal{L}_1$ adaptive control architecture for dynamic configuration

same as in the previous configurations. One key difference was the simultaneous acquisition of displacement and force plots. A portion of the power generated by the SMA contributed to displacement, while the remainder was used for force generation. During Step 1 of Configuration 3, displacement tracking was performed.

As in Configurations 1 and 2, the $\mathcal{L}_1$ adaptive controller showed better tracking performance compared to the PID controller. Temperature and supply voltage were also measured, as in the previous setups. According to Figure 2.29, the $\mathcal{L}_1$ tracking results demonstrate superior performance in terms of faster response and robustness. The force profiles reflect the internal forces generated during SMA actuation, and the temperature plots confirm that the SMA wire remained within safe temperature limits in all tracking scenarios.

For **Step 2** of the dynamic force Configuration 3 experiments, the $\mathcal{L}_1$ adaptive controller was implemented as a force tracking controller based on Equation (2.28) in Section 2. The SMA wire was actuated using force feedback from the load cell. The tracking results for both $\mathcal{L}_1$ and PID force tracking, along with the corresponding SMA displacement, temperature, and supply voltage, are shown in Fig. 2.30.

These results follow a similar format to the plots in Fig. 2.28. The parameters were defined consistently with those used in previous experiments; that is, they were tuned for the initial desired value and then kept constant to track the remaining desired trajectories for both the $\mathcal{L}_1$ and PID

controllers. Notably, the $\mathcal{L}_1$ controller demonstrated faster convergence to the desired force levels compared to the PID controller, highlighting its effectiveness in dynamic force tracking scenarios.

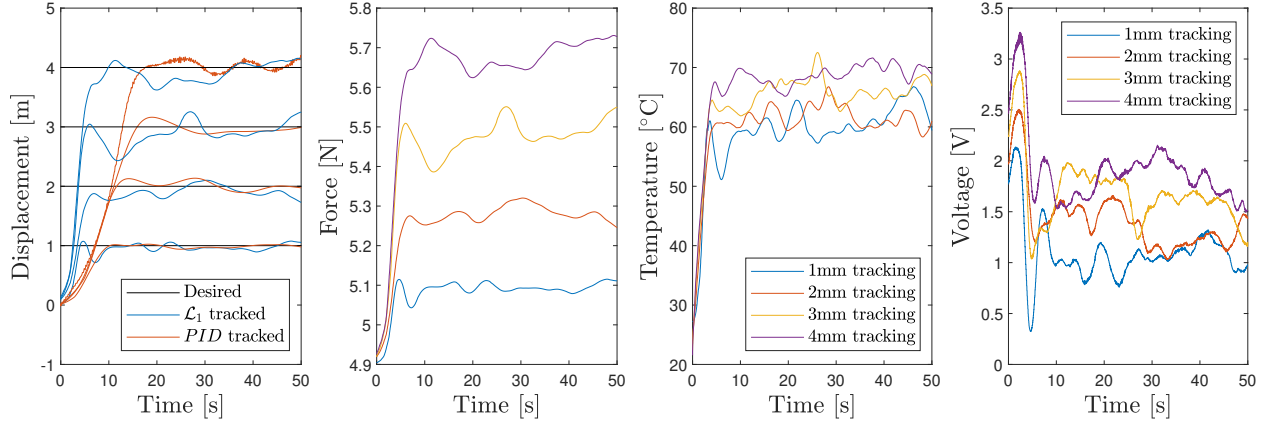


Figure 2.30. SMA force tracking validation using $\mathcal{L}_1$ adaptive control architecture for dynamic configuration

In conclusion, experimental validation confirms that the proposed $\mathcal{L}_1$ robust adaptive controller consistently outperforms the conventional PID controller across all three SMA configurations—constant stress, constant strain, and dynamic force. In each scenario, the $\mathcal{L}_1$ controller demonstrated faster response, higher tracking accuracy, and improved robustness, particularly under dynamic and nonlinear conditions. This is especially critical for smart materials like SMAs, which are sensitive to thermal and mechanical stress.

Whether tasked with tracking displacement or force, the $\mathcal{L}_1$ controller adapted quickly and maintained performance using the same set of tuned parameters across varying setpoints, an essential feature for real-world applications requiring reliability and scalability. Notably, in the dynamic force configuration, the controller achieved rapid convergence to the desired force levels while keeping the SMA within safe operating temperatures.

Overall, these experiments demonstrate that the $\mathcal{L}_1$ adaptive control strategy provides a compelling solution for the precise and safe operation of shape memory alloy actuators, laying a solid foundation for their deployment in complex, real-time systems.

CHAPTER 3. ARTIFICIAL MUSCLES FOR REHABILITATION ROBOTICS APPLICATIONS

Chapter 3 extends the theoretical modeling and control frameworks developed in earlier chapters into practical applications in the field of rehabilitation robotics. The focus is on utilizing twisted and coiled artificial muscles (TCAMs) and shape memory alloy (SMA) actuators to develop innovative robotic devices for rehabilitation.

Chapter 3.1 introduces a soft exoskeleton glove for wrist rehabilitation powered by TCAMs. This section details the design and development of the exoskeleton, emphasizing the integration of TCAMs for actuation and the initial validation of the device using Electromyography (EMG) signals to monitor muscle activity and ensure effective rehabilitation.

Chapter 3.2 presents developing and controlling a variable stiffness actuator for an Ankle-Foot Orthosis (AFO) powered by carbon fiber TCAMs. The chapter covers the complete design process, including the control circuitry and architecture. The proposed circuitry effectively manages power distribution among the TCAMs, while the controller allows for individual actuation based on the rehabilitation needs. Experimental validation of the actuator's torque generation is performed for critical static angles, and the results are discussed in detail.

Chapter 3.3 extends the application of the $\mathcal{L}_1$ adaptive control framework to SMA actuators used in a finger exoskeleton glove. This section provides a comprehensive analysis of the kinematic calculations and presents the results of finger angle tracking experiments, demonstrating the efficacy of the proposed control strategy in enhancing the performance and adaptability of the rehabilitation device.

Through these applications, this chapter illustrates the potential of artificial muscles in developing advanced robotic systems for rehabilitation, highlighting their versatility, effectiveness, and ability to provide targeted assistance in diverse assistive robotics contexts.

3.1 Lightweight Bioinspired Exoskeleton for Wrist Rehabilitation Powered by TCAMs

3.1.1 Introduction

Strokes, cerebral palsy [166–168], and spinal cord injuries [169, 170] lead to impairments and disabilities with negative physiological impacts on the activities of daily living (ADLs). In particular, stroke represents one of the most common causes of limb impairment. Over 80% of stroke survivors demonstrate weakness [171], loss of dexterity, spasticity, and pain [172–175] in their upper limbs [176]. The Heart Disease and Stroke Statistics reported around 0.8 million stroke episodes in the United States [177] in 2019 and 1.6 million cases in Europe [178] in 2015. Costs of around 45.5 billion dollars for the United States and 45 billion euros for the European Union are estimated annually to cover medical expenses related to stroke rehabilitation [177, 178]. Part of these costs is related to expensive and time-consuming hospital-based rehabilitation procedures. There is a critical need to develop wearable and inexpensive rehabilitation devices to allow for home rehabilitation [179–182]. Rehabilitation robotics have made significant progress in the past few decades, proposing robotic devices able to reproduce rehabilitation exercises and help patients gain range of motion (ROM) and strength of upper limbs after a stroke [183–186]. These active rehabilitation devices can be mainly classified into two groups: end effectors (EE) and EXOskeletons (EXO) [187]. End effectors are attached to the patient at one distal point, and there is no correspondence between the degrees of freedom (DOF) of the device and those of the human body. Due to a single attachment point, providing motion to a single joint at a time is usually challenging. Figure 3.1(a) represents a schematic representation of an EE attached to a patient’s arm. For this reason, controlling the position, force, and torque is highly challenging, resulting in uncomfortable motion and uncontrollable loads. However, EE devices are easy to design and can fit a wide range of patients [30, 42, 187–193]. On the other hand, EXOskeletons are characterized by many attachment points, providing the same DOF as human joints with broader ROM compared to EE. In this case, there is a correspondence between biological joints and device joints, and the device can provide fine anthropomorphic motion and wide ROM compared to EEs. However, their

design is more complex, less adaptable, and more customizable than the EE [30, 42, 189–195]. Due to the presence of many joints, EXO devices provide better control motion [192] compared to EE. Among the EXOs, further classification is possible considering structural connections between the patient and an external frame. When the EXO is linked to an external frame, it is called grounded [196]; otherwise, it is called ungrounded. Figure 3.1(b) shows a grounded EXO, while Figure 3.1(c) represents an ungrounded EXO.

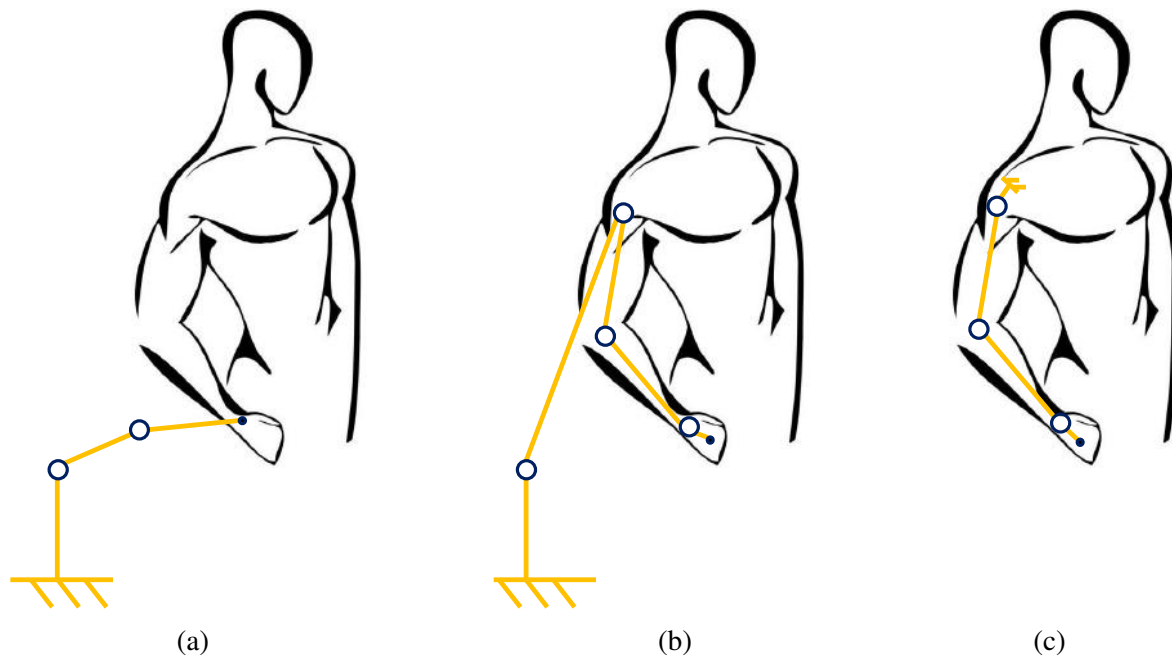


Figure 3.1. (a) Schematic representation of end effector, where the connection with the patient is provided by just one point; (b) grounded exoskeleton. The device joints match the human ones with a connection to an external frame; (c) ungrounded exoskeleton. The device has no attachments with an external frame.

Over time, many passive and active assist EXO devices have been developed to assist and aid the wrist in recovering mobility during rehabilitation [197, 198]. Passive-assist devices provide motion to all the joints, while the user does not provide any active muscle contraction. On the other hand, active-assist devices assist the users with “extra strength” while they are actively performing a rehabilitation exercise by compensating for muscular/joint weakness.

In 2012, Pehilvan et al. developed “Rice Wrist-S” for wrist rehabilitation after spinal cord injury and stroke [199]. Rice Wrist-S is a grounded EXO device with one DOF for forearm supina-

tion/pronation and two DC motor-actuated active DOFs for wrist flexion/extension (F/E) and ulnar/radial deviation (U/R). It is composed of a serial mechanism of revolute joints. The ROM device can provide wrist F/E and U/R deviation of 120° and 70° , respectively. The patient grips a handle that is coupled through a linear joint at the device end to prevent misalignments of the device joints. Higher torque output, increased ROM, and low cost due to a serial mechanism instead of a DC motor are some of the pros of this Rice Wrist-S.

In 2013, Martinez et al. [200] developed the “Wrist Gimbal”, a 3-DOF device for wrist and forearm rehabilitation. This grounded device has a serial kinematic configuration of three revolute joints as Rice Wrist-S. The frame mainly comprises 3D-printed ABSplus (Acrylonitrile, Butadiene, and Styrene) components and aluminum bars to reach a sturdy but lightweight structure. Three DC motors power the device for each three-active DOF, and the moment is delivered to the handle through cable drive systems. The ROM is 180° for wrist F/E and 60° for U/R. The wrist gimbal includes a grasping handle for the user’s convenience. Pro: Significant ROM makes this device outstanding among other wrist rehabilitation devices. The use of heavy-gear DC motors is a notable con of this device.

In 2014, French et al. [201] presented “MAHI EXO-II”, a grounded robotic exoskeleton for upper limb rehabilitation. The device offers four active DOFs wrist: F/E, U/R, forearm supination/pronation (S/P), and elbow flexion/extension, with shoulder abduction/adduction (Ab/Ad) as passive DOF. The passive DOF aims for better user comfort. The frame is composed of revolute joints for the elbow and forearm, while the wrist structure involves a parallel mechanism of revolute–prismatic–spherical joints. A handle is placed at the end for the patient’s grasping. A DC motor actuates each DOF except for the wrist module, which needs three DC motors. In addition, the transmission is delivered through cable systems and capstans. The wrist range of motion provided is 65° and 630° for the wrist F/E and U/R. Pros: workspace, total torque output, and bandwidth match human capabilities. Cons: Conventional actuators make the system complex and bulky, requiring a complex control system.

In 2017, Pezent et al. developed “MAHI OpenWrist” [202], an evolution of Rice Wrist-S.

It has been designed to work with the ReNeu Maestro hand exoskeleton module in the combined READAPT device for hand–wrist rehabilitation. OpenWrist, as Rice Wrist-S, is a grounded EXO device involving the same DOFs with a wide ROM of 135° for F/E and 75° U/R. The structure is composed of a serial mechanism of revolute joint, and the actuation system employs brushed DC motors and cables. Conventional DC motors make the device heavier, which can be classified as a drawback.

Yin-Yu Su et al. [203] presented a 3–DOF grounded wrist rehabilitation robot in 2019. This device performs flexion/extension ($-70^\circ \sim 70^\circ$), ulnar/radial deviation ($-50^\circ \sim 50^\circ$), and forearm supination/pronation ($-75^\circ \sim 75^\circ$) as active ROM. The device consists of revolute joints and a passive spherical mechanism to adjust the device upon user measurements [203]. The device is designed to accurately control the force and stiffness; hence, series elastic actuators (SEA) were attached to the stepper motors. Pros: lightweight, compact device. Cons: passive joint could be misaligned for patients with larger forearms.

POWROBOT is a 3-DOF grounded wrist rehabilitation device proposed in 2022 by Mayetin and Kucuk [204], which can perform wrist flexion/extension, ulnar/radial deviation, and forearm supination/pronation in ranges of ($-80^\circ \sim 80^\circ$), ($-30^\circ \sim 30^\circ$), and ($-80^\circ \sim 80^\circ$), respectively. Three geared motors actuate the rehabilitation robot. Pros: this is a portable device, easily manufacturable, and low-cost for materials and control systems. Cons: ROM is slightly less than existing devices yet sufficient for rehabilitation exercises.

Goncalves et al. [205] proposed a grounded, single-DOF servo motor-operated wrist rehabilitation device in 2020. It has an active joint for flexion/extension motion. Due to the nature of its design, the remaining two passive motions make it a 3-DOF device. The ROM for each motion is as follows: wrist flexion/extension ($-85^\circ \sim 85^\circ$), ulnar/radial deviation ($-15^\circ \sim 55^\circ$), forearm supination/pronation ($-90^\circ \sim 85^\circ$). Pros: user-friendly (due to the controller interface), low-cost, and easily operatable device. Cons: it has a limited ROM for passive DOF.

In 2019, Zhang et al. [206] developed the PWRR wearable wrist rehabilitation robot, which has two DOFs ranging ($-65^\circ \sim 65^\circ$) flexion/extension and ($-80^\circ \sim 25^\circ$) ulnar/radial deviation.

The grounded device consists of universal joints actuated by two pneumatic actuators. Xu et al. developed the wrist rehabilitation device (WReD) [207] in 2018, consisting of two DOF passive revolute joints. Only a single DC motor is used in this grounded wrist rehabilitation device. The ROM of flexion/extension is $(-90^\circ \sim 90^\circ)$, and ulnar/radial deviation is $(-90^\circ \sim 90^\circ)$. Pros: this device is compact, conveniently wearable, and flexible compared to other parallel devices. Cons: the overall system is bulky.

A cable-driven 3-DOF wrist rehabilitation robot: SEU was developed by Ke Shi et al. in 2020 with the ROM of flexion/extension $(-40^\circ \sim 40^\circ)$, ulnar/radial deviation $(-75^\circ \sim 75^\circ)$, forearm supination/pronation $(-90^\circ \sim 90^\circ)$ [208]. This is a ground device and is bulky due to its construction. Each passive motion was actuated using Bowden cables connected to the DC motor via MR clutches for each cable. Pros: better safety, high force, larger workspace than cable-driven systems. Cons: certain characteristics of cables make them less convenient to use in the clinical environment.

A lightweight shape memory alloy (SMA)-based wearable robot, named “Soft Wrist Assist (SWA),” was proposed by Jeong et al. [209] in 2019. It is a 2-DOF device for wrist F/E and U/R. The ungrounded device provides high portability and compliance for better patient comfort and consists of a wearable fingerless glove and a wearable strap. The glove is worn on the hand, while the strap is placed on the forearm. The actuation system consists of five muscle-like actuators (SMA coil springs). Three of these actuators are placed over the back of the hand, and the remaining two are over the palm. The actuators are activated via joule heating and are cooled down with water using a pump for the water motion. The ROM provided is 33.8° , 30.4° , 15.4° , and 21.4° for flexion, extension, ulnar, and radial deviation, respectively (64.2° and 36.9° degrees for the 2-DOF). The device weighs 0.151 kg for the wearable part, but the pump and radiator weigh up to around 1 kg. Pros: This is a flexible, lightweight, easier-to-wear device. Cons: certain components of the device tend to dislocate despite the anchors; in addition, the device is not flexible enough for people with larger forearms.

A pneumatic soft artificial muscle-actuated parallel wrist rehabilitation robot developed by

Wang et al. in 2021 [210] is a grounded 3-DOF device. The ROM for each motion is as follows: wrist flexion/extension ($-73^{\circ} \sim 71^{\circ}$), ulnar/radial deviation ($-19^{\circ} \sim 33^{\circ}$), forearm supination/pronation ($-71^{\circ} \sim 86^{\circ}$). Six linear actuators have been developed using pneumatic artificial muscles (PAM), and a stepper motor is available for central motions. Pros: this is a high degree of flexibility, adaptability, safety, and mobility due to the use of soft actuators compared to the existing rehabilitation devices. Cons: the control system needs to be updated from an open-loop system to a closed-loop one for better safety from a control perspective.

Table 3.1 highlights the characteristics of the devices mentioned above and compares them with the wrist's biological performances during daily living activities [211]. The actuation system mainly consists of electric motors and cable systems, except for SWA, which uses SMA as actuators. The weight for the electric motor-based device is not available, but the motor's load requires a grounded device. As a result, the device's portability is reduced. SWA using SMA-based artificial muscles represents a lightweight device with higher portability than the others. On the other hand, electric motors allow for broader ROM.

In this work, a new exoskeleton for wrist rehabilitation is proposed. It is powered by twisted and coiled artificial muscles (TCAMs), and therefore, in the following, it will be identified with the acronym "TCAMs-Exo." The new exoskeleton is a 2-DOF, soft, wearable, ungrounded, and lightweight exoskeleton powered by the recently introduced [27], low-cost twisted and coiled artificial muscles made with carbon fibers in silicone rubber (CF/SR). CF/SR TCAMs are very promising actuators that can lift to 12,600 their weight and provide tensile strokes and specific work up to 25% and 758 J kg^{-1} , respectively, with less than 0.2 V cm^{-1} of input voltage. TCAMs-Exo can monitor passive movements and perform resisted-active motions, and it can provide a ROM for wrist flexion/extension and ulnar/radial deviation of 130° and 79° , respectively. The wearable parts of the device weigh about a tenth of a kilogram. An elastic arm sleeve is used to avoid direct contact between the TCAMs and the user's skin. To measure the device rotation angles and the user's intention, inertial measurement units (IMU) [212, 213] and EMG electrodes [214] are used as feedback sensors, respectively, while a thermal camera is used to measure the real-time tem-

Table 3.1. EXO device for wrist rehabilitation main features.
Abbreviations—Grounded: *Gnd.*, Un-grounded: *Ugnd.*, Joint: *JT*.

Device Name and Reference	Joints (JTs.)	Actuation	DOFs	ROM	Weight	Type
Rice Wrist—S (2012) [199]	Series of Revolute JTs.	Electric Motors, Cable Systems	Wrist F/E Wrist U/R Forearm P/S	F/E 120° U/R 70°	-	Gnd.
Wrist—Gimbal (2013) [200]	Series of Revolute JTs.	Electric Motors, Cable Systems	Wrist F/E Wrist U/R Forearm P/S	F/E 180° U/R 60°	-	Gnd.
Mahi—Exo II (2014) [201]	Revolute—Prismatic—Spherical JTs.	Electric Motors, Cable Systems	Wrist F/E Wrist U/R Forearm P/S Elbow F/E Shoulder Ab/Ad	F/E 65° U/R 63°	-	Gnd.
Mahi Open—Wrist (2017) [202]	Series of Revolute JTs.	Electric Motors, Cable Systems	Wrist F/E Wrist U/R Forearm P/S	F/E 135° U/R 75°	-	Gnd.
Parallel spherical wrist (2019) [203]	Revolute JTs. Spherical JT.	SEA with two Stepper motors	Wrist F/E U/R Forearm P/S	F/E 140° U/R 100° P/S 150°	1.5 kg	Gnd.
POWROBOT (2022) [204]	Passive Revolute JTs.	Three-gear DC motors	Wrist F/E Wrist U/R Forearm P/S	F/E 160° U/R 60° P/S 160°	0.22 kg	Gnd.
Goncalves et al. (2020) [205]	Single Revolute JT.	One Servo motor	Wrist F/E Wrist U/R Forearm P/S	F/E 170° U/R 70° P/S 175°	2.75 kg	Gnd.
PWRR (2020) [206]	2-Universal JTs.	Two Pneumatic actuators	Wrist F/E Wrist U/R	F/E 130° U/R 105°	-	Gnd.
WReD (2018) [207]	Passive 2-Revolute JTs.	One DC motor	Wrist F/E Wrist U/R	F/E 180° U/R 180°	-	Gnd.
SEU (2020) [208]	Series of Revolute JTs.	Bowden-cables, clutches, DC motor	Wrist F/E Wrist U/R Forearm P/S	F/E 80° U/R 150° P/S 180°	-	Gnd.
Soft Wrist Assist (SWA) (2019) [209]	-	SMA Coil Spring	Wrist F/E Wrist U/R	F/E 64.2° U/R 36.9°	0.15 kg	Ugnd.
Soft Parallel Robot (2021) [210]	Soft Universal JT.	6 PAM and 1 Stepper motor	Wrist F/E Wrist U/R Forearm P/S	F/E 144° U/R 52° P/S 157°	-	Gnd.
Human Wrist [211]	Wrist	Biological Muscles		F/E 115° U/R 70°	-	-

perature of each TCAM. The user inputs from either IMU or EMG is feedback to the $\mathcal{L}_1$ adaptive controller [83, 97], which was implemented to control TCAMs robustly. The fabrication cost of the device is relatively inexpensive, with an overall materials cost of around USD 10.

All the existing rehabilitation devices for wrist rehabilitation mainly rely on electric motors, pneumatic actuators, or shape memory alloys as actuators (see Table 3.1). The novelty of our

solution relies on the use of TCAMs, that compared to the actuators mentioned above, present several advantages: contrary to electric motors and pneumatic actuators, TCAMs do not need heavy and noisy power sources to be actuated, and they overcome the limitations of shape memory alloys in terms of cost and hysteretic behavior.

The paper is organized as follows. In Subsection 3.1.2, we present the mathematical background of the TCAMs-Exo design. Subsection 3.1.3 describes the thermal model for the insulation of the TCAMs. Subsection 3.1.4 presents the results on the device manufacturing and experimental testing setup. In Subsection 3.1.5, the experimental performances of our TCAMs-Exo are described and discussed. Finally, in Subsection 3.1.6, we draw our conclusions.

3.1.2 TCAMs-Exo Design

3.1.2.1 Bioinspired Design

In the present paper, only two DOFs are considered for the wrists: flexion extension and ulnar/radial deviation (see Figure 3.2) [215]. As in the literature papers mentioned above and listed in Table 3.1, the pronation/supination DOF involving joints from the forearm is not considered. Flexion and extension movements describe the movement of bending the palm down toward the wrist and raising the hand's back, respectively. Ulnar deviation, otherwise known as ulnar flexion, is the bending of the wrist to the little finger, or ulnar bone, side. Radial deviation, otherwise known as radial flexion, is the movement of bending the wrist to the thumb, or radial bone, side [216]. As illustrated in Figure 3.2, θ will indicate the wrist flexion and extension rotation angles, while β will indicate radial and ulnar deviation rotation angles. Figure 3.2 depicts the skeletal muscles involved in the respective wrist motions.

The extension motion is produced by the Extensor Carpi Ulnaris (ECU) muscle (Figure 3.2a), the Extensor Carpi Radialis Longus muscle (ECRL) (Figure 3.2b), and the Extensor Carpi Radialis Brevis muscle (ECRB) (Figure 3.2c). The flexion motion is produced by the Flexor Carpi Ulnaris muscle (FCU) (Figure 3.2d), Flexor Carpi Radialis muscle (FCR) (Figure 3.2e), and Palmaris Longus muscle (PL) (Figure 3.2f) [217]. FCU and ECU muscles produce ulnar deviation,

and ECRL and FCR produce radial deviation. However, the ECRB muscle is activated even during radial deviation. All the muscles and tendons located in the forearm [218] are attached to the hand and elbow.

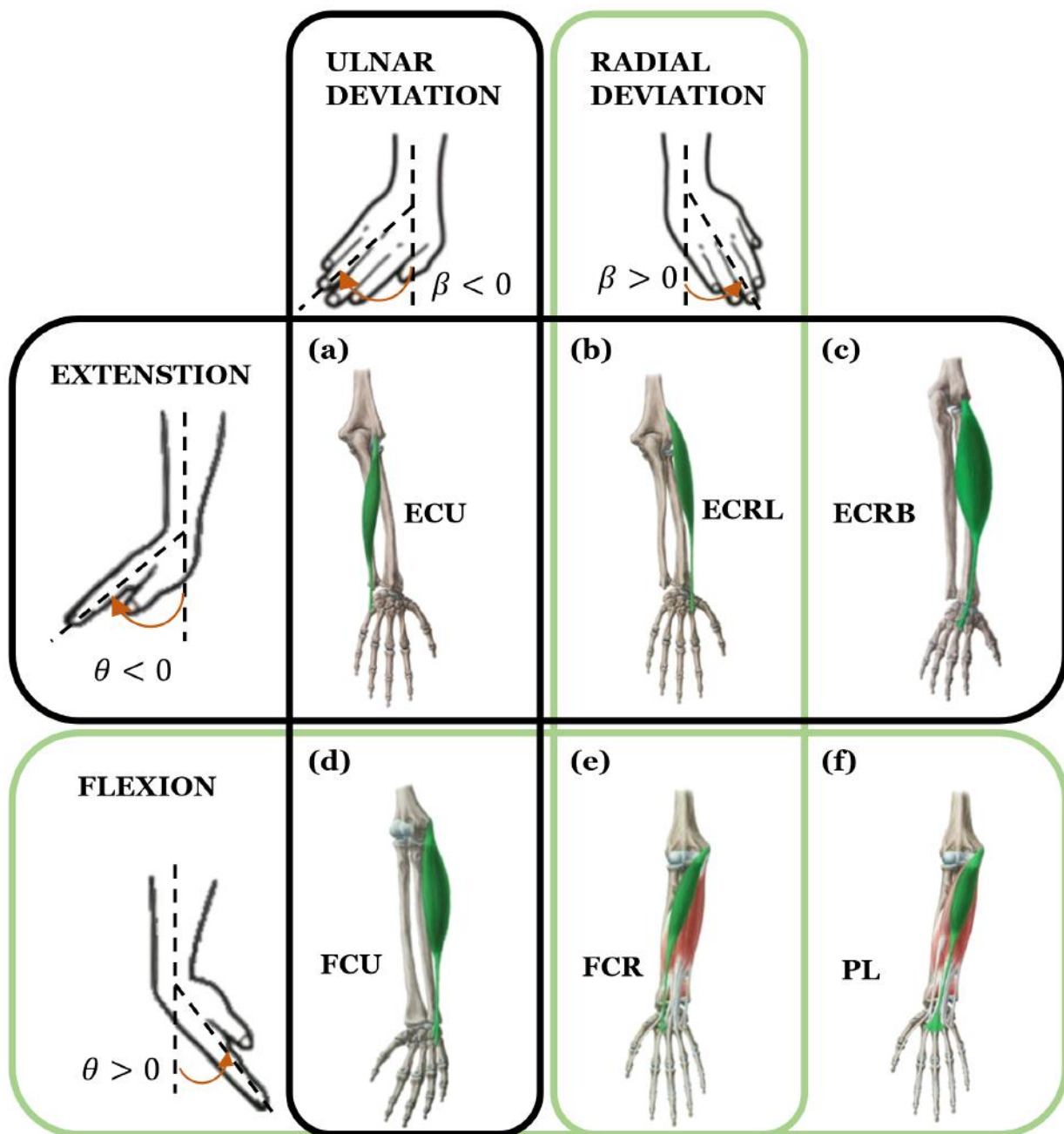


Figure 3.2. Muscles mainly involved in the wrist rotations: (a) Extensor Carpi Ulnaris; (b). Extensor Carpi Radialis Longus; (c) Extensor Carpi Radialis Brevis; (d) Flexor Carpi Ulnaris; (e) Flexor Carpi Radialis; (f) Palmaris Longus. (Adapted from Kenhub).

Natural muscles are capable of performing wide ROMs (see Table 3.1) due to their mechan-

ical performances, arrangement, and attachment points to the skeletal system. Flexion/extension and adduction/abduction can be reproduced with fidelity by mimicking the arrangement of natural skeletal muscles using artificial muscles [7]. Mechanical performances, compact size, and flexibility of CF/SR TCAMs [27] make this type of artificial muscle particularly suitable for performing this task. TCAMs lend themselves to designing and realizing, in a straightforward way, a bioinspired exoskeleton with human-like features and performance to mimic the behavior of natural muscles. Figure 3.3a illustrates a schematic representation of the device proposed by the authors. It consists of two non-deformable links (NDL) attached to eight thermally insulated TCAMs. The two NDLs fastened around the hand and the forearm, respectively, house the attachments (screws and nuts) for the TCAMs. The eight artificial muscles have the task of replacing, if necessary, even completely, the action performed by the six skeletal muscles of Figure 3.2.

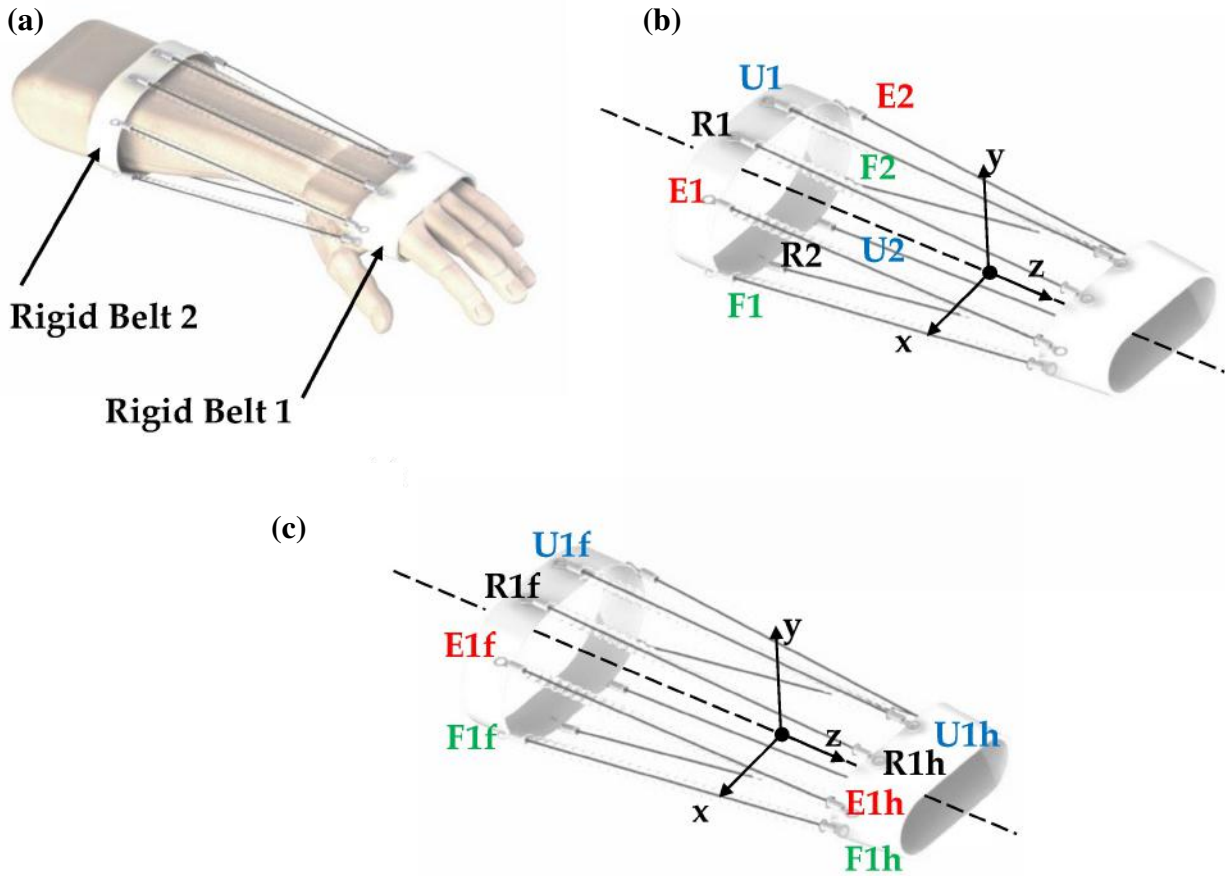


Figure 3.3. Design and components of the TCAMs-Exo: (a) schematic representation of the device; (b) the artificial muscles; (c) attachment coordinates.

The flexion motion is produced by the Flexor TCAMs (F1 and F2), the radial TCAMs (R1 and R2) produce radial deviation, and the ulnar TCAMS (U1 and U2) produce radial deviation. The extension motion is made by the extensor TCAMs (E1 and E2) Figure 3.3b). To guarantee user safety, an adequate distance between the skin and TCAMs must be considered. To this aim, a sleeve covers the human body, and the artificial muscles are thermally insulated.

3.1.2.2 Geometrical and Biomechanical Considerations

How to perform wrist rotations is a problem that nature has already solved. As depicted in Figure 3.2, the six muscles are attached to the skeletal system on the hand and forearm through the tendons. In particular, the attachment points are below the fingers and are close to the elbow. By analogy, the attachments of TCAMs are allocated near the same positions and are attached to NDLs, which perfectly adhere to the hand and the forearm (below the elbow).

The biological muscle efficiency in wrist motion can be attributed to the geometrical disposition of the skeletal muscles. The skeletal muscles cover the skeletal system but are close to the hand middle planes, resulting in high rotation with short contraction. This configuration allows for performing the wide ROM provided by skeletal muscles (see Table 3.1)). In addition, the contraction capability of the biological muscles makes the wrist ROM challenging to emulate. Then, bio-inspiration is a reasonable solution to design a device that can perform human wrist motions as closely as possible.

However, an exoskeleton can be placed over the skin instead of being attached directly to the skeletal system. This produces a less efficient system due to the increase in the moment arm. The broader distance between the artificial muscle attachment points and the skeletal system results in the more significant muscle contraction required. To avoid this issue, artificial muscles must be placed close to the skeletal muscle middle planes. To set the muscles as close as possible to the median plane, it is necessary to move the flexor and extensor muscles to the hand sides as close as possible to the hand top plane. In addition, a couple of muscles are used for each motion to provide just one rotation and to avoid secondary bending (combinations of extension/flexion

with ulnar/radial deviation). The couple is composed of two muscles, one for each side. The same considerations for the deviations, where the muscles, instead of being placed on the sides, are placed on the top and bottom of the hand as close as possible to the hand side plane. Thus, a couple of muscles are used for each motion, with an overall amount of eight muscles. The muscles are attached to the two NDLS, which emulate the biological muscle's attachment points.

3.1.2.3 TCAMs Contraction and ROM

Kinematic analysis was carried out to evaluate the relationships between artificial muscles contraction and wrist rotations. In Figure 3.4a, a schematic representation of the hand kinematic is illustrated. Consider a coordinate system $oxyz$ with the origin o in the center of the plane containing the wrist joint and the axis z oriented along the forearm axis (Figure 3.4g). The two axes x and y identify the axes of rotation of the joint in the plane yz (flexion/extension movements) and xz (adduction/abduction movements), respectively. For simplicity, this first analysis considers just one TCAM. The generic muscle is attached between the points A and B on the hand and the forearm NDL, respectively, with its longitudinal axis parallel to z axes to any distance from this last.

When the TCAM is not activated (cold status), the hand is in the resting position (see Figure 3.4a for the side view, Figure 3.4c for the top view, and Figure 3.4e for the isometric view, where TCAMs are represented in black line). When the TCAM is activated (heated status), it contracts and the hand rotates, point B assumes the position B' , (see Figure 3.4b for the side view, Figure 3.4d for the top view, and Figure 3.4f for the isometric view, where contracted TCAMs are represented in red line). The rotation of the hand in the space can be considered as the combination of two rigid movements, a rotation of θ (between hand middle side plane (HMSP) and z axis) along the x axis in the plane zy (extension), and a rotation of β (between hand middle top plane (HMTP) and z axis) along the y axis, in the plane xz (ulnar deviation).

When there is only a single TCAM present, one muscle provides two rotations at the same time (Figure 3.4g), which will challenge to generate the user-desired motion at any given moment.

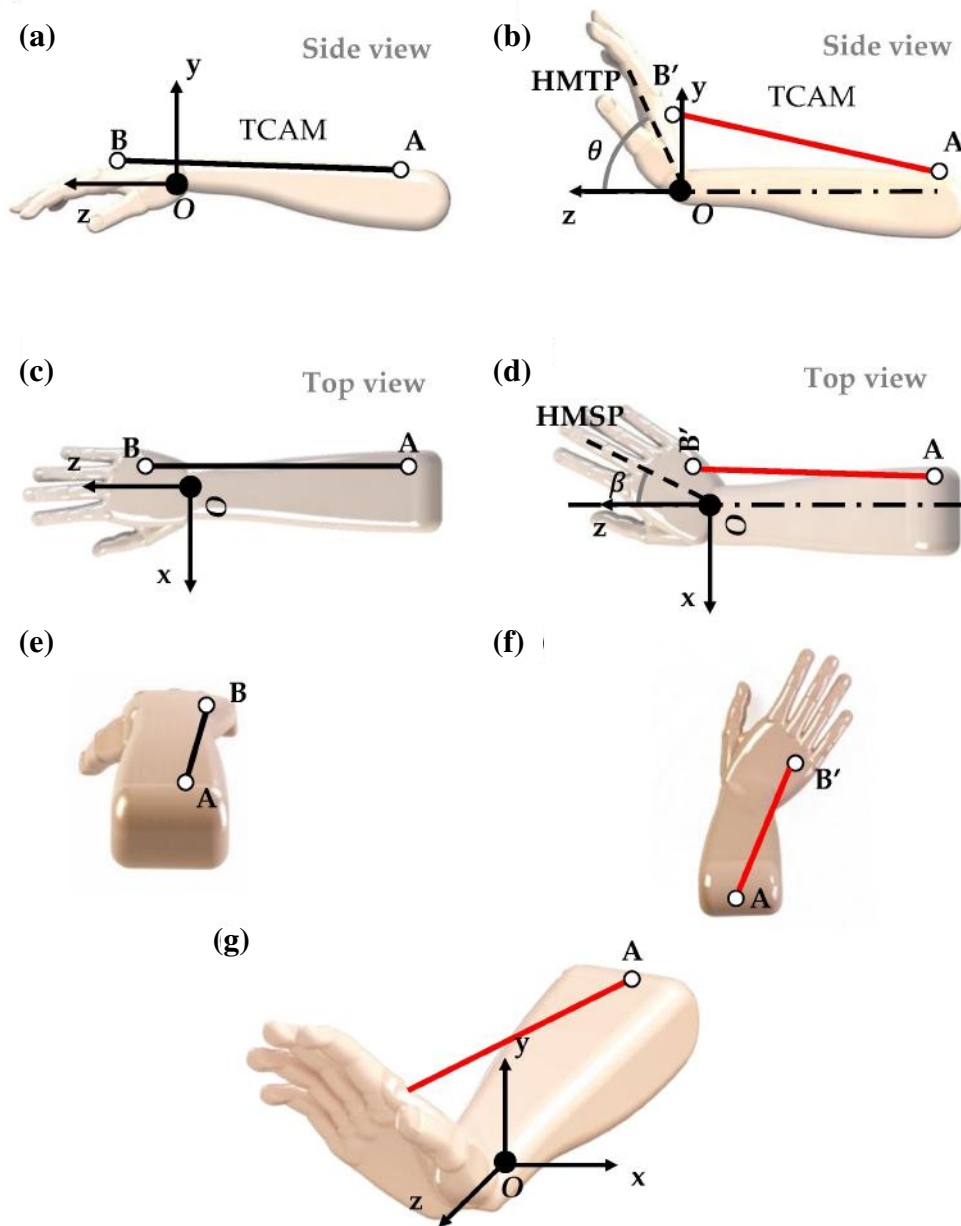


Figure 3.4. schematic representation of the kinematics of the actuation principle of the hand. (a) The generic muscle between the points A and B (black not activated, red activated) from the side plane; (b) the generic muscle contracts providing wrist extension; (c) the generic muscle between A and B attachments from the side plane; (d) the muscle contracts, providing ulnar deviation; (e) back view of the muscle in the cold status; (f) back view of the muscle in the activated status, the muscle provides extension and ulnar deviation; (g) three-dimensional view for the activated status.

The use of a geometric symmetric solution allows for overcoming this challenge.

3.1.2.4 Extensional and Flexional Motions

Consider the configuration with two equal muscles placed symmetrically around the zy plane (thumb side and little thing side) close to the back of the hand. Then, the contraction of the muscle located on the side opposed to the thumb will produce extension and ulnar deviation motions, while the second one located on the same side of the thumb will produce extension and radial deviation motion. If the electric input is the same for both TCAMs, then the number of movements are the same, with ulnar deviation and radial deviation in opposition. Consequently, the contemporary use of two muscles placed symmetrically and parallel to each other on the palm (or back) allow for obtaining only the extension (or flexion). Figure 3.5a shows the disposition of the final muscle disposition for extension (on the back) and flexion (on the palm). Whatever the disposition, TCAMs can only contract on a plane parallel to the zy plane. The contraction that must be imposed on the muscles to obtain the desired angle θ of flexion or extension depends on the initial coordinates of the attachments $A(x_A, y_A, z_A)$ and $B(x_B, y_B, z_B)$ and on the final coordinates of the attachment $B'(x_{B'}, y_{B'}, z_{B'})$. In the resting position, when $\theta = 0$, the length of the muscles is L_{AB} and the length of the muscle while under activation is $L_{AB'}$ as shown in Equations (3.1) and (3.2).

$$L_{AB} = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2 + (z_B - z_A)^2}, \quad (3.1)$$

$$L_{AB'} = \sqrt{(x_{B'} - x_A)^2 + (y_{B'} - y_A)^2 + (z_{B'} - z_A)^2}. \quad (3.2)$$

Then, the contraction c is:

$$c = \left(\frac{L_{AB} - L_{AB'}}{L_{AB}} \right) 100, \quad (3.3)$$

and considering that the coordinates of the point B change from the coordinate frame $B(x_B, y_B, z_B)$ to $B'(x_{B'}, y_{B'}, z_{B'})$ according to the rotation matrix $R_{(\theta, \beta)}$:

$$\begin{bmatrix} x_{B'} \\ y_{B'} \\ z_{B'} \end{bmatrix} = \underbrace{\begin{bmatrix} \cos \beta & 0 & \sin \beta \\ \sin \beta \sin \theta & \cos \theta & -\cos \beta \sin \theta \\ -\sin \beta \cos \theta & \sin \theta & \cos \beta \cos \theta \end{bmatrix}}_{R(\theta, \beta)} \begin{bmatrix} x_B \\ y_B \\ z_B \end{bmatrix}. \quad (3.4)$$

When $\beta = 0$, substituting Equation (3.4) to Equation (3.2) provides the extension (or flexion) $C_{F/E} = f(A(x_A, y_A, z_A), B(x_B, y_B, z_B), \theta)$ as

$$C_{F/E} = \left(\frac{\sqrt{(x_B - x_A)^2 + (y_B \cos \theta - z_B \sin \theta - y_A)^2 + (y_B \sin \theta - z_B \cos \theta - z_A)^2}}{\sqrt{(x_B - x_A)^2 + (y_B - y_A)^2 + (z_B - z_A)^2}} - 1 \right) 100. \quad (3.5)$$

It is intuitive to think that larger muscle contraction provides larger ROM (for both flexion and extension). The ROM amplitude depends on the distance between the TCAM attachments on the NDLs and the center of the joint. Smaller distance results in greater amplification. Thus, to obtain a high value of ROM (for flexion and extension) with little contractions, it is very convenient to choose the muscle attachments far from the zy plane (on the hand sides). In addition, the coordinates y_A and y_B are very low, as close as possible to the hand midplane (HMP).

3.1.2.5 Ulnar and Radial Deviation Motions

Following the same principle mentioned above, consider two equal muscles placed symmetrically around the zx plane, and if both are, for example, on the portion of the xz plane opposite to the thumb, but one is on the back and the other on the palm side, then the first TCAM will produce extension and ulnar deviation motion while the second one will produce flexion and ulnar deviation motion. With this configuration, the same electric input results with the same amount of motion, with flexion and extension in opposition. As a consequence, the contemporary use of two muscles allows for obtaining only ulnar deviation (or if the two muscles are on the portion of the plane zx near the thumb, only radial deviation). Figure 3.5b depicts the configuration for ulnar (little finger

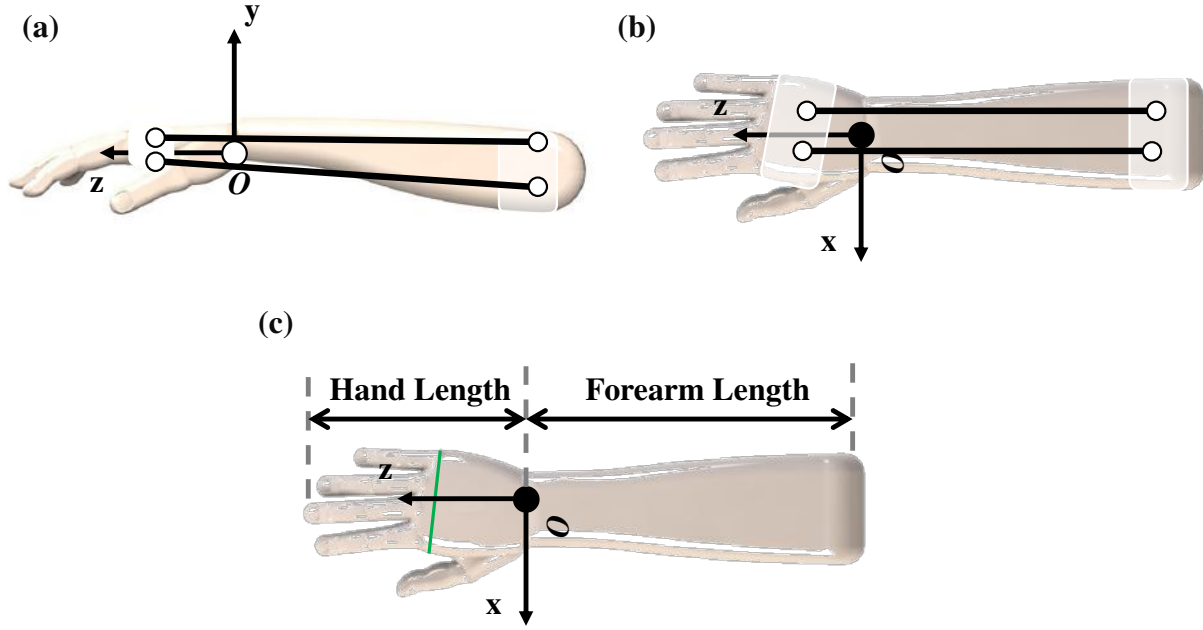


Figure 3.5. Schematic representation of the kinematic of the actuation principle of the hand. The muscles are attached to the bands placed on the hand and close to the elbow: **(a)** extensional/flexional attachments; **(b)** ulnar and radial deviations attachments; **(c)** length considered for the design.

side) and radial deviation (thumb side). Figure 3.5c shows the lengths considered for hand and forearm length calculations.

Considering these muscles' dispositions, then the contraction required to the muscles to obtain the desired angle β for radial or ulnar motion depends, similarly, to the case studied previously on the coordinates of the points A , B and B' . Contraction $C_{U/R} = f(A(x_A, y_A, z_A), B(x_B, y_B, z_B), \theta)$ can be shown as:

$$C_{U/R} = \left(\frac{\sqrt{(x_B \cos \beta + z_B \sin \beta - x_A)^2 + (y_B - y_A)^2 + (-x_B \sin \beta - z_B \cos \beta - z_A)^2}}{\sqrt{(x_B - x_A)^2 + (y_B - y_A)^2 + (z_B - z_A)^2}} - 1 \right) 100. \quad (3.6)$$

Following the same principle used before, the results are convenient to choose the muscles' attachments as close as possible to the plane zy . Multiple literature studies have reported body segment length and weight. 3.2 reports a summary of both male and female forearm seg-

Table 3.2. Length and body weight percentage for both male (M) and female (F) subjects.

	Gender	% of Height	% of Weight	Center of Gravity as % of Height (Along z-axis from the origin)	References
Hand	M	11.00	0.65	5.75	[219, 220]
	F	10.65	0.50	5.75	[219, 220]
Width	M	5.11	-	-	[220]
	F	4.80	-	-	[220]
Circumference	M	12.10	-	-	[220]
	F	11.42	-	-	[220]
Forearm	M	15.70	1.87	43.00	[219, 220]
	F	15.22	1.57	43.40	[219, 220]

ment length statistics as a percentage of total body length and weight. As mentioned above, the flexion/extension motion and ulnar/radial deviation have been treated separately. However, the simultaneous activation of flexion/extension muscles together with ulnar/radial deviation provides a combined rotation.

Table 3.3. Muscle attachment points as hand and forearm percentage for both extension/flexion and ulnar/radial deviation motions. (Abbreviations are reported in Figure 3.3 above).

Extension/Flexion								
Coordinate Axis	Hand NDL Attachment Coordinates as a percentage of Hand Length				Forearm Muscle Attachment Coordinates as a percentage of Forearm Length			
	E_{1h}	E_{2h}	F_{1h}	F_{2h}	E_{1f}	E_{2f}	F_{1f}	F_{2f}
x	16	-16	16	-16	16	-16	16	-16
y	16	16	-16	-16	11	11	-11	-11
z	16	16	16	16	-75	-75	-75	-75
Ulnar/Radial Deviation								
Coordinate Axis	U_{1h}	U_{2h}	R_{1h}	R_{2h}	U_{1f}	U_{2f}	R_{1f}	R_{2f}
x	-5	-5	5	5	-5	-5	5	5
y	16	-16	16	-16	11	-11	11	-11
z	16	16	16	16	-75	-75	-75	-75

Attachment coordinate positions are required to calculate the TCAM contraction percentages, which decide the TCAM lengths. These coordinates depend on the patient's height. 3.3 presents the attachment coordinates that are a percentage of the patient's length.

3.1.2.6 Load Capacity of TCAMs

The diameter to be attributed to each of the TCAMs of the exoskeleton presented in this paper can be evaluated through a simple load's static analysis.

The force developed by the muscle during its contraction must be able to counterbalance the weight of the hand in the two worst conditions. The latter can be identified with the wrist extension or flexion from a horizontal start position (see Figure 3.6a,b) and with ulnar or radial deviation when the hand is in a vertical plane (see Figure 3.6c,d). In both cases, the figures report just one muscle, but a muscle couple must be considered. The force value can be evaluated by imposing the balance of the moment for the axis of rotation.

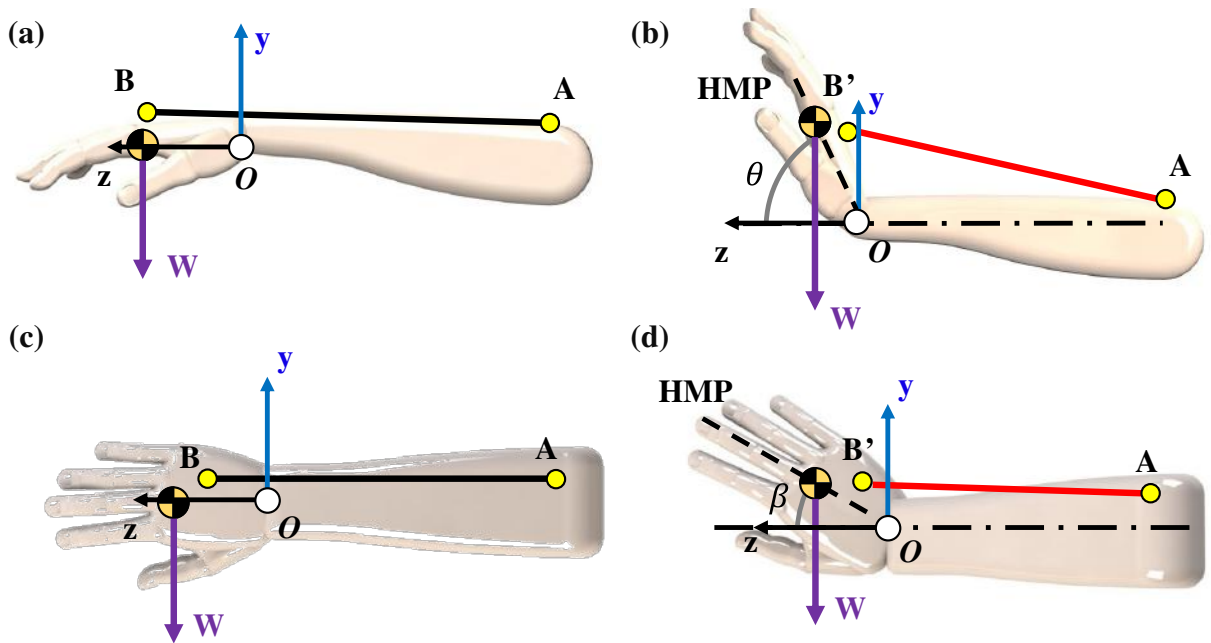


Figure 3.6. (a) Worst case for wrist extension motion. The weight is applied in the center of gravity generating the maximum bending moment around the x axis; (b) after an extension of θ , the bending moment decreases because of the relative angle between the weight and its arm OB; (c) worst case for the ulnar deviation; (d) after a rotation of β , the bending moment decreases.

According to the configurations depicted in Figure 3.6, the equilibrium is satisfied by Equation (3.7) :

$$2\vec{F} \times \vec{OB} = \vec{W} \times \vec{OCG}, \quad (3.7)$$

where $\vec{F}$ is the force generated by just one muscle, $\vec{OB}$ is the arm between the origin O and the point B, $\vec{W}$ is the hand weight, and $\vec{OCG}$ is the arm between the origin O and the hand center of gravity. The most critical situation arises when the weight force is perpendicular to $\vec{OB}$. Figure 3.6 shows this configuration for both extension and ulnar deviation. Figure 3.6a,c represents the critical configurations, where the weight W is perpendicular to $\vec{OB}$. During rotation, the arm direction changes while gravity keeps the y-direction. The angle between them decreases together with the vectorial product. Figure 3.6b,d highlights the new configuration after θ and β rotation.

For the analysis, the weight of the hand can be considered concentrated on its center of gravity. The position of the center of gravity, as well as the hand weight, can be deduced from 3.2 once the patient weight and height are known. With dimensions and the weight considered, in the worst condition, the muscle supports around 7 N. Once the force necessary to face the most critical situations has been calculated, the identification of a suitable TCAM can be accomplished. 3.4 presents necessary data for the calculation. All the dimensions listed in the table are suitable for the TCAMs. Then, the choice is based on the handling of the muscle and fabrication process. CF/SR (carbon fiber/silicone rubber) TCAMs are fabricated following the manufacturing process suggested by Lamuta et al. [27]. TCAMs are fabricated by commercial 3K carbon fiber tows (each tow contains 3000 fibers) from ACP composites, mixed with epoxy sizing to hold the fibers. Each CF n-tow is used for each n-bundle TCAM. First, the tows are dipped into a mixture of uncured Silicone Rubber Sylgard 184 Polydimethylsiloxane (PDMS) and are diluted with hexane (the PDMS:hexane weight ratio is 2:1). Hexane is used to decrease the viscosity of the PDMS and to obtain a homogeneous distribution of the PDMS inside the CF tow. After dipping, the CF/PDMS composite is pre-twisted (240 turns/m) using a drill by attaching an external weight. Then, the composite yarn with twisted CF is placed in a dry place to cure PDMS. Once the PDMS is

Table 3.4. Load applied for different muscle diameters.

Number of Bundles	Diameter (mm)	Stress (MPa)
2B	0.7	18
4B	1.0	9
6B	1.4	5
8B	1.8	3

cured, the CF/PDMS composite is twisted further until full. Considering the explained fabrication process, four-bundle (4B) TCAMs provide a good handling and load capacity.

Consider that a 4B TCAM muscle has a diameter of around 1 mm and provides a maximum strength of about 9 MPa, so that it can easily withstand a force of about 7 N. It must be observed that TCAMs are very promising artificial muscles capable of supporting up to 60 MPa of mechanical stress [27].

3.1.3 Theory of TCAM Insulation Model

The actuation of the TCAMs is based on the heating generated by the passage of current through the carbon fibers. Such heating produces a rise in the temperature of the muscle that can quickly exceed 100 °C. The high value of temperature can be very dangerous for the patients, and thus, it becomes very important to protect them by designing a proper thermal insulation system. To this aim, a stretchable polymeric tube surrounding the TCMA seems to represent a good solution. If correctly designed, the tube can satisfy the task of containing the temperature of the muscle and can allow it to contract freely. Between the many possibilities, the silicone rubber seems to satisfy both the thermal and the mechanical requirements. The thickness and the diameter of a tube made with this material can be evaluated by the thermal study of the physical system schematized in Figure 3.7a, where the TCAM, represented in black, is inside the cylindrical tube, represented in red. Figure 3.7b shows a cross-section in which R_1 represents the radius of the TCAM, and R_2 and R_3 are the inner and the outer radius of the silicone rubber tube, respectively. The tube is surrounded by air both inside and outside.

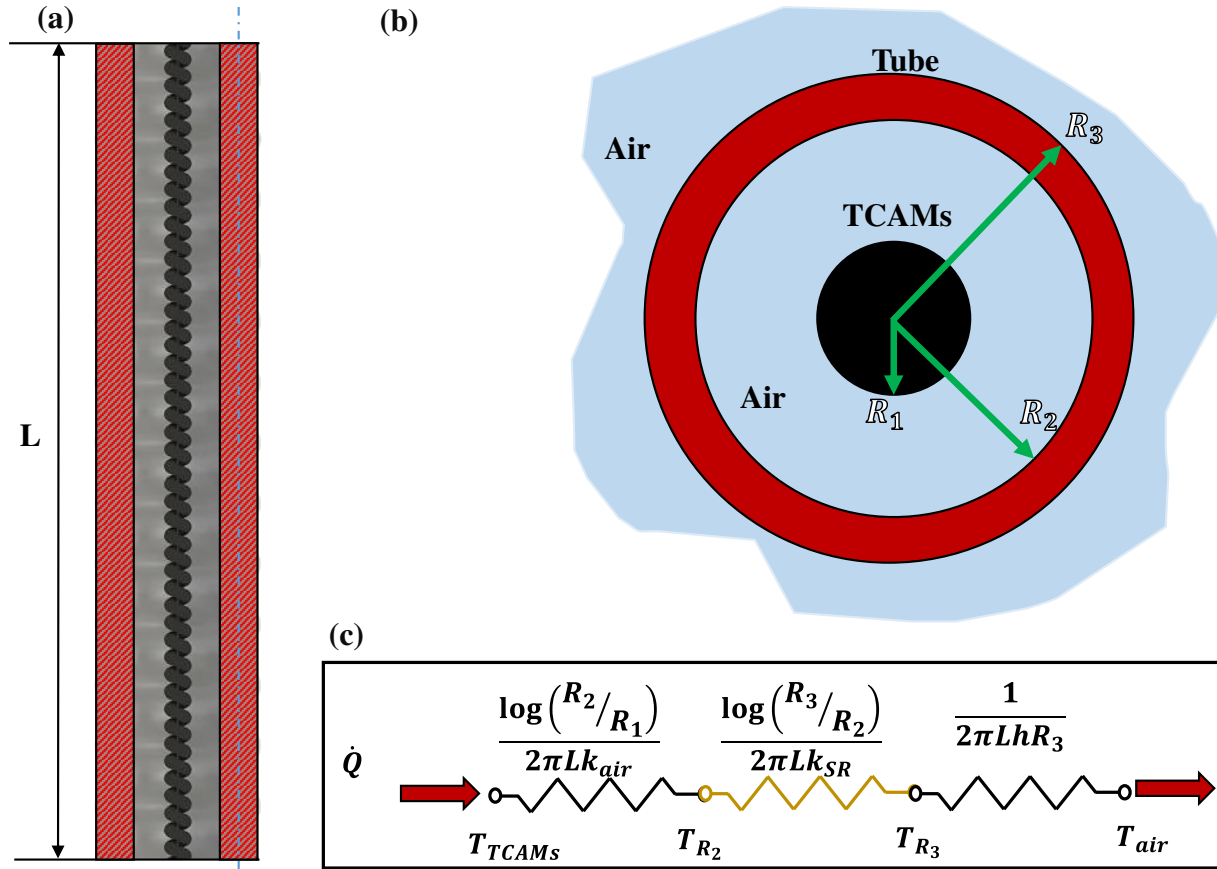


Figure 3.7. (a) Longitudinal section for muscle and insulation tube, where L is the tube length; (b) cross-section showing radius and components; (c) thermal circuit for the system considered, where k_{SR} is 0.16 W/m K.

The heat produced by the muscle for joule heating can be completely dissipated by passing through the walls of the tube and the layers of air provided; they are well sized. Crossing the insulating path, the temperature decreases by passing from the value of the activated muscle (which, for a contraction values of 5 %, can be assumed, on average, equal to about 105 °C), at the value corresponding to the external air temperature (typically 24 °C). Cooling occurs more or less quickly depending on both the material chosen for the tube and the size of the layers. The thermal power $\dot{Q}$ produced by the TCAM is first transferred through the air to the internal surface of the tube, approximatively by a conductive heat transfer mechanism, then by the same mechanism, it is transmitted to the outer surface of the tube, and finally, it is transferred to the outside where the ambient temperature is governing.

By the electrical–thermal analogy, the heat transfer analysis of the thermal system of Figure 3.7b can be studied through the electrical circuit illustrated in Figure 3.7c, [25] in which electrical resistance corresponds to thermal resistance, voltage corresponds to temperature difference, and the electrical current corresponds to heat transfer rate. Under steady-state conditions, we have:

$$\dot{Q} = \frac{\Delta T}{R_{tot}}. \quad (3.8)$$

Thus, if $\dot{Q}$ and ΔT are known, the value of R_{tot} can be computed. It can be shown that R_{tot} depends on the three variables R_2 , R_3 , and T_3 . Therefore, diagrams similar to that reported in Figure 3.8 can be built, which are very useful for determining an optimized solution. In Figure 3.8 are plotted the contour lines for a 4B TCAM developing about 1 watt of thermal power, and for the temperatures T_3 , as a function of the tube thickness and the inner tube diameter. A tube of diameter R_2 of 1.75 mm and thickness of 0.5 mm has a predicted temperature of around 51 °C under steady-state conditions.

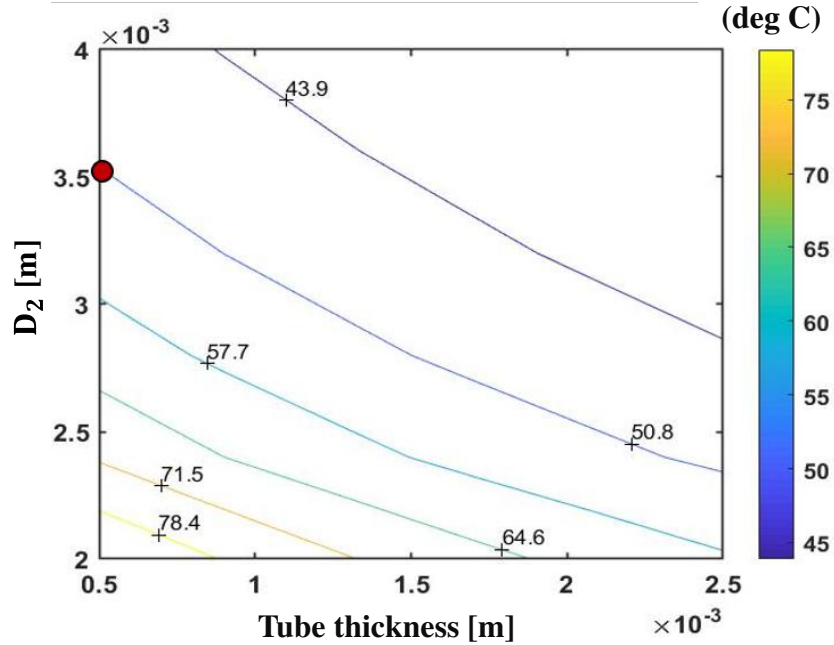


Figure 3.8. Contour lines for the temperature T_3 as a function of D_2 and the tube thickness for a steady-state condition. The temperature of the TCAM is 105 °C.

The heat exchange between the tube's outer layer and the air is convection heat transfer.

The thermal resistance is the convection resistance of the surface. It can be expressed as

$$R_{Conv} = \frac{1}{2\pi L h R_3}. \quad (3.9)$$

The free convective heat transfer coefficient h is calculated based on Equation (3.10).

$$h = \frac{k_{air}}{2R_i} \left(0.68 + \frac{0.67 Ra_{2R_i}^{1/4}}{\left(1 + (0.492/Pr)^{9/16} \right)^{4/9}} \right), \quad (3.10)$$

where $k_{air} = 0.026 \text{ W m}^{-1} \text{ K}^{-1}$ is the thermal conductivity of the air, Ra_{2R_i} is the Rayleigh number evaluated at $2R_i$, with i as the generic radius, and Pr is the Prandtl number = 0.71 for air [221]. The Rayleigh number is evaluated through the following equation:

$$Ra_{2R_i} = Pr Gr_{2R_i}. \quad (3.11)$$

Here, Gr_{2R_i} is the Grashof number evaluated at $2R_i$ by

$$Gr_{D_i} = \frac{g \Delta T \beta (2R_i)^3}{\nu_{air}^2}, \quad (3.12)$$

where g is the gravity acceleration ($9.81 \text{ m}^2 \text{ s}^{-1}$), ΔT is the temperature difference between the external surface of the tube and room temperature 24°C , β is the coefficient of thermal expansion $\left(\frac{1}{T_{room}} \right)$, and ν_{air} is the kinematic viscosity of the air ($1.562 \times 10^{-5} \text{ m}^2 \text{ s}^{-1}$). Finally, L is the length of the TCAM (10 mm). All the variables, such as kinematic viscosity and thermal conductivity, are evaluated at room temperature because Grashof and Rayleigh's numbers are local parameters, and they depend on both the diameter and the operating temperature.

In particular, inside the internal layer of air, for the TCAM operating temperature and the diameter $2R_2$, the resulting Rayleigh number is in the order of 1×10^3 , which lies in the range of the conduction regime instead of convection [222]. In addition, the tiny air layer inside the tube seems a reasonable heat exchange regime. In light of what has been mentioned, the thermic resistance governing this layer can be considered a conduction resistance, and therefore:

$$R_{Cond} = \frac{\log \frac{R_2}{R_1}}{2\pi L k_{air}}. \quad (3.13)$$

Finally, the heat transfer through the wall of the tube is conductive, and the thermal resistance can be expressed as:

$$R_{Cond} = \frac{\log \frac{R_3}{R_2}}{2\pi L k_{sr}}. \quad (3.14)$$

Therefore, the total resistance is:

$$R_{tot} = \frac{1}{2\pi L h R_3} + \frac{\log \frac{R_2}{R_1}}{2\pi L k_{air}} + \frac{\log \frac{R_3}{R_2}}{2\pi L k_{sr}}. \quad (3.15)$$

It is preferable to use corrugated tubes shaped like bellows to obtain a better adaptation between the deformation of the walls of the insulating tube and to move the muscles and increase the heat exchange surface. The procedure for designing tubes with this kind of geometry does not differ much from previous ones; however, what was just described can be used with a good approximation. The procedure is shown in the flow chart in Figure 3.9. The length of the insulation tube is based on the algorithm described in Figure 3.9. This procedure was followed to fabricate the insulation tubes for the TCAMs-Exo prototype.

Figure 3.9. Step-by-step procedure to simulate the trend temperature T_3 as a function of the inner radius R_2 and the thickness of the silicone rubber tube.

3.1.4 Experimental Setup

To verify the feasibility and functionality of the new wrist rehabilitation system, the authors created a prototype suitable for an adult man weighing 70 kg and at a height of 1.74 m (corresponding to the stature of the first author of the paper). However, using our models, and considering the scalability and ease of manufacturing of TCAMs [104, 223], the device can be easily designed for different individuals, with different weights and heights.

For the experimental study, a test bench was equipped with a 3D plastic hand model coupled with a forearm and a TCAM–Exo prototype complete with sensors and all the devices necessary for supplying power and for controlling and measuring the signals. A brief description of the various components will be given below.

3.1.4.1 Three-Dimensional Prototype of the TCAM-Exo

In Figure 3.10a, the two pieces (hand and forearm) making up the artificial right arm used for the experimentation are shown. The arm was made by additive manufacturing. The 3D printer RAISE3D Pro2 Plus dual extruder, by Raise3D (Irvine, CA, USA), was adopted for the fabrication of the two parts in standard PLA (polylactic acid) plastic material. The weight of the printed hand was increased to a human hand weight by using steel disks embedded in the plastic at the center of gravity of the hand (orange plates in the figure). The CAD model was modified according to the purpose of the work. In particular, the wrist joint was replaced with a printed 2–DOF revolute joint able to simulate both wrist motions (for ulnar/radial deviation, see Figure 3.10b, and for flexion and extension, see Figure 3.10c). To reduce joint friction, appropriate lubricant was applied. The exoskeleton is composed of eight CF/SR TCAMs with a respective isolating tube, two non-deformable NDLs, two IMU, one sleeve, and surface electromyography (EMG) electrodes (Figure 3.10a).

The TCAMs were fabricated following the manufacturing process suggested by Lamuta et al. [27]. All the muscles were produced with the same material, the same geometry, and the same number of bundles. The diameter of the generic TCAM was determined through the load analysis, described in the previous section of this paper. For the analysis, the weight and the length of the hand of the patient are provided through Table 3.2 (70 kg and 1.74 m), and the gravity center and the distance between the muscle and gravity center were determined, as shown in Table 3.3. For the more critical conditions, the equation results in a force of 14 N, and Table 3.4 suggests for this load to use a couple 4B with a diameter of 1 mm. Furthermore, the muscle is oversized and can also be used for hands that are heavier or longer.

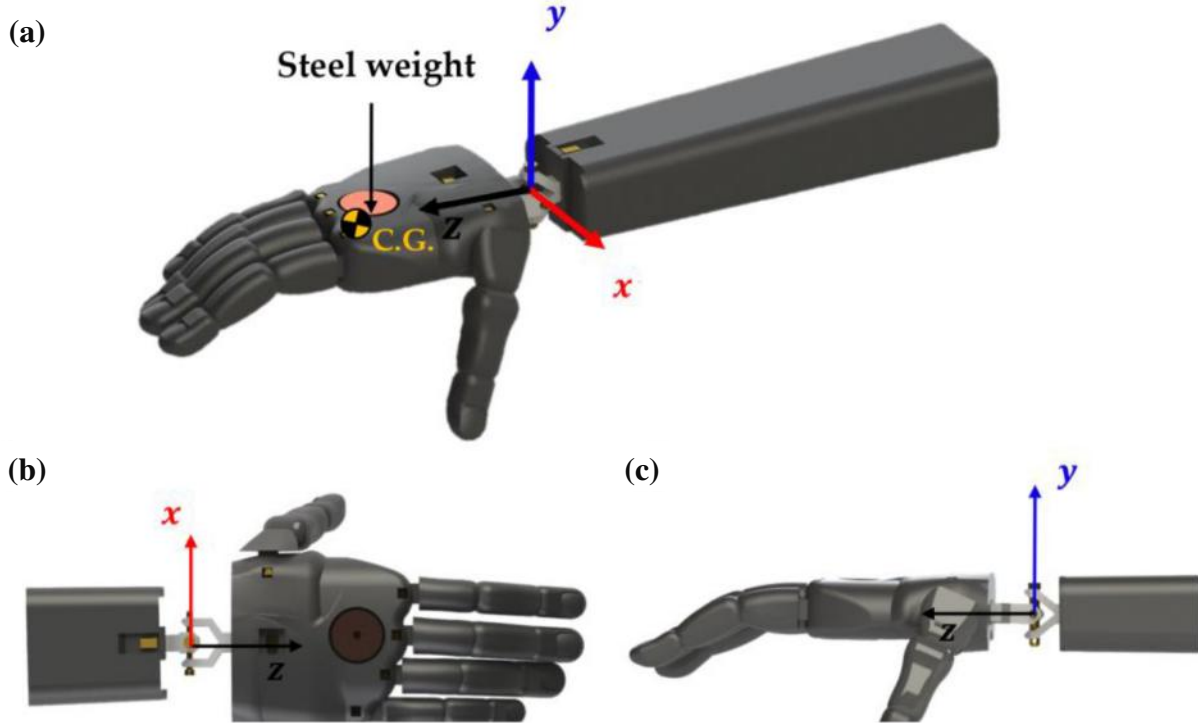


Figure 3.10. (a) Three-dimensional view of the 3D-printed hand and forearm. The center of gravity is given by the steel weights; (b) top view of the 2-DOF revolute joint; (c) side view of the 2-DOF revolute joint.

The manufacturing process involves the use of commercial 3K carbon fiber tows from ACP composites, mixed with epoxy sizing to hold the fibers. Four CF tows (with an initial length of 0.5 m) were used for each four-bundle TCAM. First, the tows are dipped into a mixture of uncured Silicone Rubber Sylgard 184 PDMS and diluted with hexane (the PDMS:hexane weight ratio is 2:1). Hexane is used to decrease the viscosity of the PDMS and to obtain a homogeneous distribution of the PDMS inside the CF tow. After dipping, the CF/PDMS composite is pre-twisted (240 turns/m) using a drill by attaching an external weight of 0.2 kg. Then, the composite yarn with twisted CF is placed in a dry place to cure PDMS. Once the PDMS is cured, the CF/PDMS composite is twisted (105 turns) further until fully coiled (120 coils). The final length of 0.23 m fully coiled TCAMs were used for the wrist flexion/extension motion generation. The same TCAMs length was used for the wrist ulnar/radial deviation motion.

3.1.4.2 The NDL

The NDLs were made by casting a quick-setting urethane elastomer with high resistance to tearing, impact, and wear, in a 3D-printed mold, suitably shaped to the patient's wrist. The mold is shown in Figure 3.11. SimpackTM 85A urethane was found to be stiffer than EcoFlex 00–30 and is thus preferred to it. SimpackTM urethanes pour easily, and the complete cure time is 48 h at room temperature. However, a release agent was applied to the mold to easily separate the mold and urethane. The TCAMs are connected to the NDLs by standard steel screws and nuts.



Figure 3.11. Three-dimensionally printed mold where liquid urethane is poured and cured.

3.1.4.3 Thermal Insulating Tubes

Tubes for thermal insulation were fabricated by casting. The casting molds were designed and 3D printed using standard PLA filament. Figure 3.12a depicts the printed molds for the fabrication of the insulation tubes. These are fabricated following the manufacturing process suggested by the manufacturer (Smooth-On, Macungie, PA, USA). The EcoFlex 00–30 silicon rubber is mixed with the same amount of A and B components. The prepared silicone rubber mixture is poured inside the mold. Figure 3.12b shows the molds with the silicone inside during the curing process. After the silicone is cured, it can be ejected from the molds. Figure 3.12c shows the insulated tubes separated from the mold. The TCAMs, together with the insulation tubes, are attached through two non-deformable bands or NDLs to the arm.

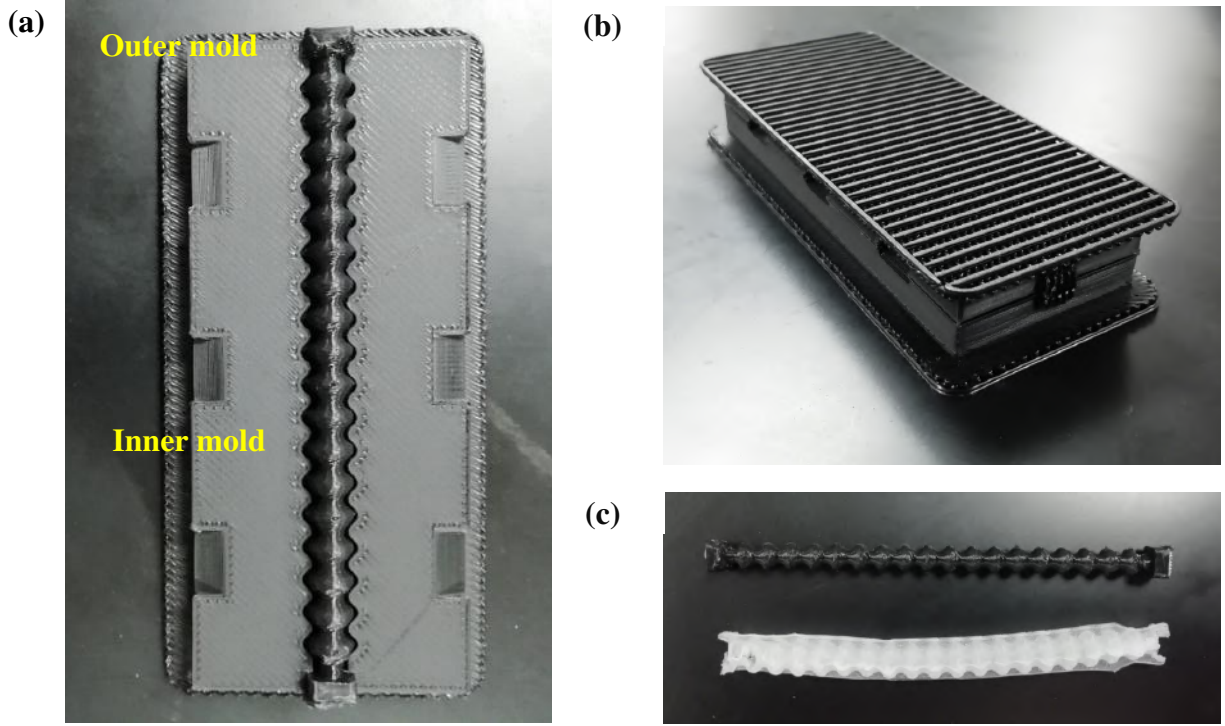


Figure 3.12. (a) Three-dimensionally printed outer and inner mold for the tube manufacturing (material: polylactic acid (PLA)); (b) silicone cast inside the molds; (c) a manufactured tube and the mold.

3.1.4.4 The $\mathcal{L}_1$ Adaptive Controller for Robust Actuation of the TCAMs

In this preliminary version of the device, a controller on the TCAMs actuation has been implemented. The controller is able to provide voltage to each TCAM according to the desired and experimental angle measured via IMUs. The same controller also guarantees device safety by turning off the input voltage when a large actuation angle is measured. Specifically, the robust and adaptive output feedback controller for electro-thermally actuated TCAMs introduced and described in [83] was adopted. The control system algorithm depends on the IMU orientation measurements. The angle between the forearm and the hand is measured using two inertial measurement unit (IMU) sensors (Model: MPU6050, TDK, San Jose, CA, USA), which are placed on the NDLs. They are powered with a sensor for a three-axis accelerometer and three-axis gyroscopes. Real-time data from IMU is essential to rotate the hand toward the desired orientation. Six-degrees-of-freedom IMUs (model: MPU 6050) are used along with microcontrollers (model: Arduino Nano) to acquire

position data. The controller is implemented on MATLAB/SIMULINK. The controller sends voltage commands through the ROS (robot operating system) to a microcontroller. ROS is used as the middleware to communicate between the control algorithm and the hardware system. The output voltages to TCAMs are supplied using motor drivers (model: Pololu Dual G2 High-Power Motor Driver 24V18 Shield for Arduino) (Refer Figure 3.13 soft robotic glove box for the Arduino microcontroller and the Pololu motor drive images).

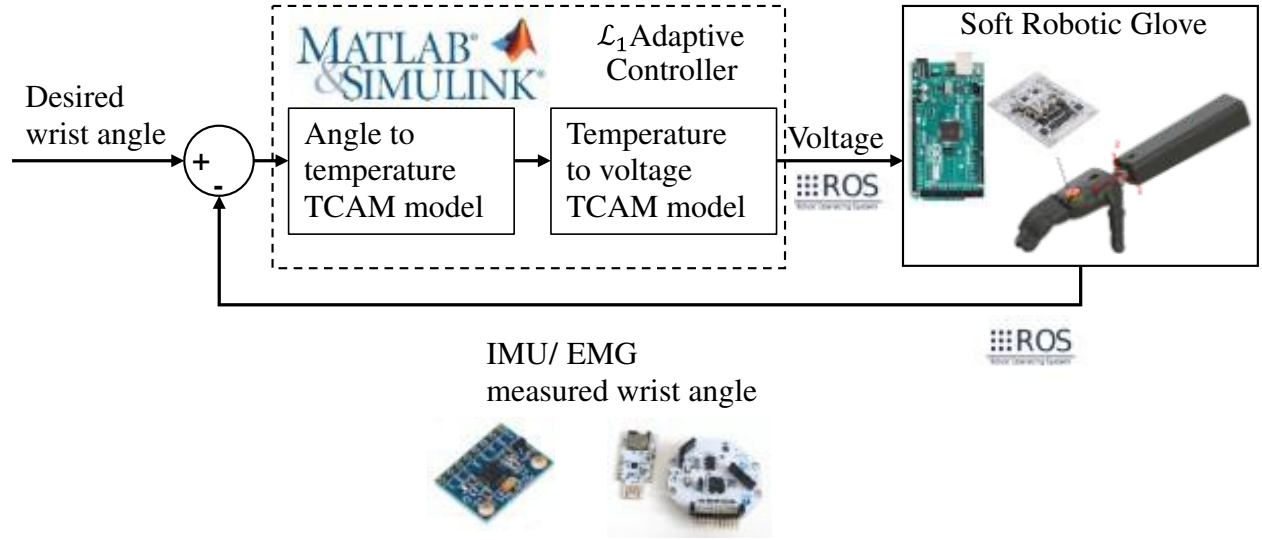


Figure 3.13. The block diagram of $\mathcal{L}_1$ adaptive control algorithm architecture.

The complete block diagram of the $\mathcal{L}_1$ adaptive controller is shown in Figure 3.13. The user-desired input signal is provided by surface EMG electrodes (openBCI Cyton,) which are under the sleeve in contact with human skin. The sensors are placed in the couple over each biological muscle described in the skeletal muscle section. Each couple of EMGs record the voltage along with the muscle. Then, by recording the signal on the six muscles, it is possible to understand which motion is desired.

A common sleeve can be used to avoid direct contact between the muscle and the skin of the patient. The sleeve gives comfort during therapy and protects against any unpleasant contact with hot parts. A thermal camera (Model: FLIR E95) is used to measure the real-time temperature. The tests were performed below 150 °C temperature to prevent degrading the PDMS [81] at higher temperatures.

3.1.5 Results and Discussion

Four different kinds of experimental tests were carried out to verify the functionalities, limits, and potentialities of the exoskeleton. The first three tests were performed by applying the prototype to the artificial arm and the last one was permitted to verify its applicability to a human. In this section, the results of the thermal insulation system assessment are, firstly, illustrated. Next, the limits of applicability in performing passive and active rehabilitation protocols are discussed. Finally, an interesting potentiality of an integrated human machine is presented.

3.1.5.1 Thermal Insulation System Assessment

Some tests were carried out to evaluate the effects of the proposed thermal insulation system. Figure 3.14a shows the variation with the time of the medium temperature, measured by a thermal camera, on the external surface of a TCAM without thermal insulation. TCAM is subjected to a constant contraction of 5 %, under an external attached load of 0.2 kg, for about 30 s. The inset shows the temperature profile along the TCAM. It can be observed that high values of temperature, above 100 °C, were registered.

Figure 3.14b shows the measured outside temperature when a corrugated silicon rubber tube with $D_2 = 3.5$ mm and a thickness of 0.5 mm is used as a thermal insulating system. The dimensions of the tube were determined on the base of the thermal analysis described in the previous dedicated section and, in particular, by the analysis of the contour lines reported in Figure 3.8. The solution taken into account is indicated on the graph by a red point. The measured temperature under the steady-state condition (after 3 minutes) is closer to 49 °C, which is very near to that estimated by the thermal analysis (see Figure 3.8). The insulation tube has provided more than 50 % of temperature reduction in this condition. However, a single repetition of a rehabilitation exercise section is usually quicker than 3 minutes (see the next section). Figure 3.14c shows that the temperature decreases to 36 °C during the rehabilitation exercise tested. Therefore, the insulation system is more than suitable for rehabilitation applications such as for a protocol that provides a wrist extension of $\pm 30^\circ$.

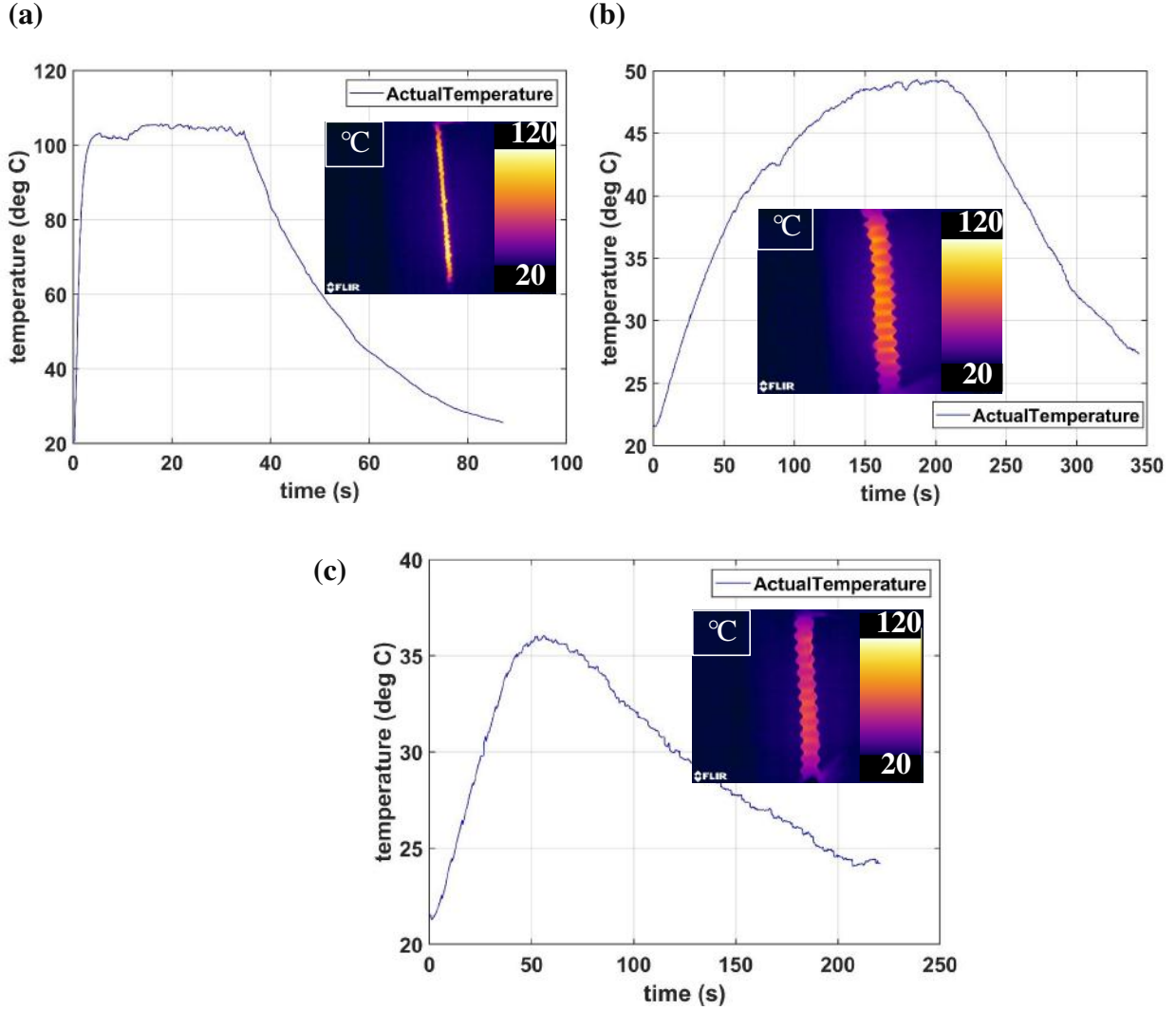


Figure 3.14. **(a)** Temperature trend for a four-bundle TCAM activated with 0.2 kg of load and 5 % of contraction $R_1 = 0.9$ mm. The inset shows the temperature along with the muscle; **(b)** temperature trend for the tube. The inset is the temperature along the tube surface in steady-state condition; **(c)** temperature trend for the tube. The inset is the temperature along the tube surface for 30 s of actuation. A FLIR E95 thermal camera is used for temperature measurements.

3.1.5.2 Passive Rehabilitation Exercise

In the passive rehabilitation exercise, the motion is generated just from the device. The wrist rehabilitation protocol suggested by Slutsky et al. [224] and the Summit Medical Group [225] was considered for the experiment. The protocol states to use a dynamic/static splint if the rotations are lower than 30° .

There, contractions of about 8 % and 5 %, respectively, were imposed to generate extensional/flexional and ulnar/radial deviations as static wrist motions of 30° (Equations (3.3) and (3.4)). To generate extensional/flexional movements, the muscles E1/F1 and E2/F2 (Figure 3.3) were activated simultaneously, while for generating ulnar/radial deviation motions, the simultaneous activation of the muscles R1/U1 and R2/U2 were performed. The angles of the wrist rotation were verified in real time by the couple of IMUs placed on the NDLs. The desired angles for each motion were selected from the work by Crisco et al. [226] in 2011. The prototype well responds to the electric input, even if the targets were not completely achieved, in particular, a range of motion of $\pm 25^\circ$, for wrist extension and flexion, with angles of 23° and 27° for radial and ulnar deviation, respectively.

Although the amplitude of the angles could be increased by raising the level of the electrical input, it was preferred not to force further. High temperatures give rise to problems of a thermal nature, such as the degradation of the polymer. However, the maximum performances were still tested. Such tests are depicted in Figure 3.15 [226]. An angle of 65° was reached for extension and flexion, as shown in Figure 3.15b, c, respectively. Figure 3.15d depicts an angle of 23° for radial deviation, and 56° is instead reached for ulnar deviation in Figure 3.15e. The maximum angles reached for each motion are plotted on the wrist ROM mean value plot in Figure 3.15f.

The failure to reach the target angles is almost certainly due to the arrangement of the muscles and, in particular, to the accuracy by which the coordinate of the attachment points was replicated on this first prototype. To these errors, which can certainly be eliminated in the next prototype by optimizing the geometry of the attachments, the errors due to the scarce resolution of the angular measurement system adopted must be added (this too is a source of error that can be eliminated).

Further experiments were carried out to verify the possibility of carrying out dynamic passive rehabilitation protocols. For example, the extension is represented by the continuous red curve in Figure 3.16a (a sinusoidal half-wave lasting about 80 s). In this case, an extensional rotation of the wrist between 0° to 10° is desired, a cycle starting from the resting position (0°), reaching

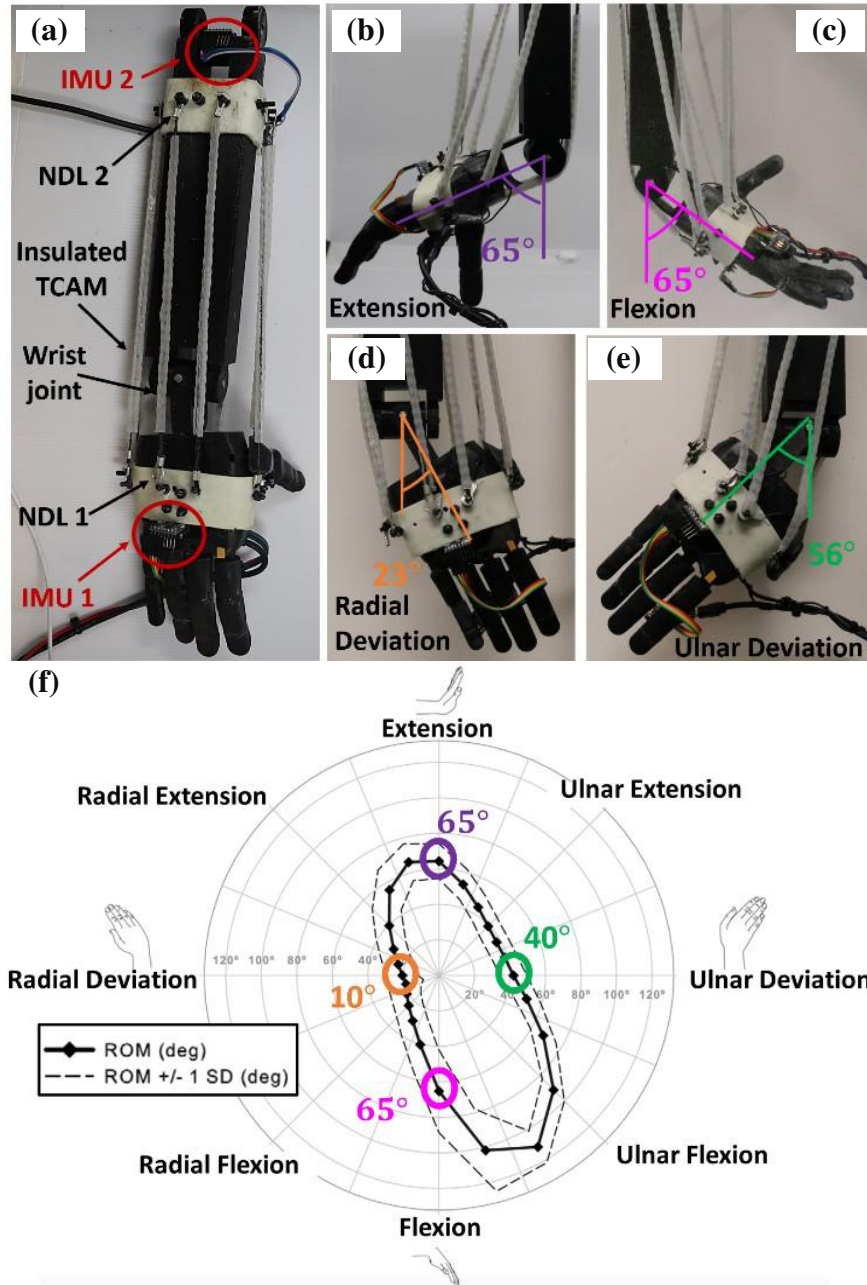


Figure 3.15. (a) Design of the glove composed of TCAM and tube, NDL 1 and 2, IMU 1 and 2, a sleeve, and the EMGs design of the hand-forearm used as a test bench; a revolute joint substitutes the wrist, and several steel weights are used to reach the CG and the weights of a human hand; (b) motion performed with the artificial wrist: extension angle performed 65°; (c) flexion angle performed 65°; (d) radial deviation angle performed 23°; (e) ulnar deviation angle performed 56°; (f) maximum wrist motion angle on a ROM polar plot.

the maximum position (10°) after 40 s and returning to the resting position after about 80 s. To replicate such a movement, a controller is necessary (see the dedicated section) for transform-

ing the desired angle signal in a contraction of the muscles and to compare it with the measured angle (from IMU system). The controller intervenes in case of a mismatch. The blue curve in Figure 3.16a represents the measured angle versus the time. It can be observed that the measured angle follows well enough the profile of the desired angle, except at the start and end of the stroke, where the system seems to suffer from some inertial delay. The average error measured between the desired and experimental angle for the passive mode rehabilitation exercises shown in Figure 3.16 is 0.9896° .

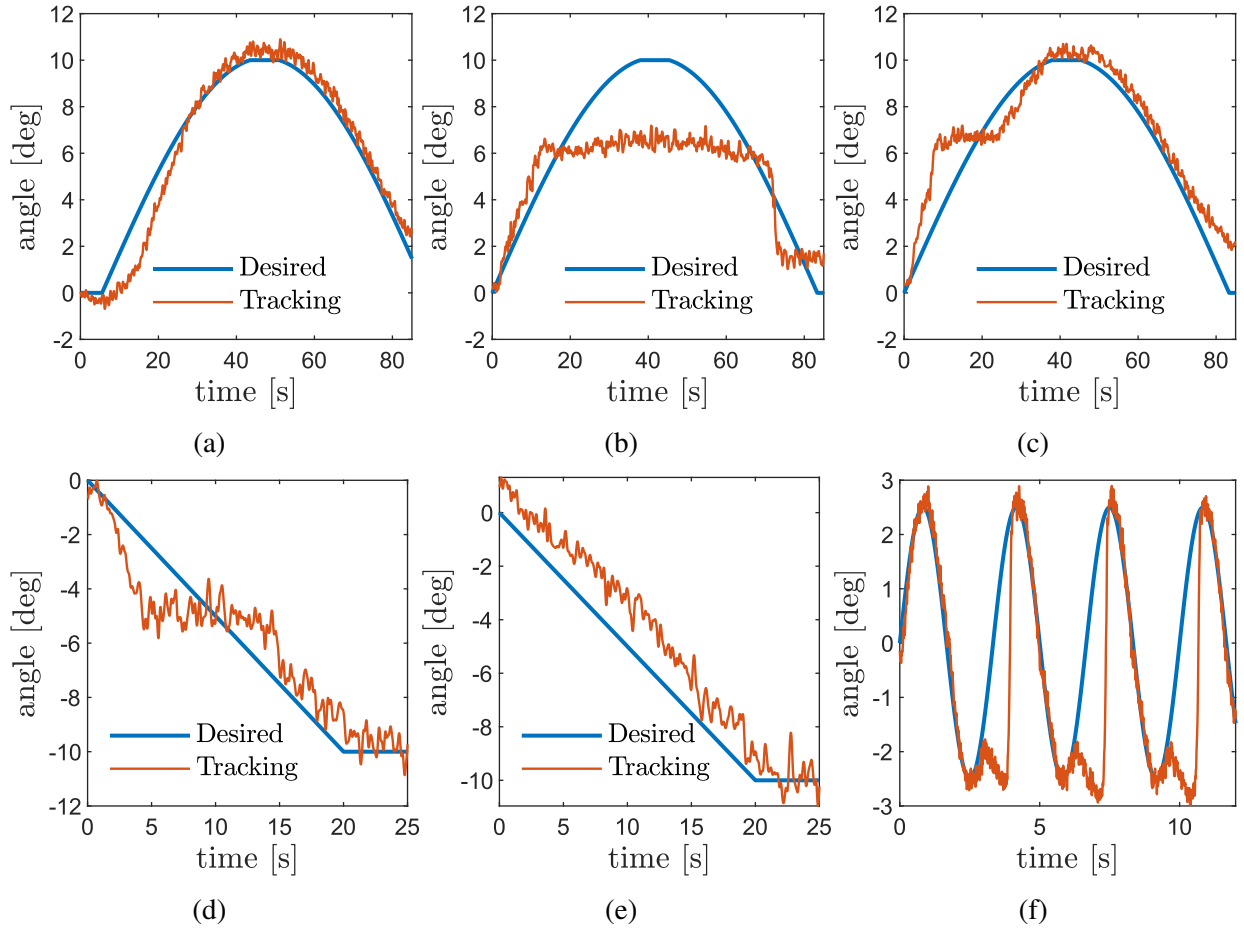


Figure 3.16. Simulations of rehabilitation exercise. The motion considered in the plot (a–c) are for extension. (a) Rehabilitation exercise performed in passive mode; (b) a rehabilitation exercise to emulate the weak hand; (c) a rehabilitation exercise of a weak hand together with TCAMs-Exo (active-assisted mode); (d) velocity for flexion and extension; (e) velocity for ulnar and radial deviation; (f) ulnar-radial deviation cycle performed at high frequency.

3.1.5.3 Active Rehabilitation Exercise

After the test campaign for the validation of the prototype in the field of passive rehabilitation, a second test campaign was carried out to validate the prototype for active rehabilitation. In this case, the device has the task of helping the patient who tries to carry out the rehabilitative exercise without succeeding.

To verify the potentiality of the device for active rehabilitation, the action of a patient with a weak wrist was simulated. The experiments were carried out simulating the action of the patient with a weak wrist by using a stepper motor that rotates while pulling a wire attached to the printed hand.

The device is asked to perform the protocol of Figure 3.16b, (red curve) with the controller deactivated. It tries to perform the desired movement, but it is too weak, and it can raise the hand to 6° and does not go further. The blue curve highlights this behavior. From the graph of Figure 3.16b, it can be seen the great difference between the measured angle and the desired angle. A second attempt is requested from the simulated patient; this time, the controller is activated at the same time. Figure 3.16c shows how the controller intervenes to help the patient as soon as the desired rotation is below the measured one. The device allows the patient to perform the imposed work cycle correctly (blue and red curves overlap). The experiment shows the active-assisted motion capability of the device.

Figure 3.16d, e shows the variation of the velocity measured versus the time for flexion/extension and ulnar/radial deviation, respectively. The speeds measured lie in the range of rotation rates for many individuals [227]. Figure 3.16f depicts the response of the TCAMs-Exo to a fast cyclic ulnar–radial deviation, which is further evidence of the functionality of the device for passive rehabilitation.

3.1.5.4 Human–Machine Interaction

The last experiments were carried out to test the possibility of using the new device on a human patient. Firstly, the active and passive rehabilitation experiments, similar to that carried out on

the mannequin arm, were successfully repeated on a human arm. Subsequently, some tests were performed to verify the possibility of making the rehabilitation device interact directly with the patient. The idea is to allow the patient (obviously, in cases where the EMG signals are reliable and the patient does not have total muscular weakness) to directly control the action of the TCAMs-Exo with the electrical signals emitted by their own muscles. These signals can be picked up through typical surface electrodes (EMGs) and transformed into electrical signals able to provide

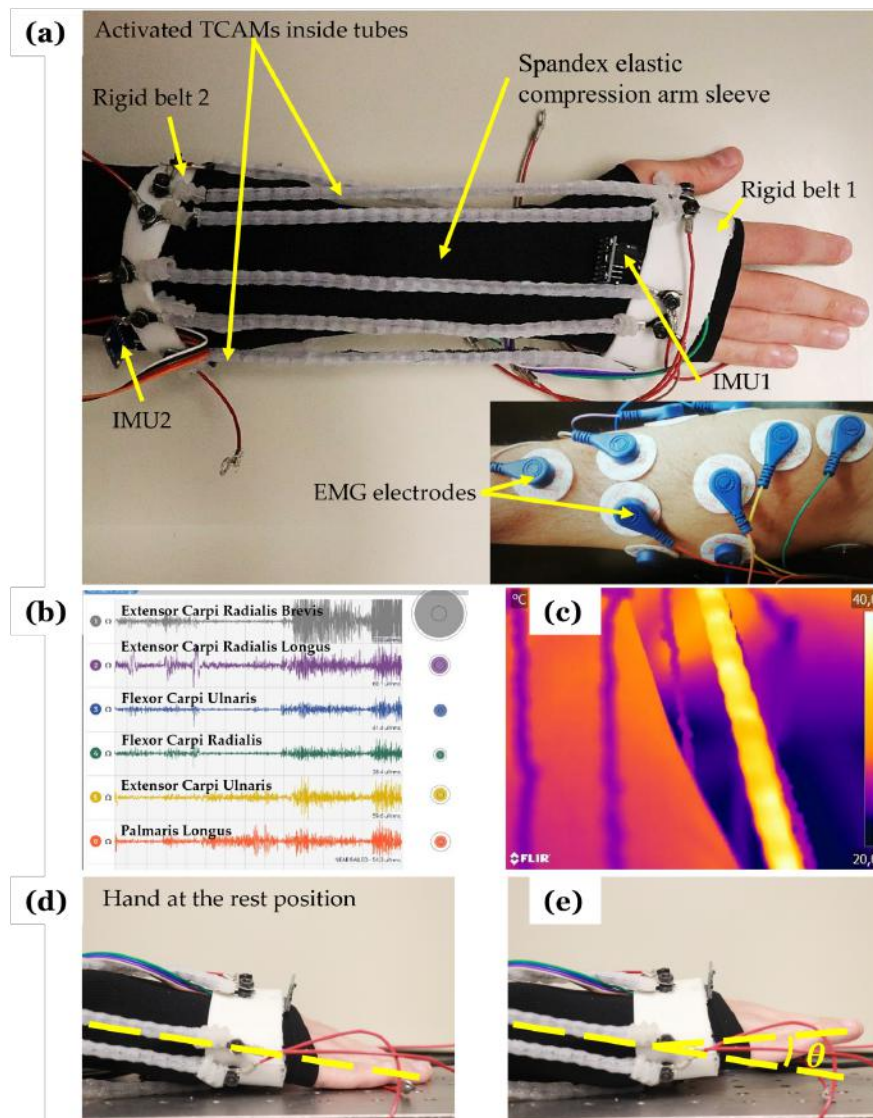


Figure 3.17. Tests with the EMGs; (a) worn glove, with EMG placed below the sleeve; (b) measuring panel for the voltage recorded by the EMGs; (c) maximum temperature along the tube during the test; (d) resting position of the hand; (e) the exercise results in a 10° rotation of the hand.

the contraction of the artificial muscles.

With this philosophy, the patient who accuses problems and cannot perform the exercise can control the machine without intermediaries. The picture in Figure 3.17a shows the human arm wearing the TCAMs-Exo. In the box at the bottom right of the same figure, the arrangement of the electrodes attached on the patient's arm can be seen. The sleeve covers the electrodes. Figure 3.17b shows the control panel of the voltage data of the surface EMG signal. In this case, the voltage plot reflecting the predominant muscles contribute to the wrist extension motion.

Once the input from EMGs is recorded, then the exercise starts. The muscles activated are shown in Figure 3.17a. Finally, Figure 3.17c shows the maximum temperature reached during the test, which is lower than 40 °C. Figure 3.17d,e represents the angle performed during the test (around 10 °). The motion follows the path described in Figure 3.16a. It is clear from Figure 3.17b that the maximum EMG signal is recorded from the Extensor Carpi Radialis Brevis muscle, mainly responsible for wrist extension. Our algorithm is able to detect the skeletal muscle with the biggest EMG signal and then trigger the actuation of the correspondent TCAM on the device (shown in Figure 3.17a) to aid the motion performed. This allows the user to then trigger specific TCAMs according to the desired motion.

3.1.6 Summary

A wearable and lightweight glove for wrist flexion and extension rehabilitation was proposed herein. The device is powered by novel twisted and coiled artificial muscles (TCAMs), able to overcome the limitations of bulky and noisy conventional actuators. The device, which weighs only 0.135 kg, emulates the arrangement and working mechanism of biological muscles in the upper extremities, and it is able to perform wrist flexion/extension and ulnar/radial deviation of 130° and 79°, respectively. The range of motion and the force provided by the exoskeleton were designed using simple kinematic and dynamic theoretical models, while a thermal model was used to design a thermal insulation system for TCAMs during actuation and to guarantee the wearer's safety. By implementing control algorithms, the device's ability to perform passive and active-

resisted wrist rehabilitation exercises and EMG-based actuation was also demonstrated. Exercises were performed with an average error of only 0.9896° between the desired and experimental angles. These promising results represent a first step toward the development of lightweight and low-cost TCAM-based human-assistive robotic devices.

3.2 Modelling and Control of Ankle-Foot-Orthotics powered by TCAMs

3.2.1 Introduction

An orthosis is an externally applied device designed to compensate for impairments in the neuromuscular and skeletal systems. Among various types, an Ankle-Foot Orthosis (AFO) specifically supports the ankle joint and the entire or partial foot, aiding in pain relief, managing deformities, enhancing or restricting movement, protecting tissues, or promoting healing [228,229]. AFOs are essential for assisting individuals with gait abnormalities caused by conditions like stroke, cerebral palsy, spinal cord injuries, multiple sclerosis, or ankle osteoarthritis, where muscle weakness, spasticity, motor control deficits, or balance issues may occur [230,231].

AFOs support ankle movements such as dorsiflexion and plantarflexion (Figure 3.18 – [232]), which are crucial for managing conditions like foot drop, where there is an uncontrolled lowering of the foot due to dorsiflexor weakness or limited push-off power caused by plantar flexor weakness [233,234].

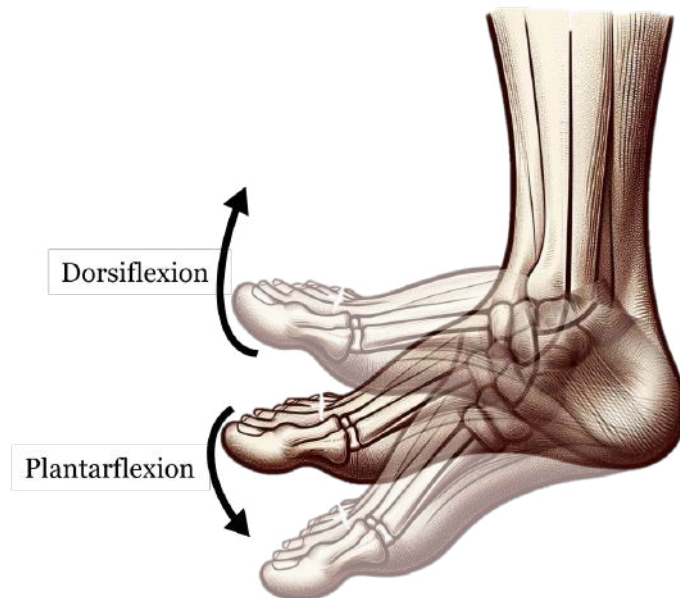


Figure 3.18. Ankle dorsiflexion and plantarflexion motions.

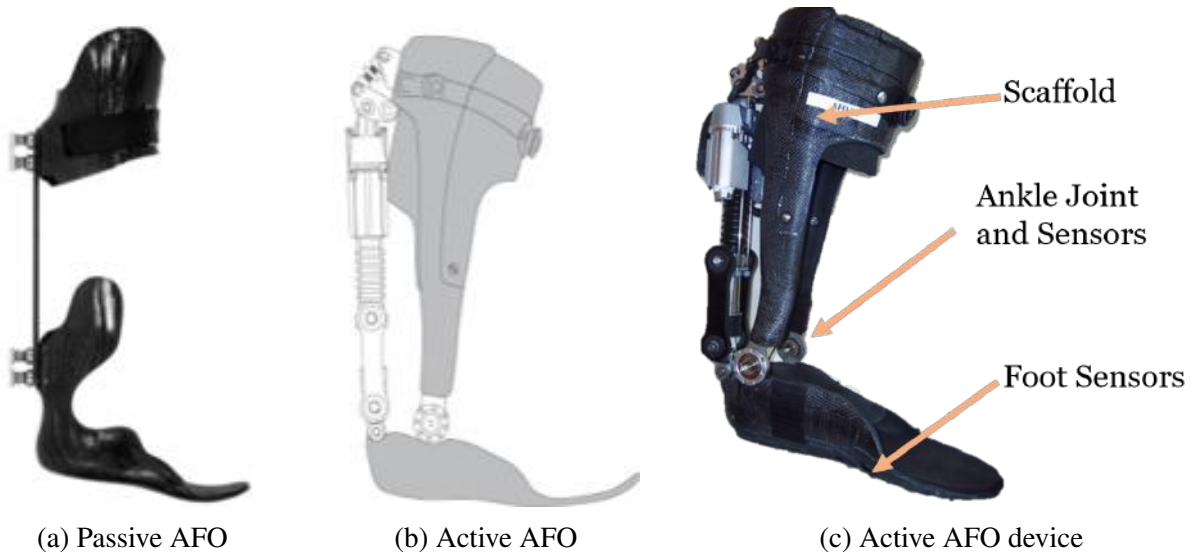


Figure 3.19. a) Passive AFO, b) Active AFO schematic device, and c) Powered active AFO.

Over the years, AFOs have evolved significantly from bulky and uncomfortable designs to more advanced devices using modern materials and technologies to enhance comfort, functionality, and user compliance. Materials such as carbon fiber, thermoplastics, and lightweight metals allow for lightweight and strong AFOs, customizable to meet each user's specific anatomical and functional requirements [235]. As it shown in Figure 3.19, AFOs are typically categorized into passive and active types [236]. Passive AFOs use their material and geometry to provide support and are suitable for offering stability without dynamic assistance [236, 237]. Active or powered AFOs employ motors or powered elements to aid movements like dorsiflexion and plantarflexion, making them suitable for users with significant muscle weakness or paralysis [238].

Despite advancements, current AFOs have notable limitations. Passive AFOs often lack adaptability, with fixed stiffness determined by their materials and design, which may not meet all users' needs. On the other hand, active AFOs, while providing dynamic assistance, tend to be bulky and require substantial power, limiting their practicality, particularly outside controlled environments. Additionally, most existing AFOs fail to offer the flexibility to adjust stiffness dynamically, making them less effective in adapting to different activities or changes in the user's gait.

This study proposes a novel Ankle-Foot Orthosis, termed the POWAFO, which features a

variable stiffness design using twisted and coiled artificial muscles (TCAM) as lightweight actuators. This approach aims to overcome the limitations of conventional bulky actuators by minimizing power requirements and reducing weight [60]. The proposed POWAFO is designed to adapt its stiffness in real-time, providing customized support and assistance, thereby eliminating the need for multiple orthoses tailored to different users.

The development of the POWAFO involves integrating advanced technologies, including sensors, microcontrollers, and artificial muscles, to create a highly adaptive device that can monitor the user's gait and adjust its stiffness or assistance level accordingly [239–242]. This work outlines the comprehensive design and development of a TCAM-actuated variable stiffness AFO, including the control strategies and validation for critical ankle angles during the gait cycle. The proposed device aims to enhance mobility and improve the quality of life for users with a wide range of impairments by providing a practical and efficient solution for user-specific needs.

This advancement represents a significant step forward in orthotic technology, offering the potential for improved functionality, comfort, and adaptability in clinical and real-world environments.

3.2.2 PowAFO Design Requirements

3.2.2.1 Anatomical Requirements

The ankle and foot comprise various muscles along with skeletal elements like bones and cartilage, all interconnected by complex tissue structures. Active movements of the ankle, such as plantarflexion and dorsiflexion, are produced by the contractions of striated, voluntary muscles. These movements are facilitated by the intricate architecture of the ankle joint, which primarily includes two sub-joints: the talocrural and subtalar joints. Both joints support plantar flexion and dorsiflexion, with their rotational axes oriented obliquely.

Biomechanical studies have demonstrated that the talocrural joint axis is inclined at approximately 80° to the sagittal plane and 84° to the frontal plane. Meanwhile, the subtalar joint axis is inclined at around 41° to the horizontal plane and 23° to the sagittal plane. To enhance the

compatibility of an Ankle-Foot Orthosis (AFO) with the natural movements of the human ankle, it is essential to consider an anthropomorphic design approach that aligns both kinematically and ergonomically with the human anatomy.

3.2.2.2 Biomechanical Requirements

During walking, the ankle joint and foot perform complex triplanar movements, which include a combination of dorsiflexion, eversion, external rotation (pronation), plantar flexion, inversion, and internal rotation (supination). These interdependent movements occur at different phases of the gait cycle. Pronation, involving dorsiflexion, eversion, and external rotation, typically occurs during the stance phase. This normal pronation helps to absorb compressive, tensile, rotary, and shear forces at the ankle and foot. However, prolonged restriction or impairment of these movements can lead to abnormal stress, potentially causing damage to connective tissues.

Conversely, supination, characterized by plantar flexion, inversion, and internal rotation, occurs toward the end of the stance phase, enabling the foot to become a rigid lever for effective propulsion. To facilitate these triplanar movements from a biomechanical perspective, it is essential for an Ankle-Foot Orthosis (AFO) to allow natural pronation and supination.

To ensure safe and effective use, the range of motion (ROM) of an AFO should closely align with the natural ROM of the ankle. Kinesiology studies indicate that the ankle is capable of achieving plantar flexion/dorsiflexion of approximately -50/20 degrees, inversion/eversion of -35/25 degrees, and internal/external rotation of 25/20 degrees about the neutral position. The AFO's capacity to meet these kinematic requirements, such as accurate positioning and speed, is crucial for maintaining agility.

Ankle stability relies on passive, active, and neuromotor control mechanisms, which are vital for sustaining dynamic balance during walking. Modulating ankle stiffness or impedance allows the foot to adapt to different surfaces, respond to variations in gait (e.g., stride length, width, rate), and regulate body momentum during the single-support phase. The passive stabilizers (ligaments) and active stabilizers (muscles and tendons) of the ankle (see Fig. 2(b)) can benefit

from the external support provided by the AFO, helping to maintain stability and reduce the risk of sprains and strains.

Moreover, the ankle and foot contribute significantly to shock absorption at heel strike and provide structural support during the stance phase. Articular cartilage in the ankle and the fibrofatty tissue beneath the calcaneus help to dampen and distribute loads, preventing musculoskeletal damage to the lower extremity's kinematic chain. An AFO with effective shock dampeners can support these natural processes by enhancing shock absorption capabilities.

3.2.3 Mechanical Design and Hardware

The primary objective of an Ankle-Foot Orthosis (AFO) is to control the range of motion while assisting the patient by providing torque during the gait cycle. Achieving this objective is challenging, and many existing AFOs use external motors to vary stiffness based on force feedback [243–248]. This work employs twisted and coiled artificial muscles (TCAMs) to replace conventional DC motors. TCAMs can generate significantly higher forces while occupying less space within the AFO and requiring a relatively smaller power source. Previous research from our lab has demonstrated that TCAMs can produce sufficient stress during the gait cycle [60]. Therefore, this study focuses on developing a variable stiffness linear actuator powered by carbon fiber TCAMs. The actuator is integrated into a standard medical-grade AFO, and its efficacy has been validated.

The variable stiffness AFO is designed to adjust stiffness within a range of 0.5–5.0 N m/° and provide a range of motion of 10°, consistent with typical devices [249]. According to the literature, the total weight of the device must be less than 1.25 % of the user's body weight [250, 251]. The device's circuitry, wiring, and sensors were designed and selected to make it portable and lightweight.

Furthermore, patient/user-oriented factors such as plantar flexor weakness, passive plantar flexor stiffness, and walking speed need to be addressed. The impact of variable stiffness on these factors can be experimentally validated using a benchtop testing device called a Stiffness

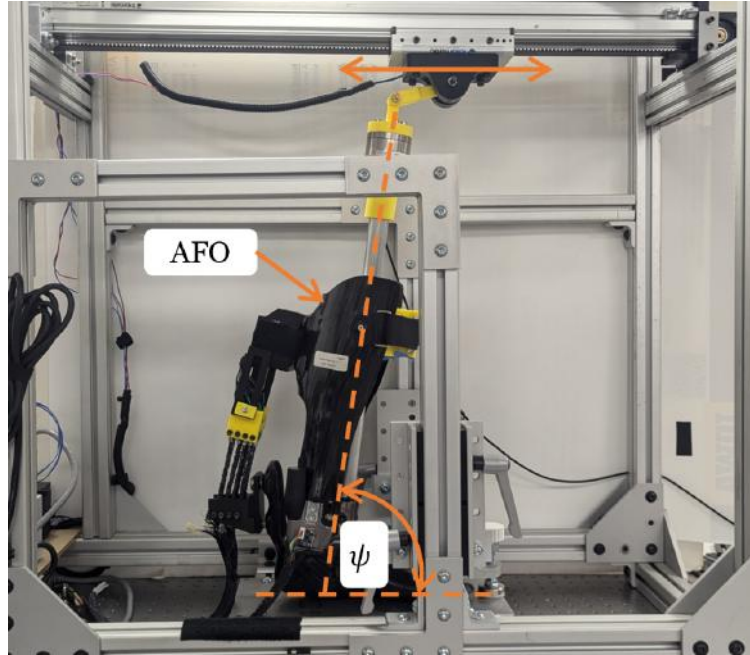


Figure 3.20. Stiffness Measurement Apparatus (SMAp) outline

Measurement Apparatus (SMAp). For the experiments, SMAp was used to fix the angle of the AFO shank to ψ as in Figure 3.20.

The CAD models of the PowAFO are shown in Figure 3.21. Figure 3.21(b) and 3.21(c)

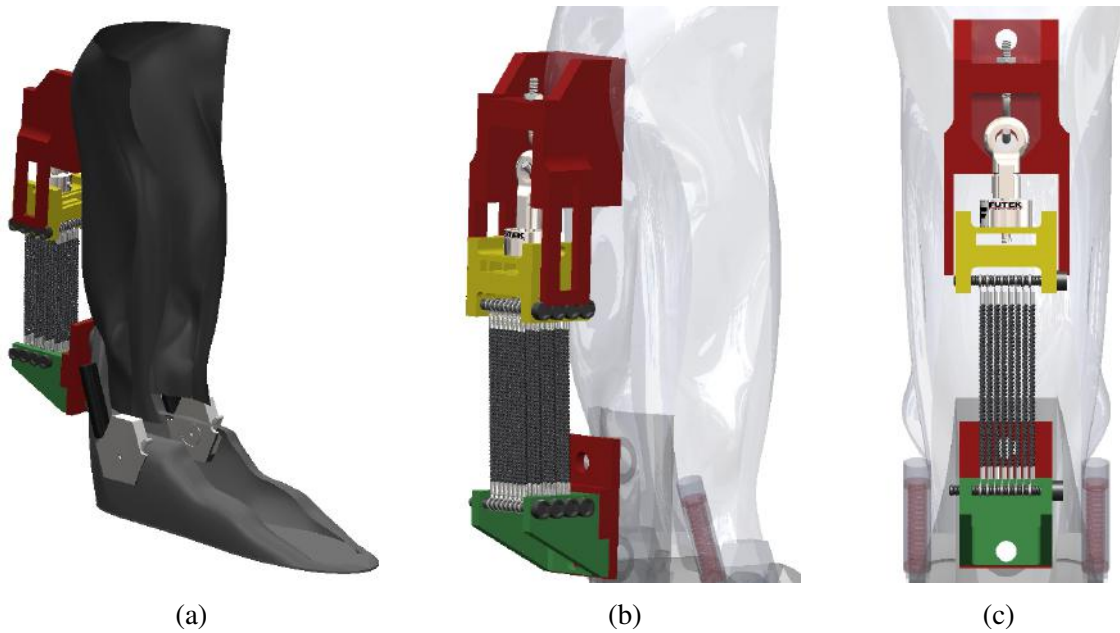


Figure 3.21. CAD models of the PowAFO

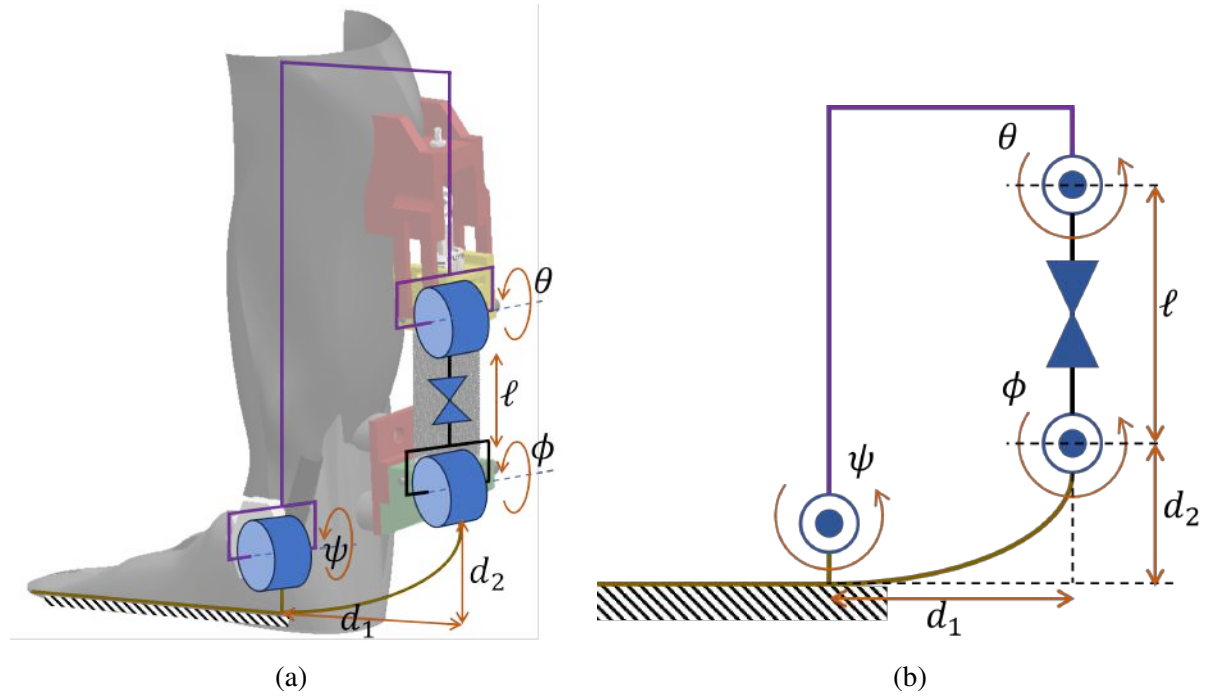


Figure 3.22. Link segments: Kinematic diagrams of the PowAFO. **(a)** Isometric view, **(b)** Sagittal plane view

show the variable stiffness actuator. This actuator consists of a force sensor to measure the torque and an Inertial Measurement Unit (IMU) to measure the angle of the shank and foot. The actuator consists of 32 4-bundle TCAMs. Each requires around 0.8 A–1.5 A current.

The kinematic diagram of the PowAFO is shown in Figure 3.22(a), and the schematic is shown in Figure 3.22(b). The TCAMs are modeled as prismatic joints connected with 2-revolute joints.

3.2.4 PowAFO Actuation Circuitry Design

The circuitry shown in Figure 3.23 is designed to actuate 32 TCAMs simultaneously. This works as a hat for Arduino. Each transistor can control up to 8 A of current, which is sufficient for this application. This system reduced the complexity of the wiring and organized the system.

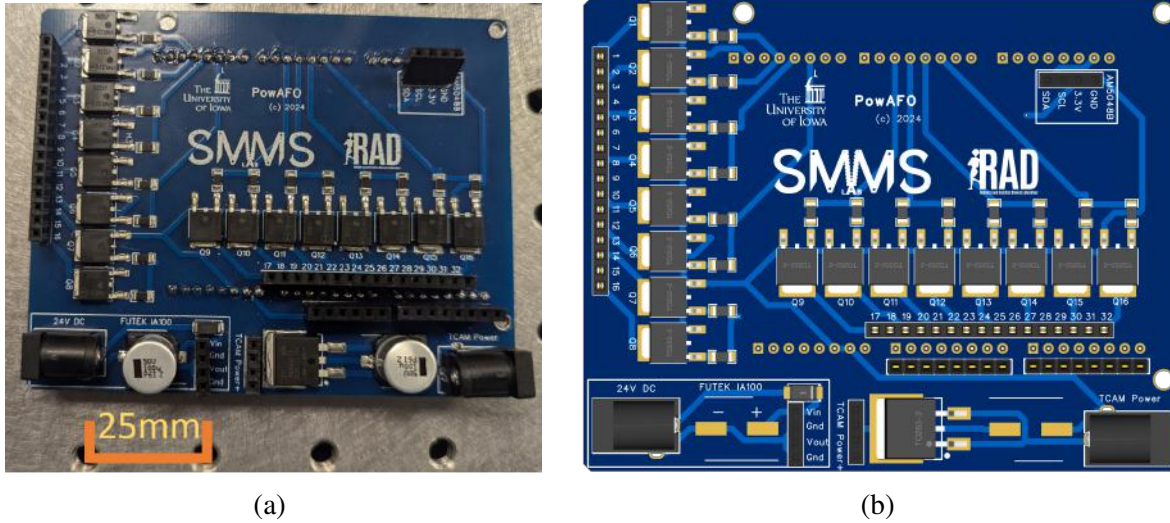


Figure 3.23. Controller hardware circuit

The control architecture of the PowAFO is shown in Figure 3.24. As it can be seen. The system can be controlled based on the desired torque or angle. ROS based hardware system is used to acquire sensing data in realtime. An external DC power supply is used to actuate the TCAMs. The $\mathcal{L}_1$ adaptive controller is implemented on MATLAB/SIMULINK, an Arduino is used to power the transistor and A National Instruments DAQ (Model: NI-MyDAQ) was used to acquire the force sensor data.

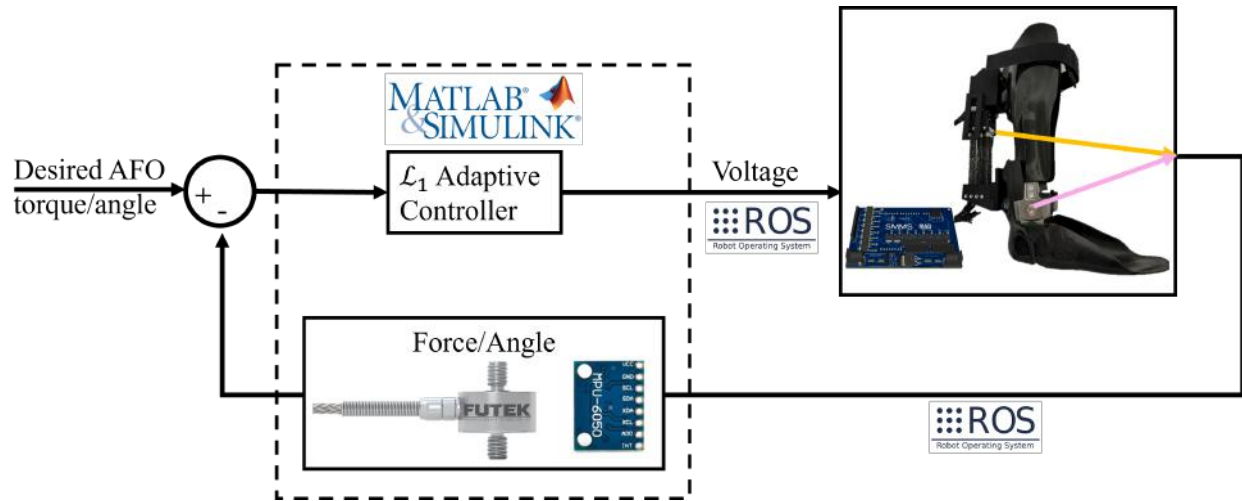


Figure 3.24. Controller diagram overview

3.2.5 PowAFO Experimental Validation

The torque tracking results are shown in Figure 3.25(a). The experiments were carried out for 10 Nm to 50 Nm torque range. During these experiments, the angled shank was fixed to 5° and tightened the springs. Then, the variable stiffness was actuated to obtain the tracking capabilities. The related average temperature plots are shown in Figure 3.25(b), and the supply voltage plots are shown in Figure 3.25(c).

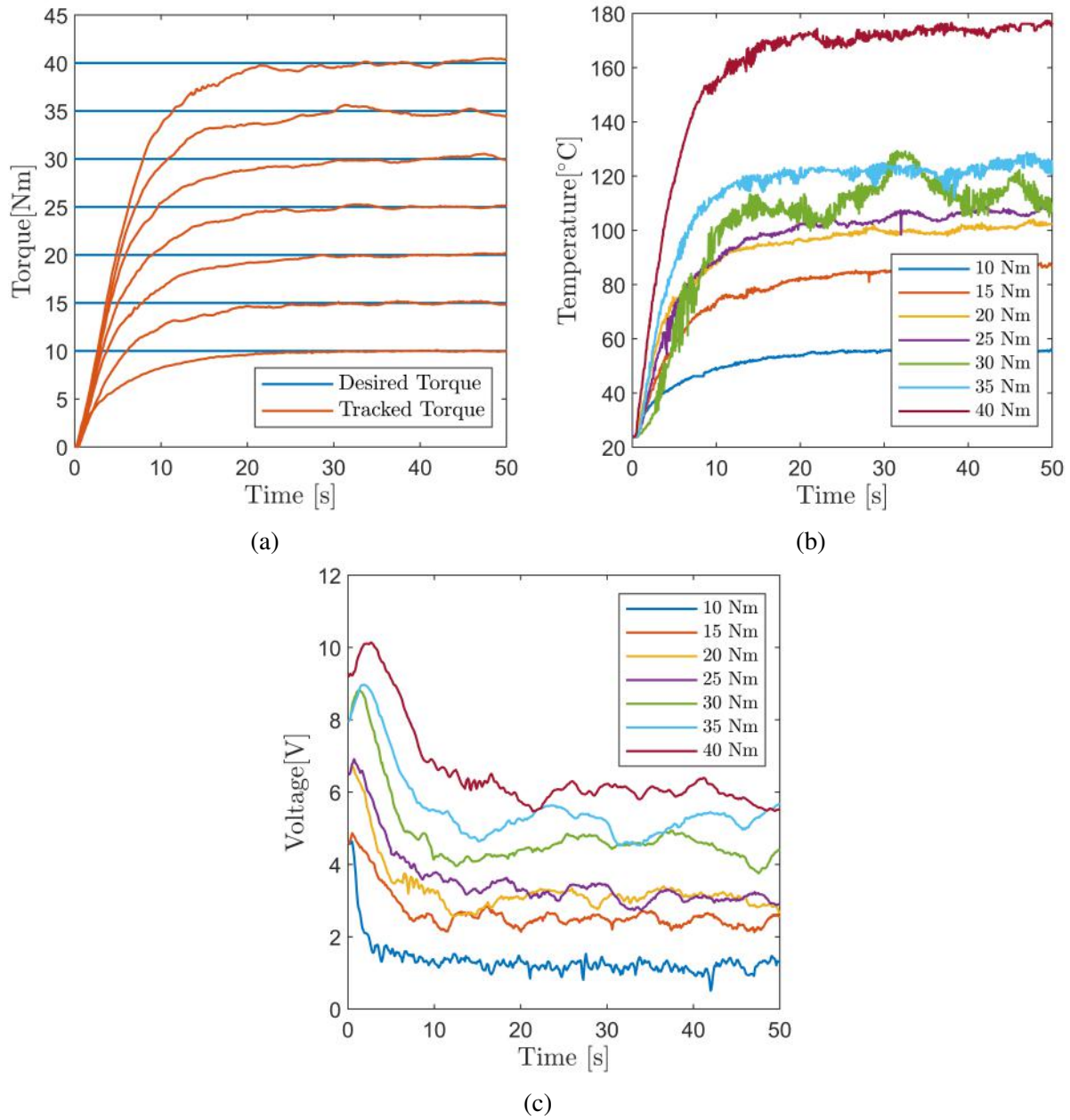


Figure 3.25. Tracking plots for **(a)** Torque tracking using $\mathcal{L}_1$ adaptive controller, and relevant **(b)** TCAM temperatures and **(c)** voltage plots

3.3 Design and Control of Rehabilitation Glove powered by Shape Memory Alloys (SMAs)

This section presents the design, development, and control of a rehabilitation glove powered by Shape Memory Alloys (SMAs). The theoretical model and control law for the SMA actuators are presented in Chapter 2.3.

Researchers have extensively explored hand exoskeletons for rehabilitation and assistance [252–257], but existing solutions face significant limitations. A novel approach proposes using SMA wires directly on the hand, addressing issues of bulk and complexity found in current designs. This innovative concept aligns with natural hand movements by utilizing SMA wires as both actuators and sensors, eliminating the need for bulky wrist-mounted components. The design leverages existing bone structures and incorporates a soft, glove-like structure made of silicone. Although promising, this approach requires careful engineering to ensure precise motion, force generation, and safety, particularly concerning heat management. The proposed solution represents a significant advancement in hand exoskeleton technology, offering a more streamlined, efficient, and user-friendly alternative to traditional methods.

The proposed solution is illustrated in Figure 3.26. The device consists of modular components called “rings” that come in various diameters to adapt to different hand sizes. Each ring fits snugly around a phalanx and is connected to adjacent rings by a steel lamina. This steel lamina serves a dual purpose: it connects the rings to each other and acts as an antagonistic element for the SMA wire. When activated, the SMA wire contracts to produce the desired finger movement, while the steel lamina provides the necessary restoring force to return the finger to its original position once the SMA wire cools down.

Currently, the device includes four fingers connected to a main ring that encircles the palm. This configuration allows for individual control of each finger while maintaining a compact and integrated design. The thumb is not included in the current prototype due to its more complex range of motion; however, future iterations may incorporate a solution for thumb movement.

The modular nature of the ring system offers several advantages. It allows for easy cus-

tomization to fit different hand sizes and finger lengths, enhancing the device's versatility and potential for widespread use. Additionally, this design facilitates the maintenance and replacement of individual components without requiring a complete overhaul of the entire exoskeleton.

The use of SMA wires as actuators within this ring-based structure represents a significant innovation in hand exoskeleton design. By placing the actuators directly on the fingers, the device achieves a level of compactness and natural movement that is difficult to attain with traditional motor-driven systems. This approach not only reduces the overall bulk of the exoskeleton but also improves its responsiveness, enhancing the user's ability to perform delicate tasks.

3.3.1 Prototyping

To perform the flexion and extension of the fingers, a design incorporating SMA wires embedded into the structure was developed. This device is innovative because the smart actuators are integrated and run alongside the fingers, resulting in a highly compact and portable design. The change in SMA physical parameters during contraction makes them useful as self-sensing actuators. Additionally, thanks to the compact hardware, the entire device can be worn on the hand or forearm. All CAD models were created using Fusion 360 and Blender, based on the dimensions of a male

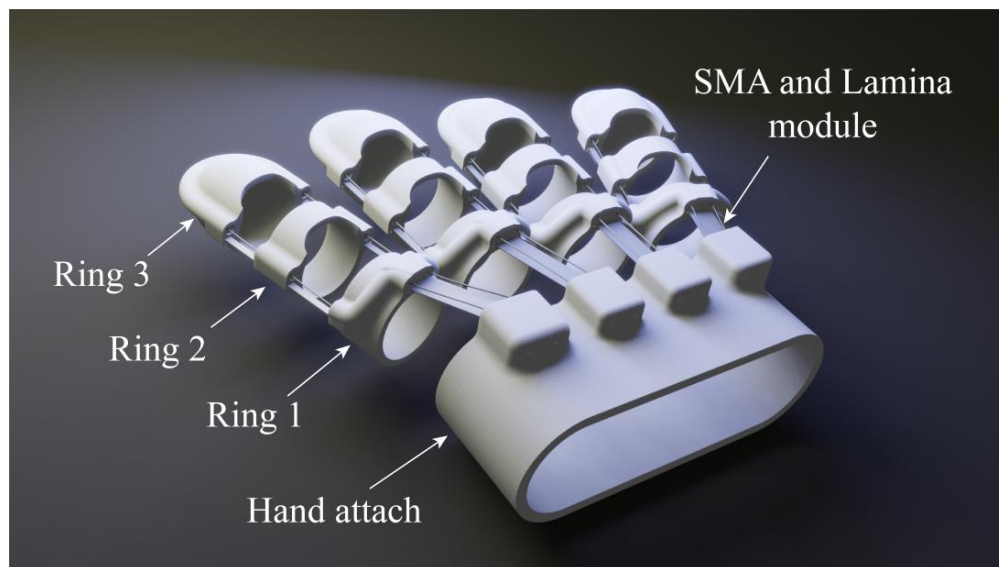


Figure 3.26. Prototype Render

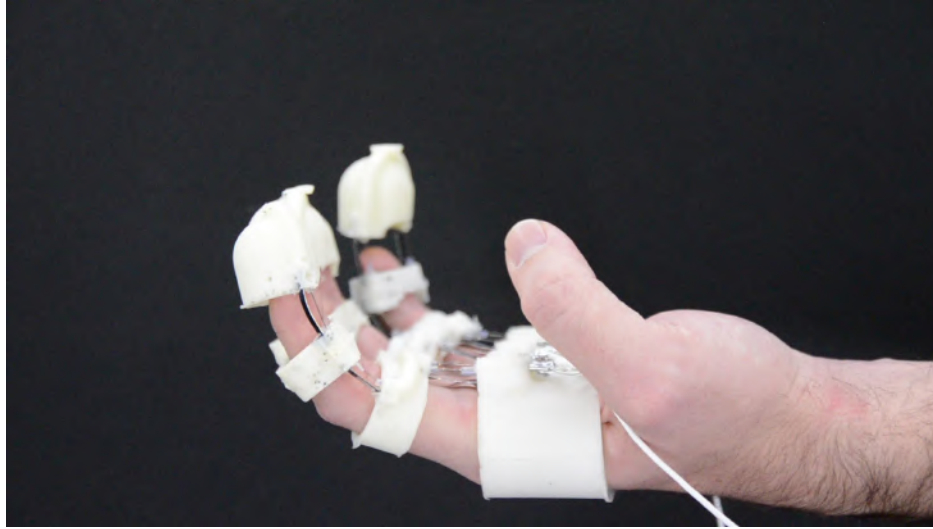


Figure 3.27. Real prototype

hand, as reported in Table 3.5. To produce the physical components, molds were designed and 3D printed. The molds were fabricated using the SLA additive manufacturing technique and used for casting the parts. A mold release agent was applied to facilitate the easy extraction of pieces from the molds. Simpect™ 85A urethane material was used for the rings and the hand attachment module. For the lamina module, a spring metal sheet was machined to the required shape. The lamina modules were then connected to the rings using adhesive.

Each part of the device is modular and can be adjusted to fit different finger dimensions in Figure 3.28. The primary goal during the design process was to create a system that is both adaptable and easy to assemble. To achieve this, each part includes an attachment section composed of two prismatic holes for inserting spring metal sheets, which can vary in shape or cross-section. Ad-

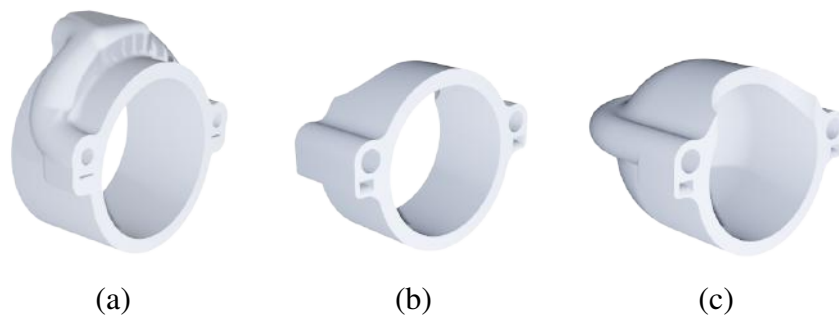


Figure 3.28. Rings for the a) proximal b) middle and c) distal phalanx

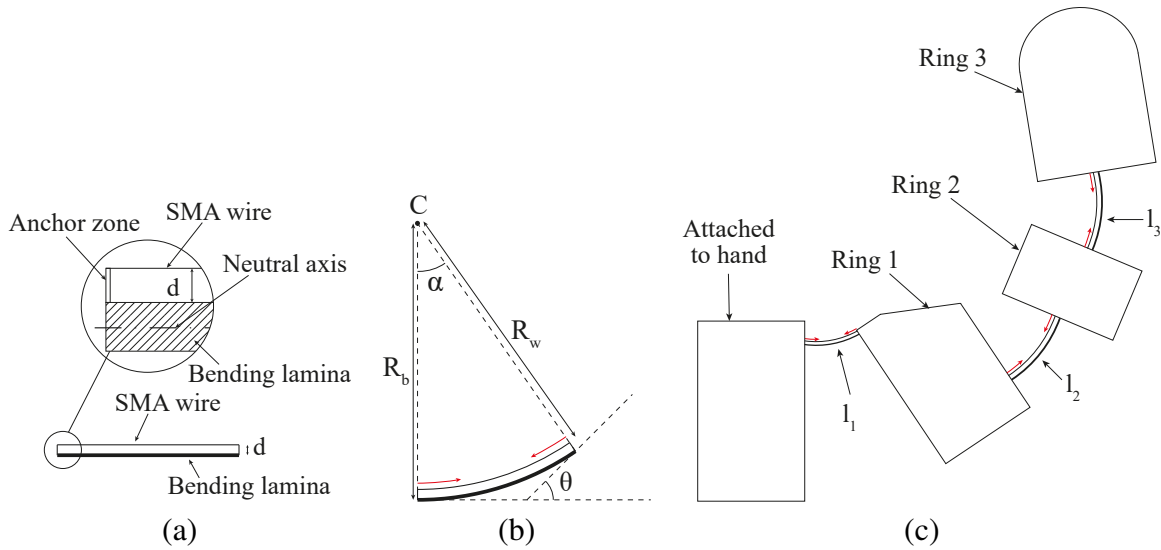


Figure 3.29. Prototype schematics in: (a) straight configuration; (b) bent configuration.

ditionally, one or two circular channels are incorporated to accommodate the SMA wires. Within these channels, small PTFE tubes (2 mm external diameter and 1 mm internal diameter) are used to guide the wires. Each finger module is connected to the main hand attachment ring and can be actuated individually or linked in series for synchronous movement.

In the side view shown in Figure 3.27, the assembly and coupling of the entire mechanism with the human finger can be observed. Specifically, the wire passes through the holes in Ring-1, Ring-2, and Ring-3, and is represented by the red and blue colors. In the blue sections, the wire slides without causing any shape change. Conversely, in the red sections, the underlying lamina deforms and rotates, generating movement.

Table 3.5. Adopted phalanx length

proximal phalanx [mm]	middle phalanx [mm]	distal phalanx [mm]
55	33	25
15	10	25

3.3.2 Soft Finger Exoskeleton Powered by SMA

To estimate the position of the rings and the phalanges, a system model can be defined. A simplified representation of the exoskeleton is shown in Figure 3.29. An analytical model was implemented to consider the laminae deflection. It is worth noting that the laminae and the wire are covered by a silicone material, similar to [258], this means that the distance (d in Figure 3.29a) between the laminae and the wire remains constant, and it is predefined via experimental tests as a convenient trade-off between achieving a low temperature at the skin surface and a fast cooling down of SMA wires. Given the above assumptions, a geometrical approach was used to define the analytical model. In fact, the contracted wire and the curved laminae have to be geometrically constrained and they share same centre of curvature as shown (C in Figure 3.29b). The two curvatures radius (R_b , for the lamina and R_w , for the wire, in Figure 3.29b) differ just by the fixed pre-defined distance from the lamina to the wire.

The following formulas were defined to calculate the position and the orientation of the bent extremity of the lamina as based on the above-mentioned assumptions. In all equations the i subscript refers to the i -th lamina.

$$ls_i = l_i - (\varepsilon \cdot l_i \cdot w_i) \quad (3.16)$$

In Equation (3.16), ls_i represents the length of the SMA wire calculated for a given deformation ε , resulting in a contraction of the SMA wire. The contraction is calculated by subtracting the product of the initial length of the SMA wire (l_i) by the i -th weight coefficient (w_i). The weight coefficient of the SMA wire is used to distribute the contraction of the whole wire by considering the portion of the wire over the lamina. The weight coefficient is evaluated using Equation (3.17).

$$w_i = \frac{l_i}{l_1 + l_2 + l_3} \quad (3.17)$$

The radius of curvature of the i -th bending lamina under the SMA wire is calculated with the

Table 3.6. Prototype main dimensions for a finger

Lamina	Length [mm]	Ring	Length [mm]	Diameter [mm]
ℓ_1	15.50	r_1	13.85	24.00
ℓ_2	12.08	r_2	12.30	22.00
ℓ_3	15.13	r_3	28.00	20.00

Equation (3.18).

$$R_{b,i} = \frac{d}{1 - (ls_i/l_i)} \quad (3.18)$$

where d is the known distance between the SMA wire and the lamina. The angle α_i is calculated using the equation for the arc of a circle, Equation (3.19).

$$\alpha_i = \frac{l_i}{R_{b,i}} = \theta_i \quad (3.19)$$

α_i is equal to the angle θ_i due to the similarity of triangles. The coordinates x_i and y_i of the endpoint of the lamina are calculated as Equation (3.20).

$$x_i = R_{b,i} \cdot \sin \alpha_i ; \quad y_i = R_{b,i} \cdot (1 - \cos \alpha_i) \quad (3.20)$$

To calculate the coordinates of next endpoints of the lamina relative to the previous one, the transformation matrix $\mathbf{T}$ (3.21) is used.

$$\mathbf{T} = \begin{bmatrix} \cos \alpha_i & -\sin \alpha_i \\ \sin \alpha_i & \cos \alpha_i \end{bmatrix} \quad (3.21)$$

Table 3.6 provides detailed dimensions of the rings and laminae used in the prototype (Figure 3.29 c)), which are essential elements of the prototype design under investigation. The same dimensions are used in the simulation.

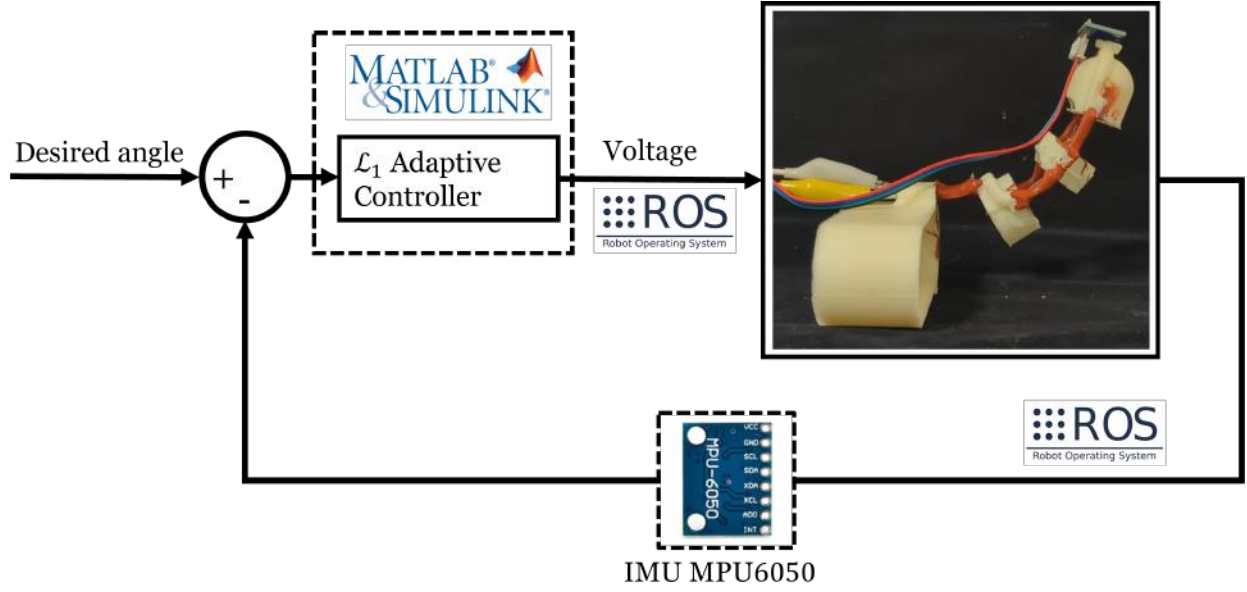
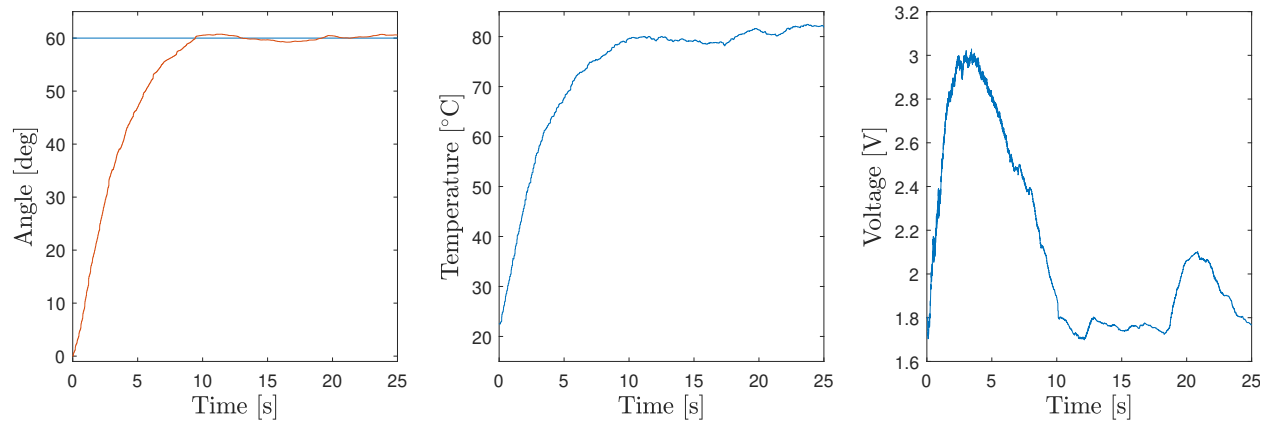


Figure 3.30. The implementation of $\mathcal{L}_1$ adaptive controller for SMA finger angle tracking

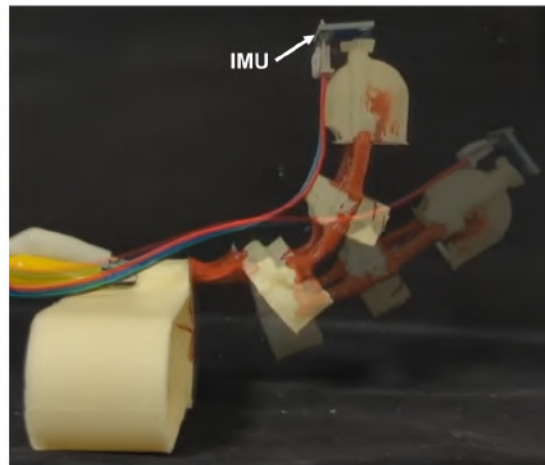
3.3.3 Experimental Validation

The motion capability of the device was evaluated using the $\mathcal{L}_1$ adaptive position tracking controller. The control algorithm was implemented in MATLAB/SIMULINK, and the desired angle was set. The voltage output published to the Arduino (Model: Mega 2560) through ROS (Robot Operating System) as in Figure 3.30. The SMA wire was powered through the external power source, and the Arduino signal controlled the desired voltage output. During the actuation, the angle of the device's fingertip was measured using an Inertial Measurement Unit (IMU) (Model: mpu6050 6-axis MEMS sensor), and the feedback was sent back to the controller using ROS.

During the experiment shown in Figure 3.31, the fingertip was bend up to 60° in 10 s and maintain the position for up to 25 s. The temperature of the SMA wire is shown in Figure 3.31 b) and the voltage output provided by the $\mathcal{L}_1$ adaptive controller is shown in Figure 3.31 c). This experiment reflects the real-time tracking capability of the proposed SMA model, the controller, and the hardware system presented in this dissertation. The initial stage of the actuation as well as the final stage is shown in Figure 3.31 b)



(a)



(b)

Figure 3.31. SMA actuated finger module angle tracking using $\mathcal{L}_1$ adaptive controller: (a) angle, temperature, and voltage plots, (b) initial and final instances of the fingertip

CHAPTER 4. ARTIFICIAL MUSCLES FOR UNDERWATER ROBOTICS

Artificial muscles hold great potential to become the leading underwater actuators. Their bio-inspired actuation mechanisms make them suitable for a variety of biomimetic robotic applications. Chapter 4 provides an in-depth exploration of underwater applications of artificial muscles.

In Section 4.1, we explore the actuation and control of Twisted and Coiled Artificial Muscles (TCAMs) in powering muscular hydrostats, enabling them to replicate contraction, extension, bending, and torsion motions underwater. This section presents both the theoretical foundation and practical insights into the implementation and control strategies for TCAM-driven hydrostats. These hydrostats were tested and controlled in real-world underwater scenarios, such as waves and currents. Additionally, this section covers data acquisition methods and the hardware circuitry implemented for underwater robots.

Expanding on the applications of artificial muscles, Section 4.2 introduces the actuation and control of Twisted and Spiraled Artificial Muscles (TSAMs) as vortex generators for drag reduction in small unmanned aerial vehicles (UAVs). Preliminary wind tunnel experiments using an airfoil demonstrate the effectiveness of TSAM-powered deployable vortex generators in reducing drag and optimizing aerodynamic efficiency. The extended work tested the TSAM-powered vortex generators in underwater environments. While this section inherits the sensing, data acquisition, and controller implementations from the previous section, the modeling techniques are unique due to the use of a different type of artificial muscle.

4.1 Octopus-Inspired Muscular Hydrostats Powered By Twisted and Coiled Artificial Muscles

4.1.1 Introduction

Conventional or traditional robots are designed to operate in a structured environment for the repetition of tasks with precision and superhuman speed. While these robots can be incredibly powerful and precise, their rigid structure restricts their range of motion, limiting their application to the domains of manufacturing and industrial automation [2]. However, as the field of robotics expands into realms such as health care, underwater exploration, and cooperative human assistance, robots are evolving to be more compliant, actively interacting with their environment. The development of continuum robots made of soft compliant materials, capable of undergoing large deformations and exhibiting stiffening mechanisms, has led to the emergence of the burgeoning field of soft robotics [259].

Due to their structural compliance, soft robots can conform to surfaces or objects, absorb energy to maintain stability, and adapt to unpredictable changes in their environment [4]. However, the compliant nature of soft robots also introduces new challenges in terms of design, manufacturing, modeling, and controls [260]. In addressing these challenges, researchers have turned to nature for inspiration, finding solutions by studying animals that combine soft tissues with stiffening mechanisms to maneuver and interact with complex, unpredictable environments [261]. Among the animals providing rich inspiration, octopodes' arms have been extensively studied [262–268]. Octopodes' arms are continuum structures without a rigid backbone (Figure 4.1(a)), providing them with a versatile repertoire of movements for locomotion, camouflage, hunting, fighting, and mating [269, 270].

The unique capabilities of octopodes' arms are derived from a muscular arrangement known as muscular hydrostats [271]. Muscular hydrostats are biomechanical systems composed of muscle and soft tissues, lacking rigid skeletal support for the muscles to act upon. In octopodes' arms, muscular hydrostats are arranged as thin layers around an axial nerve cord (AN in Figure 4.1(b)). Each

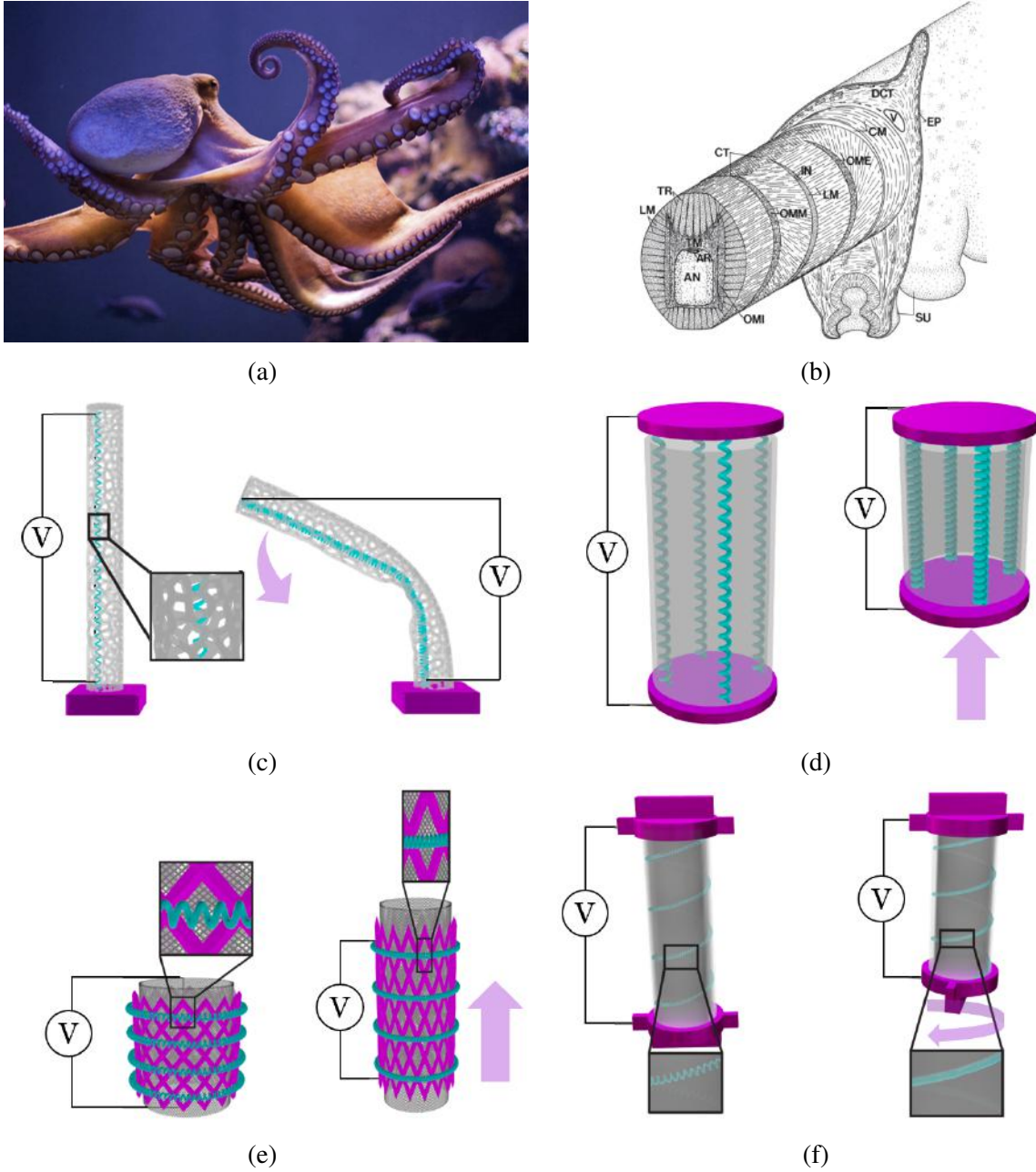


Figure 4.1. Bioinspiration and TCAM Arrangement for Proposed Hydrostats: **(a)** Octopus limbs exhibit extraordinary degrees of freedom with fine control. **(b)** Schematic diagram of octopus limb muscular anatomy with oriented fiber muscle layer. Schematic illustration of the proposed: **(c)** Bending hydrostat (BH), **(d)** Contraction hydrostat (CH), **(e)** Extension hydrostat (EH), and **(f)** Torsion hydrostat (TH).

layer contains tightly packed muscle fibers oriented in different directions, including longitudinal muscles (LM), transverse muscles (TM), and internal, external, and medial oblique muscles (OMI, OME, and OMM, respectively), as depicted in Figure 4.1(b). Other abbreviations in Figure 4.1(b)

are as follows: AR–artery; CM–circumferential muscle layer; CT–connective tissue; DCT–dermal connective tissue; EP–epidermis; IN–intramuscular nerve; SU–sucker; TR–trabeculae; V–vein. Activation of one or more of these muscle fibers drives the four primary motions of an octopode’s arm: shortening, torsion, bending, and extension. Numerous successful designs, such as the OCTOPUS Integrating Project [272], the OctArm manipulator [273, 274], and the STIFFness controllable Flexible and Learnable manipulator for surgical operations [275], have been proposed to achieve the versatile range of motion exhibited by octopodes’ arms.

However, these designs employ conventional actuator technologies such as pneumatic actuators, cable-driven systems, and electrical motors, which still present the main limitations of traditional robots in terms of low power/weight ratios and compliance. Therefore, there is a strong need for new actuator technologies that can generate large power/weight ratios within a soft and compliant structure.

Artificial muscles represent a new breed of lightweight actuators with a high power/weight ratio, capturing the attention of scientists for soft robotic applications [7, 276–278]. In a previous work, we explored the benefits and shortcomings of different classes of artificial muscles [7]. Unlike multicomponent conventional actuators, artificial muscles can reversibly contract, expand, or rotate within a single component when excited by an external input [7].

With regard to muscular hydrostats inspired by octopodes’ arms, researchers have employed artificial muscles such as shape-memory alloys (SMAs) [277–280], dielectric elastomers (DEs) [281–283], and McKibben actuators [284–287] to drive soft robotic manipulators. However, these artificial muscles present significant limitations. McKibben actuators exhibit high-power density, speed, and strain values but require heavy and noisy air compressors for actuation [288]. SMAs can generate high stress values, but they are expensive, difficult to control, and exhibit significant hysteresis during contraction [289]. DEs exhibit high strain values, but they require high voltages (kilo-megavolt range) and pretraining for their operation [290].

In this work, we propose octopus-inspired muscular hydrostats powered by an emerging class of artificial muscles called twisted and coiled artificial muscles (TCAMs) [99, 291–294].

TCAMs are fabricated by twisting and coiling inexpensive fibers, capable of sustaining stresses up to 60 MPa and providing tensile strokes of nearly 50 % with $<0.2 \text{ V cm}^{-1}$ of input voltage [112]. Additionally, we recently demonstrated that TCAMs can offer the same active and passive force profile of skeletal muscles within a single component [60]. Cyclic testing of nylon 6 TCAMs has been performed in a range of environmental conditions, demonstrating less than a 1% change in stroke length over 5000 cycles. The TCAMs maintain their ability to withstand a consistent stress, with a constant response rate, a 2.2 s activation time, and a 46.2 s cooling period. These performances overcome the limitations of shape-memory alloys (SMAs), McKibben actuators, dielectric elastomers (DEs), and other artificial muscles in terms of cost, power, and portability [223], leading to the development of truly bioinspired muscular hydrostats.

Four different configurations of muscular hydrostats with TCAMs arranged in different orientations are designed to emulate the four main behaviors of octopodes' arms: bending (Figure 4.1(c)), shortening (Figure 4.1(d)), extension (Figure 4.1(e)), and torsion (Figure 4.1(f)). This work marks the initial step toward the development of a lightweight, octopus-inspired soft tentacle capable of exhibiting a broad repertoire of complex movements.

4.1.2 Semi-Automated polymer fishing-line TCAM manufacturing system

Nylon 6 TCAMs are manufactured on a custom-built automated twister is shown in Figure 4.2. Manufacturing begins by crimping hook terminals to the ends of a strand of precursor fiber 3.25 times the length of the desired actuator and mounting to the machine. Figure 4.2(a) shows the heating wire spool for guiding through wire carriage and wrapping precursor fiber. The wire is run down through the wire carriage and back up to the proximal hook terminal and secured with tape, and any slack in the wire is removed. In Figure 4.2(b), Wire carriage lead screw, wire carriage, bearing housing, and wire motor (M2) assembly guides conductive wire around precursor fiber during twisting. Arduino Uno, motor drivers, and DC controller circuit for driving the motors M1-M3 is shown in Figure 4.2(c).

The detailed PCB schematic is available in Appendix A.1. The full machine setup enables

the incorporation of muscles with varying lengths and fiber types, as well as both self-coiled and mandrel-coiled processes [295], as depicted in Figure 4.2(d). This comprehensive setup ensures versatility in the fabrication of muscular hydrostats tailored to specific requirements.

In Figure 4.2(e), the top twist motor M1, driven by a DC motor, facilitates smooth rotation of the fiber during wire wrapping and self-coiling processes. This critical component ensures precise control over the twisting mechanism, crucial for achieving optimal performance of the muscular hydrostats.

Transversely mounted mandrel-coiling motor, illustrated in Figure 4.2(f), along with self-

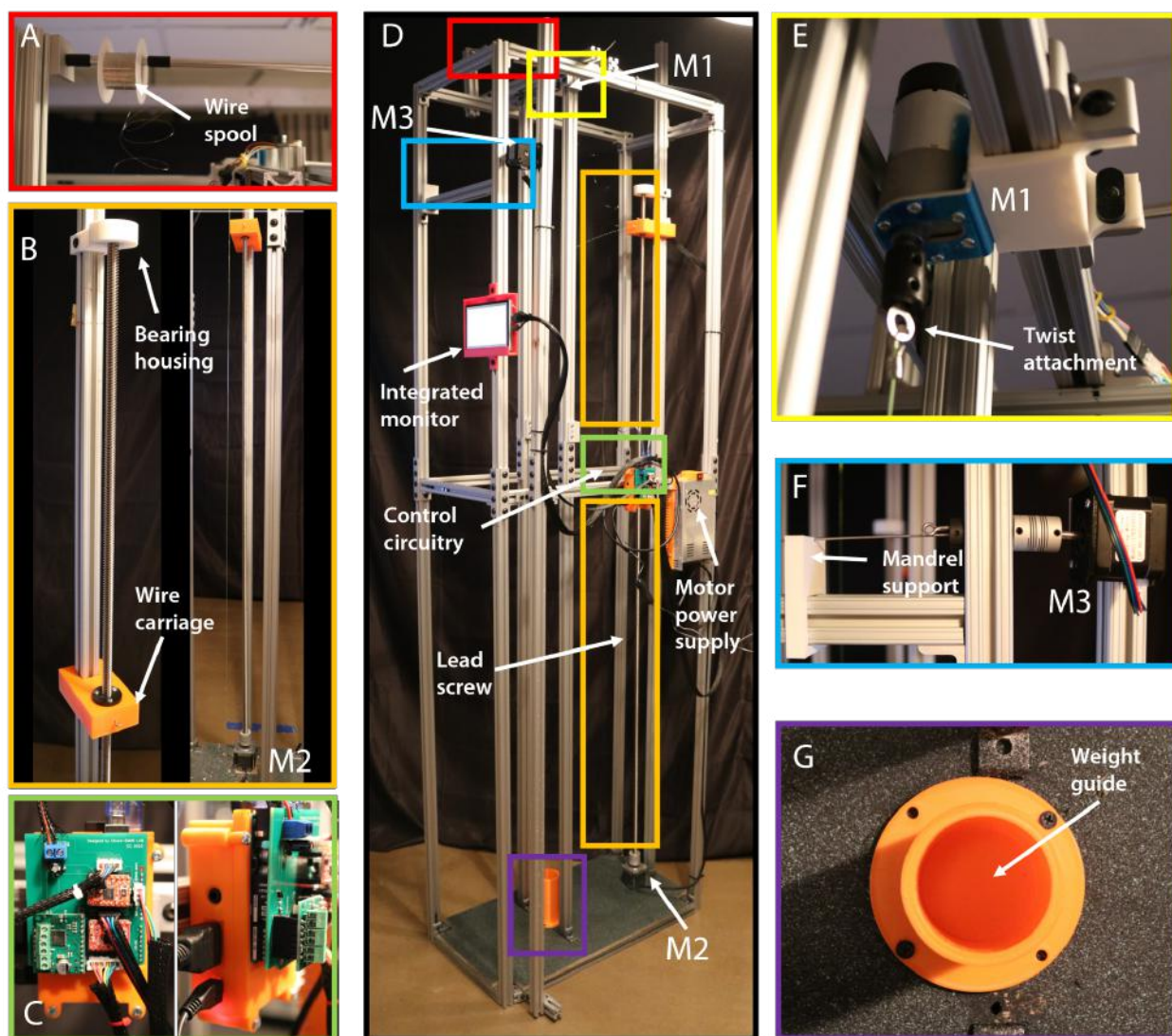


Figure 4.2. Automated manufacturing machine components and steps

driving mandrel support, automates the mandrel-wrapping process with precision. This automation streamlines the manufacturing process, enhancing efficiency and consistency in the fabrication of TCAMs.

Figure 4.2(g) showcases the weight guide tube mounted to the base of the machine. This apparatus facilitates the vertical travel of the tensing load during the wire-wrapping process while ensuring coaxial alignment with the fiber-twist motor. Additionally, a secondary fiber is strategically connected between the 1 kg tensioning load and the precursor fiber, enabling complete wrapping of the heating wire without premature coil formation in the muscle.

Moving to the manufacturing process, Figure 4.3(a) illustrates the initial steps, where the conductive wire is meticulously wrapped around the nylon fishing line during the twisting process. Subsequently, in Figure 4.3(b), the coiling process unfolds, showcasing the TCAM spontaneously coiling into its characteristic shape.

Transitioning to the post-coiling phase, Figure 4.3(c) captures the coiled TCAMs positioned on a rack, ready for transfer to the oven for annealing. These TCAMs are meticulously

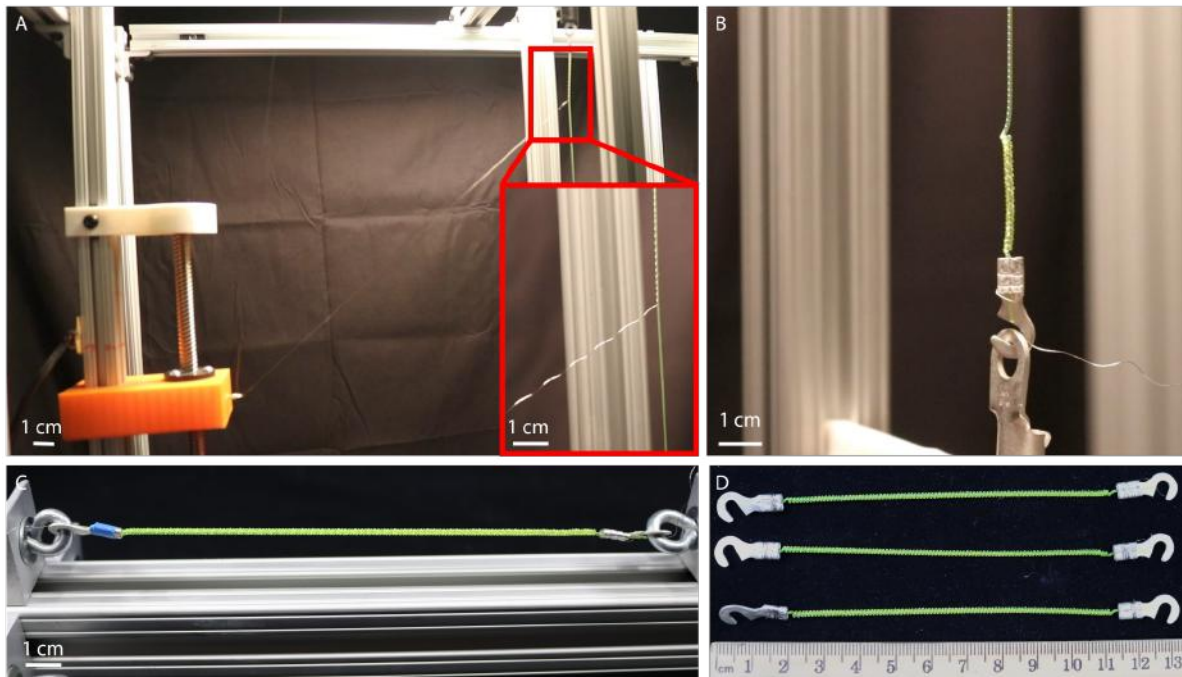


Figure 4.3. (a) Manufacturing process for self-coiled muscles; (b) Spontaneous coiling TCAM; (c) Completed self-coiled TCAM on annealing rack; (d) Completed self-coiled TCAM

placed under tension to prevent untwisting and ensure the integrity of the coiled structure during the annealing process. Upon completion of annealing, as depicted in Figure 4.3(d), the TCAMs demonstrate their ability to maintain their shape without uncoiling, underscoring the effectiveness of the annealing process in stabilizing the structure and properties of the muscular hydrostats.

4.1.3 Contraction muscular hydrostat: Actuation modeling and validation

In octopodes' arms, muscles (LM in Figure 4.1(b)) that drive the shortening mechanism are oriented parallel to the longitudinal axis of the arm. The contraction hydrostat (CH) used for emulating the shortening mechanism of octopodes' arms is depicted in the schematic of Figure 4.1(d) and Figure 4.4(a). The CH consists of four TCAMs attached to two three-dimensional (3D) printed holders and wrapped with a thin silicone sheet. The total weight of the device is 99 g.

Figure 4.4(b) and 4.4(c) depict the CH before and after TCAM actuation, respectively. A maximum contraction of 40 % is depicted in Figure 4.4(c). The shortening of the CH is derived directly from the radial expansion of TCAMs. The TCAM actuation can be directly related to the geometry, loading conditions, and mechanical properties of the TCAM materials with simple mechanical models we developed. TCAMs are fabricated by overtwisting Nylon 6 fishing lines under a tensile load until spontaneous coiling (Figure 4.4(d)). In this work, we fabricated TCAMs using an automated twister setup and electrothermally actuated using a direct current power supply through a microcontroller and transistor. The plot in Figure 4.4(e) depicts the validation of our electrothermal model, while Figure 4.4(f) shows validation of the dynamic model for the CH displacement.

The shortening of the CH is derived directly from the radial expansion of TCAMs. The TCAM actuation can be directly related to the geometry, loading conditions, and mechanical properties of the TCAM materials with simple mechanical models we developed [27, 55, 81, 118]. TCAMs are fabricated by overtwisting Nylon 6 fishing lines under a tensile load until spontaneous coiling (Figure 4.4(d)). In this work, we fabricated TCAMs using an automated twister setup (Figure 4.2) and electrothermally actuated using a direct current power supply through a micro-

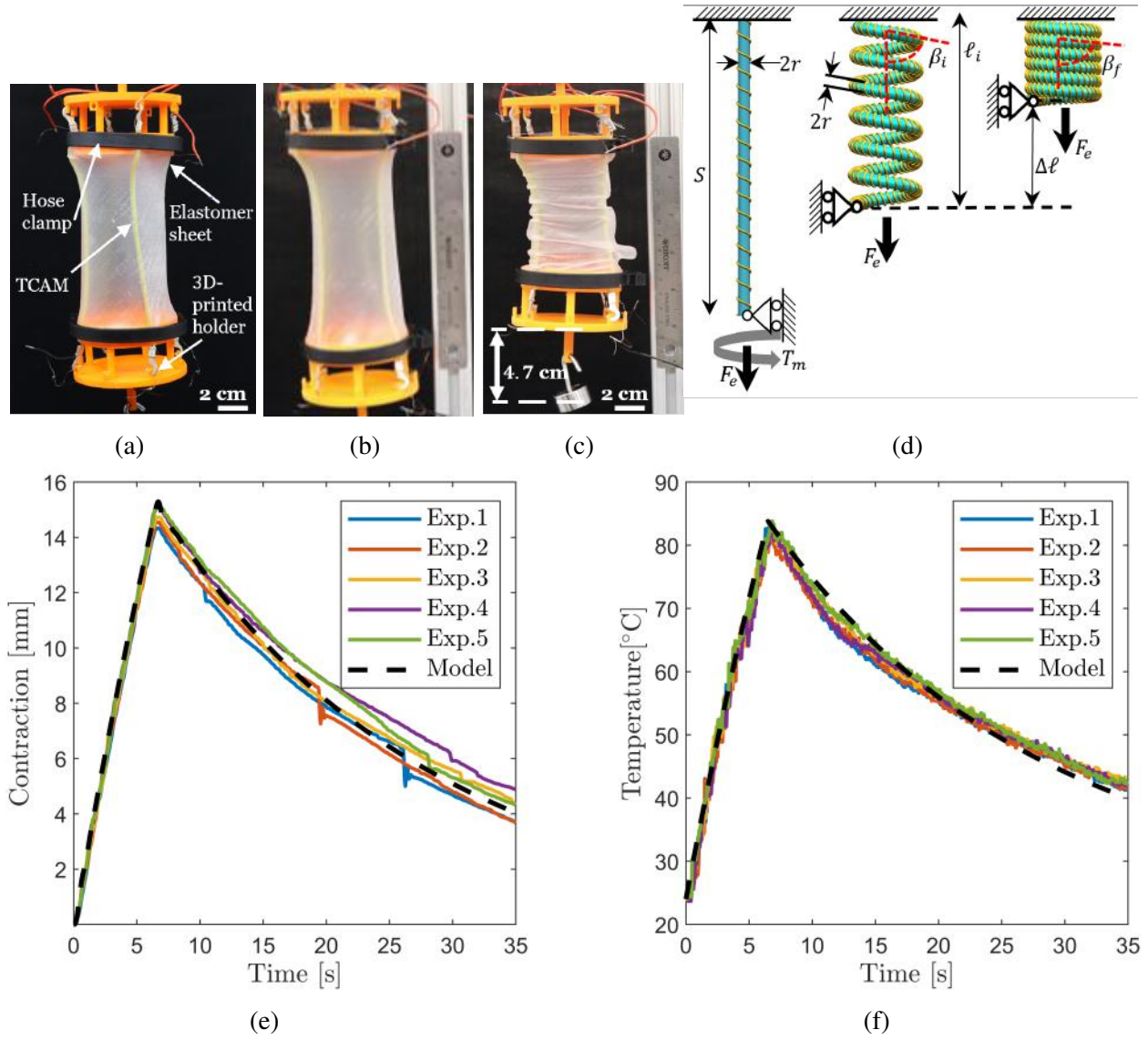


Figure 4.4. (a) CH containing 4 TCAMs attached to 3D printed holders. A silicone elastomer sheet is attached to the holders using hose clamps. (b) CH before and (c) after electrothermal actuation of TCAMs using an input power of 62.4 W for 12 s. A maximum shortening of 4.7 cm (40 %) is achieved on TCAM contraction. (d) Working principle of nylon fishing line TCAMs. Validation of the (e) electrothermal and (f) dynamic model for TCAMs employed in our CH. 3D, three-dimensional.

controller and transistor. The plot in Figure 4.4(e) depicts the validation of our electrothermal model, while Figure 4.4(f) shows validation of the dynamic model for the CH displacement.

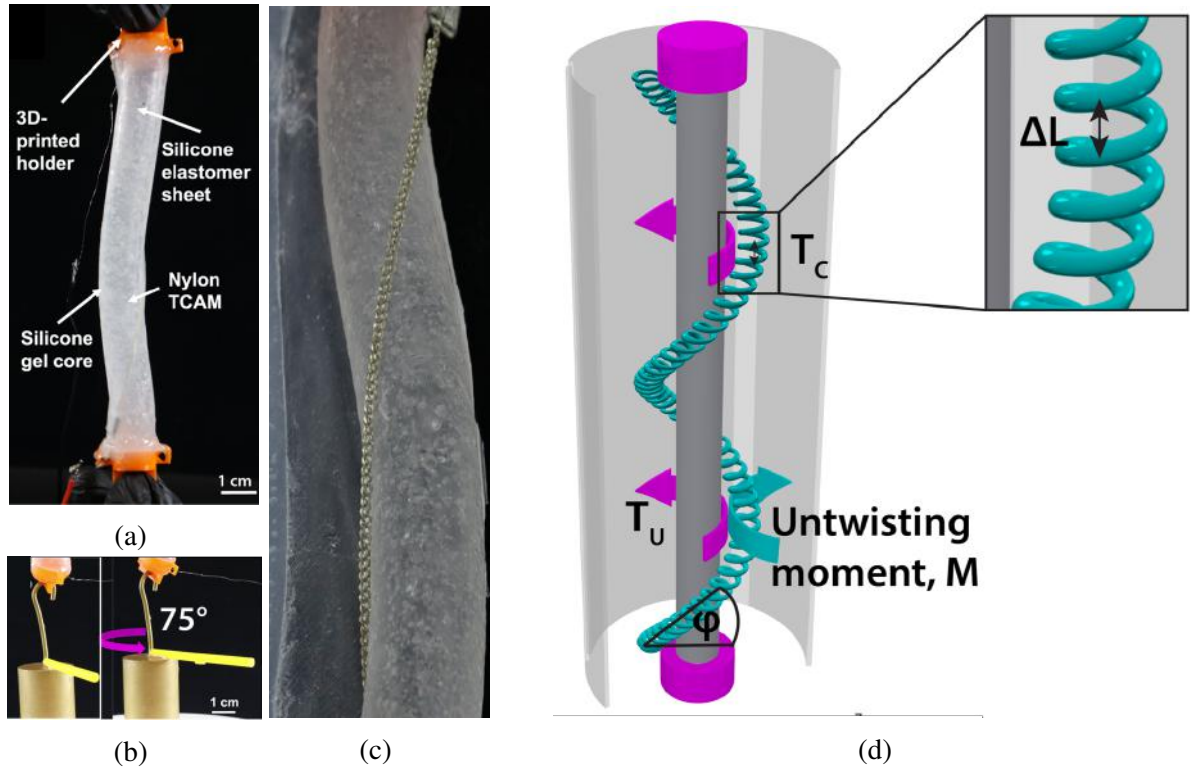


Figure 4.5. TH with theoretical model. **(a)** TH device. **(b)** Current TH can achieve up to 75° of rotation. **(c)** Detail of TH showcasing the gel core, TCAM winding, and opened external skin. **(d)** Expanded view of torsion device schematic highlighting dual contribution of TCAM actuation and the closure of intercoil spacing along the length of the TCAM can be viewed as an arc length along the circumference of the gel core.

4.1.4 Torsional muscular hydrostat: Actuation modeling and validation

Limb torsion is achieved by the octopus through alternating layers of muscle fibers arranged obliquely to the axial nerve core (OMI, OME, and OMM fibers in Figure 4.1(b). [263] In our soft robotic analog (Figure 4.5(a)) we replicate the AN with a silicone gel core with 3D printed caps at the ends serving as attachment points for the associated TCAM. The TCAM is helically wound around the gel core and secured tightly along the entire length of the hydrostat by an outer skin of silicone elastomer. The proximal cap is used to mount the device to a frame for suspension and 2 N of pretension is applied to the TCAM at the distal cap. The combined device weight is 38 g and can demonstrate a maximum twist of 75° in current iterations (Figure 4.5(b)). The direction of motion is determined by the direction of twist insertion into the muscle during manufacturing, and

the magnitude of the rotation can be maximized by matching this twist direction with the direction of muscle wrapping around the central core. Details of the hydrostat showing the layered arrangement of the core, TCAM, and skin can be seen in Figure 4.5(c). We developed a theoretical model to predict the angle of twist achieved by the torsional hydrostat (TH) device, which is dictated by the geometric properties of the TCAM and the gel core. The central gel core mimics the AN of the in vivo octopus' limb, and we treat its behavior as a hyperelastic solid rod operating within a linear elastic region. The torsion angle of a linear elastic solid rod can be written as follows:

$$\alpha = \frac{LT}{JG} \quad (4.1)$$

where α is the angular deflection of the shaft in radians, L is the length of the shaft, T is the applied external torque, J the polar moment of inertia of the shaft, and G is the shear modulus of the shaft.

The first and dominant contribution is from the untwisting of the TCAM fiber as it is heated and radially expands. The gel core has a highly tacky surface, which enables transfer of the untwisting moment from the fiber to the core when the fiber is held tightly against it by the outer skin (Figure 4.5(c), 4.5(d)). This untwisting moment generates a torque on the gel core that contributes to the torsional motion of the cylinder.

The direction of rotation for the overall cylinder is determined by the direction of twist inserted into the artificial muscle during manufacturing. If the fiber is twisted in the clockwise (cw) direction during manufacturing, then upon heating the fiber untwists in the counterclockwise (ccw) direction. This ccw rotation held against the compliant gel core causes the cw rotation of the entire device due to the high friction between the TCAM and the gel core. The magnitude of rotation can be maximized by matching the direction of twisting of the muscle with the direction of winding of the muscle around the gel core and is reduced when the winding and twisting directions are opposed. This effect occurs because of the second twisting motion imposed by the TCAM.

The second component of motion occurs as a result of the linear contraction of the TCAM upon actuation (Figure 4.5(d)). Helical winding of the actuator around the gel core means that a

linear shortening of the muscle acts to draw the termini of the device closer together following the helical path. The shortening of the TCAM can be calculated using our theoretical models we developed previously [81, 83, 148].

4.1.5 Bending muscular hydrostat: Actuation modeling and validation

In octopodes' arms, bending is achieved through selective contraction of LM fibers along one side of the arm in combination with activation of TM fibers that resist longitudinal shortening [263]. To maximize bending actuation and minimize the number of TCAMs in our bending hydrostat (BH), we exploited mechanical instabilities to convert the contraction of a single TCAM into a bending motion (Figure 4.6(a)). The device is designed by applying a Voronoi tessellation algorithm [296] to a primitive cylinder with a diameter of 2 mm and a length of 17.5 cm. We tailored the density of Voronoi cells in the primitive cylinder to allow for nonuniform topological distribution of the lattice points for controlled bending. The Voronoi cells were distributed with decreasing density toward the center of the cylinder along the radial direction to ensure muscle contraction was converted into bending instead of pure shortening (Figure 4.6(b), 4.6(c)). Hydrostat bending depicted in Figure 4.6(d) is achieved by actuating the TCAM using an input power of 10 W for 3 s. A maximum bend of 72° is depicted in Figure 4.6(d).

The BH was fabricated through a stereolithography technique using a flexible photopolymer resin. A single TCAM attached to the Voronoi cylinder through 3D printed holders and eyebolts was used to initiate bending. The total weight of the BH was 27 g.

In order to demonstrate the $\mathcal{L}_1$ adaptive controller in the context of a practical device and extend past academic tests, we developed and controlled a soft robotic manipulator actuated by a nylon TCAM. The device consists of a 3D-printed elastomer cylinder (Elastic 50A Resin by formlabs was used as resin) characterized by a mesh of unit cells with different sizes. The mesh, also known as Voronoi tessellation [297], is designed, so that bigger unit cells are placed in the center of the cylinder to induce mechanical instability and then maximize bending when a nylon TCAM is placed inside the cylinder is actuated. The total weight of the device is 15 g, and the

length is 17 cm. The goal of our experiment was to reach a target bending angle of 13 degrees in order to close an electric circuit (by contacting a metal pin placed on the top extremity of the device) and power a flywheel (with the University of Iowa Hawkeye logo) and a LED (see Figure 4.7(a)). The desired angle is defined from Matlab/SIMULINK with a geometric relationship between the angle and TCAM displacement and sent through ROS to a microcontroller (Model: Arduino Mega2560) and external power supply system to actuate the TCAM. Then, an Inertial Measurement Unit (IMU Model: mpu9250 9-axis MEMS sensor) measures the soft manipulator orientation, and the real-time sensor data feedback is sent through ROS to the $\mathcal{L}_1$ adaptive controller implemented in Matlab/SIMULINK. Figure 4.7(b) shows the desired and measured angle and 4.7(c) shows the voltage generated by the $\mathcal{L}_1$ adaptive controller. One can observe as the device is able to reach the desired angle in 30 seconds and power the flywheel and the LED.

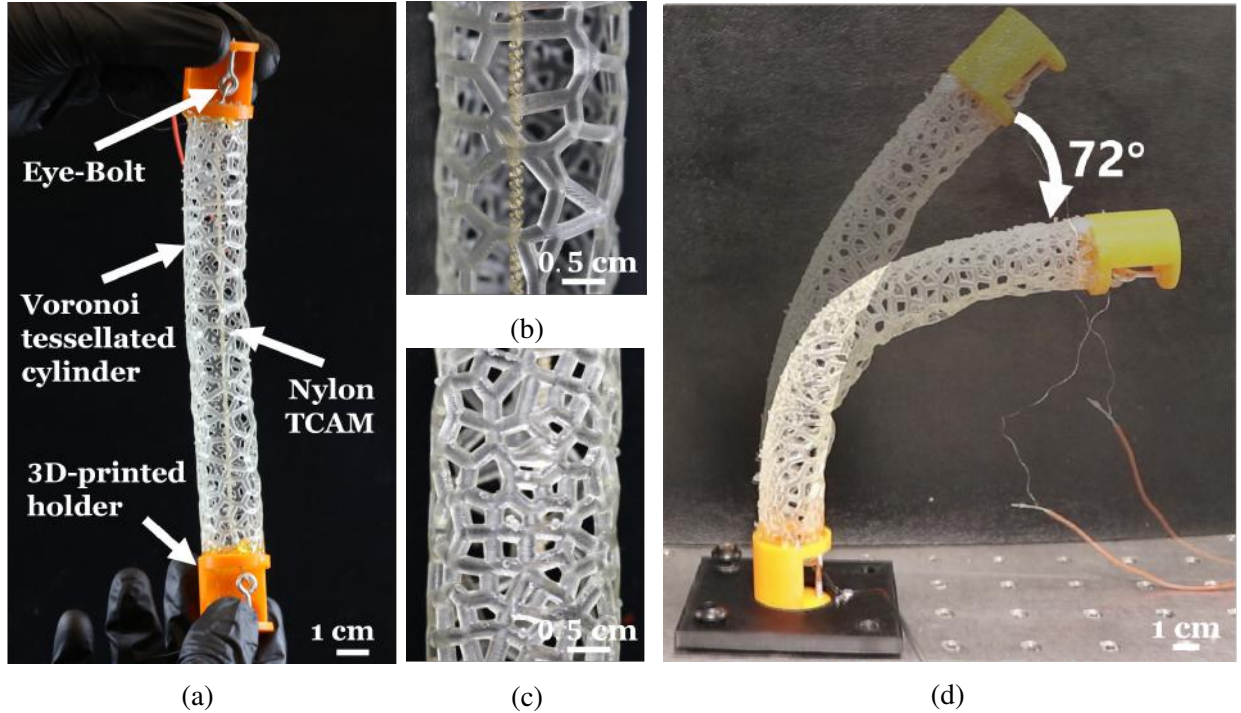


Figure 4.6. Images, actuation, and FE simulations of BH. **(a)** BH consisting of a Voronoi tessellated cylinder with a mechanical instability and a single TCAM attached to the cylinder through 3D-printed holders and eyebolts. **(b)** Larger less dense Voronoi cells. **(c)** Small dense Voronoi cells reinforce the back of the cylinder to control the direction of bending. **(d)** Hydrostat undergoing bending due to TCAM contraction using an input power of 10 W for 3 s. A maximum bend of 72° is achieved on TCAM actuation.

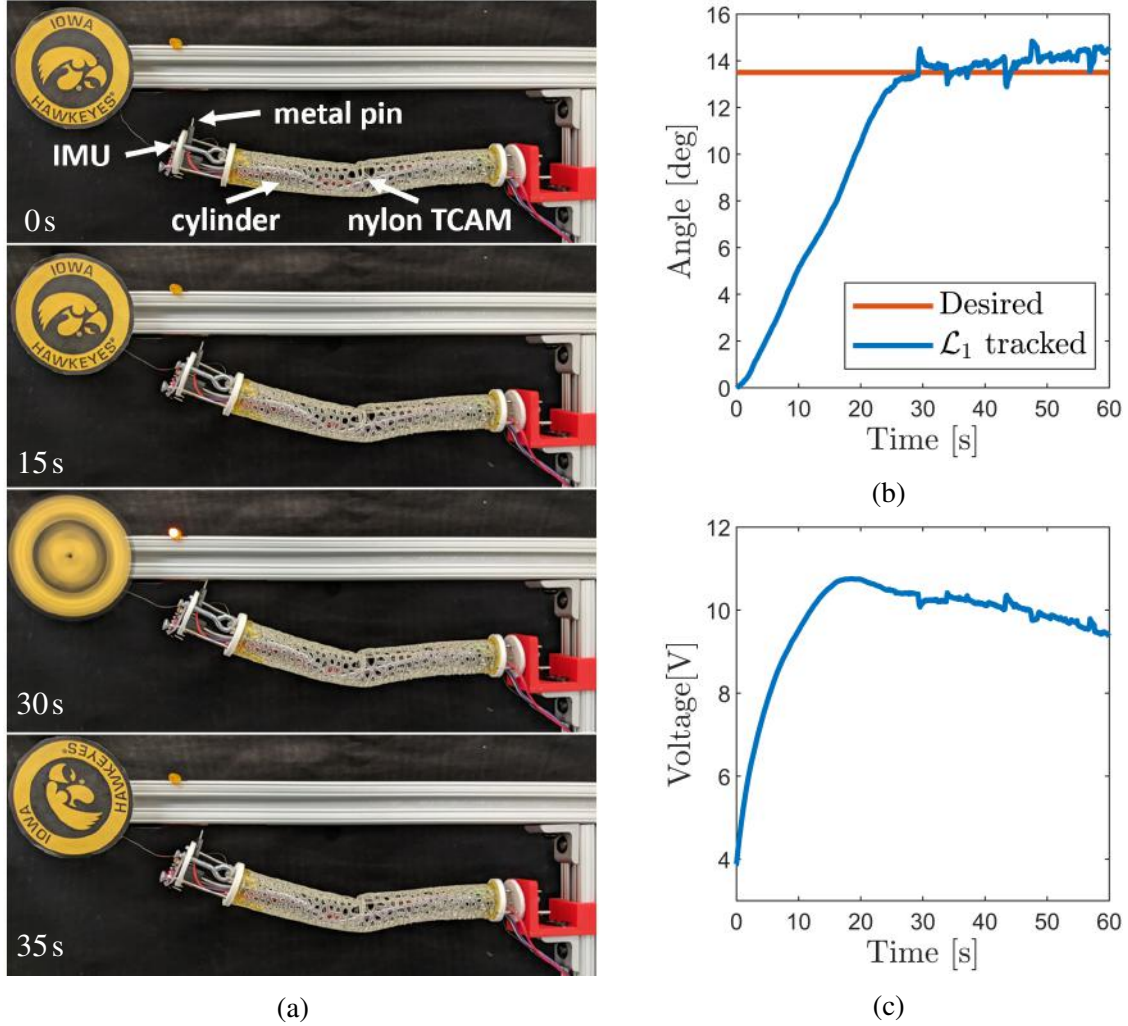


Figure 4.7. Application of TCAMs- Soft Bending Manipulator. a) Time-lapse diagram of bending manipulator motion, and b) orientation tracking and voltage provided by the $\mathcal{L}_1$ controller.

4.1.6 Extension muscular hydrostat: Actuation modeling and validation

Limb extension is achieved by octopodes through the selective contraction of TM fibers along its length [263]. In accordance with the isovolumetric nature of the arm, any decrease in the cross-sectional area will result in an extension of the limb, with the magnitude of the limb extension dictated by the degree of contraction of the TM fibers as well as the number recruited along the length of the limb. This selective contraction can be highly localized for fine control and positioning of the limb.

In the proposed extension hydrostat (EH), circumferentially arranged TCAMs have been

preferred over radially arranged TCAMs to develop a more compact device with a small number of actuators and with an empty core, which will foster the assembly of different muscular hydrostat layers for soft tentacles.

In our hydrostat, we achieve elongation using a combination of the circumferentially arranged TCAMs and a composite mesh array with a large Poisson's ratio. In our EH design, circumferential TCAMs contract to impinge upon the composite mesh array, which then converts the radial compression into an axial elongation. The composite mesh array comprises a polyethylene terephthalate (PET) braided sleeve with a coating of silicone elastomer and a red silicone mesh (Figure 4.8(a), 4.8(b)). The unit cell of the red silicone mesh consists of struts with a deformable hinge that allows extension of the cell when a radial contraction is applied (Figure 4.8(c), 4.8(d)).

4.1.7 Hardware layout and transfer function for the underwater actuation and control of muscular hydrostats

The proposed muscular hydrostats were actuated and controlled

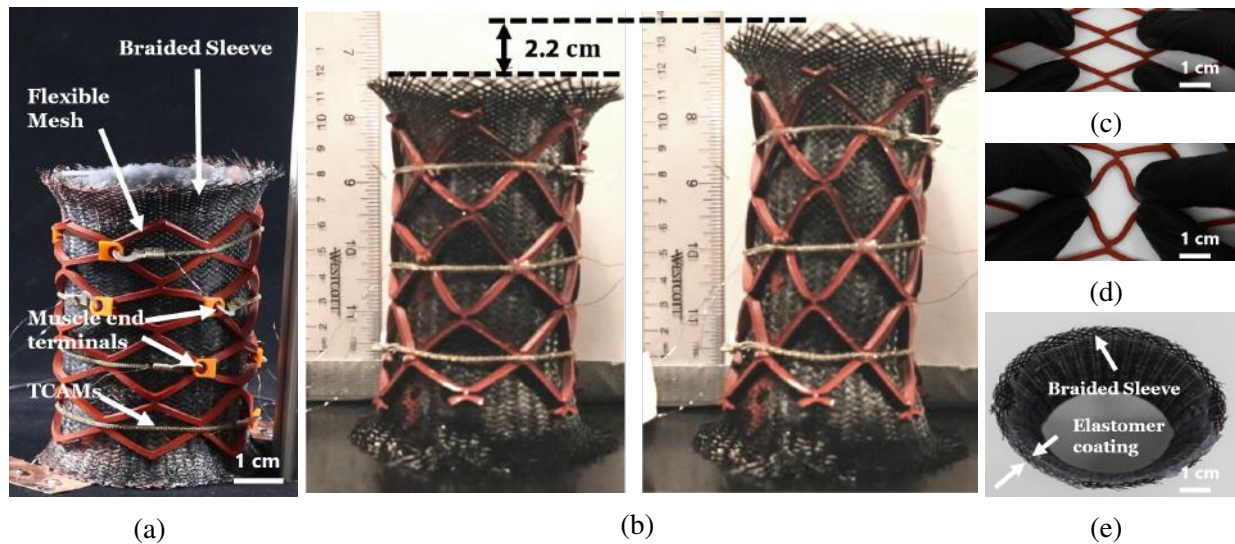


Figure 4.8. Images, actuation, and FE simulations of extension muscular hydrostat. **(a)** EH prototype comprising a PET braided sleeve with an elastomer coating, a flexible mesh, 4 TCAMs, and TCAM end terminals; **(b)** EH after TCAM contraction depicting an extension of 18.3% (2.2 cm). TCAMs are electrothermally actuated using an input power of 40 W for 6 s; Unit cell of the red silicone mesh **(c)** before and **(d)** after extension; **(e)** Braided sleeve frozen into an initial compressed state using a thin layer of silicone elastomer.

The S^4 muscular hydrostats were experimentally validated for their tracking capabilities. The following three experiments were carried out:

1. S^4 devices in air and underwater using the $\mathcal{L}_1$ adaptive controller
2. S^4 devices in air and underwater using both $\mathcal{L}_1$ adaptive and PID controllers
3. S^4 devices inside the towing tank against high-frequency high-amplitude waves;
4. S^4 devices inside the towing tank against high current waves

4.1.7.1 S^4 devices in air and underwater using the $\mathcal{L}_1$ adaptive controller

Conducting experiments underwater presented challenges in feedback sensing. One viable candidate for feedback sensing was an underwater-compatible camera. The video data captured by the camera during hydrostat actuation was transmitted to MATLAB for adaptive controller analysis. To initiate the data acquisition process, marker points $O = (O_x, O_y)$ and $\bar{P} = (\bar{P}_x, \bar{P}_y)$ were defined through pixel selection. Throughout actuation, the point P was monitored as $P(t) = (P_x, P_y)$. The MATLAB Image Processing Toolbox facilitated the definition of tracking points, camera calibration, and linearization, while the MATLAB Computer Vision Toolbox was employed for marker point tracking during actuation.

The actuation of each device was quantified by tracking marker points with the camera. The camera view for the contraction MH is depicted in Figure 4.9a. The initial point is denoted by $\bar{P} = (\bar{P}_x, \bar{P}_y)$, and the point during any actuation phase is denoted by $P(t) = (P_x, P_y)$. The marker point P is illustrated in Figure 4.9b. Similarly, the camera view for the torsional MH is presented in Figure 4.9c, where the yellow rod rotates as the torsion MH actuates. This approach remained consistent across all four setups shown in Figure 4.9a-h. The bending MH angle was determined by measuring the position of point P relative to the initial points $\bar{P}$ and (O_x, O_y) .

The bending angle ($\theta_b(t)$) and torsion angle ($\theta_t(t)$) are calculated using the following formula:

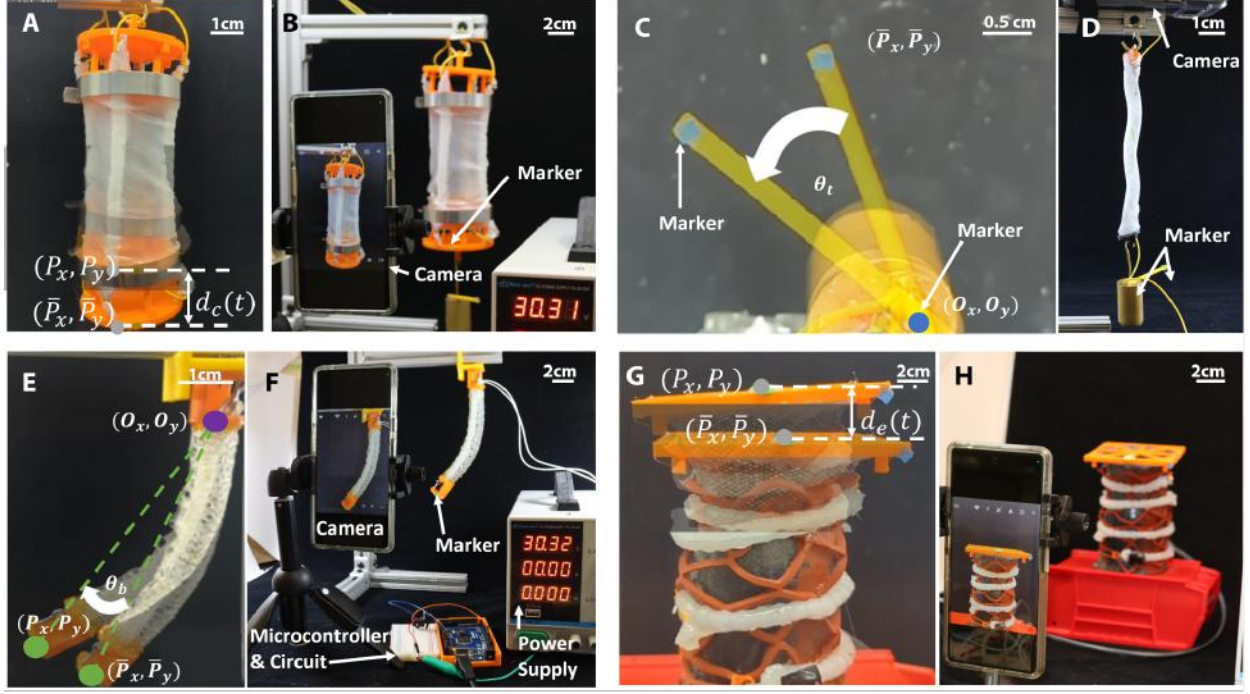


Figure 4.9. Angle/displacement measurement of each muscular hydrostat. Camera detection markers for the (a) CH, (c) TH, (e) BH, and (g) EH, respectively. Real-time experimental measurement system for the (b) CH, (d) TH, (f) BH, and (h) EH, respectively.

$$\theta_b(t) = \theta_t(t) = \cos^{-1} \left(\frac{\ell_1^2 + \ell_2^2(t) - \ell_3^2(t)}{2\ell_1\ell_2(t)} \right) \quad (4.2)$$

Here,

$$\begin{aligned} \ell_1 &= \sqrt{(\bar{P}_x - O_x)^2 + (\bar{P}_y - O_y)^2} \\ \ell_2(t) &= \sqrt{(P_x - O_x)^2 + (P_y - O_y)^2} \\ \ell_3(t) &= \sqrt{(\bar{P}_x - P_x)^2 + (\bar{P}_y - P_y)^2}. \end{aligned}$$

Similarly, contraction ($d_c(t)$) and extension ($d_e(t)$) of the hydrostats are measured using the following equations.

$$d_c(t) = d_e(t) = \sqrt{(\bar{P}_x - P_x)^2 + (\bar{P}_y - P_y)^2}. \quad (4.3)$$

The computed measurements, generated by the MATLAB script, are transmitted to and

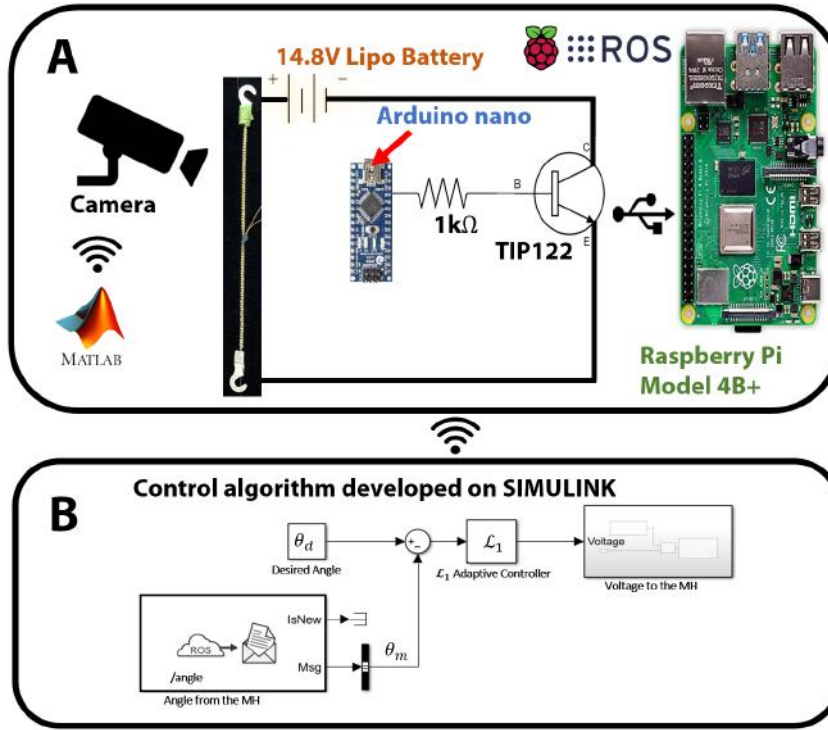


Figure 4.10. Schematic layout of the onboard hydrostat control and actuation system. (a) The hardware architecture of the onboard actuation system. (b) The $\mathcal{L}_1$ adaptive control architecture implemented on SIMULINK.

subscribed by the controller SIMULINK script through the ROS (Robot Operating System) middleware suite. The voltage, as estimated by the controller algorithm, is then published to the Microcontroller-SIMULINK script. The implementation of this script utilizes the Simulink Support Package for Arduino Hardware.

The schematic layout in Figure 4.10 depicts the hardware configuration and control algorithm governing the onboard actuation and control of the muscular hydrostats. The power electronic circuit, composed of the Arduino Nano and TIP 122 Darlington transistor, served as the core of the onboard system.

The control algorithm was implemented using MATLAB/SIMULINK and wirelessly transmitted to the microprocessor through the Robot Operating System (ROS) communication package.

A wireless camera provided real-time video of device actuation for angle/displacement measurement. The controller subscribed (received) from the ROS node to the Simulink controller, and the calculated voltage was published (transmitted) to the ROS node.

4.1.8 Control and underwater actuation of TCAM-powered hydrostats

In this study, we utilized a previously developed $\mathcal{L}_1$ adaptive controller [83, 148] for closed-loop control of the hydrostats. Let TCAMs in S^4 MHs model as a second-order system of

$$m\ddot{\ell}(t) + b\dot{\ell}(t) + k\ell(t) = cT_{\Delta}. \quad (4.4)$$

Here, m , b , k , ℓ , and T_{Δ} represent the mass of the system, damping coefficient, stiffness coefficient, current length, and temperature of the TCAM at any given time t , respectively. The state space model for the system can be described as follows:

$$\begin{aligned} \dot{x}(t) &= A_p x(t) + B_p(u(t) + f(t, x(t))), \\ y(t) &= C_p x(t), \end{aligned} \quad (4.5)$$

and $f(t, x(t))$ represents time-varying model uncertainties and disturbances.

$x(t) = [d(t), \dot{d}(t), T_{\Delta}(t)]^{\top}$, and the input is the voltage squared, $u = V^2$, so the matrices (A_p, B_p, C_p) take the form

$$\begin{aligned} A_p &= \begin{bmatrix} 0 & 1 & 0 \\ -\frac{\bar{k}}{m} & -\frac{\bar{b}}{m} & \frac{c}{m} \\ 0 & 0 & -\frac{\bar{\lambda}}{C_t} \end{bmatrix}, \\ B_p &= \begin{bmatrix} 0 \\ 0 \\ \frac{1}{C_t R} \end{bmatrix}, \quad C_p = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}. \end{aligned}$$

Figure 4.11a, 4.11b, and Figure 4.10 illustrate the untethered waterproof hardware system employed for the underwater control and actuation of the hydrostats. The hardware system features a microcontroller that transmits the Pulse Width Modulation (PWM) voltage signal for the electrothermal actuation of the TCAM from a Raspberry Pi microprocessor through a power electronic circuit coupled with a Darlington pair transistor (Figure 4.11b). Connecting to the microprocessor via a universal serial bus hub, the microcontroller is powered by a 5V rechargeable battery. The power needed for TCAM actuation is supplied by a 14.8 V Lithium Polymer battery pack. The actuation of the CH and BH is depicted in Figure 4.11c and 4.11d, respectively.

The implementation of our $\mathcal{L}_1$ adaptive control algorithm on the proposed hydrostats is illustrated in Figure 4.12(a)–(d) and Figure 4.13(a)–(d). Comprehensive testing of all muscular hydrostats was conducted both in air and underwater to achieve the desired actuation. The dynamics of the BH, TH, and EH were grounded in a constant-strain configuration, while the CH dynamics were based on a constant-stress configuration. Figure 4.12(a),(b) depicts the position tracking of CH in air and underwater, showcasing the robustness of the $\mathcal{L}_1$ adaptive controller. In Figure 4.12(c), (d), the angle tracking of TH in air and underwater is presented. This device, being very lightweight, exhibited delicate motions, resulting in slightly fluctuating underwater tracking results, primarily due to the impact of waves generated within the small tank during actuation. Compared to TH, BH displayed greater rigidity, hence yielding better angle tracking results in both air and underwater environments (Figure 4.13(a),(b)). However, the EH, being very lightweight and complex in construction, posed challenges in achieving the desired displacements. Consequently, maintaining the desired trajectory proved challenging, resulting in slight fluctuations and converging tracking plots.

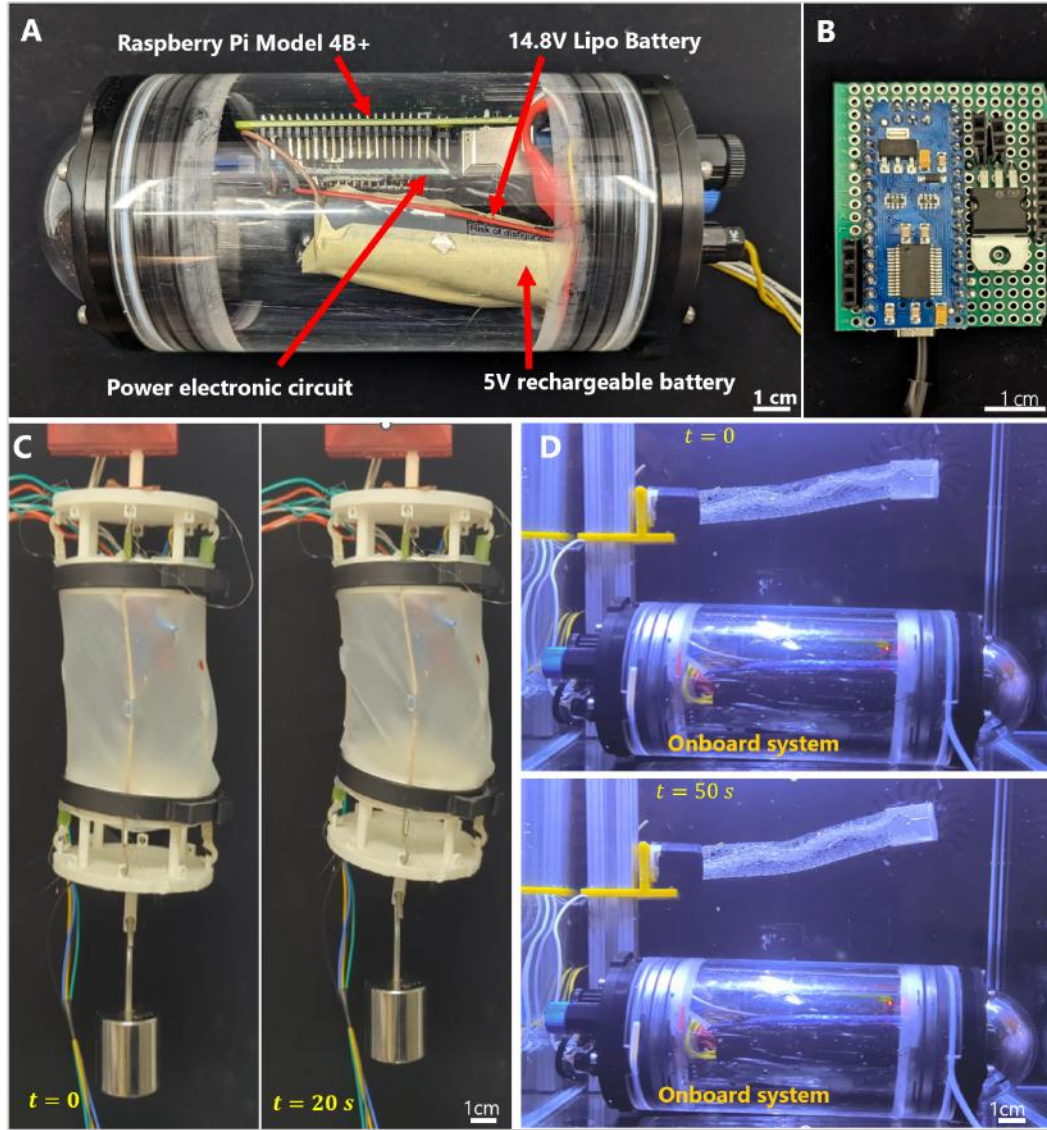


Figure 4.11. Control and underwater actuation of hydrostats. (a) The hardware configuration of the underwater onboard system. (b) The power electronic circuit for TCAM actuation (c) The CH before and during actuation. (d) The BH during the actuation.

4.1.9 Underwater Actuation of MHs for Waves and Currents

To demonstrate how well the proposed muscular hydrostat designs can actuate in real-world underwater applications with waves and current, we conducted a series of experiments. The experiments were carried out using the towing tank facility in the Fluids Fundamental Lab (1070 SC) at the University of Iowa. The facility is shown in Figure 4.14. The validation of the muscular hydrostat tracking was carried out using the $\mathcal{L}_1$ adaptive controller.

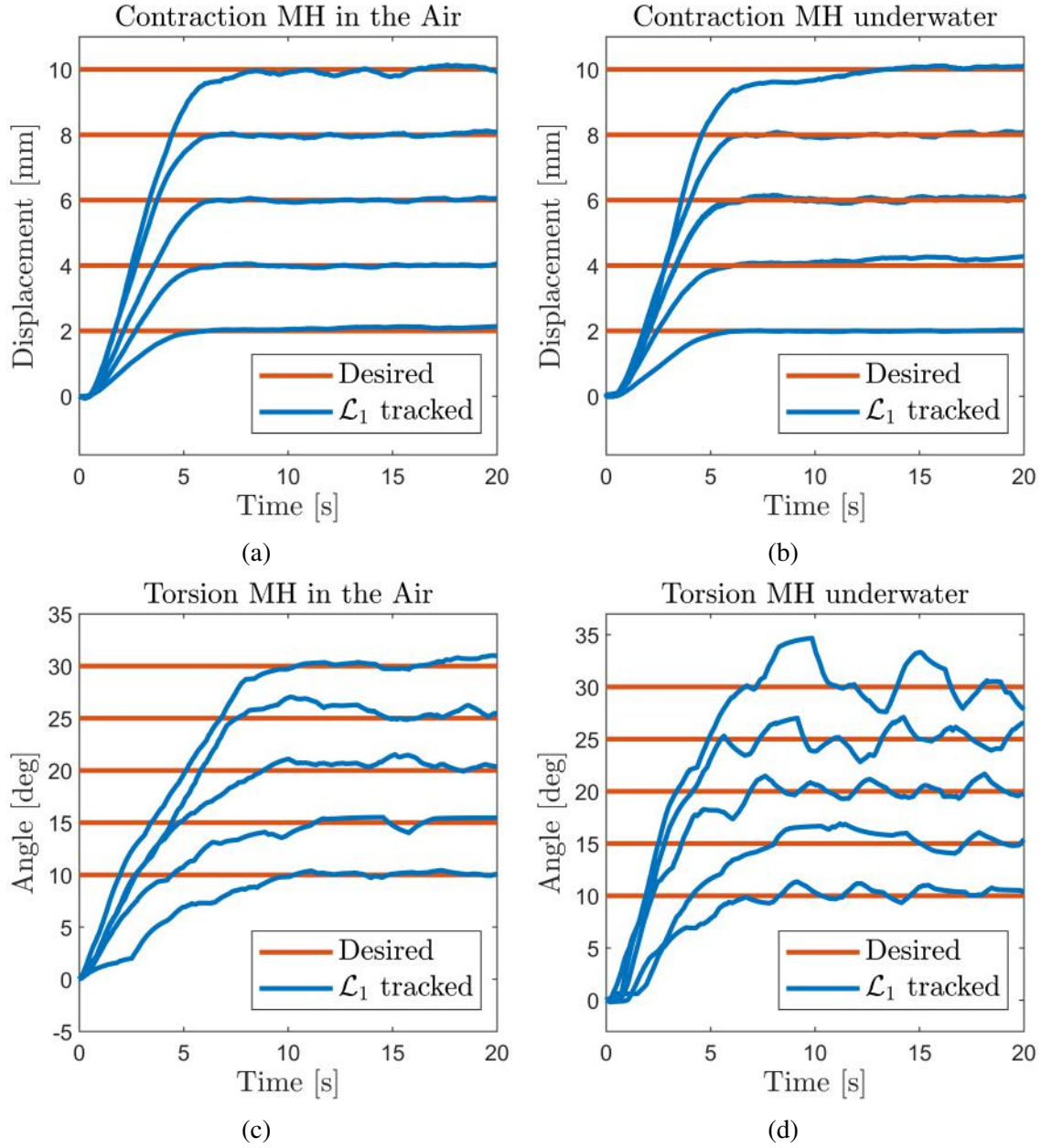


Figure 4.12. Displacement/Angle tracking for muscular hydrostats using the $\mathcal{L}_1$ adaptive controller. CH angle tracking in (a) air and (b) underwater. TH angle tracking in (c) air and (d) underwater.

The controller structure is depicted in Figure 4.15. For enhanced image clarity, we utilized a high-resolution video streaming and underwater-compatible camera (model: GoPro Hero 11), connected to MATLAB/SIMULINK via USB tethering. The camera was securely housed within a waterproof enclosure (model: BlueRobotics watertight Enclosure), alongside microcontrollers

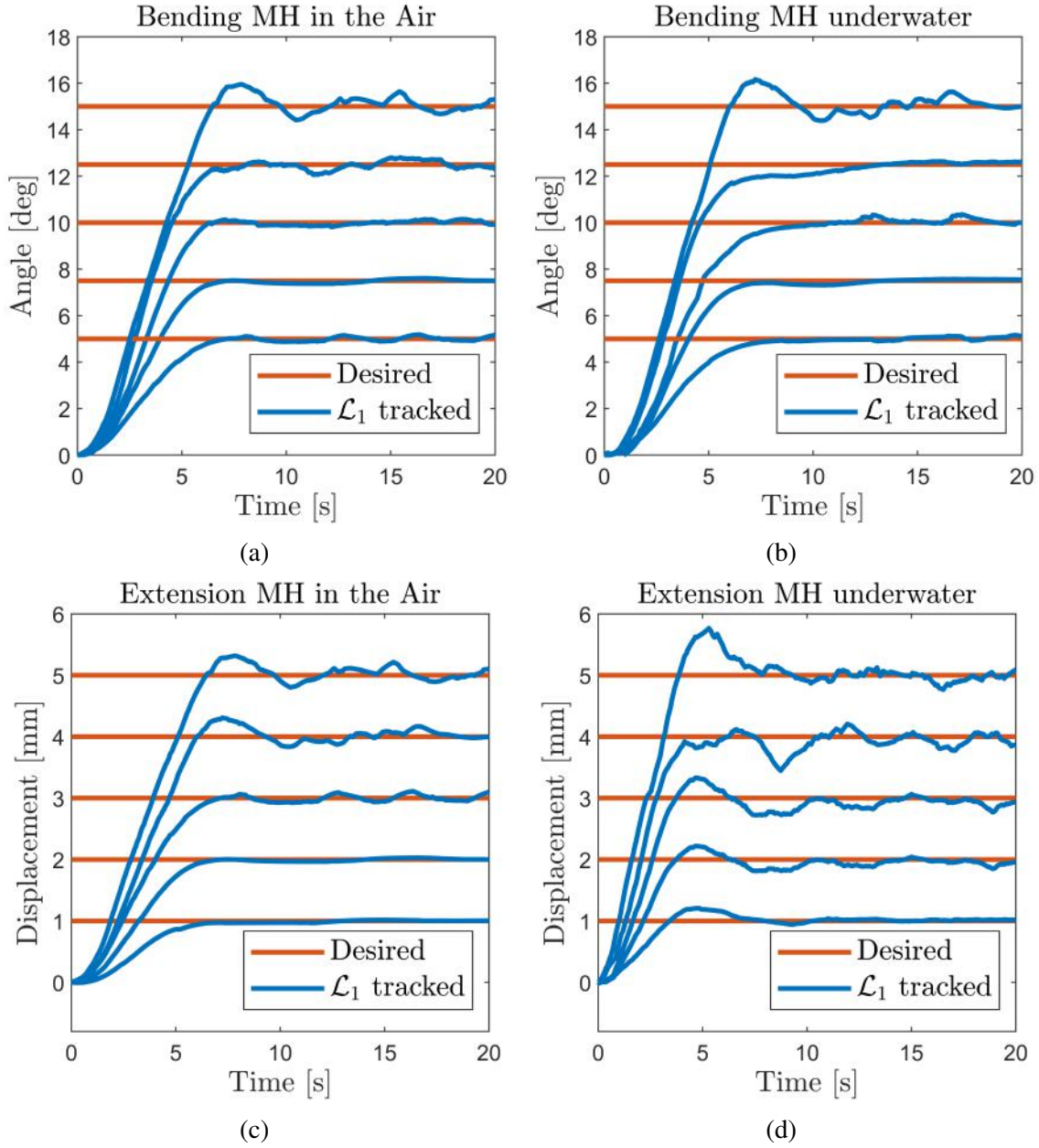


Figure 4.13. Displacement/Angle tracking for muscular hydrostats using the $\mathcal{L}_1$ adaptive controller. BH angle tracking in (e) air and (f) underwater. EH angle tracking in (g) air and (h) underwater. MH, muscular hydrostats.

(model: Arduino Uno) and other circuitry. Communication between the hardware and the controller was established using ROS (Robot Operating System), ensuring seamless integration and control.

The following presents the tracking actuation of muscular hydrostats under different condi-

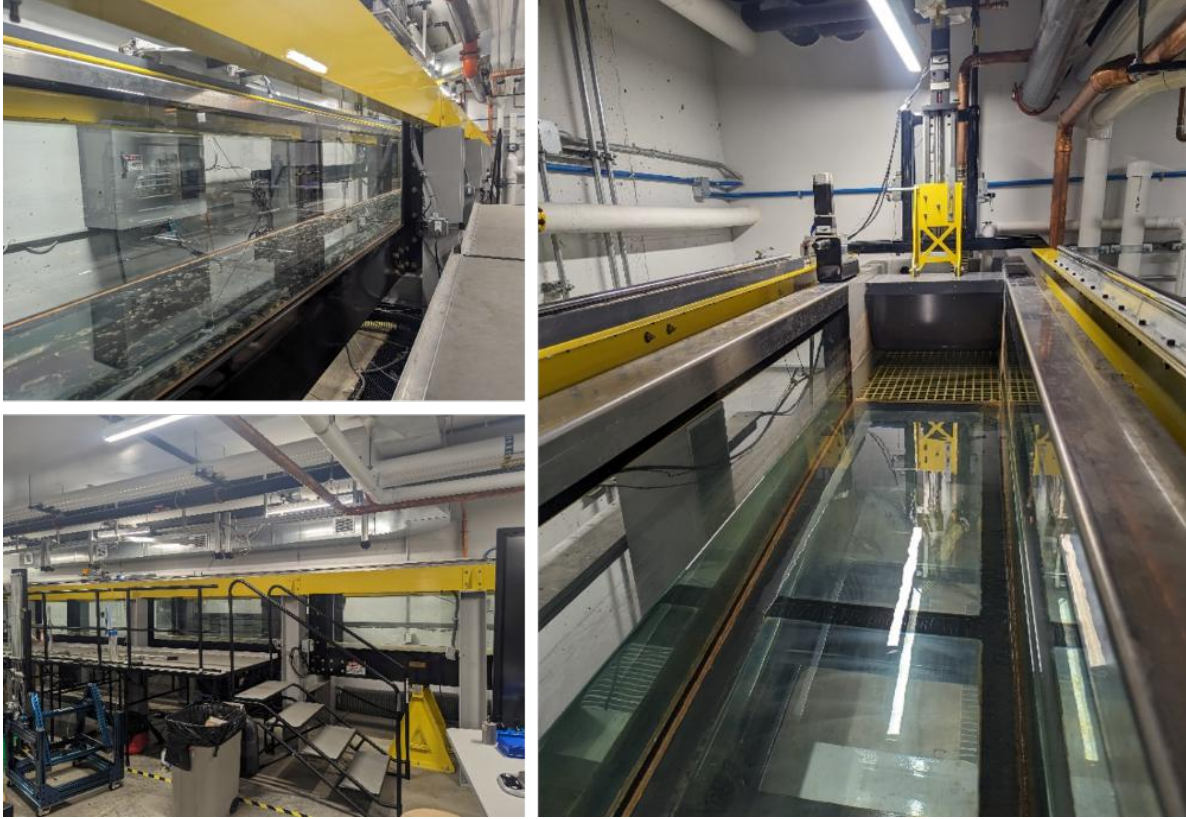


Figure 4.14. University of Iowa Fluid's lab towing tank facility.

tions: (a) waves with an amplitude of 20 cm and a frequency of 2 Hz, (b) current flow at a speed of 1 knot, and (c) current flow at a speed of 2 knots. Figure 4.16 shows the position tracking plots for the contraction muscular hydrostat under these three different wave/current types. As the current speed increases, the tracking plots tend to fluctuate more, although they still reach the desired po-

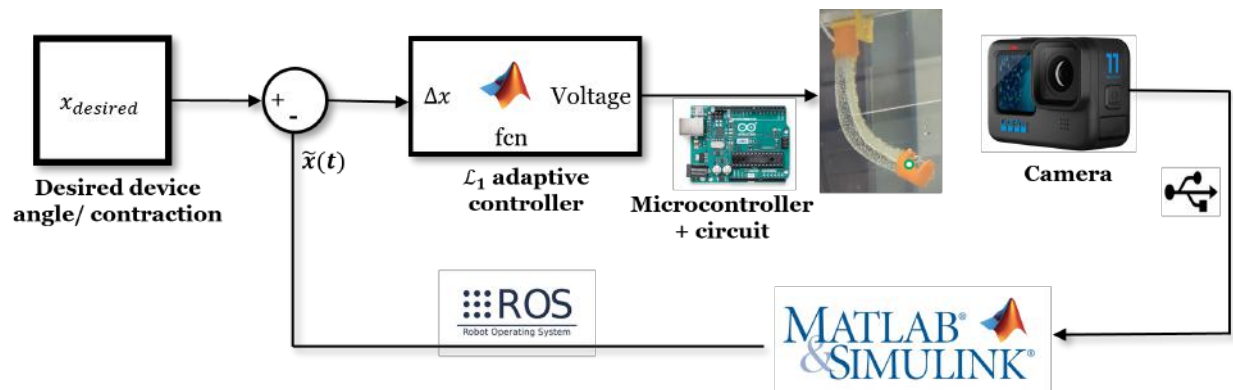


Figure 4.15. The hardware and the control architecture for the real-time towing tank testing.

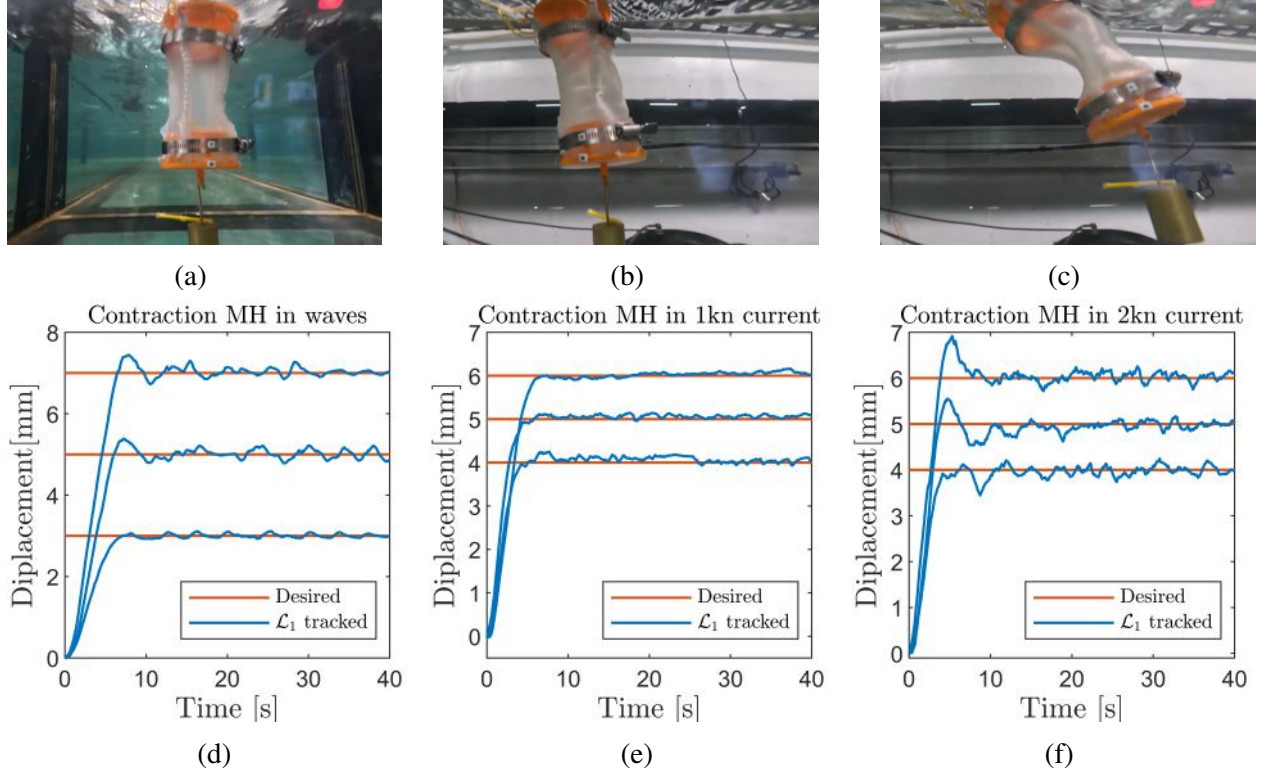


Figure 4.16. Tracking plots for contraction muscular hydrostats.

sitions. Similarly, Figure 4.19, Figure 4.17, and Figure 4.18 show the $\mathcal{L}_1$ tracking plots for torsion, bending, and extension muscular hydrostats, respectively.

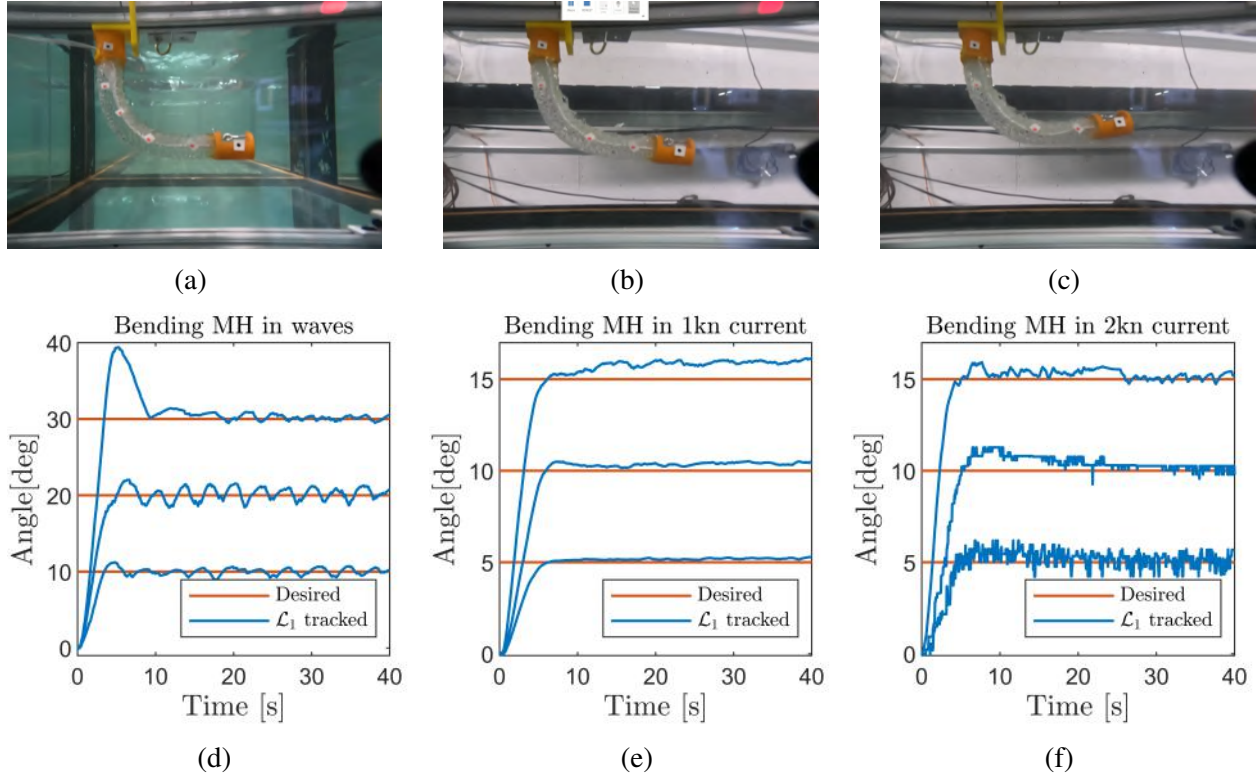


Figure 4.17. Tracking plots for Bending muscular hydrostats.

4.1.10 Rube Goldberg obstacle course

To demonstrate the combined actuation of all the devices underwater, we constructed a subaquatic obstacle course for the different configurations to manipulate a glass marble. The hydrostat prototypes are powered by Nylon 6 TCAMs, waterproofed using flexible tubes (Figure 4.20). The course itself is constructed from an aluminum T-channel frame with 3D-printed obstacles attached. The four hydrostat devices are mounted in place to the frame, and a custom-built control circuit utilizing an Arduino Uno microcontroller is wired into the devices and mounted to the frame for actuation.

4.1.10.1 Fabrication of Rube Goldberg setup

One sample of each class of device (CH, TH, BH, and EH) was manufactured according to the methods described previously for mounting to a frame constructed from 11" aluminum t-channeling. The extension hydrostat was mounted with a platform and wire track on its top surface

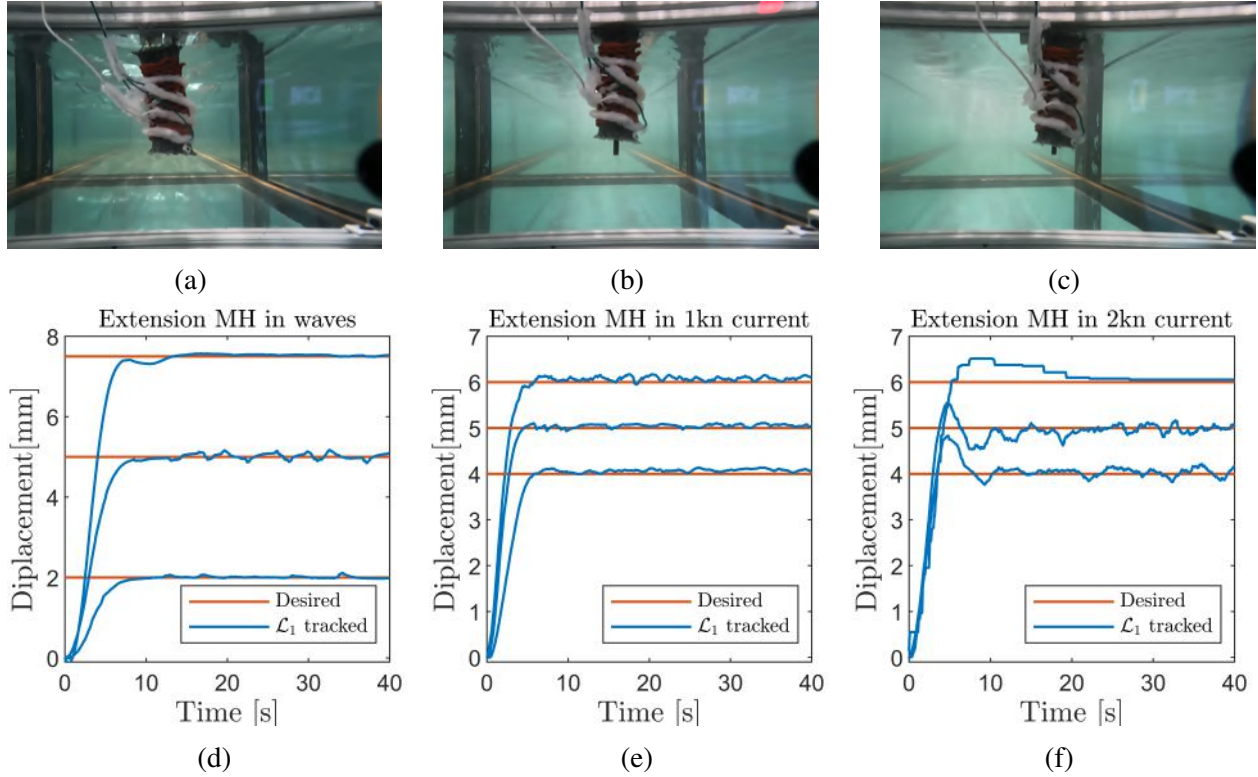


Figure 4.18. Tracking plots for Extension muscular hydrostats.

and the torsion hydrostat had the addition of a flag on its distal cap to act as a swing gate across the marble track. A tracked “basket” was added to the lower compression holder, and a wired pin was added to the bending cylinder. Two separate ramps were 3D printed for securing to the frame, and a control circuit was constructed from an Arduino Uno microcontroller and TIP-122 transistors. Mounting everything to a single rigid frame allowed for easy immersion and removal from a 30-gallon acrylic aquarium tank. To prepare the devices for underwater actuation TCAMs for CH, BH, and EH were enclosed in EcoFlex 00-30 molded tubes and sealed with DragonSkin FX-Pro prior to device attachment.

4.1.10.2 Insulation of TCAMs for underwater actuation and Rube Goldberg demonstration

To prepare the various devices for underwater actuation the muscles and electrical connections were insulated with EcoFlex 00-30 and EcoFlex DragonSkin FX-Pro silicone elastomer. The TCAMs are wrapped in 00-30 insulation tubes made by 3D printing a PLA mold and pouring



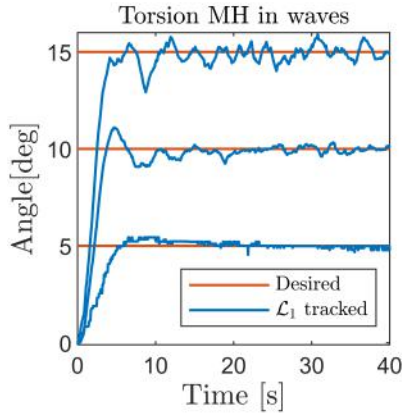
(a)



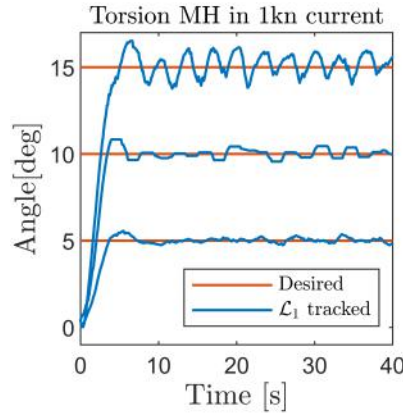
(b)



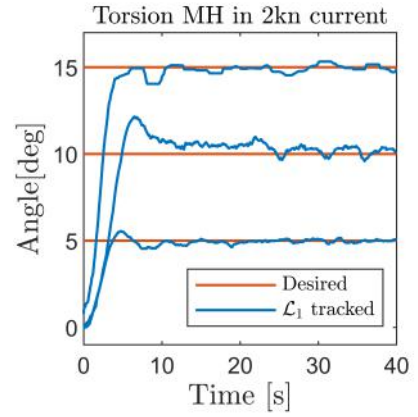
(c)



(d)



(e)



(f)

Figure 4.19. Tracking plots for Torsion muscular hydrostats.

partially cured elastomer inside (Figure 4.20(a)). A rod is set into the mold plate and two identical halves are joined together while the elastomer is highly viscous but not set. After curing for four hours the molds are separated and the rod and elastomer removed. The tube is trimmed and split down one side to remove the central rod and place the TCAM within. The tube is then sealed at the seam and ends using DragonSkin FX-Pro fast-curing silicone (Figure 4.20(b)). To ensure all electrically conductive components are sealed against ingress the wire leads and any thin points in the tubing are painted over with another coat of DragonSkin FX-Pro and allowed to cure before

mounting to the appropriate devices. This process is not used for the torsion hydrostat due to its sealed outer skin which provides insulation as well as the interference with the dominant component of twisting (TCAM untwist) that would occur with an isolated muscle. The sealed TCAM is depicted in Figure 4.20(c).

The marble begins atop the EH at rest (Figure 4.21(a)). Actuation of the EH causes the marble to roll down the first ramp until it is blocked by a swing arm attached to the distal cap of the torsion device (Figure 4.21(b)). The TH is actuated to rotate the blockade, allowing the marble to continue down the ramp into the basket on the base of the compression hydrostat (Figure 4.21(c)). The CH for this case has the TCAMs wired in two sets, a front and a back, which allow the basket to rise when actuated in tandem and tilt when actuated individually. The marble then rolls into its final position where the BH is actuated. A pin attached to the tip of the BH closes the circuit powering a light-emitting diode lamp and illuminates the chest as illustrated in Figures 4.21(d) and 4.21(e). Inset photos Figure 4.21(f)–(i) provide enlarged views of the hydrostat–marble interactions with the extension and contraction devices, respectively. The circuit schematic and the PCB diagrams are available in Appendix A.2.

4.1.11 Summary

Inspired by the musculature of the arms of octopodes, we designed muscular hydrostats powered by bioinspired actuators called TCAMs. Four different muscular hydrostats that exhibit the pri-

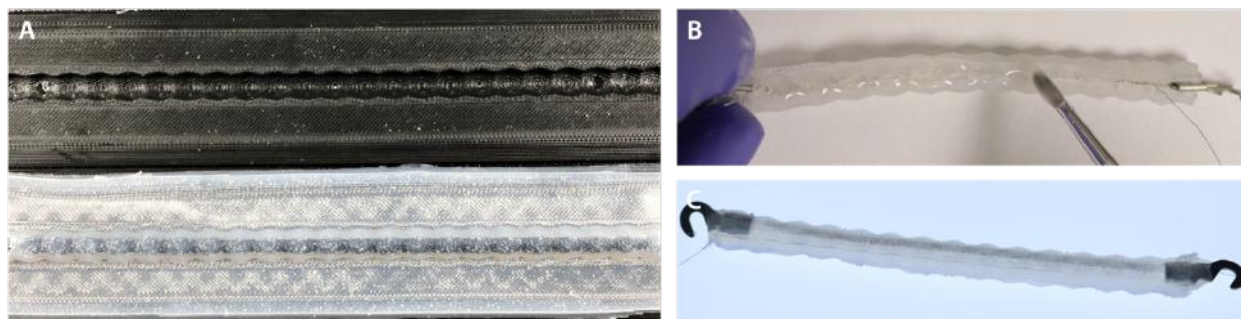


Figure 4.20. Insulation process for TCAMs. (a) Tube mold, central rod, and cured unprepared insulation tube. (b) Split tube with TCAM placed within being painted with DragonSkin FX-Pro. (c) Insulated TCAM ready for use.

mary motions of octopodes' arms, i.e., shortening, torsion, bending, and elongation, are designed and fabricated by altering the orientation and arrangement of TCAMs. The CH is solely driven by the anisotropic volume expansion of electrothermally actuated TCAMs and demonstrated a maximum compression of 40 %. For the torsion hydrostat, the helical arrangement of a single TCAM around a compliant core converts TCAM contraction into a torsional motion. A maximum twist of 75° was achieved using the TH. Additionally, we described the motion of the torsion hydro-

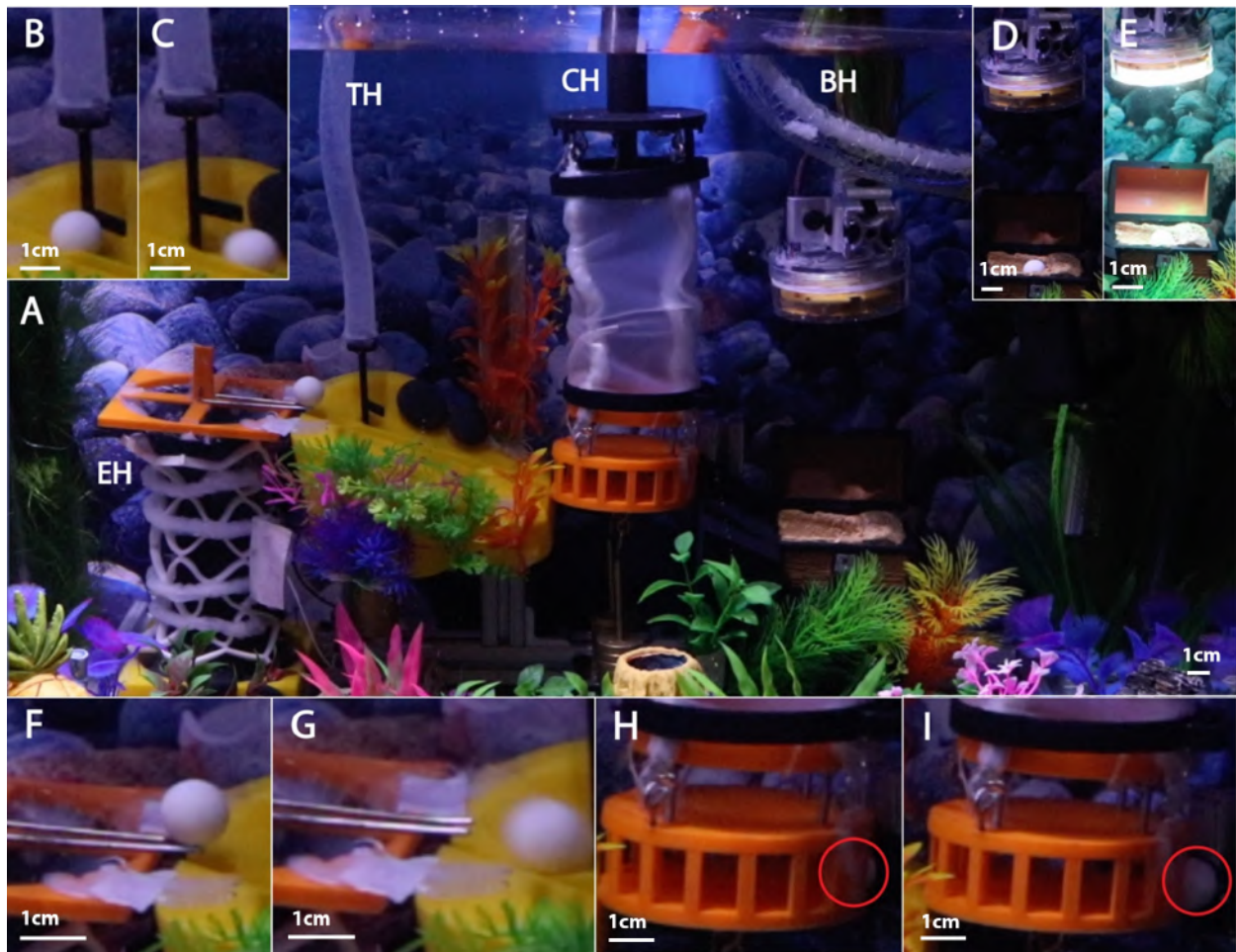


Figure 4.21. Rube Goldberg underwater obstacle course demonstration. (a) Complete obstacle course at the ready, (b) Marble blocked by torsion gate, (c) Actuated torsion gate allows passage, (d) Marble in chest awaiting illumination, (e) Actuated bending cylinder shines the spotlight on the marble, (f) Marble in its initial position atop the extension ramp, (g) Actuated extension cylinder tilts the ramp and sends the marble down the path, (h) Marble is caught in the contraction cylinder basket as indicated by the red circle, (i) Marble is tipped out of the basket and into the treasure chest again highlighted by the red circle.

stat using a theoretical model that considers both the untwisting moment and contraction of the obliquely oriented TCAM. For the BH and EH, to convert the contraction of the TCAM into a different degree of freedom, we employed mechanical instabilities and anisotropic material properties of compliant materials, respectively. The BH utilized mechanical instabilities generated by localized distribution of Voronoi cells to convert TCAM contraction into a maximum localized bending of 72° . The EH exploited the large Poisson's ratio of braided cable sleeving and a specially designed compliant mesh to transform TCAM contraction into a maximum elongation of 18.3 %. The weight of the CH, TH, BH, and EH is only 99 g, 38 g, 27 g, and 58 g, respectively.

All the developed muscular hydrostats can be independently actuated or assembled to develop waterproof soft robots with potentially infinite degrees of freedom. Future steps involve the development of these individual hydrostats into unit cell-based structures to achieve multi-modal motion while preventing interference from each other. In addition, we are investigating the interplay between contraction and extension as an alternative route to achieve bending without requiring the structural support provided by the current Voronoi hydrostat. Like the muscle fibers in the arms of octopodes, our TCAMs can exhibit motion within one single component and are manufactured from inexpensive polymer fibers using a simple low-cost fabrication technique we developed. As a first step toward the development of an integrated system, we also assembled a completely untethered waterproof device with on-board control, sensing, actuation, and a power source for driving our hydrostats underwater. We used an $\mathcal{L}_1$ adaptive controller for closed-loop control of the hydrostats. The hydrostats were controlled by tracking marker points on each device using a camera. Last, we built a Rube Goldberg setup that deploys the four hydrostats in tandem in an underwater setting. This demonstration was the first step in addressing the challenging future work of combining the four different hydrostats in a single modular structure for truly emulating the repertoire of movements of octopodes' arms. This work opens the door to the development of compliant and truly bioinspired robots that can be used to explore the underwater world and perform complex tasks in harsh and dangerous environments.

4.2 Control of TSAM Powered Vortex Generators

4.2.1 Introduction

Small unmanned aerial vehicles (UAVs) have gained widespread interest in recent years owing to their versatility, maneuverability, and endurance. They are particularly valuable for real-time reconnaissance in high-risk areas and operations in challenging environments such as border patrol, wildlife management, and search and rescue missions [298–300]. These UAVs usually operate at high altitudes and low Reynolds numbers, which are often less than 200,000 as shown in Figure 4.22(a) [301]. At low Reynolds numbers, the main challenge for small UAVs is the occurrence of laminar flow separation. When Reynolds numbers fall below 200,000, the laminar boundary layers close to the leading edge of the airfoil tend to separate due to the formation of an adverse pressure gradient. This leads to the formation of a laminar separation bubble (LSB) between the separation and reattachment points, which adversely affects the aerodynamic behavior of the airfoil. Laminar flow separation significantly reduces lift and increases drag, which results in a decrease in the lift-drag ratio and, in some cases, aerodynamic stall [302].

One common method to enhance the performance of airfoils operating at low Reynolds

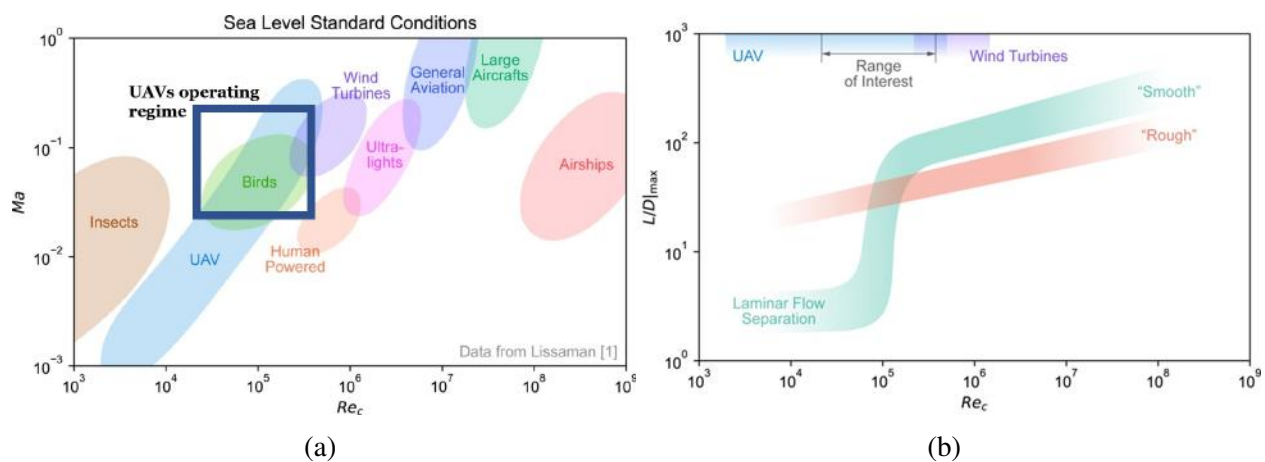


Figure 4.22. (a) Reynolds number operating regime for various aircraft, including UAVs, under sea level standard conditions. (b) Effect of Reynolds numbers on maximum L/D ratio for smooth and rough airfoils. *Adapted with permission from Lissaman, P B S Annual Review of Fluid Mechanics, 31 Dec 1982, Vol. 15, Issue 1, pages 223–239. Copyright 1983 by Annual Reviews Inc.

numbers is to induce turbulent flow by tripping the boundary layer. Turbulent flow promotes reattachment of the flow after separation by mixing low and high momentum flow [303]. Figure 4.22(b) [301] illustrates the aerodynamic performance of ‘smooth’ and ‘rough’ airfoils at low Reynolds numbers. The graph shows that ‘smooth’ airfoils experience a low lift/drag (L/D) ratio when the Reynolds number is less than 100,000 due to the presence of the LSB [298–300]. On the other hand, ‘rough’ airfoils do not exhibit the same behavior because the flow transitions to a turbulent regime, which re-energizes the flow and resists separation.

In order to mitigate the problems of flow separation, static vortex generators (VGs) have been widely used to promote boundary-layer transition [304]. The use of VGs is widespread because they offer cost-effective and easy-to-design solutions [305]. VGs can be fabricated in different shapes, including rectangular, triangular, trapezoidal wishbone, doublet, wedge, ramp, and numerous others [306]. Different shapes of VGs can yield to different separation suppression modes due to the varying strength of their vortices.

The geometry and position of VGs need to be optimized to suite different flow conditions. Specifically, the efficiency of VGs is greatly influenced by their position with respect to the leading edge of an airfoil, spanwise spacing, and their height in relation to the boundary layer thickness [307]. The chordwise position of a VG can be expressed using the dimensionless parameter, x/c , where x is the position of the VG along the chord and c is the chord length. When VGs are placed too close to the leading edge, they can generate vortices that disrupt incoming flow thereby causing premature flow separation and reduce lift generation. Conversely, when the VGs are too far downstream, the flow might separate before reaching the VGs. Generally, a value of x/c between 0.2 and 0.3 has been found to provide the best performance in terms of lift increment and stall delay [308, 309]. VG height must be chosen to avoid drag increase [307, 310, 311]. Conventional VGs are usually characterized by $h/\delta \geq 1$ where h is the VG height and δ is the thickness of the boundary layer. Finally, concerning VGs orientation, they can be placed in a co-rotating manner such that each VG is oriented in the same direction, or a counter-rotating manner where VGs in a pair have different orientations [304, 312].

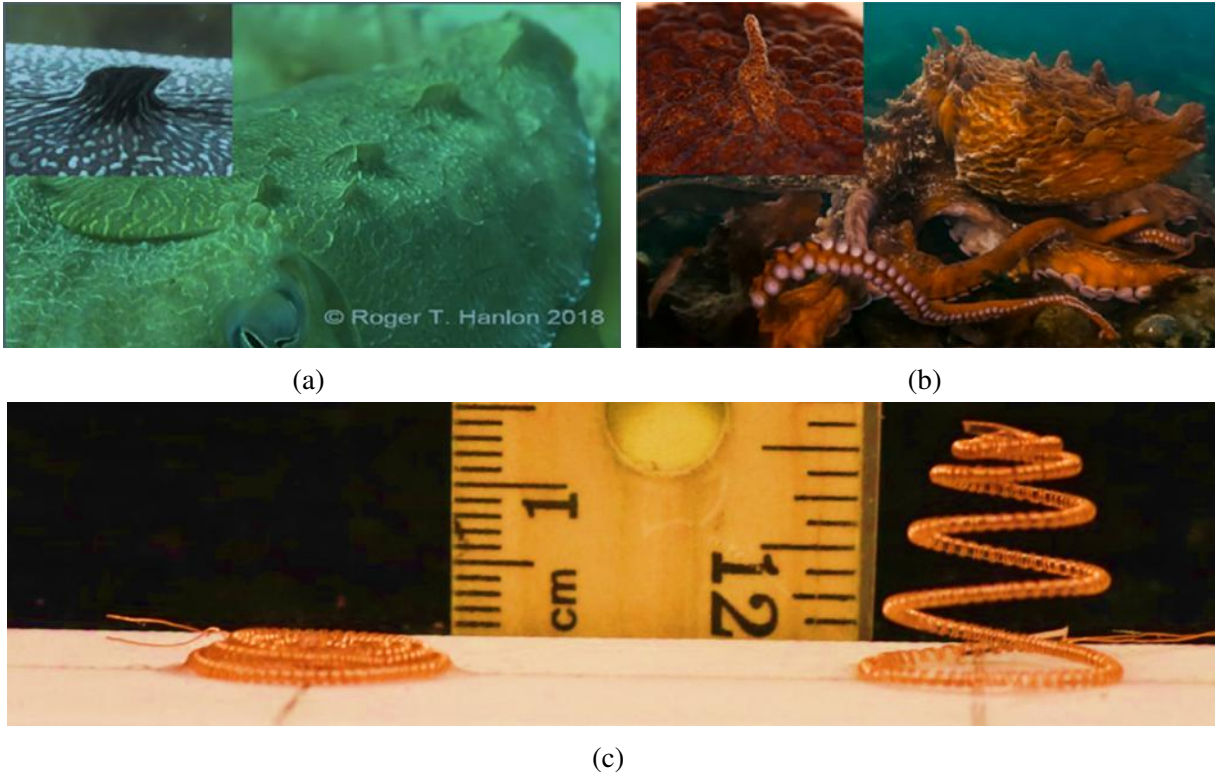


Figure 4.23. **(a)** Cuttlefish with actuated papillae muscles (inset showing protrusion of a single papillae). **(b)** Octopus with actuated papillae muscles (inset showing protrusion of a single papillae). **(c)** Out of plane vertical actuation of a TSAM (left: flat configuration before actuation; right: actuated configuration upon electrothermal input).

It has been demonstrated that each VG orientation offers several advantages over the other [313–315]. The counter-rotating VG orientation generates vortices that promote more efficient mixing of fluid streams compared to co-rotating vortices. Since counter-rotating vortices are characterized by two or more vortices rotating in opposite directions, when such vortices are formed, the opposing rotation directions of the vortices result in the formation of higher-pressure regions between them, which results in the generation of additional lift. The co-rotating VG orientation tends to generate vortices that are more coherent than the counter-rotating VG orientation. In terms of VG layout, single or multiple-row arrays of VGs have been adopted in several studies. The advantages of single or multiple rows however depend on specific flow conditions [316–318].

Since conventional static VGs are passive flow control devices, they are not able to adapt to different flow regimes, characterized by large variations in the Reynolds numbers. Several ac-

tive devices for flow controls have been proposed in the literature. This includes synthetic jet and plasma actuators, fluid oscillators, jet flaps, microelectromechanical systems and pulse blowing devices [319–322]. Active devices can adjust to changes in flow conditions, reduce parasitic drag, and delay stalls while increasing the lift coefficient. Synthetic jet actuators and dielectric barrier discharge (DBD) plasma actuators represent the most researched devices for active flow controls. The application of synthetic jet actuators is mainly limited by their power consumptions since they rely on oscillating membranes that require consistent supply of power for actuation. Moreover, piezoelectrically driven synthetic jets are characterized by low orifice velocities which are undesirable for applications where significant flow manipulations are required [323]. DBD plasma actuators have been widely employed in controlling flow separation. Apart from high voltage requirements, they are also limited to relatively low Reynolds number range, due to weak induced flow generated by the plasma [324]. In general, the need for multiple and bulk components for operation, and high-power requirements represent common drawbacks for all active devices for flow controls proposed so far, by hampering their integration into the wings of small UAVs.

Besides purely passive and active VGs, deployable VGs have also been proposed in the literature [325–327]. These solutions deploy static VGs on demand, using hydraulic actuators [325], or shape memory alloy actuators [326,327]. Although stall delay was demonstrated in these works, the proposed solutions are still limited in terms of portability and controllability, considering the high volume and weight of hydraulic actuators, and the highly hysteretic behavior of shape memory alloys, respectively.

In this study, deployable VGs powered by bioinspired twisted spiral artificial muscles (TSAMs) are proposed for improving the aerodynamic performance of small UAVs operating at low Reynolds numbers (i.e., lower than 200,000). TSAMs can mimic the dynamic behavior of the papillae muscles of cephalopods under electrothermal actuation [55]. Figure 4.23(a) shows the activation of papillae in the cuttlefish [328, 329], while Figure 4.23(b) shows papillae in the octopus [330]. Cephalopods use papillae to change the texture of their skin to camouflage or adapt their hydrodynamic performance while swimming [59, 330, 331]. While TSAMs do not mimic the materials

structure and working mechanism of papillae muscles (based on the interplay between muscular structure and hydrostatic pressure [59, 330, 331]), they take inspiration from cephalopods' papillae to produce a large out-of-plane vertical displacement upon actuation. They are manufactured by twisting inexpensive polymer fishing lines and are coiled to form Archimedean spirals. When electrothermally actuated, TSAMs undergo reversible extensions exceeding 2000% strain (see Figure 4.23(c)) within a few seconds, with an input voltage of only 0.2 V/cm. In previous works, we proposed theoretical models [55, 118] and different manufacturing methods [55, 56, 59] for TSAMs, and we also demonstrated that TSAMs are promising candidates for the development of fouling release devices [332]. In this work, we demonstrate how TSAMs can be used to actively deploy VGs, by emulating cephalopods' papillae in terms of flow controls. Thanks to their ease of manufacturing, low specific cost, low actuation voltage requirements, large output displacement, high specific power, non-hysteretic behavior, and their compact, lightweight, and single-component nature [7], TSAMs can effectively address the limitations of existing active flow control and deployable VGs devices in terms of power requirements, portability, ease of integration, and controls.

In this study, rectangular vane VGs are deployed on-demand by TSAMs actuation in response to changes in the flow conditions. The fabrication and design of the proposed setup is first described. Lift measurements are then reported for the airfoil operating at two different Reynolds numbers with and without passive VGs. Finally, passive VGs are mounted to electrothermally actuated TSAMs and the corresponding lift measurements are compared with the purely passive case to demonstrate that TSAMs can deploy different VGs configurations on demand, to provide optimal performance under different flow conditions. This approach stands out due to its affordability, minimal component requirements, ease of integration, and low power requirement. Our solution can be considered a hybrid technology between static and dynamic devices for flow control, and it is able to combine the advantages of static and dynamic solutions by proposing adaptive and compact deployable VGs.

4.2.2 Manufacturing and Actuation Modelling of TSAMs

The manufacturing process for the copper wound TSAMs unfolds as follows: Initially, a nylon fishing line (sourced from KastKing Clear Premium Monofilament Fishing Line, KastKing Garden City, NY, USA) with a fiber diameter of 0.8 mm is affixed to the chuck of a drill bit at one end, while the other end is secured to a load using a horizontal bar to prevent untwisting. Concurrently, an insulated copper wire of 36 AWG is folded in half and firmly held within the chuck of the drill bit. The drill then sets in motion, rotating and twisting the fishing line while the copper wire is manually maneuvered towards it. This orchestrated movement results in the helical wrapping of the copper wire around the fishing line, forming an electrically conductive composite. Twisting continues until the fibers start coiling spontaneously. The spiral shape was retained by annealing the TSAMs at 150 °C for 3 hours. For further insights into the intricate manufacturing process, refer to [55].

4.2.3 TSAM Model

In this section, we consider a dynamic model for TSAMs developed and evaluated in our previous work of [118]. The TSAM is fixed to the surface, and a vortex generator is attached, as is shown in Figure 4.24(a).

The following equation can express the temperature variation due to Joule heating caused by the applied voltage V .

$$\dot{T}_{\Delta}(t) = \frac{V^2}{C_t R} - \frac{\lambda(T_{\Delta})}{C_t} T_{\Delta}(t) \quad (4.6)$$

where R is the TSAM electrical resistance, C_t is the thermal mass, and λ is the absolute thermal conductivity. The equations for C_t , λ and CTE can be found in work [118]. $T_{\Delta}(t) = T(t) - T_{amb}$. The fiber radial expansion due to Joule heating can be expressed by the following equation.

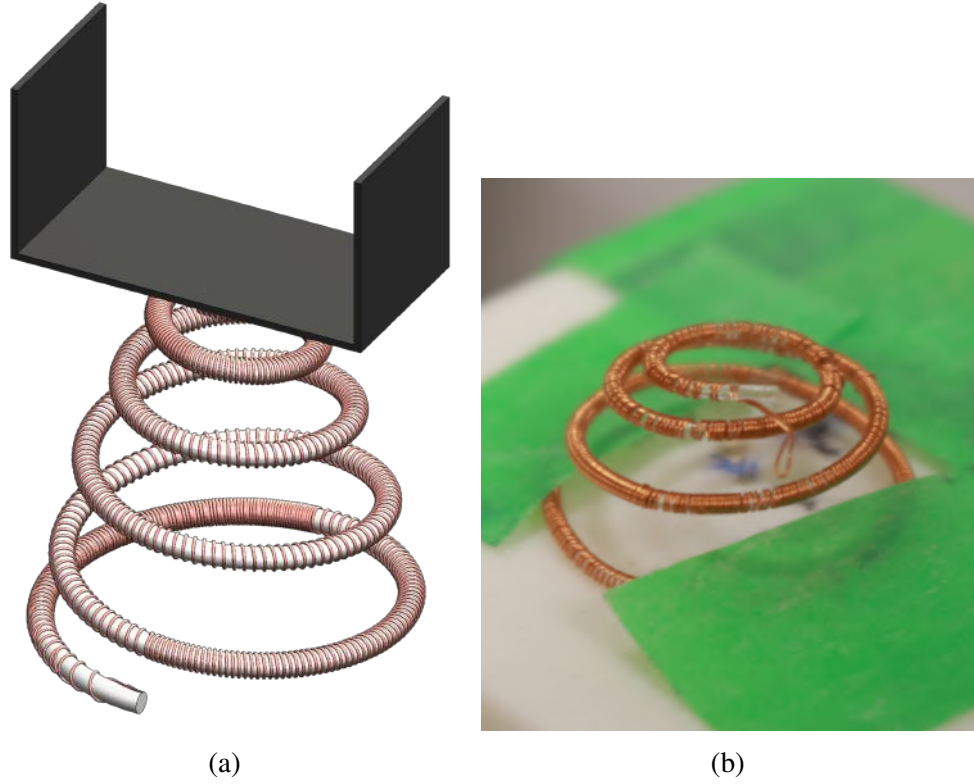


Figure 4.24. **(a)**TSAM with a vortex generator attached on the top and **(b)** image of a TSAM.

$$\frac{d_i(t) - d_0}{d_0} = CTE \cdot T_{\Delta}(t). \quad (4.7)$$

TSAMs can be considered conical springs, with the exception that they initially have a ground configuration, and, on actuation, an internal untwisting generates a tensile load that leads to a vertical displacement of the spiral. Since TSAMs experience an elongation during actuation, without facing any nonlinear reaction from the ground (present during compression), the relationship between the tensile load P_i and the vertical displacement $\Delta\ell_i$ is considered to be linear [333]. The tensile load generated by the TSAM during actuation is calculated from the internal shear stress τ_1 using the equations of the mechanics of compression conical springs [334] and is given by:

$$P_i = \frac{\pi \tau_1 d_i(t)^2}{8CK_w} \quad (4.8)$$

where, C is the spring index, and K_w is the Wahl's curvature correction factor, calculated as follows:

$$C(t) = \frac{D_2}{d_i(t)}, \quad (4.9)$$

$$K_w(t) = \frac{4C(t) - 1}{4C(t) - 4} + \frac{0.615}{C(t)}. \quad (4.10)$$

while τ_i is the internal shear stress due to untwisting, generated by anisotropic thermal expansion, and can be calculated using the single helix theory [335]

$$\tau_i(t) = \frac{d_i(t)Gn_0}{sl} \left(1 - \frac{d_0}{d_i(t)} \right), \quad (4.11)$$

where, G is the shear modulus of the nylon fishing line, $l = \frac{\pi n_a}{2}(D_1 + D_2)$ is the total length of the straight twisted fiber, n_0 is the number of inserted turns during the twisting process of TSAMs, d_0 is the initial diameter of the twisted fiber, and $d_i(t)$ is the fiber diameter at time t , during thermal expansion.

Considering the stiffness of a conical spring given by [333]:

$$k_i(t) = \frac{Gd_i(t)^4}{2n_a(D_1^2 + D_2^2)(D_1 + D_2)} \quad (4.12)$$

Ideally, the TSAM can be actuated till it uncoils completely to form a straight twisted fiber. In this case, the number of spirals decreases with an increase in temperature. However, the complete uncoiling of the TSAM is limited by the maximum temperature the TSAM can attain during electrothermal actuation. The temperature reached by the TSAM during actuation is maintained below the glass transition temperature (70°C) [336] to ensure that the TSAM returns to its flat spiral shape on cooling. In our tests we observed that when a TSAM is heated to temperatures below the glass transition temperature it experiences a partial uncoiling, and the number of spirals remains constant during actuation.

The vertical displacement of the TSAM can be calculated (assuming a linear behavior) as:

$$x(t) = \frac{P_i}{k_i(t)}. \quad (4.13)$$

By substituting Equations (4.8), (4.11), and (4.12) in Equation (4.13) and simplifying we obtain the following equation of the TSAM vertical displacement:

$$x(t) = \frac{(D_1^2 + D_2^2)n_0}{4D_2K_w} \left(1 - \frac{d_0}{d_i(t)} \right) \quad (4.14)$$

The following second-order dynamic equation governs the TSAM motion:

$$m\ddot{h}(t) + b\dot{h}(t) + kh(t) = F_e(t). \quad (4.15)$$

This static height model will be used to estimate the height from given input temperature of TSAMs obtained through Thermistors in our experimental work describe in later sections.

In Equation (4.15), $F_e(t)$ is an external compression force applied on the top of the TSAM, m is the total mass of the TSAM, b and k are the damping and spring coefficients of the TSAM, respectively. The highly twisted fiber's spring coefficient k can be expressed as a function of $T_\Delta(t) = T(t) - T_{amb}$, i.e. $k \triangleq k(T_\Delta)$, where $T(t)$ is the temperature of the TSAM at time t , and T_{amb} is ambient temperature. Then we have

$$k(T_\Delta) = \frac{mg + F_e}{\frac{(D_1^2 + D_2^2)n_0}{4D_2K_w} \left(1 - \frac{d_0}{d_i(t)} \right) - \frac{2n_a(D_1^2 + D_2^2)(D_1 + D_2)F_e}{Gd_i(t)^4}}, \quad (4.16)$$

where $D_1, D_2, K_w, n_0, n_a, d_0, G$ are the inner spiral diameter, the outer spiral diameter of the TSAM, the Wahl curvature correction constant, the number of twisted turns, the number of spirals, fishing line fiber diameter, and the shear modulus of the polymer fiber, respectively. Similarly, the damping coefficient b can be expressed as a function of $b \triangleq b(T_\Delta)$

$$b(T_\Delta) = \frac{\frac{\pi d_i(t)^4}{4} \eta}{\frac{(D_1^2 + D_2^2) n_0}{4 D_2 K_w} \left(1 - \frac{d_0}{d_i(t)} \right) - \frac{2 n_a (D_1^2 + D_2^2) (D_1 + D_2) F_e}{G d_i(t)^4}} \quad (4.17)$$

Where η is the viscosity of the nylon fishing line. In [118, 147, 148, 337], it is shown using the theoretical and experimental observation that the increase of artificial muscle temperature increases the stiffness and damping coefficient k and b . The internal untwisting and uncoiling results in vertical displacement of the TSAM. Let vertical displacement $d(t) = h(t) - \bar{h}$, where $h(t)$ is the change of TSAM height and $\bar{h} \geq 0$ is the initial TSAM height when $T_\Delta = 0$. Then $\bar{h} = \frac{mg}{k(0)}$. Then, Equation (4.15) can be rewritten as

$$m\ddot{d}(t) + \bar{b}\dot{d}(t) + \bar{k}d(t) = F_e - b_\Delta \dot{d}(t) - k_\Delta d(t) - k(T_\Delta) \bar{h} \quad (4.18)$$

where $k(T_\Delta) = \bar{k} + k_\Delta$ and $b(T_\Delta) = \bar{b} + b_\Delta$. Let's consider the following assumption to linearize Equation (4.18),

$$cT_\Delta \approx F_e - b_\Delta \dot{d}(t) - k_\Delta d(t) - k(T_\Delta) \bar{h}. \quad (4.19)$$

The coefficient $c > 0$. These terms consist of uncertainties that will be handled by equations proposed in the next section. To mathematically formulate our control problem, let's define the state vector $x(t)$, input vector $u(t)$, and the output vector $y(t)$ as follows.

$$x(t) = \begin{bmatrix} d(t) \\ \dot{d}(t) \\ T_\Delta(t) \end{bmatrix}, \quad u(t) = V^2(t), \quad y(t) = d(t).$$

Considering Equation (4.19), the equations (4.18) and (4.6) can be simplified into

$$\begin{aligned} \dot{x}(t) &= A_p x(t) + B_p(u(t) + f(t, x(t))), \\ y(t) &= C_p x(t), \end{aligned} \quad (4.20)$$

where

$$A_p = \begin{bmatrix} 0 & 1 & 0 \\ -\frac{\bar{k}}{m} & -\frac{\bar{b}}{m} & \frac{c}{m} \\ 0 & 0 & -\frac{\lambda}{C_t} \end{bmatrix},$$

$$B_p = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{C_t R} \end{bmatrix}, C_p = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix},$$

and $f(t, x(t))$ represents time-varying model uncertainties and disturbances.

4.2.4 Controller Implementation, Hardware and Circuitry

The Control algorithm was implemented in MATLAB/SIMULINK, and the hardware communication was set up using ROS. Figure 4.26 shows the $\mathcal{L}_1$ adaptive controller algorithm. Height tracking using displacement was challenging with TSAMs, even during the preliminary experiments, because some TSAMs tended to move sideways instead of vertically elongation. During the preliminary experiments, the tip/top edge of the TSAM was colored (as denoted in Figure 4.25), and the change in height was measured using a camera that tracked that colored point. This was an image-tracking method. The detailed code is available in Appendix: TSAM Image Processing based Height Measuring. The experimental setup is shown in Figure 4.25. As can be seen, the temperature was measured using the thermal camera (model: FLIR–E95), and the height change was measured using the camera (a mobile phone). The temperature and displacement plots showed a proportional relationship. Based on these results, a physics based model was implemented and validated. The model was published in Kotak et. al. [118].

For applications involving TSAMs, accurately measuring the height of an array of TSAMs is challenging. Therefore, thermistors were chosen to obtain feedback. A temperature-to-height model, developed in our previous work [118], was used to convert the measured temperature into height. The control algorithm flow is shown in Figure 4.26. The measured resistance of the ther-

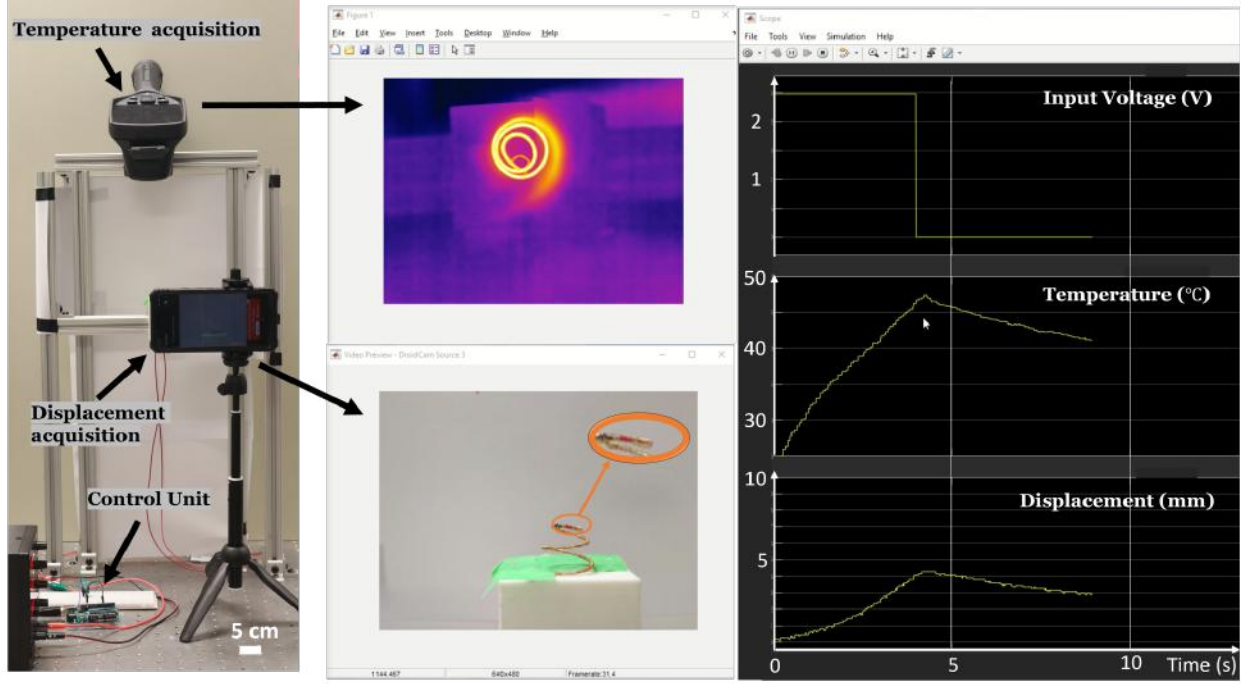


Figure 4.25. TSAM position tracking using a camera

mistor, $\tilde{R}(t)$, is converted to $\tilde{T}(t)$ using a voltage divider circuit. This is followed by an Arduino analog read, which measures the voltage and then converts it to resistance. The generic Steinhart–Hart thermistor equation,

$$\frac{1}{T} = A + B \ln R + C (\ln R)^3 \quad (4.21)$$

is then used to calculate the temperature, where A , B , and C are constants. These parameters were tuned experimentally. After attaching the thermistors to the TSAM, it was heated, and the resistance was measured via the thermistor, while the actual temperature was recorded using a thermal camera. Using multiple sets of resistance vs. temperature data, a basic machine learning model was developed to optimize the constants.

After converting the resistance to temperature, the TSAM model-based equations were used to calculate the static height values for the given temperatures as shown in Equation (4.14) above. The extremely small thermistors require a very compact voltage divider circuit, making them an ideal sensor type for complex TSAM systems.

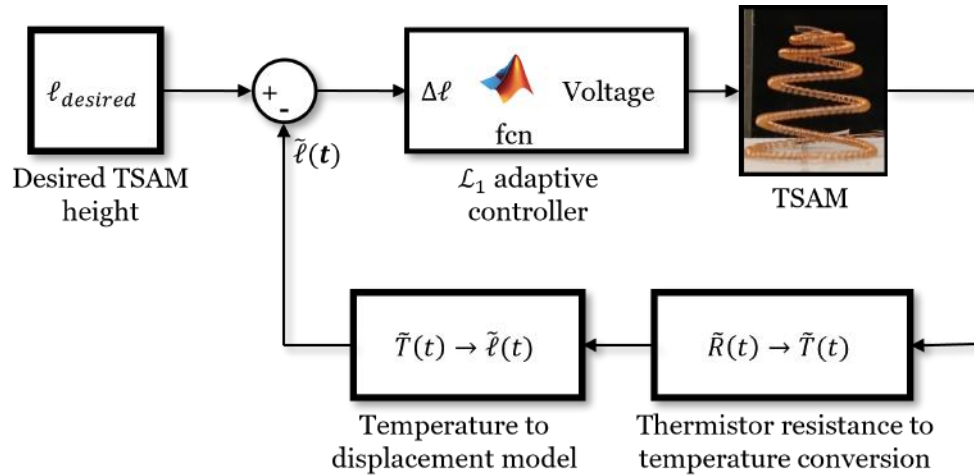


Figure 4.26. Control architecture for TSAM actuation system

Other than that, each TSAM consists of power regulating transistor (Model: Tip122) to actuate upon user desire. These transistors are biased using Arduino pulse width modulation (PWM) signals. The schematics and the flow diagram of the hardware circuitry is shown in Figure 4.27.

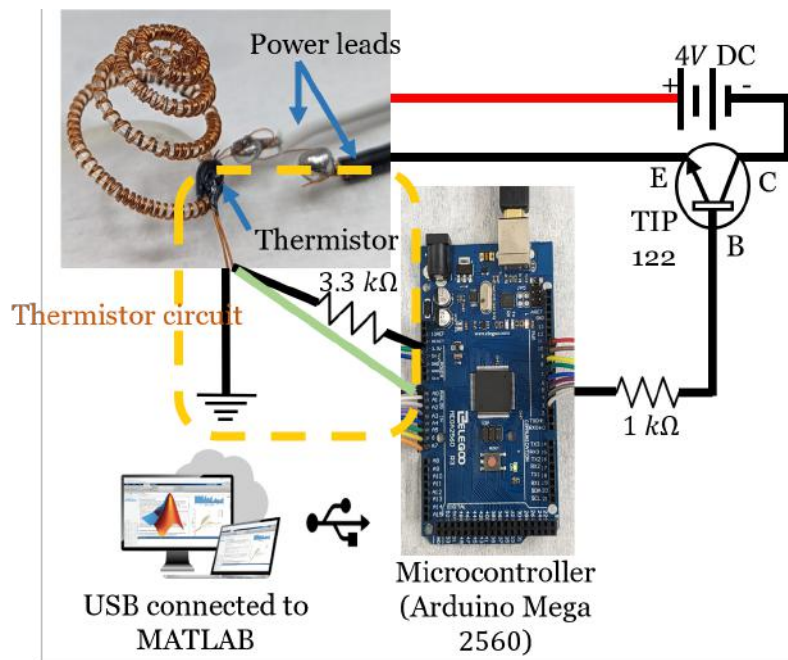


Figure 4.27. Circuitry diagram for the TSAM actuation system

4.2.5 Wind Tunnel actuation and control of deployable VGs powered by TSAMs

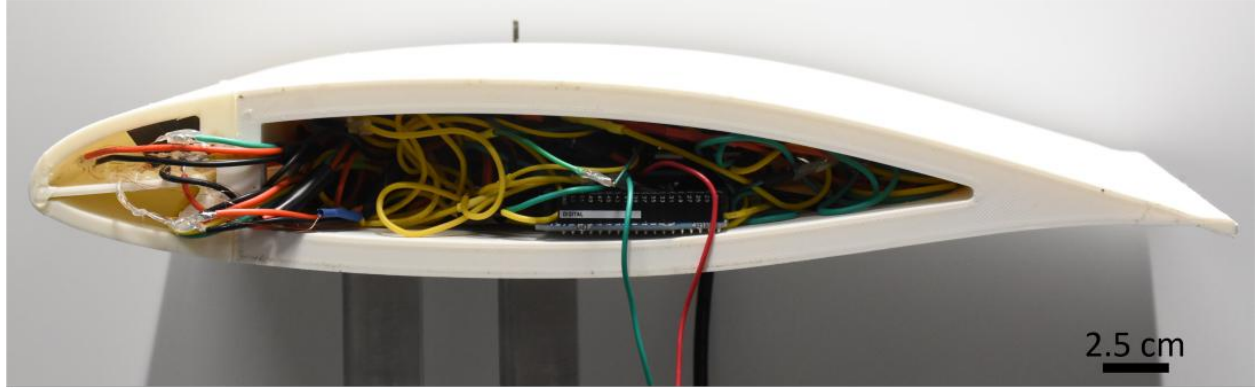
The primary objective of this study is to determine if the on-demand deployment of the TSAM-powered VGs can match the peak performance of different VG configurations corresponding to different angles of attack. In this setup, each TSAM powers two VGs. Since there are 8 VGs in each row, 4 TSAMs are placed in each row of the insert plate (Figure 4.28(b)) at the same distance from the leading edge corresponding to the VG configuration. For a given Reynolds number, the TSAM-powered VGs were selectively actuated to match the peak values of the coefficient of lift provided by different VGs configuration for different angles of attack. Tables 1 and 2 depict the corresponding VG configurations deployed at each angle of attack when testing the dynamic device at Reynolds number of 100,000 and 140,000, respectively. Each VG configurations was deployed at the maximum height of the VG element (i.e., 2 mm or 8 mm).

To obtain the maximum lift throughout the entire angle range, the powered VGs were deployed. The hardware setup is shown in Figure 4.28(a). The top is shown in Figure 4.28(b). As it can be seen, each TSAM connects to 2 power leads and 2 more wire for thermistors. Hence, the

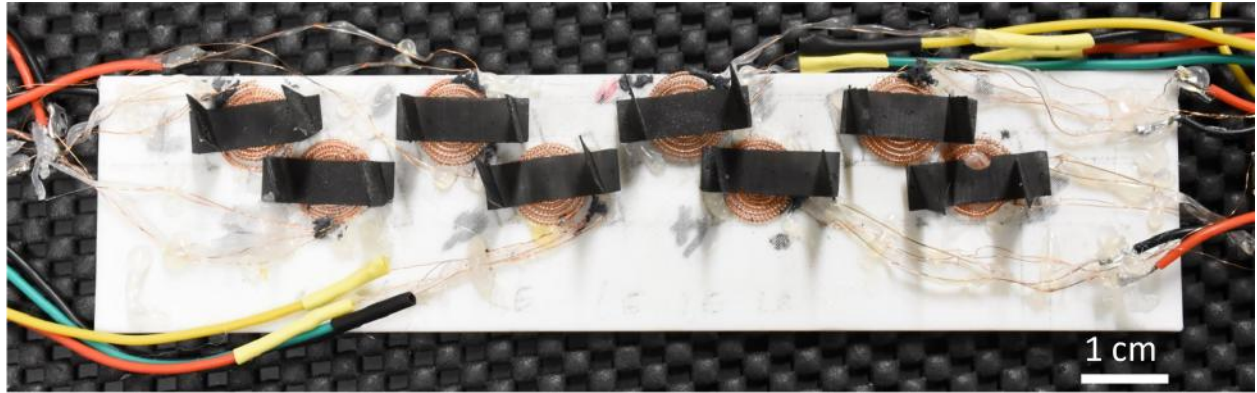
Table 4.1. Angles of attack Vs. VG configuration deployed at $Re = 100,000$ and $Re = 140,000$.

Angle of attack in degrees (°)	Configuration for $Re = 100,000$	Configuration for $Re = 140,000$
0	VG4	VG1
2	VG4	VG1
4	VG4	VG2
6	VG5	VG4
8	CA	VG1
10	CA	VG5
12	CA	VG4
14	CA	VG2
16	VG2	VG2
18	VG2	VG5
20	VG2	VG5

CA–Clean Airfoil



(a)



(b)

Figure 4.28. Manufactured airfoil with TSAMs-powered VGs. **(a)** Airfoil section with the deployable VGs (shown in the inset) and electrical setup for actuating TSAMs. **(b)** Top view of the TSAMs powered VGs mounted on an insert plate that can be slotted into the nose of the airfoil.

entire system has 32 wires. Since the system was tested in air, there was no need to covering up or insulate to protect from water. A microcontroller (Model: Arduino MEGA 2560) was used to data acquisition and PWM signal generation to bias the power transistors.

The VG configurations are shown in the following Table 4.2. These configurations were obtained through series of static experiments conducted by coauthors of this work. (Published in [151])

Figure 4.29(a) shows the measured temperatures of 4 TSAMs in a row–1, during their actuation of 8 mm. All the TSAMs were heated up between 75 °C and 95 °C. Simultaneously, row–2 was actuated to 2 mm. The TSAMs were heat up between 50 °C to 60 °C. These plots reflects the use of controller by the plateau. This is because the controller control the height of the TSAM

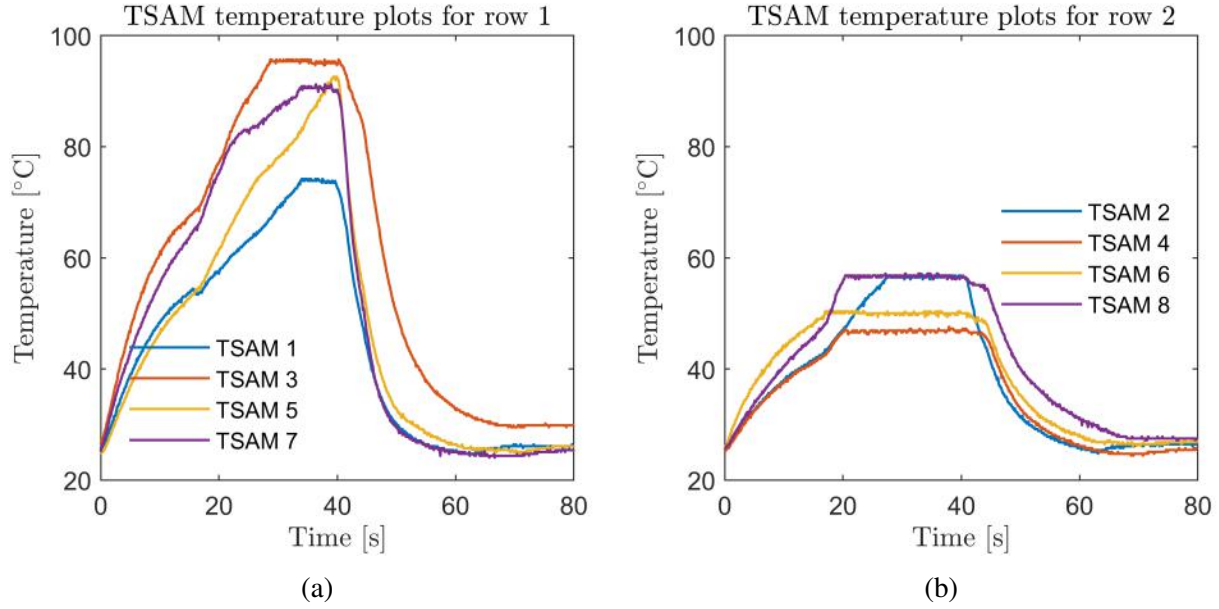


Figure 4.29. Temperature plots of VG in (a) row1 = 8 cm and (b) row2 = 2 cm.

and there is a proportional relationship between height and temperature. Figure 4.30 shows the actuation of 2 TSAMs for 2 different heights of 8 mm and 2 mm. Figure 4.30(b) and 4.30(e) shows the actuation temperatures. These TSAMs were operated under very high wind speeds of $Re=100,000$ and $140,000$. Hence the cooling was faster and the controller had to provide its maximum saturated voltage to maintain the desired height. This voltage saturation can be seen in voltage plots of Figure 4.30(c) and 4.30(f).

As shown in Figures 4.31(a) and 4.31(b), the black curve depicts the performance of the

Table 4.2. Airfoil vortex generator configurations.

VG configurations	Distance from leading edge ($\frac{x}{c}$)	Number of rows
VG1	0.06 for VG_B	Single row
VG2	0.06 for VG_B and 0.1 for VG_S	Double row of VG_B and smaller VG_S
VG3	0.1 for VG_S	Single row VG_S
VG4	0.06 and 0.1 for VG_B	Double rows of VG_B
VG5	0.06 for VG_S and 0.1 for VG_B	Double rows of VG_S and VG_B

VG_B : Big VGs with dimensions $8 \text{ mm} \times 0.4 \text{ mm} \times 8 \text{ mm}$; VG_S : Small VGs with dimensions $6 \text{ mm} \times 0.4 \text{ mm} \times 2 \text{ mm}$.

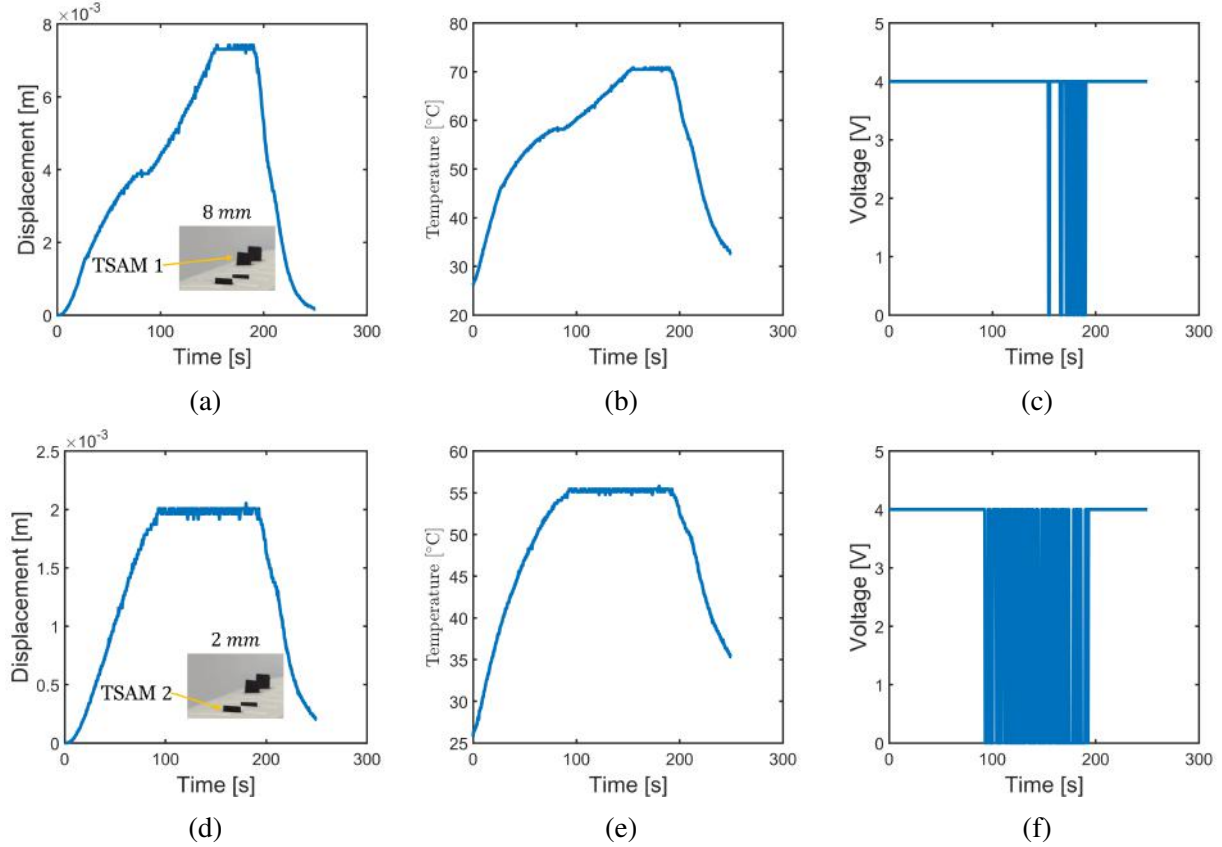


Figure 4.30. (a), (d) Displacement, (b), (e) temperature, and (c), (f) voltage plots for two TSAMs actuated to and maintained at different heights using the $\mathcal{L}_1$ adaptive controller.

TSAM-powered VGs. The effects of the active VGs powered by TSAMs at a Reynolds number of 100,000 are shown in Figure 4.31(a). It can be observed from the black curve that the TSAM-powered VG configuration follows the path of maximum lift for the entire range of angles of attack tested. Specifically, this curve was obtained by deploying VG4 for angles of attack between 0° and 4° , VG5 at 6° , no VGs between 8° and 14° (this corresponds to the clean airfoil condition), and VG2 between 16° and 20° , as listed in Table 4.1.

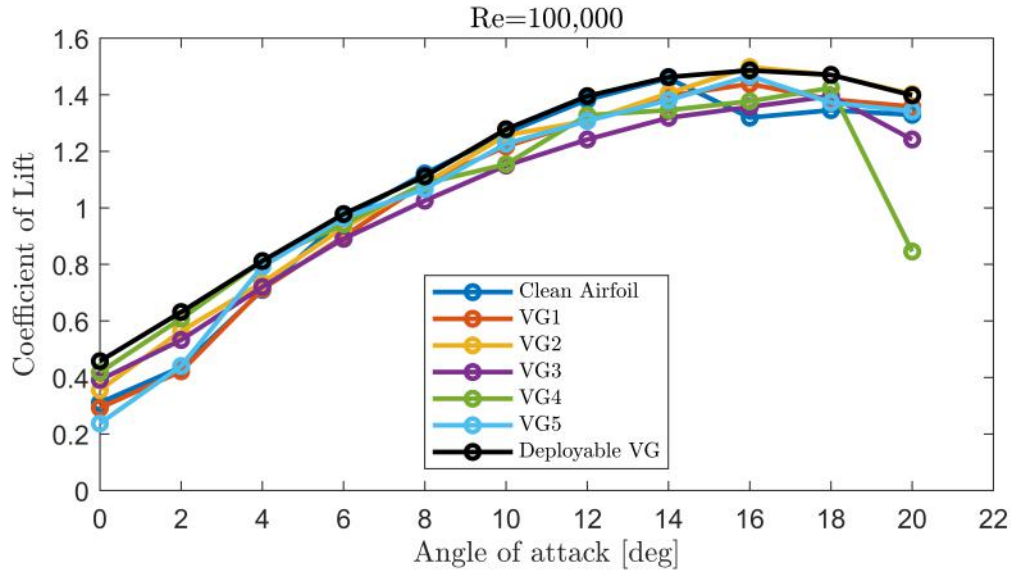
By deploying different VG configurations in terms of VG height and distance from the leading edge, the lift on the wing can then be increased. Figure 4.31(b) shows the active VG configuration's effects on the airfoil's lift coefficient at a Reynolds number of 140,000. Like Figure 4.31(a), it can be observed that the active device can always deploy different VG configurations to achieve the maximum value of the coefficient of lift. Specifically, the black curve was obtained

by deploying VG1 for angles of attack between 0° and 2° , VG2 at 4° , VG4 at 6° , VG1 at 8° , VG5 at 10° , VG4 at 12° , VG2 at angles of attack between 14° and 16° , VG5 at 18° , and VG4 at 20° , as listed in Table 4.1. Results in Figure 4.31 show that our device can always provide the maximum lift coefficient for different values of Reynolds number and angles of attack by deploying different desired VG configurations.

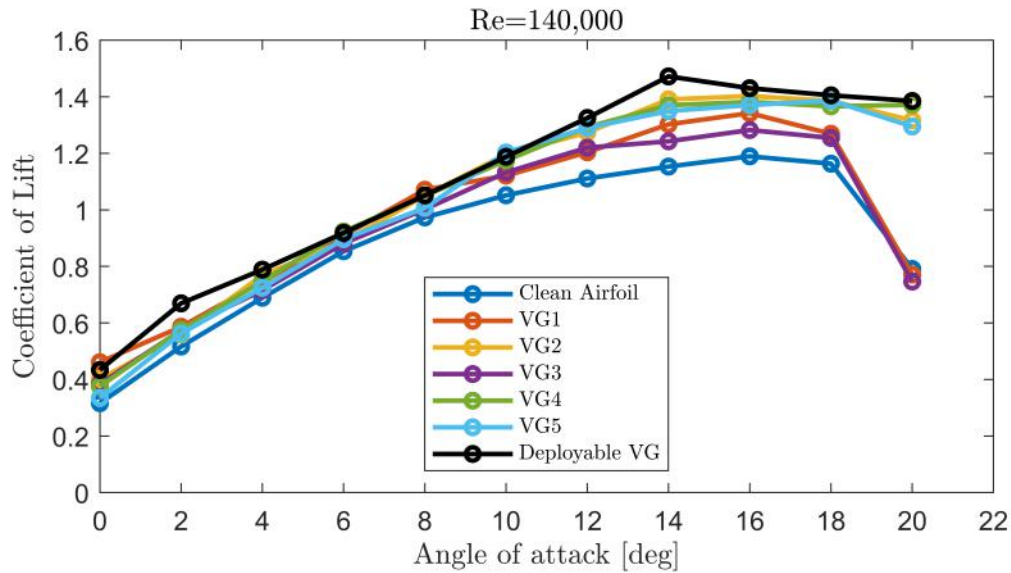
Experimental tests using static passive VGs were first performed to identify the best VGs configuration for different flow conditions. Figure 4.31(a) shows a plot of lift coefficient versus angles of attack at $Re = 100,000$. The lift coefficient C_L is defined as:

$$C_L = \frac{L}{\frac{1}{2}\rho U^2 A} \quad (4.22)$$

where ρ is the air density, U is the free stream velocity, A is the wing area. For the clean airfoil (dark blue curve in Figures 4.31(a) and 4.31(b)) the onset of stall can be observed at an angle of 14° and all passive VGs configurations seem to delay the onset of stall to larger angles of attack. The VG 1, VG 2, and VG 5 configurations appear to delay stall until 16° while significant delay was recorded for VG3 and VG4 at 18° . In terms of lift generation, it was observed that VG2, VG3, and VG4 generates more lift than the clean airfoil at angles of attack between 0 and 6° while the clean airfoil generates more lift between 6° up until the stall angle.



(a)



(b)

Figure 4.31. Effects of VGs deployed by electrothermally actuated TSAMs on the lift characteristics of the airfoil (a) Reynolds number 100,000 (b) Reynolds number of 140,000 in air.

4.2.6 Underwater actuation and control of deployable VGs powered by TSAMs

Underwater vehicles can be subjected to hydrodynamic stall and flow separation, especially when operating at high angles of attack. Hydrodynamic stall occurs when flow separates from surfaces due to adverse pressure gradient at high angles of attack. The nature of the stall can either be

dynamic or static. Dynamic stall results from flow separation due to rapid change in the lifting surface. This results in the formation of vortex cores at the leading edge of the body which grow and move over the surface and eventually shed from the surface at the trailing edge thereby causing a sudden drop in lift and an increase in drag. Under steady conditions, static stall is also possible when the angle of attack is gradually varied, resulting in the detachment of the boundary layer from the leading edge of the surface at a critical level of incidence angle. Stalls are undesirable because turbulence and pressure fluctuations associated with stalls can result in structural damage. It can also result in accidents due to loss of control and maneuverability.

In this section, the airfoil was tested underwater (using a towing tank that can change the flow rate as in Figure 4.32) to evaluate the impact of deployable VGs on the lift coefficient. The experiment was conducted at $Re=100,000$ and $Re=120,000$. From a control perspective, the algorithm remained the same as in the previous tests. Since the entire system needed to be submerged, complex custom-made printed circuit boards (PCBs) were designed to prevent potential water leakage and reduce the need for complex wiring, as shown in Figure 4.28(a). A rectangular PCB was designed, and most of the wires were embedded in the plate in the form of copper lines. The side-

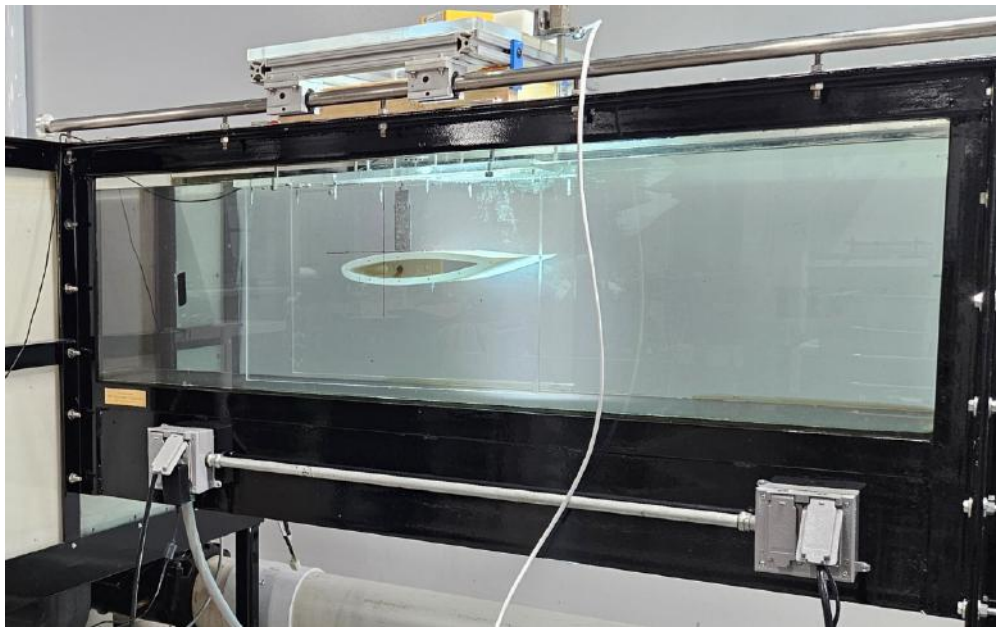
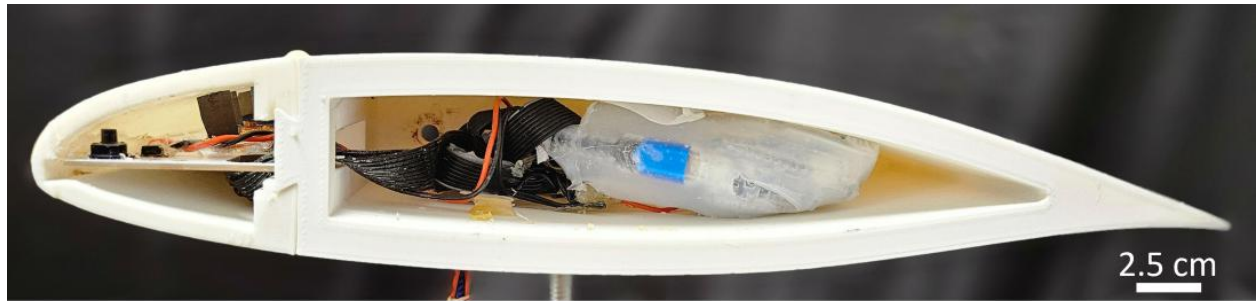
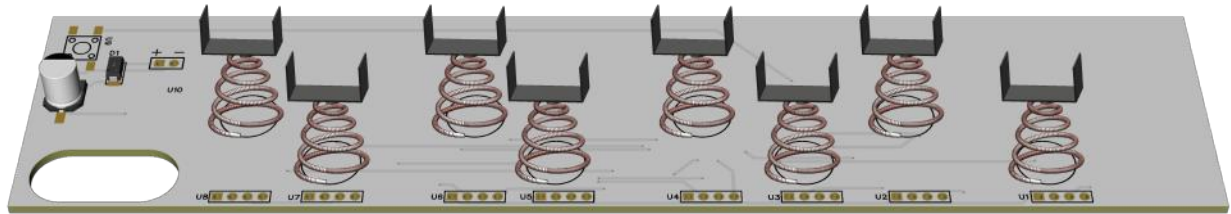


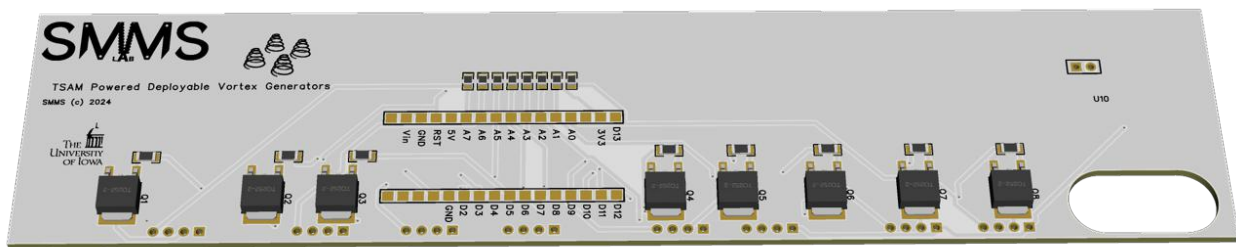
Figure 4.32. Airfoil inside the towing tank



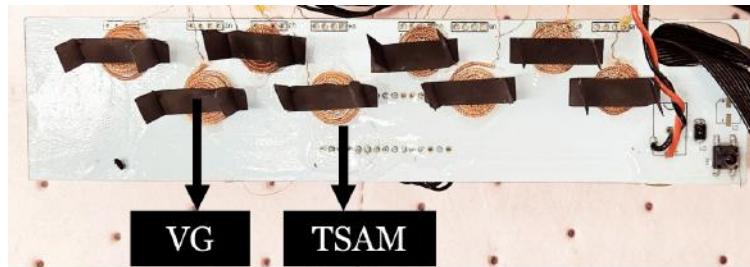
(a)



(b)



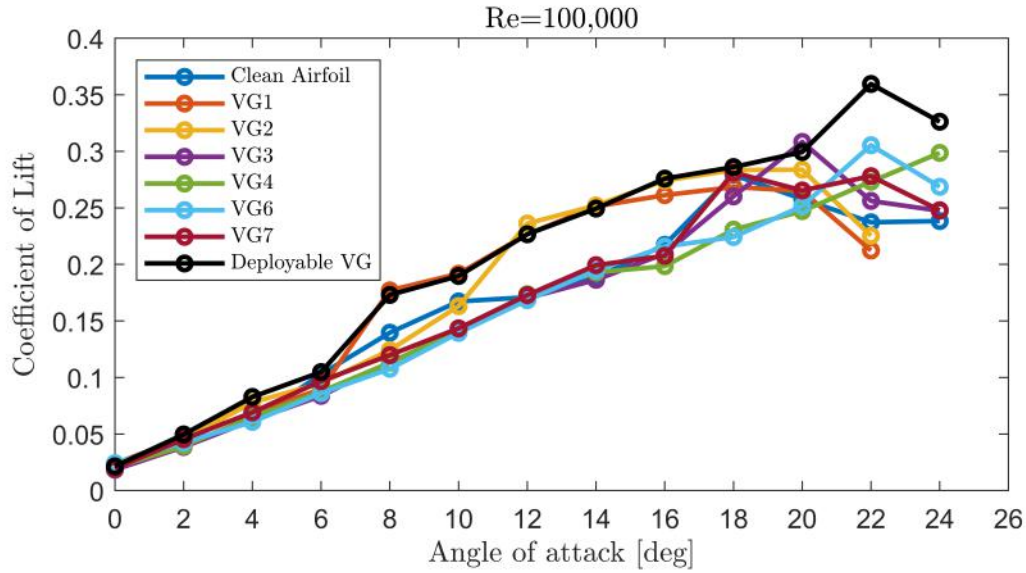
(c)



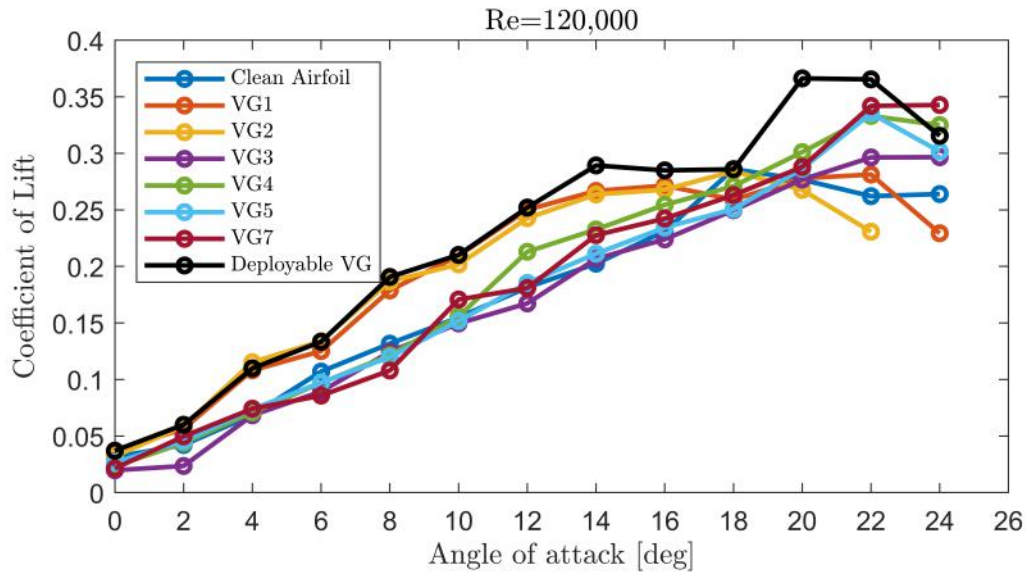
(d)

Figure 4.33. Manufactured airfoil with TSAMs-powered VGs. **(a)** Airfoil section with the deployable VGs (shown in the inset) and electrical setup for actuating TSAMs. **(b)** Graphical view of the top side, **(c)** the bottom side, and **(d)** Actual top view of the TSAMs powered VGs mounted on an insert plate that can be slotted into the nose of the airfoil.

view of the device can be seen in Figure 4.33(a). A graphical image of top view in Figure 4.33(b) show the TSAM placement. The actual image of the plate in Figure 4.33(d) depicts that the upgraded system is suitable for underwater testing.



(a)

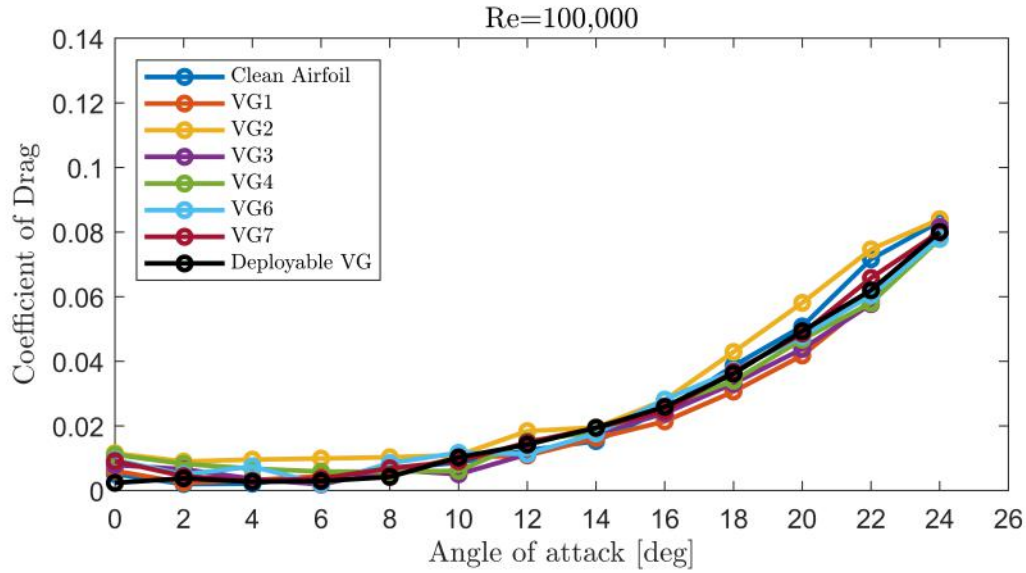


(b)

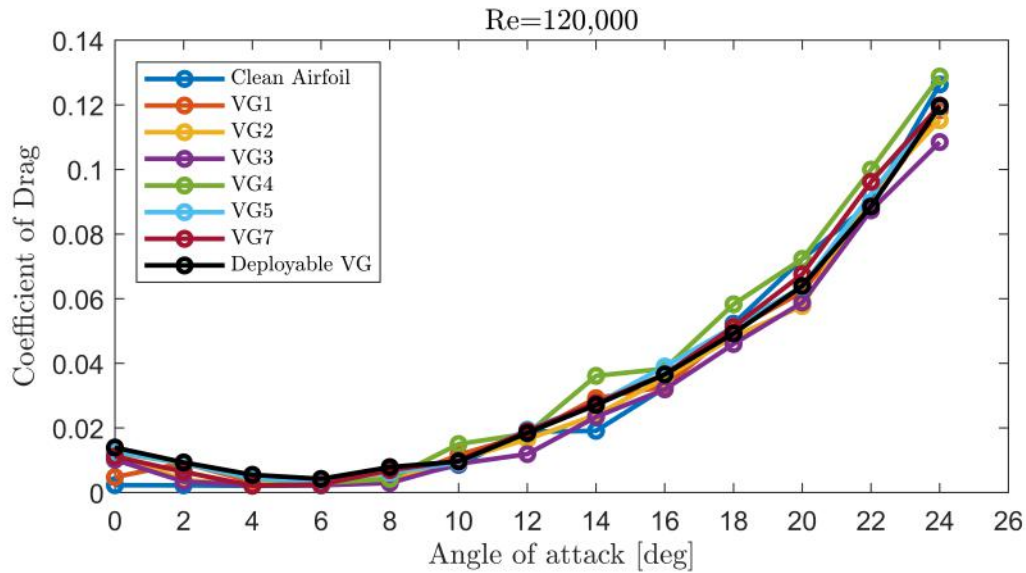
Figure 4.34. Effects of VGs deployed by electrothermally actuated TSAMs on the lift characteristics of the airfoil **(a)** Reynolds number 100,000 **(b)** Reynolds number of 120,000 underwater.

Figure 4.34 shows the lift coefficient plots for 2 different Reynolds numbers.

Figure 4.35, 4.34 shows the outcomes of the tests with the active VGs (i.e., deployed on demand by TSAMs according to flow conditions) compared with the static device. The aim is to ascertain that upon electrothermal actuation of the active device at each angle of attack, the



(a)



(b)

Figure 4.35. Effects of VGs deployed by electrothermally actuated TSAMs on the drag characteristics of the airfoil **(a)** Reynolds number 100,000 **(b)** Reynolds number of 120,000 underwater.

TSAM-powered VGs can match the best performing static configuration (corresponding to maximum values of coefficient of lift) at those angles of attack. As depicted in Figure 4.34(a), the coefficient of lift of the active VGs matches the peak performance of each configuration up to 20°. A considerable spike in lift coefficient was observed at 22° and 24°, which could partly be

attributed to the presence of VGs in the groove, which could facilitate the trapping of vortices in the groove that might affect the lift coefficient. The presence of trapped vortices in an airfoil with a groove has been reported to cause vortex shedding and limit the performance of the wing. A similar trend was observed at $Re = 120,000$. However, the stall sets in at 24° in contrast to the VG7 configuration (see Figure 4.34(b)). However, a vortex can also form near the upstream edge of the groove and if the groove geometry and position is not well optimized, the groove will not be able to trap the vortex, which then gets shed at the trailing edge of the groove. The un-trapped vortex can cause additional vortex shedding into the boundary layer and affect the hydrodynamic performance of the wing. The drag penalties were further investigated (see Figure 4.35(a) and 4.35(b)) and it was observed that the drag coefficient of the active VGs was consistent with other configurations at $Re = 120,000$. However, at $Re = 100,000$ the coefficient of drag of the active VGs was high compared to other configurations between 16° to 24° angles of attack.

CHAPTER 5. ARTIFICIAL MUSCLE POWERED SEMI-AUTONOMOUS SURGICAL ROBOT

This chapter presents the design of a surgical robot to automate the paracentesis procedure. The project has two phases: Phase I: Design of the surgical robot powered by conventional actuators which is described in Section 5.1; Phase II: Design of a surgical robot powered by Artificial muscle actuators and nonferrous materials that can be used inside an MRI machine described in Section 5.2. Each section presents the design and simulations of proposed surgical robots.

5.1 Design of a Surgical Robot

5.1.1 Introduction

Healthcare costs in the United States (US) have shown a consistent upward trajectory [338], exacerbated by the prevailing and projected shortages in healthcare personnel [339–342]. These converging factors are compelling stakeholders to explore innovative avenues for managing healthcare expenses without compromising patient care quality. While automation has revolutionized efficiency and cost-effectiveness in various economic sectors, its integration into healthcare remains comparatively limited. Although robotic systems have found specific applications in certain medical domains [343–345], their predominant role lies in augmenting human capabilities rather than significantly impacting time or cost dynamics.

A notable gap exists in leveraging robotics for low to mid-level complexity medical procedures [346], where autonomous systems with minimal human intervention hold immense potential to enhance efficiency, reduce expenses, and improve patient outcomes. By facilitating precise and consistent procedures, robotic interventions can mitigate complications and streamline healthcare delivery. Recognizing this opportunity, this research endeavors to develop a semi-autonomous robotic platform adaptable to a spectrum of medical procedures.

This work presents the design and simulation of a surgical robot to automate the paracentesis procedure. The project has two phases: Phase I: Design of the surgical robot powered by

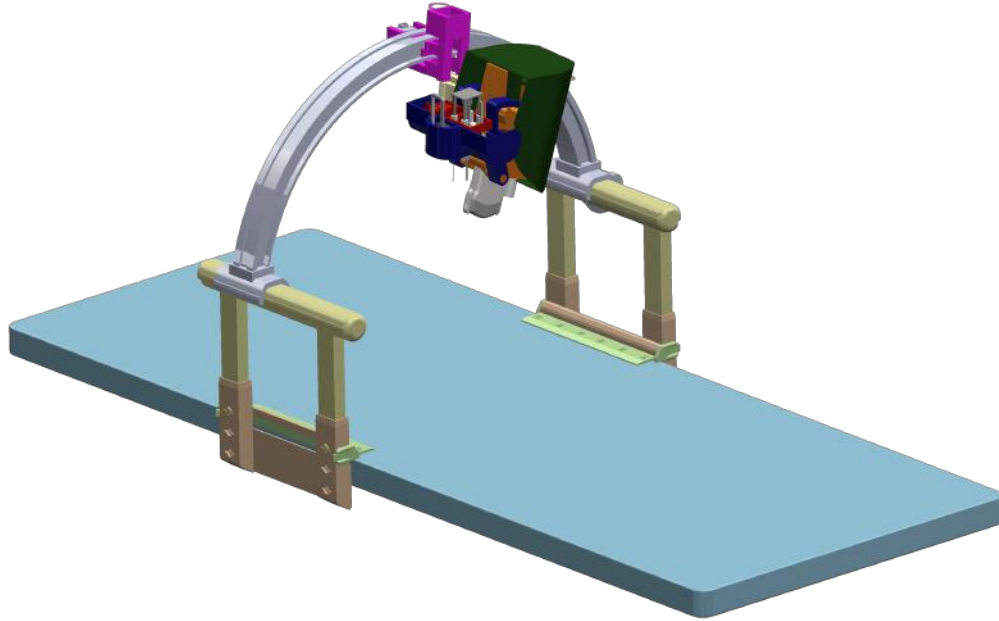


Figure 5.1. Overview of Phase I design powered by DC motors

conventional actuators; Phase II: Design of a surgical robot powered by TCAMs that can be used inside of a Magnetic Resonance Imaging (MRI) machine. The kinematic and dynamic analysis, workspace, and control methodology will be proposed for each device. Figure 5.1 shows the completed design of the Phase I device, and Figure 5.2 shows the design of the Phase II.

The author's major contribution to this work is designing the devices. The following subsections detail the mechanical design, kinematics, and control system implementation.

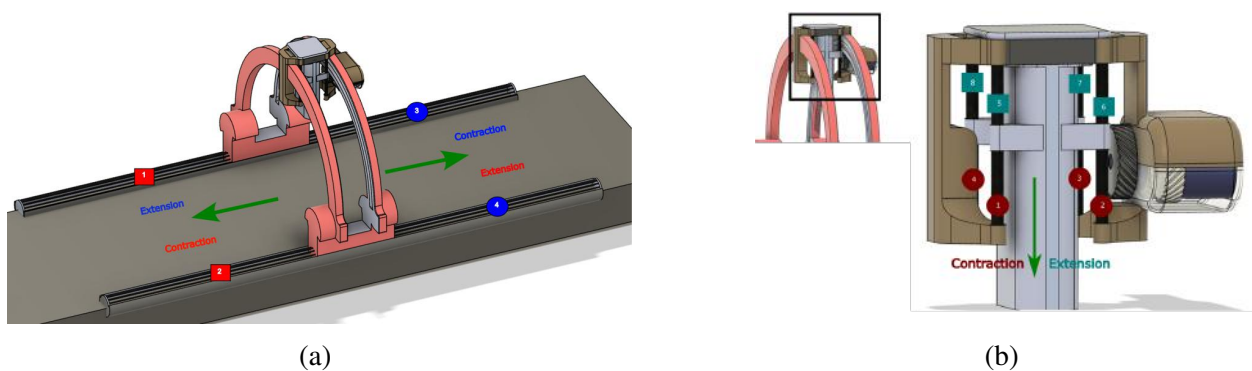


Figure 5.2. Overview of Phase II design powered by TCAMs

5.1.1.1 Mechanical Design

As shown in Figure 5.3, a prismatic joint exists between the bed handles and the gantry. Let the origin be on the bed, and the gantry can move along the z_1 direction with the aid of stepper motors. The gantry slider mount can slide along the gantry arc with the maximum angle of θ_2 .

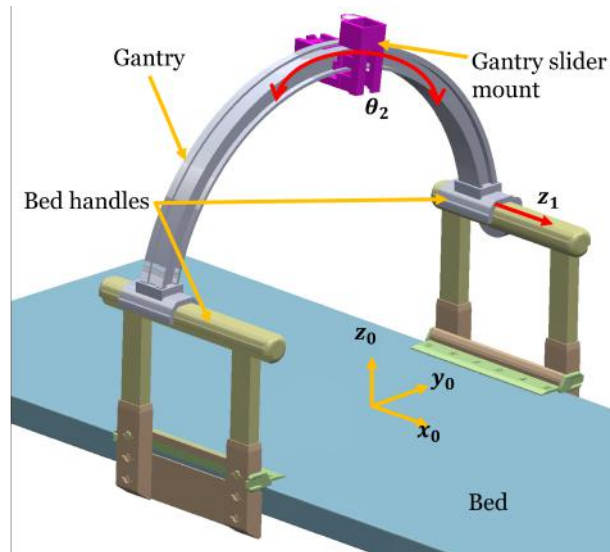


Figure 5.3. Design overview of bed and the gantry

The vertical slider mount consists of a prismatic joint as shown in Figure 5.4. This will enable the system to lower and higher. The motion direction is indicated by z_3 . The ultrasound

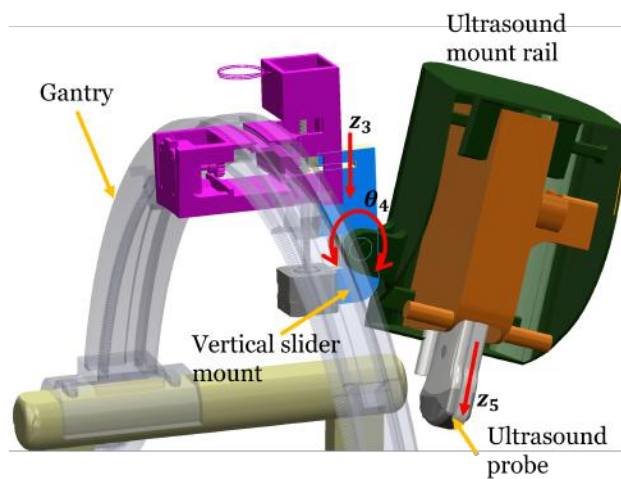


Figure 5.4. Design overview of ultrasound rail

mount rail can rotate on the vertical slider mount at an angle of θ_4 . The ultrasound probe can translate along z_5 with respect to the ultrasound mount rail.

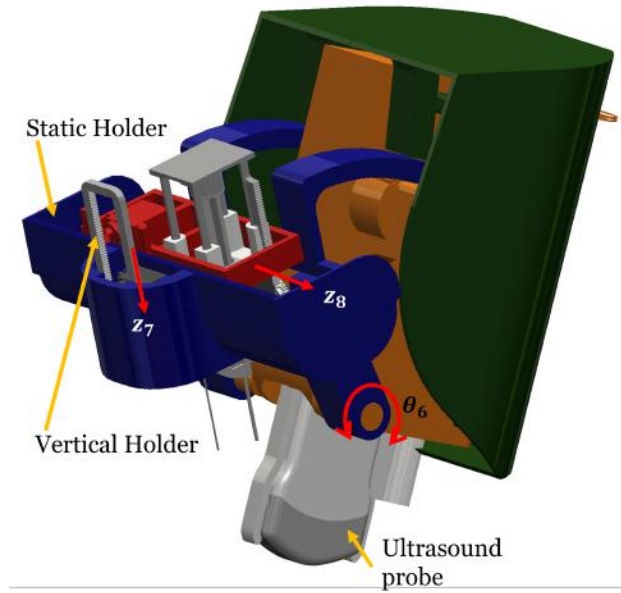


Figure 5.5. Design overview of the mechanism

The static holder can rotate with respect to the ultrasound rail at an angle of θ_6 . Vertical holder shown in Figure 5.5 enables 3 probes to vertically translate in z_7 . The horizontal moving holder can slide at z_8 .

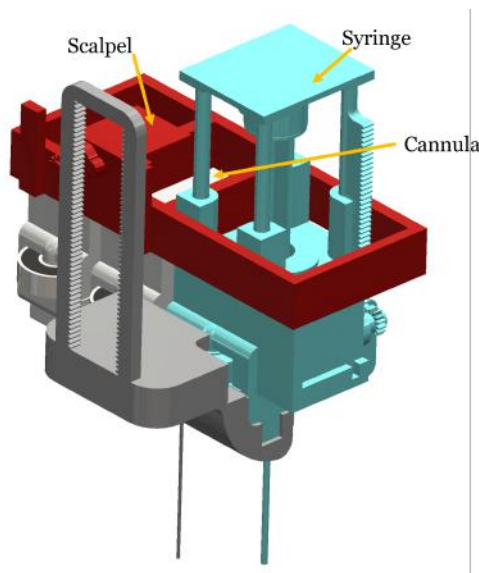


Figure 5.6. Design overview of the slider

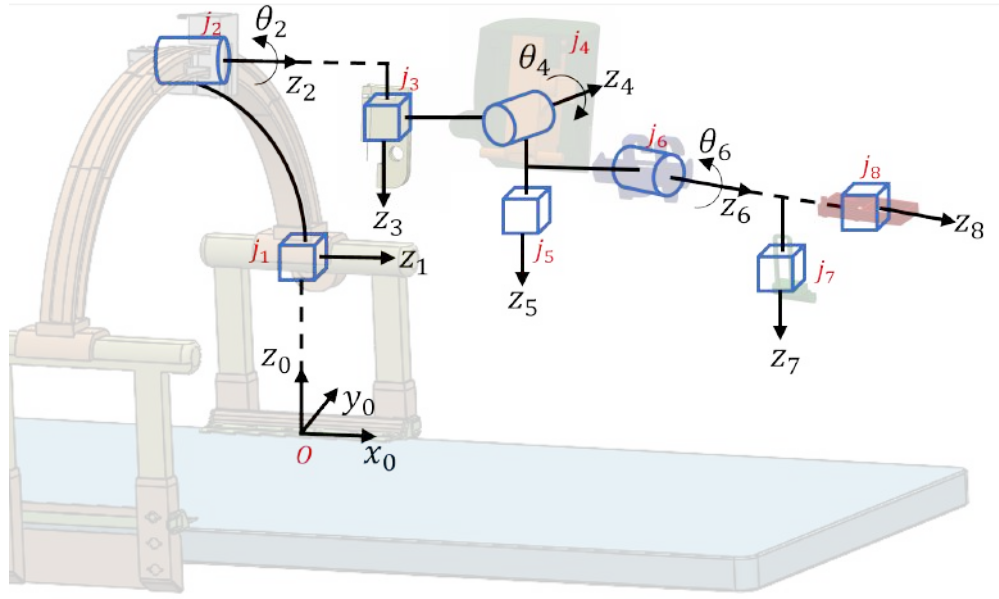


Figure 5.7. Kinematic diagram of the device

Figure 5.6 shows detailed illustration of 3 components actuating using a single vertical holder. The horizontal moving holder enables the slot to in contact with the vertical holder. Then, the vertical holder moves the syringe, cannula, and scalpel upon requirement. A precise z_8 horizontal holder actuator is critical for repeatable, accurate actuation system.

Figure 5.7 shows the kinematic diagram of the complete device. Based on the kinematics implemented on the CAD model (SOLIDWORKS), the model was converted to a kinematic model using MATLAB/Simscape. Figure 5.8 shows the overview of the Simscape model. This model consists of all the revolute and prismatic joints of the system denoted in the kinematic diagram at Figure 5.7. To actuate this model, a simple PID control-based simulation was developed following the design to implement the real-time controller.

The simulation was implemented using SIMULINK/SIMSCAPE and the plots for each actuator is shown in Figure 5.9. Later this simulation CAD models ports of the simulink code can be replaced by actual hardware feedback sensory data. Therefore, this model will be used to tune the controller parameters. The screenshot of the PID control scheme is available in Appendix B.3.

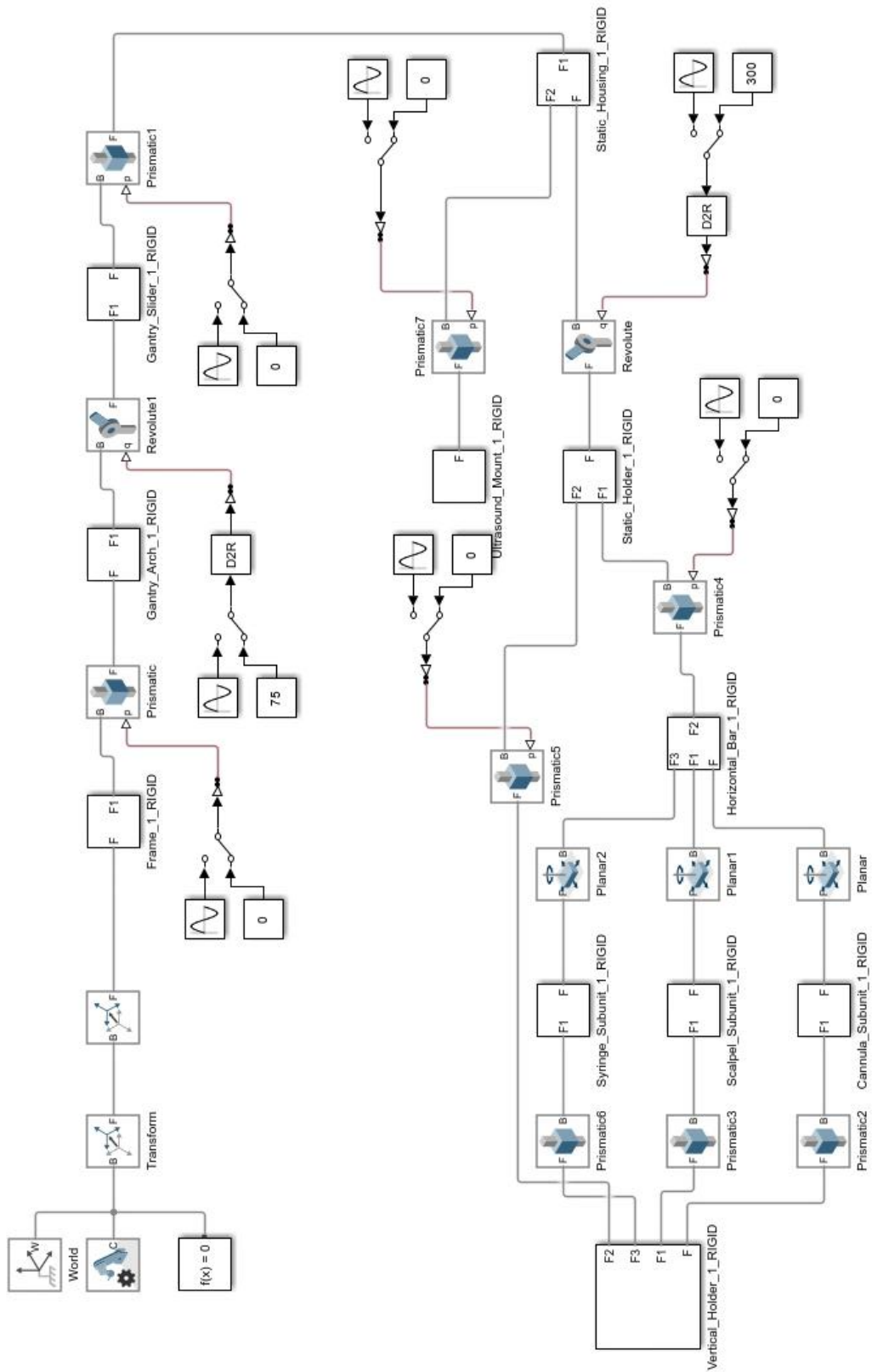


Figure 5.8. Yantra 1 - Simscape Model

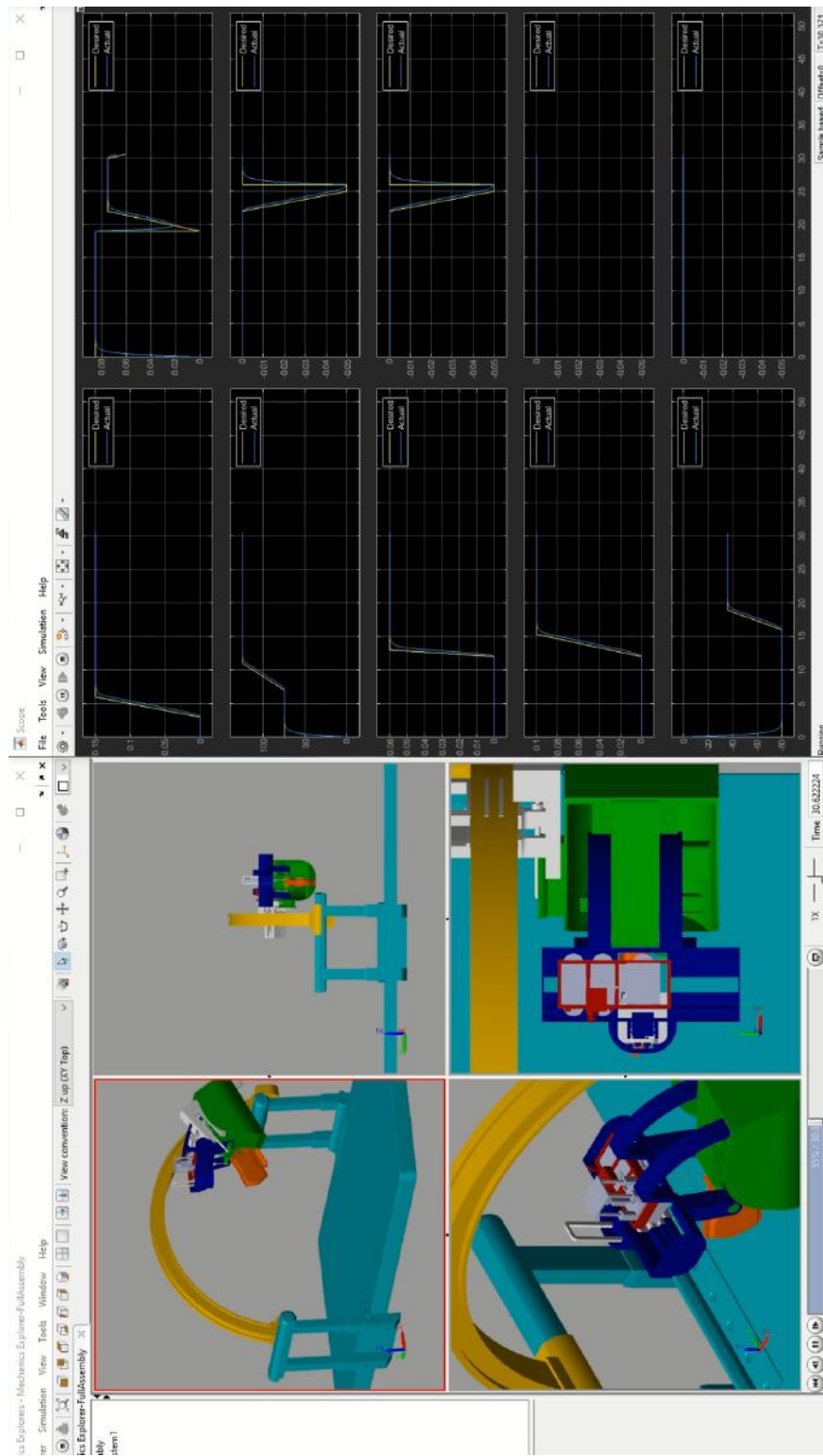


Figure 5.9. SIMULINK SIMSCAPE Control Model and plots

5.2 Design of an Artificial Muscle Powered Surgical Robot

This section demonstrates the preliminary design work of a surgical robot powered by Twisted and Coiled Artificial Muscles. Since the primary goal was to use this surgical robot inside the MRI machine, the design approach was to replace conventional electrical actuators with TCAM-based actuator mechanisms. Figure 5.10 shows the top view of the bed, and Figure 5.11 shows the side view. The gantry can slide along the bed using TCAMS, which are linearly attached to the system.

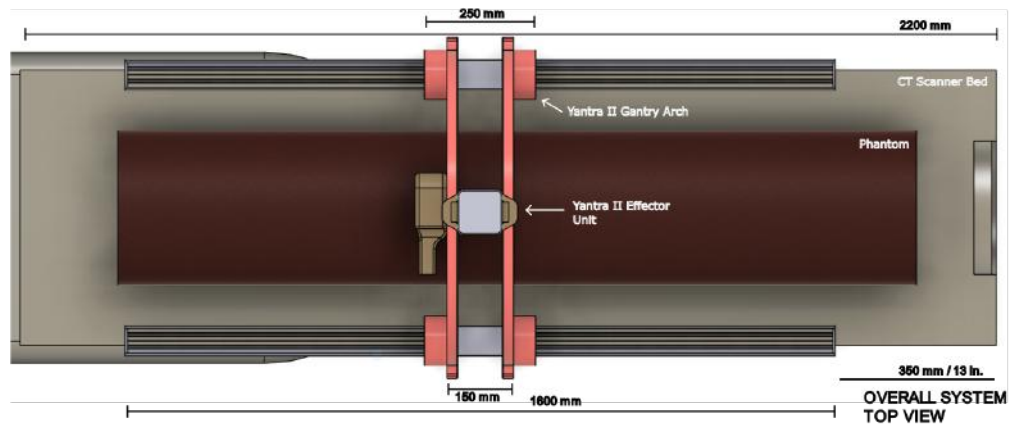


Figure 5.10. Yantra 2 - Design Overview 1

Figure 5.12 shows the syringe, scalpel, and cannula sub-unit system that can be actuated using TCAMs. This concept modeled using SIMSCAPE to identify the motion range (Figure 5.13).

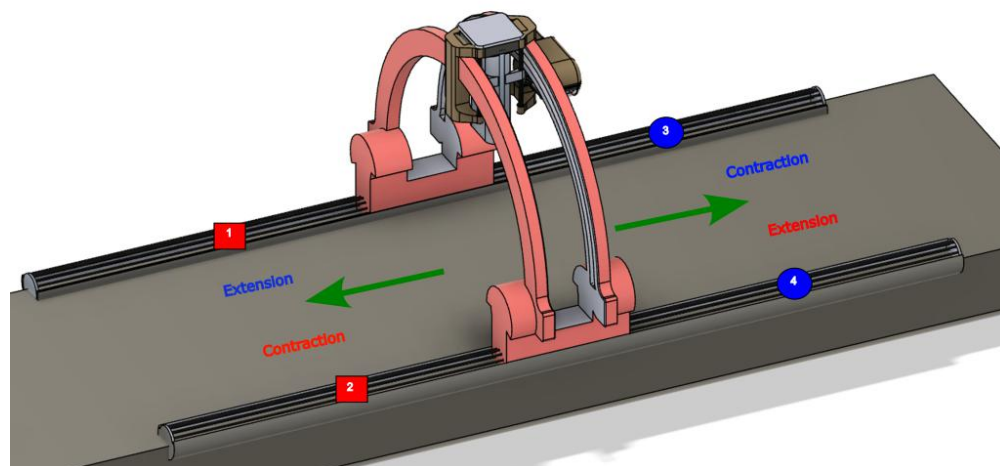


Figure 5.11. Yantra 2 - Design Overview 2

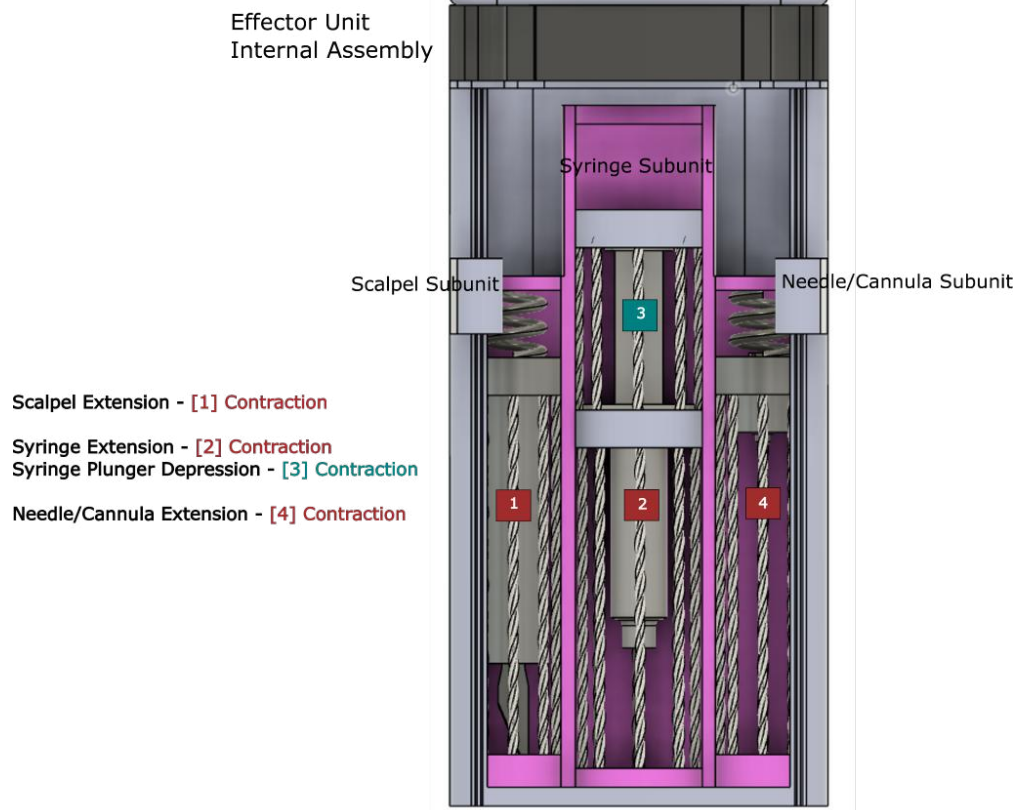


Figure 5.12. Yantra 2 - Design Overview 3

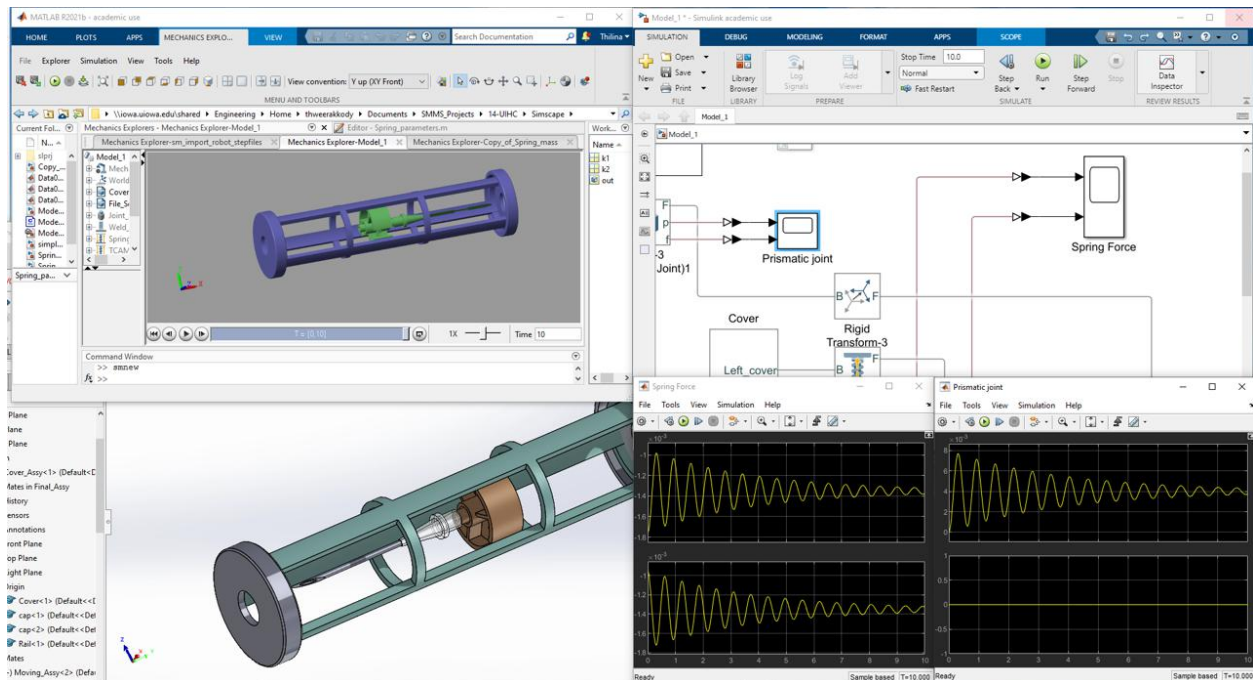


Figure 5.13. Yantra 2 - Needle device Simulation

CHAPTER 6. CONCLUSIONS AND FUTURE WORK

The final chapter provides conclusions to the present work and discusses potential future directions. The primary goal of this dissertation has been to develop hardware and control methods for the real-time control of artificial muscles in soft robotic devices. While many research groups focus on the development of artificial muscles, only a few have successfully implemented control algorithms for creating end-user products. Due to their high nonlinearity, many existing artificial muscles are difficult to control in soft robotic applications to achieve desired motions. Moreover, the limitations of existing theoretical frameworks further complicate control algorithm implementation.

This dissertation addresses these challenges by implementing a theoretical model for Twisted and Coiled Artificial Muscles (TCAMs), followed by the development of robust and adaptive controllers for both TCAMs and Shape Memory Alloys (SMA). These models and controllers were then applied to rehabilitation, underwater, and medical applications to assess their effectiveness in real-world scenarios.

6.1 Conclusions

TCAMs are a novel, lightweight actuator type with significant potential for use in soft robotics applications, thanks to their inexpensive manufacturing process, high stress-strain generation capability, high specific power, and low specific cost. Given these advantages, a generalized theoretical model is essential for predicting the actuation force and displacement of TCAMs in practical applications. The proposed physics-based model relates the output force and displacement to the physical, thermal, electrical, and mechanical properties of TCAMs. Moreover, the robust position and force $\mathcal{L}_1$ adaptive controller ensures stability and performance against disturbances and highly nonlinear TCAM model uncertainties. The experimental results obtained for different types of TCAMs validate the proposed theoretical model and demonstrate the efficacy of the $\mathcal{L}_1$ adaptive controller for both position and force control in real-time applications. Furthermore, the effectiveness of the $\mathcal{L}_1$ adaptive controller compared with a PID controller to prove the use to such

complicated controller over a conventional control scheme in terms of robustness.

The $\mathcal{L}_1$ robust adaptive controller was extended to control a highly nonlinear yet widely used type of artificial muscle known as Shape Memory Alloys (SMA). The $\mathcal{L}_1$ adaptive controller was implemented for generalized SMA configurations, and its performance in position and force tracking was experimentally validated and compared to PID tracking.

In the following chapters, this dissertation addresses the control applications of artificial muscles. For rehabilitation applications, it presents a TCAM-powered wrist rehabilitation glove, a TCAM-powered variable stiffness actuator for an Ankle Foot Orthosis, and an SMA-powered hand rehabilitation glove for fingers. In each case, hardware systems were developed to acquire real-time sensory data and apply the $\mathcal{L}_1$ adaptive controller for the respective applications.

For underwater robotics applications, this thesis presents a TCAM-powered muscular hydrostat robot capable of bending, compressing, extending, and twisting, as well as Twisted and Spiraled Artificial Muscle (TSAM)-powered deployable vortex generators to reduce lift coefficients. These devices were tested in both still air and underwater environments. The contributions span hardware circuitry development, sensory feedback system design, real-time data acquisition, and control system implementation.

This dissertation also addresses the use of TCAMs in the development of medical robotic devices aimed at automating surgical procedures. Carbon fiber TCAMs, composed of non-ferrous materials, are ideally suited for use inside MRI machines. This work extends to the design of a TCAM-based surgical robot for industry-focused applications.

In summary, this dissertation contributes to the field of artificial muscle-powered soft robotics through novel mathematical models, control theory, and experimental validation with hardware systems. The use of offline tracking algorithms demonstrates the practicality of these models and theories in real-world applications. As soft robotics continues to expand across various fields, we believe the work presented in this dissertation will be directly relevant to practical implementations and product development.

6.2 Original Contributions

This dissertation presents several original contributions, including the development of generalized physics-based models for Twisted and Coiled Artificial Muscles (TCAMs) and the implementation of the $\mathcal{L}_1$ adaptive controller for various artificial muscles such as TCAMs, Twisted String Actuators (TSAMs), and Shape Memory Alloys (SMAs). Moreover, the practical application of these models and controllers has been demonstrated in both rehabilitation devices and underwater robotics.

Additionally, this work introduces the design of specialized circuitry and hardware systems, alongside the use of observer-based sensors for real-time robotic actuation. The design of these circuits has been critical to success. All artificial muscles in the study were actuated through Joule heating, with position and force tracking achieved via fast voltage/current variation. Thus, careful selection of electronic components, precise control of high DC voltage/current, and minimizing communication delays were vital for real-time actuation and the prevention of TCAM burnout. The circuits were designed as a high-voltage/low-voltage system duo, where the high-voltage circuit provided the required current (0.5 A to 2 A per muscle) and the low-voltage side controlled transistor switches that powered the high-voltage side (with currents typically less than 0.2 A). Feedback systems were integrated within the same circuit to ensure rapid sensor feedback.

As the number of artificial muscles increased (e.g., the AFO setup with 32 TCAMs, Vortex Generator setup with 8 TSAMs), the system architecture was expanded using custom PCB shields to minimize wiring complexity. Each PCB unit included its own microcontroller (ATmega 2560), designed to communicate with a central microcontroller via ROS, facilitating control and coordination. The hardware design employed in this system addresses a significant gap in soft robotics and the control of such systems. Additionally, certain circuits, such as those for TSAM and muscular hydrostat actuation, were specifically designed for underwater operation.

The modular design of the data acquisition system allowed for the integration of various sensors into the network, adaptable to specific application requirements. Controller tuning and

extensive experimental testing were carried out across multiple scenarios.

A major contribution of this thesis is the successful real-time control of artificial muscles in rehabilitation devices. Artificial muscle-powered systems have the potential to be portable and require a smaller power source compared to conventional exoskeletons. Other advantages of these systems include their lightweight nature, ease of fabrication, and ability to generate higher stress compared to traditional actuators. This work paves the way for future development of artificial muscle-powered devices.

The underwater application of artificial muscles demonstrated their efficacy as miniature actuators for submerged or floating robots. The custom-designed underwater-compatible hardware systems highlighted their utility in challenging environments, effectively withstanding real-world conditions such as waves, currents, and high-speed winds. These experiments provide evidence of the robustness and reliability of the hardware and control strategies developed in this thesis.

Furthermore, a novel surgical robot, designed specifically for clinical environments, was introduced. This device was developed based on input from medical professionals and represents an entirely original contribution to the field of surgical robotics.

6.3 Future Work

Much of the work carried out during this PhD has evolved over time. However, the following future directions could further advance control investigations of artificial muscles. The current generalized TCAM models do not account for the dynamics of mandrel-type or multi-ply TCAMs. Since mandrel muscles show promising load-bearing, antagonistic, and larger strain capabilities, developing models that address these characteristics would be valuable for future research.

Flexible dynamics play a significant role in soft robotics. The use of energy-based flexible dynamic equations could help achieve more reliable control dynamics, particularly during initialization stages. At present, many studies scarcely utilize flexible dynamic methods to address challenges in soft or continuum robots. Techniques such as Finite Element Analysis (FEA), Rod theory, and Piecewise Constant Curvature (PCC) are emerging as promising methods for evaluating

soft robot dynamics.

Physics-informed machine learning models could also be effective for modeling the dynamics of artificial muscles. These methods are already gaining popularity in other fields, and material scientists will likely adopt them for dynamic modeling in the near future.

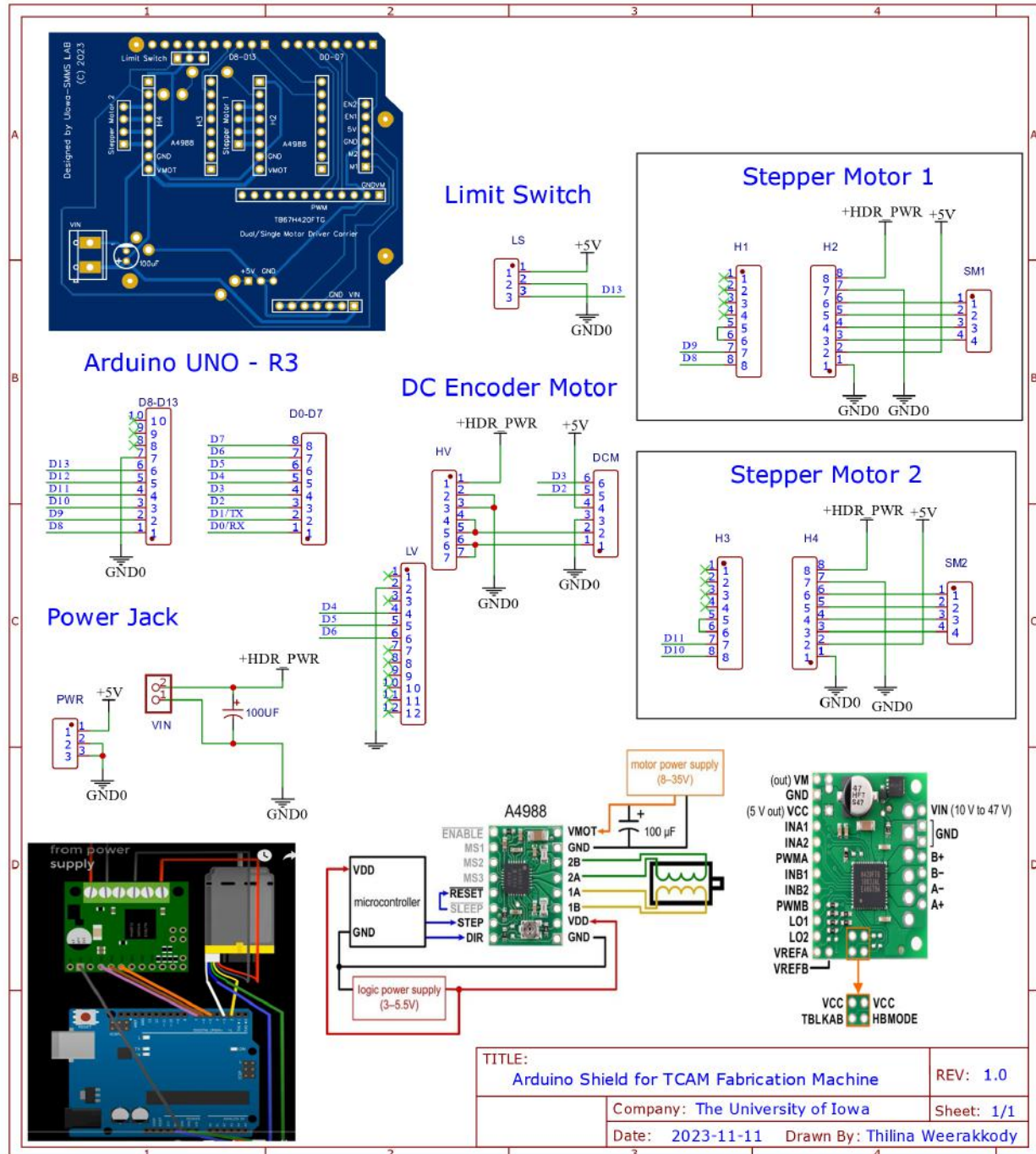
The developed rehabilitation devices could be fully converted into portable systems by incorporating Lithium-polymer batteries and embedded systems. These devices can also be adapted to electromyography (EMG) or electroencephalogram (EEG)-based feedback systems, significantly increasing their appeal within the medical community. The effectiveness of the variable stiffness actuator in Ankle Foot Orthosis (AFO) devices can be further improved by integrating mechanical components. Additionally, AFOs could function as energy-harvesting devices by incorporating spring-based systems along with TCAMs.

Underwater muscular hydrostats could be transformed into self-navigating robots by introducing novel low-frequency communication protocols. This would conserve energy while enabling efficient communication in underwater (near-surface) applications.

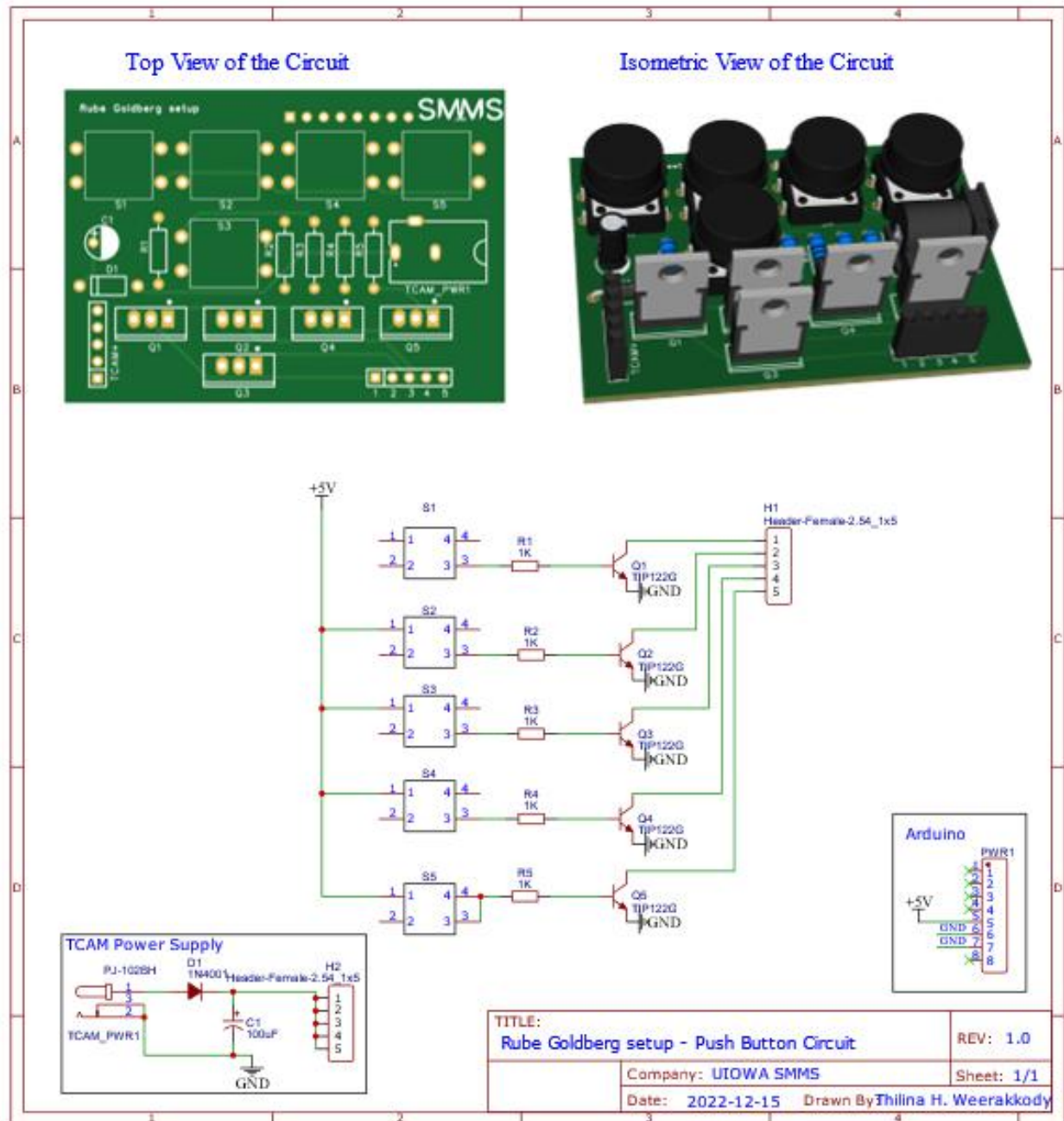
Finally, vision-based manufacturing systems could greatly increase the productivity of artificial muscle production. Integrating computer vision with machine learning-based automated systems would be the next step for scaling up TCAM/TSAM manufacturing.

APPENDIX A: HARDWARE AND CIRCUITRY

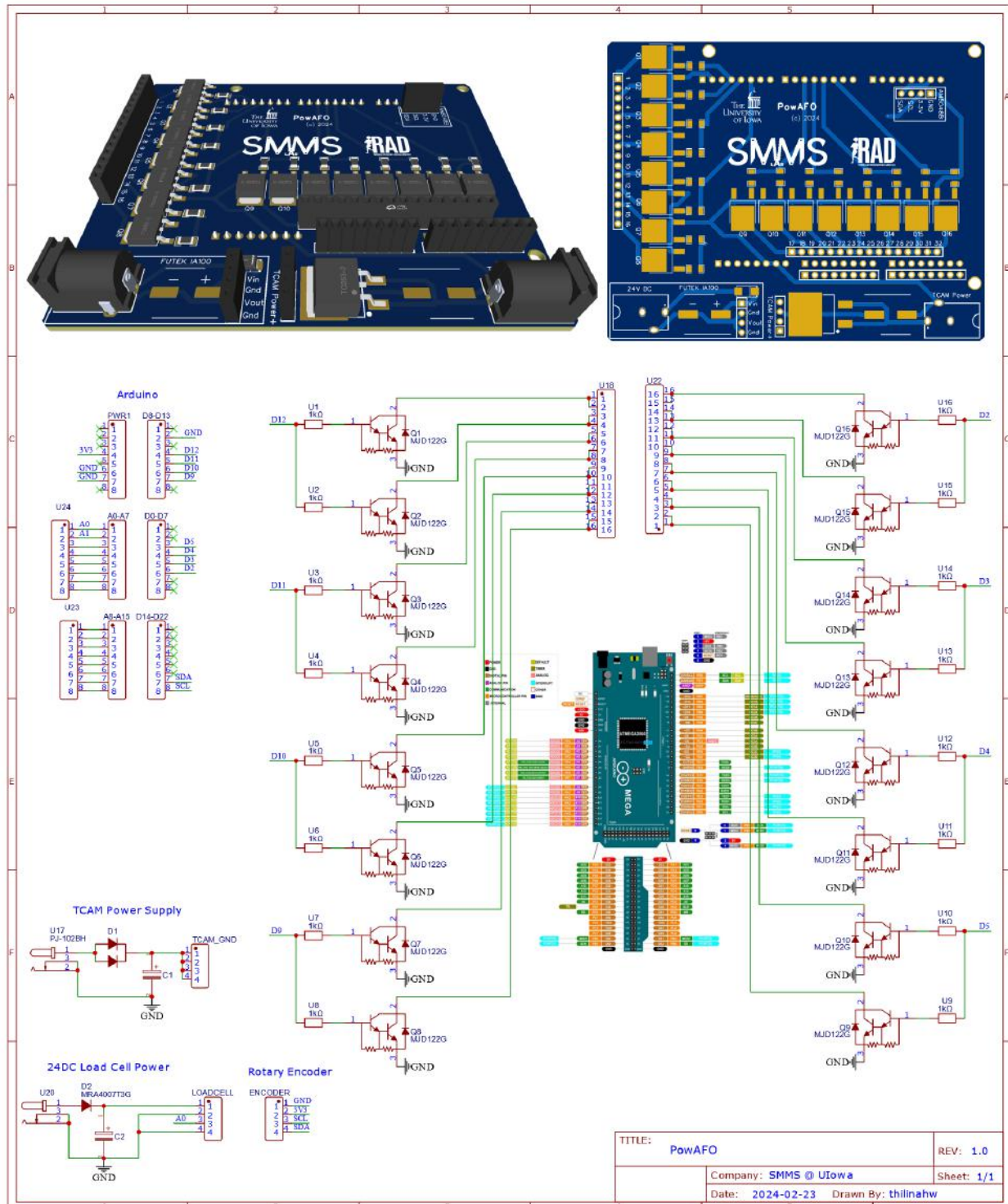
A.1 Automated TCAM manufacturing system circuitry



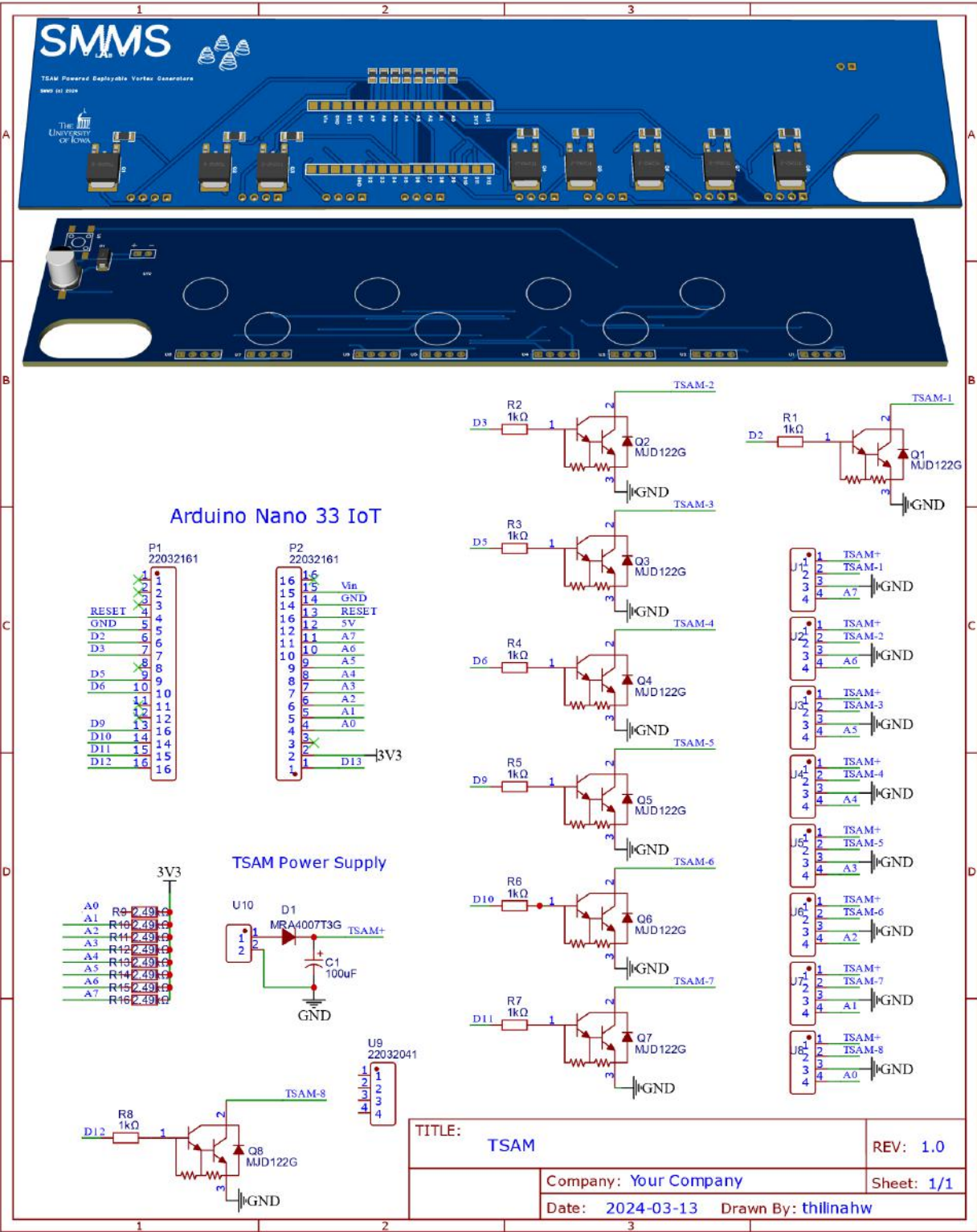
A.2 Rube Goldberg setup actuation circuitry



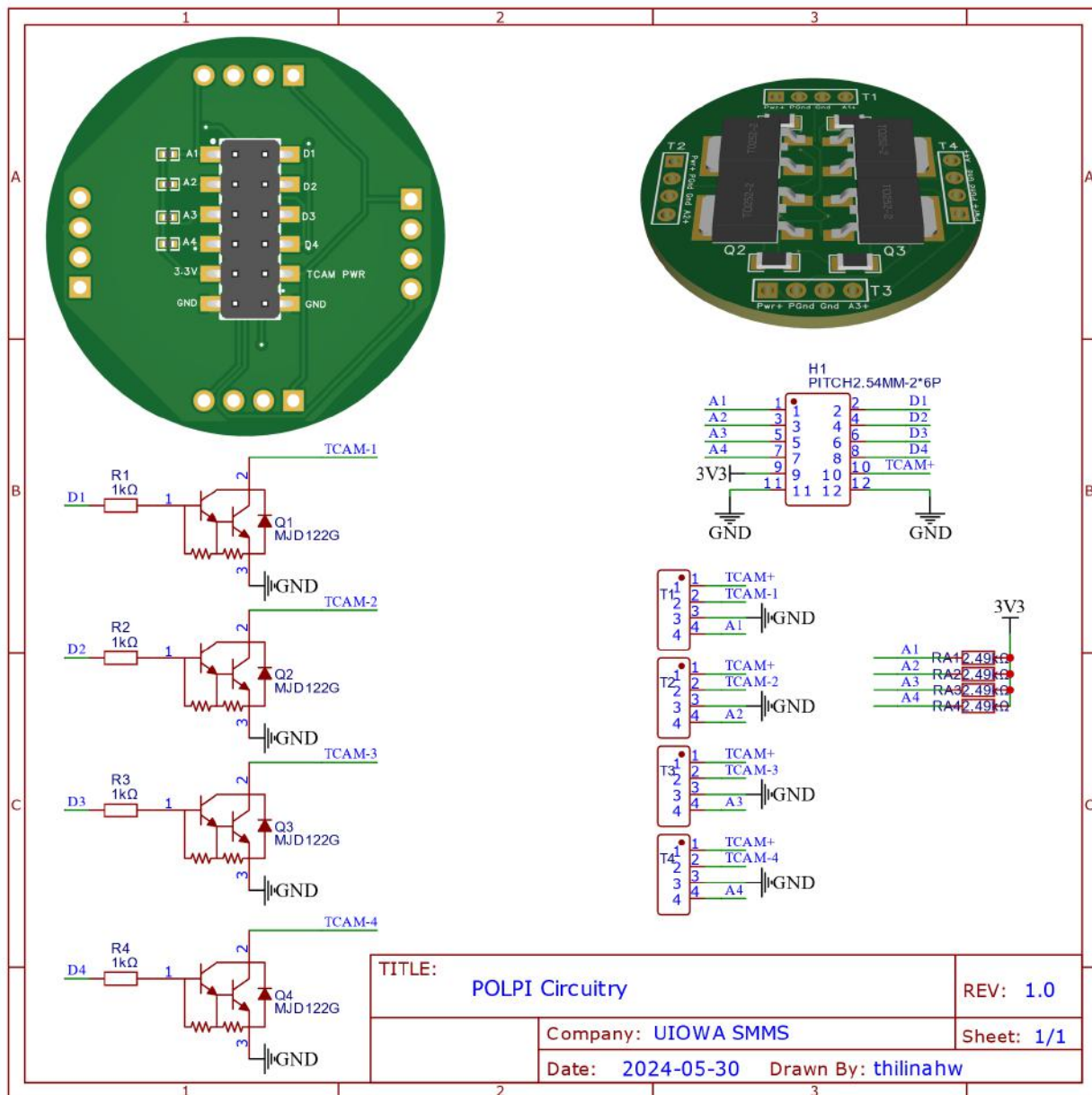
A.3 PowAFO actuation circuitry - 32 TCAMs



A.4 TSAMs with VG underwater actuation circuitry



This is a 3.5 cm diameter circuitry for mandrel TCAM actuation. Each board can control 4 TCAMs and multiple boards will be connected to the main microprocessor (Raspberry PI) using Arduino ATmega2560 chips with ROS.



APPENDIX B: MATLAB AND SIMULINK CODES

B.1 Temperature Tracking Using FLIR Thermal Camera

```
1  % FLIR E95 Camera Data Acquisition FLIR Atlas MATLAB SDK to ROS Program
2  %% Disconnect the ImStream connection only if it exists
3  try
4      if(ImStream.IsConnected)
5          ImStream.Disconnect()
6      end
7  catch
8  end
9  rosshutdown
10
11 %% Define the time (seconds) for camera search
12 T_SEARCH = 3;
13
14 %Define the two points of the line
15 x1 = 180;
16 y1 = 100;
17 x2 = 350;
18 y2 = 275;
19
20 %% Run ROS for communication
21 try
22     rostopic list
23 catch
24     rosinit
25 end
26 publisher = rospublisher("temperature", "std_msgs/Float64");
27 msg = rosmessage(publisher);
28
29 %% Add Assembly Flir.Atlas.Live.dll
30 atPath = getenv('FLIR_Atlas_MATLAB');
31 atLive = strcat(atPath, 'Flir.Atlas.Live.dll');
32 asmInfo = NET.addAssembly(atLive);
33
34 % init camera discovery
35 temp = Flir.Atlas.Live.Discovery.Discovery;
36
37 % search for cameras for T_SEARCH seconds
38 disp(['Searching for camera for ', num2str(T_SEARCH), ' seconds...'])
39 disc = temp.Start(T_SEARCH);
40 disp('Searching complete.')
41 assert(disc.Count > 1, 'Camera not found')
```



```

42
43 %% Find the FLIR E95 camera
44 for i = 0:disc.Count-1
45     if contains(char(disc.Item(i).Name), 'FLIR E95')
46         camInfo = disc.Item(i);
47         break
48     end
49 end
50 handles.disc = disc;
51
52 %It is FlirFileFormat init a ThermalCamera
53 ImStream = Flir.Atlas.Live.Device.ThermalCamera(true);
54 ImStream.Connect(camInfo);
55
56
57 disp('Waiting for image stream...')
58 while ~ImStream.IsGrabbing, end
59 disp('Image stream found')
60
61 disp('ROS topic is now live!')
62 rate = rosrate(100);
63
64 %save the stream
65 handles.ImStream = ImStream;
66
67 %set the Iron palette
68 pal = ImStream.ThermalImage.PaletteManager;
69 ImStream.ThermalImage.Palette = pal.Iron;
70
71 img = ImStream.ThermalImage.ImageArray;
72 %convert to Matlab type
73 X = uint8(img);
74 % [m00,x1]=max(X(y1,:,1));
75 % [m01,x2]=max(X(y2,:,1));
76 % spot = ImStream.ThermalImage.Measurements.Add(System.Drawing.Point(x1, y1));
77 rect = ImStream.ThermalImage.Measurements.Add(System.Drawing.Rectangle(x1, y1,x2, y2));
78 % line = ImStream.ThermalImage.Measurements.AddLine(x1, y1, x2, y2);
79
80 pause(1);
81 try
82     while 1
83         %get the colorized image
84         img = ImStream.ThermalImage.ImageArray;
85         %convert to Matlab type
86         X = uint8(img);
87
88         %show image with Matlab imshow

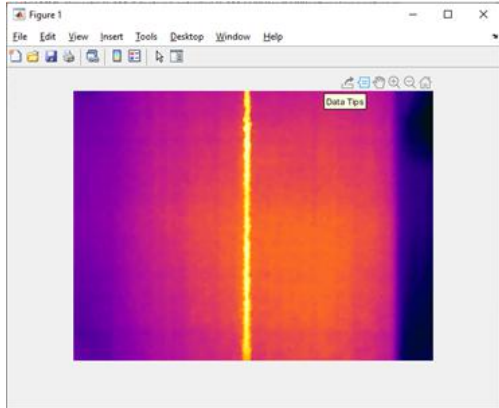
```

```

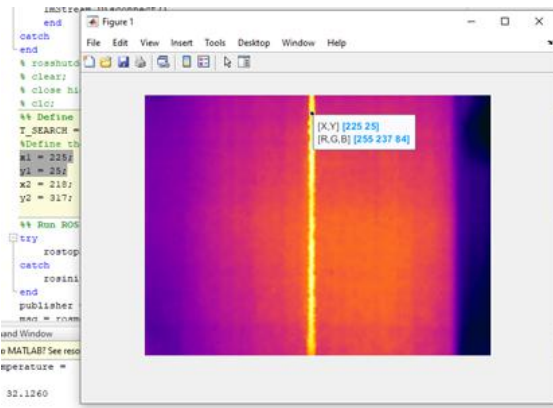
89     imshow(X);
90     %Read temp from spot
91     %Temperature = line.Max.Value;
92     Temperature = rect.Max.Value;
93     fprintf('%f\t%i\t%i\n',Temperature, x1, x2)
94     msg.Data = Temperature;
95     send(publisher, msg);
96     waitfor(rate);
97 end
98 catch
99     %Disconnect the Camera
100    disp('Disconnecting from thermal camera...')
101    ImStream.Disconnect()
102    rosshutdown
103 end

```

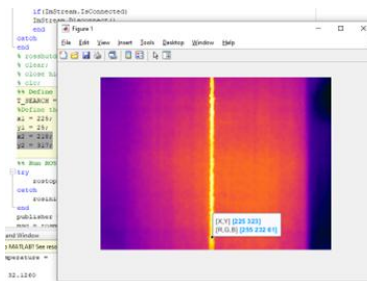
1. Connect the FLIR E-95 to the computer using the USB cable
2. Align the camera. Make sure the Camera is fixed.
3. Get the points as in Figure B.1 and run "ReadFLIRRealtimeTemperature.m".



(a)



(b)



(c)

Figure B.1. Steps to Configure the Thermal Camera.

B.2 Realtime Multiple Points Tracking in Air and Underwater

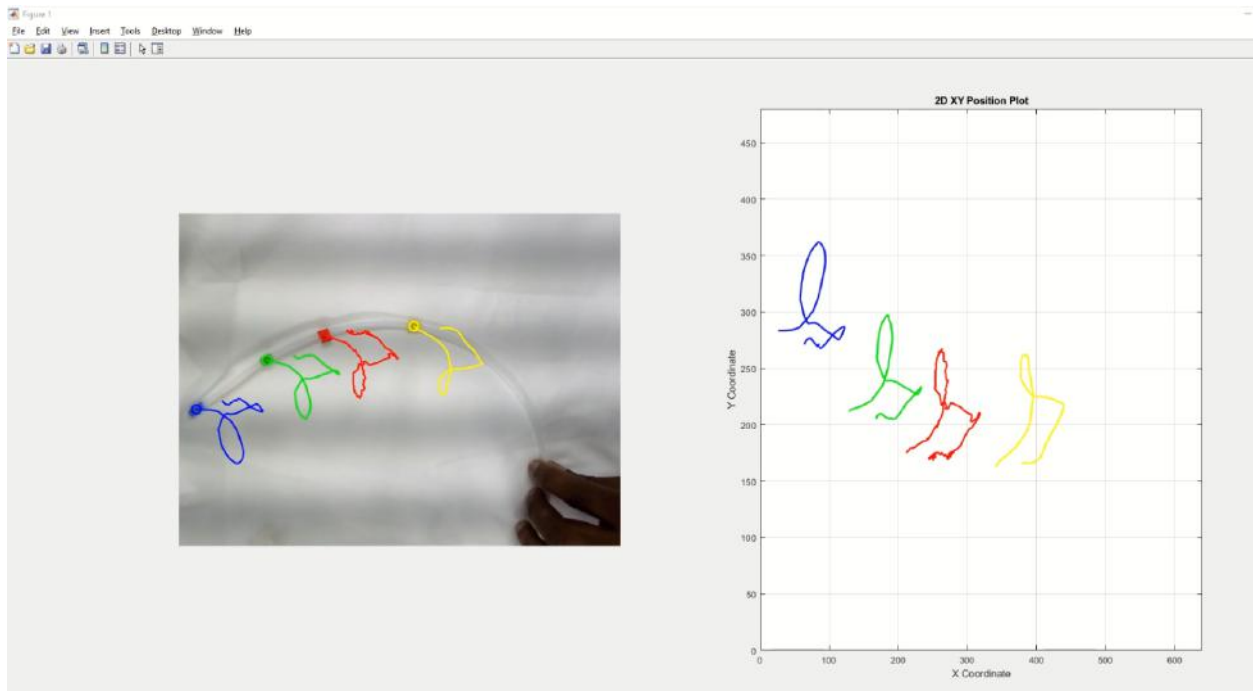


Figure B.2. Point Tracking in air and underwater.

```
1  clear all
2  close all
3  clc
4
5  % Initialize ROS
6  try
7      rostopic list;
8  catch
9      rosinit;
10 end
11
12 % Create ROS Publishers for each color
13 orangePublisher = rospublisher('/orange_coordinates', 'geometry_msgs/Point');
14 yellowPublisher = rospublisher('/yellow_coordinates', 'geometry_msgs/Point');
15 greenPublisher = rospublisher('/green_coordinates', 'geometry_msgs/Point');
16 bluePublisher = rospublisher('/blue_coordinates', 'geometry_msgs/Point');
17 orangeMsg = rosmessage(orangePublisher);
18 yellowMsg = rosmessage(yellowPublisher);
19 greenMsg = rosmessage(greenPublisher);
20 blueMsg = rosmessage(bluePublisher);
21
```

```

22  % Initialize Webcam
23  cam = webcam(1);
24  frameSize = size(snapshot(cam));
25
26  %% coordinates of the rectangle [x0, y0, x_end, y_end] 640 x 480
27  cropRegion = [255, 40, 510, 473];
28
29  x = cropRegion(1);
30  y = cropRegion(2);
31  width = cropRegion(3)-cropRegion(1)+1;
32  height = cropRegion(4)-cropRegion(2)+1;
33  %%
34
35  % Tracking Path Storage
36  orangePath = [];
37  yellowPath = [];
38  greenPath = [];
39  bluePath = [];
40
41  % Set up Real-Time Plot
42  figure(1);
43  subplot(1,2,1); % Left side for video tracking
44  videoPlot = imshow(snapshot(cam));
45  hold on;
46  orangeMarker = plot(0, 0, 'o', 'MarkerSize', 10, 'LineWidth', 2, 'Color', '#FFA500');
47  yellowMarker = plot(0, 0, 'yo', 'MarkerSize', 10, 'LineWidth', 2);
48  greenMarker = plot(0, 0, 'go', 'MarkerSize', 10, 'LineWidth', 2);
49  blueMarker = plot(0, 0, 'bo', 'MarkerSize', 10, 'LineWidth', 2);
50  orangeLine = plot(0, 0, '-', 'LineWidth', 2, 'Color', '#FFA500');
51  yellowLine = plot(0, 0, 'y-', 'LineWidth', 2);
52  greenLine = plot(0, 0, 'g-', 'LineWidth', 2);
53  blueLine = plot(0, 0, 'b-', 'LineWidth', 2);
54
55  subplot(1,2,2); % Right side for XY plot
56  orangeXYPlot = plot(NaN, NaN, '-', 'LineWidth', 2, 'Color', '#FFA500');
57  hold on;
58  yellowXYPlot = plot(NaN, NaN, 'y-', 'LineWidth', 2);
59  greenXYPlot = plot(NaN, NaN, 'g-', 'LineWidth', 2);
60  blueXYPlot = plot(NaN, NaN, 'b-', 'LineWidth', 2);
61  %%
62  xlim([0 width]); % Update x-axis limit based on cropping
63  ylim([0 height]); % Update y-axis limit based on cropping
64  %%
65  title('2D XY Position Plot');
66  xlabel('X Coordinate');
67  ylabel('Y Coordinate');
68  grid on;

```

```

69
70 % Real-Time Processing Loop
71 while true
72     %% Capture Frame
73     frame = snapshot(cam);
74
75     % Crop the Frame
76     croppedFrame = frame(y:y+height-1, x:x+width-1, :);
77
78     % Convert to HSV for Color Segmentation
79     hsvFrame = rgb2hsv(croppedFrame);
80
81     % Masks for each color
82     % Define the HSV tolerance ranges
83     hueTolerance = 0.05;
84     saturationTolerance = 0.2;
85     valueTolerance = 0.2;
86
87     % HSV color mask with tolerance
88     orangeMask = (hsvFrame(:,:,1) > (0.0189 - hueTolerance) & hsvFrame(:,:,1) ...
89                  < (0.0189 + hueTolerance)) & ...
90                  (hsvFrame(:,:,2) > (0.94634 - saturationTolerance) & hsvFrame(:,:,2) ...
91                  < (0.94634 + saturationTolerance)) & ...
92                  (hsvFrame(:,:,3) > (0.80392 - valueTolerance) & hsvFrame(:,:,3) ...
93                  < (0.80392 + valueTolerance));
94
95     % Define the HSV tolerance ranges
96     hueTolerance = 0.05;          % Adjust based on the range you need
97     saturationTolerance = 0.2;    % Adjust based on the range you need
98     valueTolerance = 0.2;        % Adjust based on the range you need
99
100    % HSV color mask with tolerance
101    yellowMask = (hsvFrame(:,:,1) > (0.12581 - hueTolerance) & hsvFrame(:,:,1) ...
102                 < (0.12581 + hueTolerance)) & ...
103                 (hsvFrame(:,:,2) > (0.84239 - saturationTolerance) & hsvFrame(:,:,2)...
104                 < (0.84239 + saturationTolerance)) & ...
105                 (hsvFrame(:,:,3) > (0.72157 - valueTolerance) & hsvFrame(:,:,3)...
106                 < (0.72157 + valueTolerance));
107
108
109    greenMask = (hsvFrame(:,:,1) > 0.25 & hsvFrame(:,:,1) < 0.4) & ...
110                (hsvFrame(:,:,2) > 0.4 & hsvFrame(:,:,2) < 1) & ...
111                (hsvFrame(:,:,3) > 0.4);
112
113    blueMask = (hsvFrame(:,:,1) > 0.55 & hsvFrame(:,:,1) < 0.7) & ...
114                (hsvFrame(:,:,2) > 0.4 & hsvFrame(:,:,2) < 1) & ...
115                (hsvFrame(:,:,3) > 0.4);

```

```

116
117 % Get Centroids of the Markers
118 orangeStats = regionprops(orangeMask, 'Centroid');
119 yellowStats = regionprops(yellowMask, 'Centroid');
120 greenStats = regionprops(greenMask, 'Centroid');
121 blueStats = regionprops(blueMask, 'Centroid');
122
123 % Process and Send orange Marker
124 if ~isempty(orangeStats)
125     orangeCentroid = orangeStats(1).Centroid;
126     orangePath = [orangePath; orangeCentroid];
127     orangeMsg.X = orangeCentroid(1);
128     orangeMsg.Y = orangeCentroid(2);
129     send(orangePublisher, orangeMsg);
130     set(orangeMarker, 'XData', orangeCentroid(1), 'YData', orangeCentroid(2));
131     set(orangeLine, 'XData', orangePath(:,1), 'YData', orangePath(:,2));
132     set(orangeXYPlot, 'XData', orangePath(:,1), 'YData', orangePath(:,2));
133 end
134
135 % Process and Send Yellow Marker
136 if ~isempty(yellowStats)
137     yellowCentroid = yellowStats(1).Centroid;
138     yellowPath = [yellowPath; yellowCentroid];
139     yellowMsg.X = yellowCentroid(1);
140     yellowMsg.Y = yellowCentroid(2);
141     send(yellowPublisher, yellowMsg);
142     set(yellowMarker, 'XData', yellowCentroid(1), 'YData', yellowCentroid(2));
143     set(yellowLine, 'XData', yellowPath(:,1), 'YData', yellowPath(:,2));
144     set(yellowXYPlot, 'XData', yellowPath(:,1), 'YData', yellowPath(:,2));
145 end
146
147 % Process and Send Green Marker
148 if ~isempty(greenStats)
149     greenCentroid = greenStats(1).Centroid;
150     greenPath = [greenPath; greenCentroid];
151     greenMsg.X = greenCentroid(1);
152     greenMsg.Y = greenCentroid(2);
153     send(greenPublisher, greenMsg);
154     set(greenMarker, 'XData', greenCentroid(1), 'YData', greenCentroid(2));
155     set(greenLine, 'XData', greenPath(:,1), 'YData', greenPath(:,2));
156     set(greenXYPlot, 'XData', greenPath(:,1), 'YData', greenPath(:,2));
157 end
158
159 % Process and Send Blue Marker
160 if ~isempty(blueStats)
161     blueCentroid = blueStats(1).Centroid;
162     bluePath = [bluePath; blueCentroid];

```

```

163         blueMsg.X = blueCentroid(1);
164         blueMsg.Y = blueCentroid(2);
165         send(bluePublisher, blueMsg);
166         set(blueMarker, 'XData', blueCentroid(1), 'YData', blueCentroid(2));
167         set(blueLine, 'XData', bluePath(:,1), 'YData', bluePath(:,2));
168         set(blueXYPlot, 'XData', bluePath(:,1), 'YData', bluePath(:,2));
169     end
170
171     % Update Video Plot
172     set(videoPlot, 'CData', croppedFrame);
173     drawnow;
174 end
175
176 % Cleanup
177 clear cam;
178 % rosshutdown;
179

```

B.3 Simulink SIMSCAPE model for kinematic analysis and control

The following screenshot shows the PID controller implementation of the Yantra 1 design. SIMSCAPE Multibody package used to convert the CAD Model to a kinematic model. The joints were carefully defined at the SOLIDWORKS design stage using the following methods mentioned at MathWorks Mates and Joints tutorial mathworks.com/help/smlink/ref/mates-and-joints. The CAD model transferred to SIMULINK using the instructions available at the following link. mathworks.com/help/smlink/ref/linking-and-unlinking-simmechanics-link-software-with-solidworks

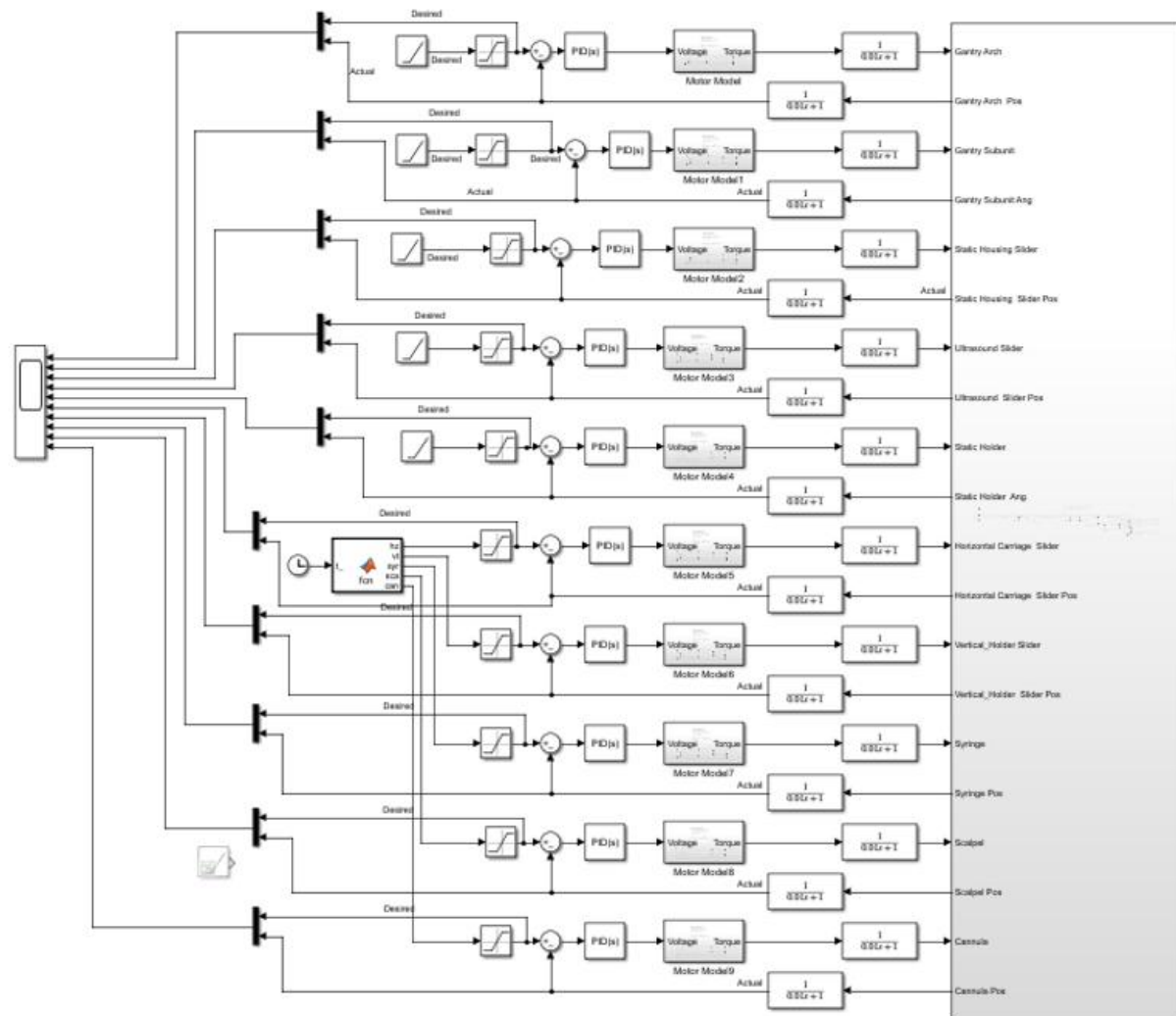


Figure B.3. SIMSCAPE PID controller implementation.

B.4 TSAM Image Processing based Height Measuring

The following code is used to track the height of TSAMs by tracking their colored marker point.

```
1  %% TSAM real-time height measuring using MATLAB
2  %% by Thilina Weerakkody - Updated: 04/21/2020
3  clear all
4  close all
5
6  cam = webcam(2);
7  preview(cam)
8
9  % take image for calibrate the 10mm line and pixels..... 75 pix = 10 mm
10 d = imixel(snapshot(cam));
11 fname1 = sprintf('TSAM_%s.csv',datestr(now,'yyyy-mm-dd-HH-MM-SS'));
12 fileH = fopen(fname1,'w'); %open file
13
14 r = mean(d(:,1));
15 g = mean(d(:,2));
16 b = mean(d(:,3));
17
18 red = img(:,:,1); green = img(:,:,2); blue = img(:,:,3);
19 out = red<r & green<g & blue<b;
20 [row, col] = find(out>0.5);
21 disp = (330-min(row))/7.3;
22
23 %% Run ROS for communication
24 try
25     rostopic list
26 catch
27     rosinit%('192.168.8.107')
28 end
29 publisher = rospublisher('height', 'std_msgs/Float64');
30 msg = rosmessage(publisher);
31 rate = rosrate(100); % disp('ROS topic is now live!')
32 try
33     while 1
34         img = snapshot(cam);
35         red = img(:,:,1); green = img(:,:,2); blue = img(:,:,3);
36         out = red<r & green<g & blue<b;
37         [row, col] = find(out>0.5);
38         disp = (330-min(row))/7.3 -1.6438 %/7.5
39         msg.Data = disp;
40         send(publisher, msg);
41         waitfor(rate);
42         if(disp~=NaN)
```

```
43         fprintf(fileH,'%f\n',disp);
44     end
45 end
46 catch
47 end
48 clear cam
```

Packages:

- Image Processing Toolbox - MATLAB
- MATLAB Support Package for USB Webcams
- DroidCam Webcam and Dev47Apps

B.5 IMU Realtime Angle Tracking - MATLAB

This code is based on the MATLAB Sensor Fusion and Tracking Toolbox. "HelperOrientationViewer.m" is used to visualize the IMU orientation in real time.

```
1  % MPU9250 IMU Real-time DAQ to Simulink via ROS
2  % by Thilina Weeakkody
3  clear all
4  close all
5  clc
6
7  Fs = 100;
8  samplesPerRead = 10;
9  runTime = 20;
10 isVerbose = false;
11 useHW = true;
12
13 a = arduino('COM18', 'Mega2560', 'Libraries', 'I2C');
14 imu = mpu9250(a, 'SampleRate', Fs, 'OutputFormat', 'matrix', ...
15     'SamplesPerRead', samplesPerRead);
16
17 % Accelerometer axes parameters.
18 accelXAxisIndex = 1; accelXAxisSign = 1; accelYAxisIndex = 2;
19 accelYAxisSign = 1; accelZAxisIndex = 3; accelZAxisSign = 1;
20
21 % Gyroscope axes parameters.
22 gyroXAxisIndex = 1; gyroXAxisSign = 1; gyroYAxisIndex = 2;
23 gyroYAxisSign = 1; gyroZAxisIndex = 3; gyroZAxisSign = 1;
24
25 % Magnetometer axes parameters.
26 magXAxisIndex = 2; magXAxisSign = 1; magYAxisIndex = 1;
27 magYAxisSign = 1; magZAxisIndex = 3; magZAxisSign = -1;
28
29 % Helper functions used to align sensor data axes.
30
31 alignAccelAxes = @(in) [accelXAxisSign, accelYAxisSign, accelZAxisSign] ...
32     .* in(:, [accelXAxisIndex, accelYAxisIndex, accelZAxisIndex]);
33
34 alignGyroAxes = @(in) [gyroXAxisSign, gyroYAxisSign, gyroZAxisSign] ...
35     .* in(:, [gyroXAxisIndex, gyroYAxisIndex, gyroZAxisIndex]);
36
37 alignMagAxes = @(in) [magXAxisSign, magYAxisSign, magZAxisSign] ...
38     .* in(:, [magXAxisIndex, magYAxisIndex, magZAxisIndex]);
39
40 %% Perform Additional Sensor Calibration
```

```

41  %% Specify Complementary filter Parameters
42  compFilt = complementaryFilter('SampleRate', Fs);
43
44  tuner = HelperOrientationFilterTuner(compFilt);
45
46  %% Run ROS for communication
47  try
48      rostopic list
49  catch
50      rosinit%('192.168.8.107')
51  end
52  publisher1 = rospublisher('pitch', 'std_msgs/Float64');
53  publisher2 = rospublisher('roll', 'std_msgs/Float64');
54  msg1 = rosmesssage(publisher1);
55  msg2 = rosmesssage(publisher2);
56  disp('ROS topic is now live!')
57  rate = rosrates(100);
58
59  while true
60      [accel, gyro, mag, t, overrun] = imu();
61      accel = alignAccelAxes(accel);
62      gyro = alignGyroAxes(gyro);
63      mag = alignMagAxes(mag);
64
65      if (isVerbose && overrun > 0)
66          fprintf('%d samples overrun ...\n', overrun);
67      end
68
69      q = compFilt(accel, gyro, mag);
70      update(tuner, q);
71      [yaw1, pitch1, roll1] = quat2angle(compact(q));
72      yaw = rad2deg(yaw1);
73      pitch = rad2deg(pitch1);
74      roll = rad2deg(roll1);
75      fprintf('%f    %f    %f\n',yaw,pitch,roll)
76
77  end

```

B.6 IMU Realtime Angle Tracking - ROS/Linux

This code is based on Arduino and ROS 1, which publish Kalman Filtered angles from MPU6050.

```
1  #include <ros.h>
2  #include <geometry_msgs/Twist.h>
3  #include <Wire.h>
4  #include <Kalman.h> // Source: https://github.com/TKJElectronics/KalmanFilter
5
6  #define RESTRICT_PITCH
7  Kalman kalmanX; // Create the Kalman instances
8  Kalman kalmanY;
9
10 ros::NodeHandle nh;
11 geometry_msgs::Twist imu2_msg;
12 ros::Publisher chatimu2("imu2", &imu2_msg);
13 float yaw1 = 0, pitch1 = 0;
14
15 /* IMU Data */
16 double accX, accY, accZ;
17 double gyroX, gyroY, gyroZ;
18 int16_t tempRaw;
19
20 double gyroXangle, gyroYangle; // Angle calculate using the gyro only
21 double compAngleX, compAngleY; // Calculated angle using a complementary filter
22 double kalAngleX, kalAngleY; // Calculated angle using a Kalman filter
23 uint32_t timer;
24 uint8_t i2cData[14]; // Buffer for I2C data
25
26
27 void setup() {
28
29     // Serial.begin(115200);
30     Wire.begin();
31     #if ARDUINO >= 157
32     Wire.setClock(400000UL); // Set I2C frequency to 400kHz
33     #else
34     TWBR = ((F_CPU / 400000UL) - 16) / 2; // Set I2C frequency to 400kHz
35     #endif
36     nh.initNode();
37     nh.advertise(chatimu2);
38
39     i2cData[0] = 7; // Set the sample rate to 1000Hz - 8kHz/(7+1) = 1000Hz
40     i2cData[1] = 0x00; // Disable FSYNC and set 260 Hz Acc filtering
41     i2cData[2] = 0x00; // Set Gyro Full Scale Range to ±250deg/s
42     i2cData[3] = 0x00; // Set Accelerometer Full Scale Range to ±2g
```

```

43 while (i2cWrite(0x19, i2cData, 4, false)); // Write to all four registers at once
44 while (i2cWrite(0x6B, 0x01, true)); // X gyroscope reference & disable sleep mode
45
46 while (i2cRead(0x75, i2cData, 1));
47 if (i2cData[0] != 0x68) { // Read "WHO_AM_I" register
48     // Serial.print(F("Error reading sensor"));
49     while (1);
50 }
51
52 delay(100); // Wait for sensor to stabilize
53
54 /* Set kalman and gyro starting angle */
55 while (i2cRead(0x3B, i2cData, 6));
56 accX = (int16_t)((i2cData[0] << 8) | i2cData[1]);
57 accY = (int16_t)((i2cData[2] << 8) | i2cData[3]);
58 accZ = (int16_t)((i2cData[4] << 8) | i2cData[5]);
59
60 #ifdef RESTRICT_PITCH // Eq. 25 and 26
61     double roll = atan2(accY, accZ) * RAD_TO_DEG;
62     double pitch = atan(-accX / sqrt(accY * accY + accZ * accZ)) * RAD_TO_DEG;
63 #else // Eq. 28 and 29
64     double roll = atan(accY / sqrt(accX * accX + accZ * accZ)) * RAD_TO_DEG;
65     double pitch = atan2(-accX, accZ) * RAD_TO_DEG;
66 #endif
67
68 kalmanX.setAngle(roll); // Set starting angle
69 kalmanY.setAngle(pitch);
70 gyroXangle = roll;
71 gyroYangle = pitch;
72 compAngleX = roll;
73 compAngleY = pitch;
74 timer = micros();
75 }
76
77 void loop() {
78     /* Update all the values */
79     while (i2cRead(0x3B, i2cData, 14));
80     accX = (int16_t)((i2cData[0] << 8) | i2cData[1]);
81     accY = (int16_t)((i2cData[2] << 8) | i2cData[3]);
82     accZ = (int16_t)((i2cData[4] << 8) | i2cData[5]);
83     tempRaw = (int16_t)((i2cData[6] << 8) | i2cData[7]);
84     gyroX = (int16_t)((i2cData[8] << 8) | i2cData[9]);
85     gyroY = (int16_t)((i2cData[10] << 8) | i2cData[11]);
86     gyroZ = (int16_t)((i2cData[12] << 8) | i2cData[13]);
87
88     double dt = (double)(micros() - timer) / 1000000; // Calculate delta time
89     timer = micros();

```

```

90
91 #ifdef RESTRICT_PITCH // Eq. 25 and 26
92     double roll = atan2(accY, accZ) * RAD_TO_DEG;
93     double pitch = atan(-accX / sqrt(accY * accY + accZ * accZ)) * RAD_TO_DEG;
94 #else // Eq. 28 and 29
95     double roll = atan(accY / sqrt(accX * accX + accZ * accZ)) * RAD_TO_DEG;
96     double pitch = atan2(-accX, accZ) * RAD_TO_DEG;
97 #endif
98
99     double gyroXrate = gyroX / 131.0; // Convert to deg/s
100    double gyroYrate = gyroY / 131.0; // Convert to deg/s
101
102 #ifdef RESTRICT_PITCH
103     // fixes the transition problem when the accelerometer angle jumps between -180 and 180
104     if ((roll < -90 && kalAngleX > 90) || (roll > 90 && kalAngleX < -90)) {
105         kalmanX.setAngle(roll);
106         compAngleX = roll;
107         kalAngleX = roll;
108         gyroXangle = roll;
109     } else
110         kalAngleX = kalmanX.getAngle(roll, gyroXrate, dt); //Kalman filter
111
112     if (abs(kalAngleX) > 90)
113         gyroYrate = -gyroYrate; // Invert rate, so it fits the restriced accelerometer reading
114     kalAngleY = kalmanY.getAngle(pitch, gyroYrate, dt);
115 #else
116     // fixes the transition problem when the accelerometer angle jumps between -180 and 180
117     if ((pitch < -90 && kalAngleY > 90) || (pitch > 90 && kalAngleY < -90)) {
118         kalmanY.setAngle(pitch);
119         compAngleY = pitch;
120         kalAngleY = pitch;
121         gyroYangle = pitch;
122     } else
123         kalAngleY = kalmanY.getAngle(pitch, gyroYrate, dt); // Kalman filter
124
125     if (abs(kalAngleY) > 90)
126         gyroXrate = -gyroXrate; // Invert rate, so it fits the restriced accelerometer reading
127     kalAngleX = kalmanX.getAngle(roll, gyroXrate, dt); // Kalman filter
128 #endif
129
130    gyroXangle += gyroXrate * dt; // Calculate gyro angle without any filter
131    gyroYangle += gyroYrate * dt;
132
133    compAngleX = 0.93 * (compAngleX + gyroXrate * dt) + 0.07 * roll; // Complimentary filter
134    compAngleY = 0.93 * (compAngleY + gyroYrate * dt) + 0.07 * pitch;
135
136    // Reset the gyro angle when it has drifted too much

```

```

137  if (gyroXangle < -180 || gyroXangle > 180)
138      gyroXangle = kalAngleX;
139  if (gyroYangle < -180 || gyroYangle > 180)
140      gyroYangle = kalAngleY;
141  Serial.print(kalAngleX); Serial.print("\t");Serial.println(kalAngleY);
142
143  yaw1    = roll;
144  pitch1  = pitch;
145  imu2_msg.linear.x = yaw1;
146  imu2_msg.linear.y = pitch1;
147  chatimu2.publish( &imu2_msg );
148  nh.spinOnce();
149  delay(100);
150 }

```


B.7 Arduino ROS1/roserial publisher for System Actuation

This code used for Publish PWM signals to Arduino from the ROS server.

```
1  #include <ros.h>
2  #include <ArduinoHardware.h>
3  #include <geometry_msgs/Twist.h>
4  #define direction1 9
5  #define direction2 10
6  ros::NodeHandle nh;
7  int pwmval1 = 0, pwmval2 = 0;
8
9  void messageCb( const geometry_msgs::Twist& msg){
10     pwmval1 = msg.linear.x;
11     pwmval2 = msg.linear.y;
12 }
13 ros::Subscriber<geometry_msgs::Twist> sub("pwm", &messageCb );
14 geometry_msgs::Twist pub_msg;
15 ros::Publisher pwm("pwm", &pub_msg);
16
17 void setup(){
18     nh.initNode();
19     nh.advertise(pwm);
20     nh.subscribe(sub);
21     pinMode(direction1, OUTPUT);
22     pinMode(direction2, OUTPUT);
23 }
24
25 void loop(){
26     analogWrite(direction1, pwmval1);
27     pub_msg.linear.x = pwmval1;
28     pwm.publish( &pub_msg);
29     analogWrite(direction2, pwmval2);
30     pub_msg.linear.y = pwmval2;
31     pwm.publish( &pub_msg);
32     nh.spinOnce();
33     delay(10);
34 }
```

B.8 TCAM manufacturing machine - Arduino Code

This code is based on Arduino to actuate 2 stepper motors and a single DC motor for TCAM manufacturing (Published the work in [295]).

```
1  // Motor Speed arrays for [DC_motor, stepper_motor, stepper_returning_speed]
2  const int wrapping_speed[2]      = {120, 1000};
3  const int coiling_speed[2]       = {60, 0};
4  const int carriageReturn_speed[2] = {0, -500};
5  const int mandreling_speed = 1000; // Mandrel Motor speed
6  // DC encoder Motor - 19:1 Metal Gearmotor 37Dx68L mm 12V with 64 CPR Encoder
7  // DC Controller    - TB67H420FTG Dual/Single Motor Driver Carrier
8  // Stepper Motors   - Nema 17
9  // Stepper Controller- A4988
10 // DC Motor Pins
11 #define ENCA 2
12 #define ENCB 3
13 #define IN1 4
14 #define IN2 5
15 #define PWM 6
16
17 // Stepper Motor Pins
18 #define dirPin1 8
19 #define stepPin1 9
20 #define dirPin2 10
21 #define stepPin2 11
22 #define limitSwitch 13 // Limit Switch pin
23
24 // External Libraries
25 #include <AccelStepper.h>
26 #define motorInterfaceType 1
27
28 AccelStepper wrappingStepper(motorInterfaceType, stepPin1, dirPin1);
29 AccelStepper mandrelStepper(motorInterfaceType, stepPin2, dirPin2);
30
31 // Actuator characteristics
32 const int stepsPerRevolution = 200; // Nema 17 200 steps/revolution
33 const int DCPins = 1200; // 1200 encoder_readings/revolution
34
35 //DC Variables
36 volatile int posi = 0;
37 long prevT = 0;
38 float eprev = 0, eintegral = 0, target_f = 0;
39 long target = 0;
40 int shutdowncounter=0, pos = 0, pwr, dir, task =0;
```

```

41  //----- PID Class -----//
42  class SimplePID{
43      private:
44          float kp, kd, ki, umax;
45          float eprev, eintegral;
46
47      public:
48          // Default initialization list
49          SimplePID() : kp(1), kd(0), ki(0), umax(255), eprev(0.0), eintegral(0.0) {}
50
51          // Set the parameters
52          void setParams(float kpIn, float kdIn, float kiIn, float umaxIn){
53              kp = kpIn; kd = kdIn; ki = kiIn; umax = umaxIn;
54          }
55
56          // Evaluate the signal
57          void evalu(int value, int target, float deltaT,int &pwr, int &dir){
58
59              // error
60              int e = target - value;
61
62              float dedt = (e-eprev)/(deltaT);
63              eintegral = eintegral + e*deltaT;
64              float u = kp*e + kd*dedt + ki*egral;
65
66              // motor power
67              pwr = (int) fabs(u);
68              if( pwr > umax ) pwr = umax;
69
70              dir = 1; // motor direction
71              if(u<0) dir = -1;
72
73              eprev = e; // store previous error
74          }
75
76          void setTarget(float t, float desiredTurns,float rpm, float deltat){
77              float positionChange = 0.0;
78              float pulsesPerTurn = 16*18.75;
79              float turnsPerS = rpm/60.0; // rev/s
80              float desiredPulses = desiredTurns*pulsesPerTurn;
81              t = fmod(t,1000); // time is in seconds
82              if(t < 1000 && target_f<desiredPulses && desiredPulses>0){
83                  positionChange = turnsPerS*deltat*pulsesPerTurn;
84              }
85              else if(t < 1000 && target_f>desiredPulses && desiredPulses<0){
86                  positionChange = -turnsPerS*deltat*pulsesPerTurn;
87              }

```

```

88     else{
89         positionChange = 0;
90     }
91     target_f = target_f + positionChange;
92     target = (long) target_f;
93 }
94 void setMotor(int dir, int pwmVal, int pwm, int in1, int in2){
95     analogWrite(pwm,pwmVal);
96     if(dir == 1){
97         digitalWrite(in1,LOW);
98         digitalWrite(in2,HIGH);
99     }
100    else if(dir == -1){
101        digitalWrite(in1,HIGH);
102        digitalWrite(in2,LOW);
103    }
104    else{
105        digitalWrite(in1,LOW);
106        digitalWrite(in2,LOW);
107    }
108 }
109 void readEncoder(){
110     int b = digitalRead(ENCB);
111     if(b > 0) posi++;
112     else posi--;
113 }
114 // PID class
115 SimplePID pid;
116 ////////// wrapping operation function
117 void twisting(long currT){
118     int turns_ = readSerial(); // read user requirement from Serial
119     printData(turns_, wrapping_speed[0], wrapping_speed[1]); // display parameters
120     int i=0;
121
122     driveDC(currT, turns_, coiling_speed[0]);
123     wrappingStepper.setCurrentPosition(0); // initialize stepper position
124     while(wrappingStepper.currentPosition() != turns_*stepsPerRevolution){ // turn until the desi
125     wrappingStepper.setSpeed(wrapping_speed[1]);
126     wrappingStepper.runSpeed();
127     i = wrappingStepper.currentPosition();
128     if ((i%200)==0){
129         Serial.println(i/200);
130         i=i+50;
131     }
132     }
133     delay(1000);
134 }

```

```

135  ////////// Twisting operation function
136  void coiling(long currT){
137      int turns_ = readSerial(); // read user requirement from Serial
138      printData(turns_, coiling_speed[0], coiling_speed[1]); // display parameters
139
140      driveDC(currT, turns_, coiling_speed[0]);
141  }
142  ////////// Mandreling operation function
143  void mandreling(){
144      int turns_ = readSerial(); // read user requirement from Serial
145      printData(turns_, mandreling_speed, 0); // display parameters
146      int i=0;
147      mandrelStepper.setCurrentPosition(0); // initialize stepper position
148      while(mandrelStepper.currentPosition() != turns_*stepsPerRevolution){ // turn until the desired
149          mandrelStepper.setSpeed(mandreling_speed);
150          mandrelStepper.runSpeed();
151          i = mandrelStepper.currentPosition();
152          if ((i%200)==0){
153              Serial.println(i/200);
154              i=i+50;
155          }
156      }
157      delay(1000);
158  }
159  ////////// Return the stepper motor powered carriage back to origin
160  void returnCarriage(){
161      while (digitalRead(limitSwitch) != 0){
162          wrappingStepper.setSpeed(carriageReturn_speed[1]);
163          wrappingStepper.runSpeed();
164      }
165  }
166  ////////// This function asks for number of turns
167  int readSerial(){
168      Serial.println("Number of turns? ");
169      while (Serial.available()) {
170          Serial.read();
171      }
172      while(Serial.available() == 0){}
173      int desiredTurns = Serial.parseInt(); //read in number of revolutions
174      return desiredTurns;
175  }
176
177  ////////// This function display the parameters
178  void printData(int turns, int speed1, int speed2){
179      Serial.print("[Desired Turns, DC Motor Speed, Stepper Motor Speed] = [");
180      Serial.print(turns); Serial.print(", "); Serial.print(speed1); Serial.print(", "); Serial.print(speed2);
181  }

```

```

182
183 ////////// DC encoder motor functions
184 void driveDC(long currT, int desiredRev, int speed_){
185     float deltaT = ((float) (currT - prevT))/( 1.0e6 );
186     long prevTarget = target;
187     prevT = currT;
188     setTarget(currT/1.0e6, desiredRev, speed_, deltaT);
189
190     noInterrupts();
191     pos = posi;
192     interrupts();
193     pid.eval(pos, target, deltaT, pwr, dir);
194     setMotor(dir,pwr,PWM,IN1,IN2);
195
196     if((prevTarget-target)==0) shutdowncounter++;
197     if(shutdowncounter>50){
198         setMotor(0,0,PWM,IN1,IN2);
199         shutdowncounter = 51;
200         exit(0);
201     }
202 }
203 void setup() {
204     wrappingStepper.setMaxSpeed(1000);
205     mandrelStepper.setMaxSpeed(1000);
206     pinMode(limitSwitch,INPUT);
207     pinMode(ENCA,INPUT);
208     pinMode(ENCB,INPUT);
209     pinMode(PWM,OUTPUT);
210     pinMode(IN1,OUTPUT);
211     pinMode(IN2,OUTPUT);
212     attachInterrupt(digitalPinToInterrupt(ENCA),readEncoder,RISING);
213     pid.setParams(1,0.1,0,255);
214
215     Serial.begin(9600);
216 }
217 void loop() {
218     task = 1;
219     if (task==1)twisting(micros());
220     else if (task==2)coiling(micros());
221     else if (task==3)mandreling();
222     else if (task==4)retunCarriage();
223     else Serial.println("Error 404 not found !!!");
224     delay(1000);
225 }

```

B.9 TSAM Scalable manufacturing system - Arduino Code

This code is based on Arduino to actuate 3 stepper motors and a single DC motor for TCAM manufacturing (Published the work in [56]). The device image is shown in Figure B.4.

```
1  const int stepPin = 12;
2  const int dirPin = 13;
3  void setup() {
4      // Sets the two pins as Outputs
5      pinMode(stepPin,OUTPUT);
6      pinMode(dirPin,OUTPUT);
7  }
8  void loop() {
9      digitalWrite(dirPin,HIGH); // Enables the motor to move in a particular direction
10     // Makes 200 pulses for making one full cycle rotation
11     for(int x = 0; x < 200; x++) {
12         digitalWrite(stepPin,HIGH);
13         delayMicroseconds(500);
14         digitalWrite(stepPin,LOW);
15         delayMicroseconds(500);
16     }
17     delay(1000); // One second delay
18     digitalWrite(dirPin,LOW); //Changes the rotations direction
19     for(int x = 0; x < 400; x++) { // Makes 400 pulses for making two full cycle rotation
20         digitalWrite(stepPin,HIGH);
21         delayMicroseconds(500);
22         digitalWrite(stepPin,LOW);
23         delayMicroseconds(500);
24     }
25     delay(1000);
26 }
```

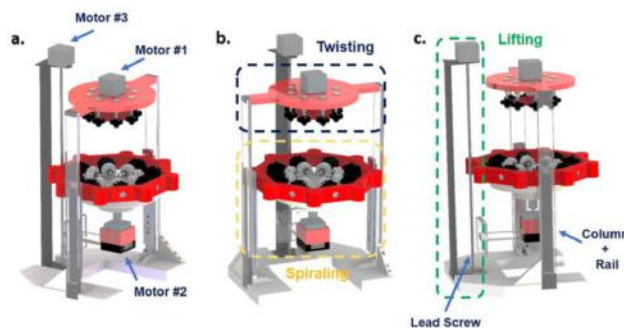


Figure B.4. TSAM Scalable manufacturing system

BIBLIOGRAPHY

- [1] A. Chen, R. Yin, L. Cao, C. Yuan, H. Ding, and W. Zhang, “Soft robotics: Definition and research issues,” in *2017 24th international conference on mechatronics and machine vision in practice (M2VIP)*, pp. 366–370, IEEE, 2017.
- [2] C. Majidi, “Soft robotics: A perspective—current trends and prospects for the future,” *Soft Robot*, vol. 1, no. 1, pp. 5–11, 2013.
- [3] D. Rus and M. T. Tolley, “Design, fabrication and control of soft robots,” *Nature*, vol. 521, no. 7553, pp. 467–475, 2015.
- [4] C. Laschi, B. Mazzolai, and M. Cianchetti, “Soft robotics: Technologies and systems pushing the boundaries of robot abilities,” *Sci Robot*, vol. 1, no. 1, p. eaah3690, 2016.
- [5] J. Zhang, J. Sheng, C. T. O’Neill, C. J. Walsh, R. J. Wood, J.-H. Ryu, J. P. Desai, and M. C. Yip, “Robotic artificial muscles: Current progress and future perspectives,” *IEEE transactions on robotics*, vol. 35, no. 3, pp. 761–781, 2019.
- [6] J. Wang, D. Gao, and P. S. Lee, “Recent progress in artificial muscles for interactive soft robotics,” *Advanced Materials*, vol. 33, no. 19, p. 2003088, 2021.
- [7] C. Greco, P. Kotak, L. Pagnotta, and C. Lamuta, “The evolution of mechanical actuation: from conventional actuators to artificial muscles,” *International materials reviews*, vol. 67, no. 6, pp. 575–619, 2022.
- [8] A. Dietrich, “Electric rod actuators vs. hydraulic cylinders: A comparison of the pros and cons of each technology,” *Mach Des*, pp. 1–13, 2016.
- [9] C. Mavroidis, C. Pfeiffer, and M. Mosley, “5.1 conventional actuators, shape memory alloys, and electrorheological fluids,” *Automation, miniature robotics, and sensors for nondestructive testing and evaluation*, vol. 4, no. 3, p. 189, 2000.

- [10] A. Hughes and B. Drury, *Electric motors and drives: fundamentals, types and applications*. Newnes, 2019.
- [11] D. Montone, *Applying Motors in Linear Motion Application*. Heydon Kerk Pittman Ametek Inc., 2013.
- [12] A. Goldhammer, “Electric motors and drives,” *Production Engineer*, vol. 44, no. 10, pp. 551–586, 1965.
- [13] S. Rosenfeld, “Advantages and drawbacks of pneumatic, hydraulic, and electric linear actuators,” *Adv Drawbacks Pneumatic, Hydraulic, Electric Linear Actuators*, pp. 3–6, 2017.
- [14] E. C. Lee, “Brushless dc: a modern approach to variable speed drives,” in *Conference Record of the 1990 IEEE Industry Applications Society Annual Meeting*, pp. 1484–1488, IEEE, 1990.
- [15] A. Mohtar, Z. Nedic, and J. Machotka, “A compact and affordable bldc motor controller for a microelectronics remote laboratory,” in *ESA*, pp. 75–80, 2008.
- [16] P. B. Zhao and Y. Y. Shi, “Design of dc motor control system based on clonal selection algorithm,” *Advanced Materials Research*, vol. 201, pp. 2343–2348, 2011.
- [17] K. Niruba and S. Boopathi, “Advanced power window motor using permanent magnet dc motor,” in *2014 POWER AND ENERGY SYSTEMS: TOWARDS SUSTAINABLE ENERGY*, pp. 1–9, IEEE, 2014.
- [18] M. Ristanović, Ž. Čojbašić, and D. Lazić, “Intelligent control of dc motor driven electromechanical fin actuator,” *Control Engineering Practice*, vol. 20, no. 6, pp. 610–617, 2012.
- [19] S. Bringsjord, *What robots can and can't be*, vol. 12. Springer Science & Business Media, 2013.

- [20] S. Salleh, M. Rahmat, S. Othman, and K. Danapalasingam, "Review on modeling and controller design of hydraulic actuator systems," *International Journal on smart sensing and intelligent systems*, vol. 8, no. 1, pp. 338–367, 2015.
- [21] J. Johnson, *Introduction to fluid power*. Cengage Learning, 2002.
- [22] H. I. Ali, S. Noor, S. Bashi, and M. H. Marhaban, "A review of pneumatic actuators (modeling and control)," *Australian Journal of Basic and Applied Sciences*, vol. 3, no. 2, pp. 440–454, 2009.
- [23] G. K. Klute, J. M. Czerniecki, and B. Hannaford, "Artificial muscles: Actuators for biorobotic systems," *The International Journal of Robotics Research*, vol. 21, no. 4, pp. 295–309, 2002.
- [24] A. Belanger and A. McComas, "Contractile properties of human skeletal muscle in childhood and adolescence," *European journal of applied physiology and occupational physiology*, vol. 58, no. 6, pp. 563–567, 1989.
- [25] S. M. Mirvakili, A. Pazukha, W. Sikkema, C. W. Sinclair, G. M. Spinks, R. H. Baughman, and J. D. Madden, "Niobium nanowire yarns and their application as artificial muscles," *Advanced Functional Materials*, vol. 23, no. 35, pp. 4311–4316, 2013.
- [26] S. M. Mirvakili and I. W. Hunter, "Artificial muscles: Mechanisms, applications, and challenges," *Advanced Materials*, vol. 30, no. 6, p. 1704407, 2018.
- [27] C. Lamuta, S. Messelot, and S. Tawfick, "Theory of the tensile actuation of fiber reinforced coiled muscles," *Smart Materials and Structures*, vol. 27, no. 5, p. 055018, 2018.
- [28] A. Mohebbi, "Human-robot interaction in rehabilitation and assistance: a review," 08 2020.
- [29] J.-H. Hsiao, J.-Y. Chang, and C. Cheng, "Soft medical robotics: clinical and biomedical applications, challenges, and future directions," *Advanced Robotics*, vol. 33, pp. 1099–1111, 10 2019.

- [30] C.-Y. Chu and R. M. Patterson, “Soft robotic devices for hand rehabilitation and assistance: a narrative review,” *Journal of neuroengineering and rehabilitation*, vol. 15, no. 1, pp. 1–14, 2018.
- [31] H. Rodrigue and J.-W. Kim, “Soft actuators in surgical robotics: a state-of-the-art review,” *Intelligent Service Robotics*, 12 2023.
- [32] S. Coyle, C. Majidi, P. R. LeDuc, and K. J. Hsia, “Bio-inspired soft robotics: Material selection, actuation, and design,” *Extreme Mechanics Letters*, vol. 22, pp. 51–59, 07 2018.
- [33] D. Trivedi, C. D. Rahn, W. M. Kier, and I. D. Walker, “Soft robotics: Biological inspiration, state of the art, and future research,” *Applied Bionics and Biomechanics*, vol. 5, pp. 99–117, 01 2008.
- [34] Z. T. H. Tse, Y. Chen, S. Hovet, H. Ren, K. Cleary, S. Xu, B. J. Wood, and R. Monfaredi, “Soft robotics in medical applications,” *Journal of medical robotics research*, vol. 03, pp. 1841006–1841006, 09 2018.
- [35] C. Laschi and M. Cianchetti, “Soft robotics: New perspectives for robot bodyware and control,” *Frontiers in Bioengineering and Biotechnology*, vol. 2, 01 2014.
- [36] T. G. Thuruthel, Y. Ansari, E. Falotico, and C. Laschi, “Control strategies for soft robotic manipulators: A survey,” *Soft robotics*, vol. 5, pp. 149–163, 04 2018.
- [37] Z. Peng and J. Huang, “Soft rehabilitation and nursing-care robots: A review and future outlook,” *Applied sciences*, vol. 9, pp. 3102–3102, 07 2019.
- [38] G. Bao, H. Fang, L. Chen, Y. Wan, F. Xu, Q. Yang, and L. Zhang, “Soft robotics: Academic insights and perspectives through bibliometric analysis,” *Soft robotics*, vol. 5, pp. 229–241, 06 2018.
- [39] L. Shi, Y. Yu, N. Xiao, and D. Gan, “Biologically inspired and rehabilitation robotics,” *Applied Bionics and Biomechanics*, vol. 2019, pp. 1–2, 03 2019.

- [40] C. M. Thalman and P. Artemiadis, “A review of soft wearable robots that provide active assistance: Trends, common actuation methods, fabrication, and applications,” *Wearable Technologies*, vol. 1, 01 2020.
- [41] M. Irshaidat, M. Soufian, A. Elkurdi, and S. Nefti-Meziani, “Soft and hard robotics for movement rehabilitation, analysis and modelling,” in *2019 12th International Conference on Developments in eSystems Engineering (DeSE)*, pp. 964–969, 2019.
- [42] P. Maciejasz, J. Eschweiler, K. Gerlach-Hahn, A. Jansen-Troy, and S. Leonhardt, “A survey on robotic devices for upper limb rehabilitation,” *Journal of Neuroengineering and Rehabilitation*, vol. 11, 01 2014.
- [43] H. M. Fard-Vatan, S. Nefti-Meziani, S. Davis, Z. Saffari, and H. El-Hussieny, “A review: A comprehensive review of soft and rigid wearable rehabilitation and assistive devices with a focus on the shoulder joint,” *Journal of Intelligent and Robotic Systems*, vol. 102, 04 2021.
- [44] R. Duffy and A. W. Feinberg, “Engineered skeletal muscle tissue for soft robotics: fabrication strategies, current applications, and future challenges,” *WIREs Nanomedicine and Nanobiotechnology*, vol. 6, pp. 178–195, 12 2013.
- [45] L. Hines, K. Petersen, G. Z. Lum, and M. Sitti, “Soft actuators for small-scale robotics,” *Advanced Materials*, vol. 29, 12 2016.
- [46] S. M. Mirvakili and I. W. Hunter, “Artificial muscles: Mechanisms, applications, and challenges,” *Advanced Materials*, vol. 30, 12 2017.
- [47] N. Minaian, Z. J. Olsen, and K. J. Kim, “Ionic polymer-metal composite (ipmc) artificial muscles in underwater environments: review of actuation, sensing, controls, and applications to soft robotics,” *Bioinspired Sensing, Actuation, and Control in Underwater Soft Robotic Systems*, pp. 117–139, 2021.

- [48] R. Wang, C. Zhang, Y. Zhang, W. Tan, W. Chen, and L. Liu, “Soft underwater swimming robots based on artificial muscle,” *Advanced Materials Technologies*, vol. 8, no. 4, p. 2200962, 2023.
- [49] C. Lamuta, S. Messelot, and S. Tawfick, “Theory of the tensile actuation of fiber reinforced coiled muscles,” *Smart Materials and Structures*, vol. 27, no. 5, 2018.
- [50] N. Li, G. M. Spinks, A. E. Aliev, J. Di, C. S. Haines, N. Li, G. M. Spinks, A. E. Aliev, J. Di, and R. H. Baughman, “Correction for: New twist on artificial muscles (Proceedings of the National Academy of Sciences of the United States of America (2016) 113 (11709–11716) DOI: 10.1073/pnas.1605273113),” *Proceedings of the National Academy of Sciences of the United States of America*, vol. 115, no. 11, p. E2663, 2018.
- [51] P. Rothemund, N. Kellaris, S. K. Mitchell, E. Acome, and C. Keplinger, “HASEL Artificial Muscles for a New Generation of Lifelike Robots—Recent Progress and Future Opportunities,” *Advanced Materials*, vol. 33, no. 19, pp. 1–28, 2021.
- [52] Y. Bar-Cohen, “EAP as artificial muscles: progress and challenges,” *Smart Structures and Materials 2004: Electroactive Polymer Actuators and Devices (EAPAD)*, vol. 5385, p. 10, 2004.
- [53] H. Christensen, N. Amato, H. Yanco, M. Mataric, H. Choset, A. Drobnis, K. Goldberg, J. Grizzle, G. Hager, J. Hollerbach, S. Hutchinson, V. Krovi, D. Lee, W. D. Smart, J. Trinkle, and G. Sukhatme, “A roadmap for us robotics – from internet to robotics 2020 edition,” *Foundations and Trends® in Robotics*, vol. 8, no. 4, pp. 307–424, 2021.
- [54] S. H. Tawfick and C. Lamuta, “Elongate fiber artificial muscles and method of fabrication,” July 13 2021. US Patent 11,060,512.
- [55] C. Lamuta, H. He, K. Zhang, M. Rogalski, N. Sottos, and S. Tawfick, “Digital Texture Voxels for Stretchable Morphing Skin Applications,” *Advanced Materials Technologies*, vol. 4, no. 8, pp. 2–7, 2019.

- [56] C. Greco, P. Kotak, J. K. Gallegos, J. Fang, A. Wilkinson, L. Pagnotta, and C. Lamuta, “Scalable manufacturing system for bionspired twisted spiral artificial muscles (tsams),” *Manufacturing Letters*, vol. 26, pp. 6–11, 2020.
- [57] A. Maziz, A. Concas, A. Khaldi, J. Stålhand, N. K. Persson, and E. W. Jager, “Knitting and weaving artificial muscles,” *Science Advances*, vol. 3, no. 1, pp. 1–12, 2017.
- [58] S. M. Mirvakili and I. W. Hunter, “Artificial Muscles: Mechanisms, Applications, and Challenges,” *Advanced Materials*, vol. 30, no. 6, 2018.
- [59] F. Fei, P. Kotak, L. He, X. Li, C. Vanderhoef, C. Lamuta, and X. Song, “Cephalopod-inspired stretchable self-morphing skin via embedded printing and twisted spiral artificial muscles,” *Advanced Functional Materials*, vol. 31, no. 46, p. 2105528, 2021.
- [60] P. Kotak, J. M. Wilken, K. M. Anderson, and C. Lamuta, “Carbon fiber-based twisted and coiled artificial muscles (tcams) for powered ankle-foot orthoses,” *Journal of biomechanical engineering*, vol. 144, no. 1, 2022.
- [61] J. Zhang, J. Sheng, C. T. O’Neill, C. J. Walsh, R. J. Wood, J. H. Ryu, J. P. Desai, and M. C. Yip, “Robotic Artificial Muscles: Current Progress and Future Perspectives,” *IEEE Transactions on Robotics*, vol. 35, no. 3, pp. 761–781, 2019.
- [62] L. Saharan, M. J. De Andrade, W. Saleem, R. H. Baughman, and Y. Tadesse, “IGrab: Hand orthosis powered by twisted and coiled polymer muscles,” *Smart Materials and Structures*, vol. 26, no. 10, 2017.
- [63] C. Tawk, G. M. Spinks, M. In Het Panhuis, and G. Alici, “3D Printable Linear Soft Vacuum Actuators: Their Modeling, Performance Quantification and Application in Soft Robotic Systems,” *IEEE/ASME Transactions on Mechatronics*, vol. 24, no. 5, pp. 2118–2129, 2019.

- [64] T. Tsabedze and J. Zhang, “Design, characterization, modeling, and comparison of helically wrapped super-coiled polymer artificial muscles,” *Sensors and Actuators A: Physical*, vol. 331, no. c, pp. 5862–5868, 2021.
- [65] K. H. Cho, M. G. Song, H. Jung, S. Y. Yang, H. Moon, J. C. Koo, J. D. Nam, and H. R. Choi, “Fabrication and modeling of temperature-controllable artificial muscle actuator,” *Proceedings of the IEEE RAS and EMBS International Conference on Biomedical Robotics and Biomechatronics*, vol. 2016-July, pp. 94–98, 2016.
- [66] K. Masuya, “Low-cost coil-shaped optical fiber displacement sensor for a twisted and coiled polymer fiber actuator unit,” *IEEE Robotics and Automation Letters*, vol. 5, no. 4, pp. 6497–6503, 2020.
- [67] F. Karami and Y. Tadesse, “Modeling of twisted and coiled polymer (TCP) muscle based on phenomenological approach,” *Smart Materials and Structures*, vol. 26, no. 12, 2017.
- [68] J. Sun and J. Zhao, “Physics-Based Modeling of Twisted-and-Coiled Actuators Using Cosserat Rod Theory,” *IEEE Transactions on Robotics*, vol. PP, pp. 1–18, 2021.
- [69] C. Wu and W. Zheng, “A modeling of twisted and coiled polymer artificial muscles based on elastic rod theory,” *Actuators*, vol. 9, no. 2, 2020.
- [70] M. C. Yip and G. Niemeyer, “On the control and properties of supercoiled polymer artificial muscles,” *IEEE Transactions on Robotics*, vol. 33, no. 3, pp. 689–699, 2017.
- [71] C. Oiwa, K. Masuya, K. Tahara, T. Irisawa, M. Shioya, T. Yamauchi, E. Tanaka, K. Asaka, and K. Takagi, *Gray-box modeling and control of torsional fishing-line artificial muscle actuators*, vol. 10594 of *SPIE Smart Structures and Materials + Nondestructive Evaluation and Health Monitoring*. SPIE, 2018.
- [72] S. Aziz and G. M. Spinks, “Torsional artificial muscles,” *Materials Horizons*, vol. 7, no. 3, pp. 667–693, 2020.

- [73] T. Luong, K. Kim, S. Seo, J. H. Park, Y. Kim, S. Y. Yang, K. H. Cho, J. C. Koo, H. R. Choi, and H. Moon, "Modeling and position control of a high performance twisted-coiled polymer actuator," *2018 15th International Conference on Ubiquitous Robots, UR 2018*, no. September 2020, pp. 73–79, 2018.
- [74] A. Abbas and J. Zhao, "A physics based model for twisted and coiled actuator," *Proceedings - IEEE International Conference on Robotics and Automation*, pp. 6121–6126, 2017.
- [75] J. E. Slightam and M. L. Nagurka, "Theoretical dynamic modeling and validation of braided pneumatic artificial muscles," *Journal of Dynamic Systems, Measurement, and Control*, vol. 142, no. 3, p. 031008, 2020.
- [76] J. Van Der Weijde, B. Smit, M. Fritschi, C. Van De Kamp, and H. Vallery, "Self-sensing of deflection, force, and temperature for joule-heated twisted and coiled polymer muscles via electrical impedance," *IEEE/ASME Transactions on Mechatronics*, vol. 22, no. 3, pp. 1268–1275, 2017.
- [77] L. Saharan, L. Wu, and Y. Tadesse, "Modeling and Simulation of Robotic Finger Powered by Nylon Artificial Muscles," *Journal of Mechanisms and Robotics*, vol. 12, no. 1, 2020.
- [78] F. Karami, L. Wu, and Y. Tadesse, "Modeling of One-Ply and Two-Ply Twisted and Coiled Polymer Artificial Muscles," *IEEE/ASME Transactions on Mechatronics*, vol. 26, no. 1, pp. 300–310, 2021.
- [79] Q. Yang and G. Li, "A top-down multi-scale modeling for actuation response of polymeric artificial muscles," 2016.
- [80] A. E. H. Love, *A treatise on the mathematical theory of elasticity*. Cambridge university press, 2013.

- [81] V. Giovenco, P. Kotak, V. Cichella, C. Maletta, and C. Lamuta, “Dynamic model for the tensile actuation of thermally and electro-thermally actuated twisted and coiled artificial muscles (TCAMs),” *Smart Materials and Structures*, vol. 29, no. 2, 2020.
- [82] N. Charles, M. Gazzola, and L. Mahadevan, “Topology, Geometry, and Mechanics of Strongly Stretched and Twisted Filaments: Solenoids, Plectonemes, and Artificial Muscle Fibers,” *Physical Review Letters*, vol. 123, no. 20, p. 208003, 2019.
- [83] M. Hammond, V. Cichella, T. Weerakkody, and C. Lamuta, “Robust and Adaptive Sampled-Data Control of Twisted and Coiled Artificial Muscles,” *IEEE Control Systems Letters*, vol. 6, pp. 1232–1237, 2022.
- [84] B. Tondu, “Single linear integral action control for closed-loop positioning of a biomimetic actuator with artificial muscles,” in *2015 European Control Conference (ECC)*, pp. 3585–3590, IEEE, 2015.
- [85] L. Zhao, L. Ge, and T. Wang, “Position control for a two-joint robot finger system driven by pneumatic artificial muscles,” *Transactions of the Institute of Measurement and Control*, vol. 40, no. 4, pp. 1328–1339, 2018.
- [86] D. Schindele and H. Aschemann, “Ilc for a fast linear axis driven by pneumatic muscle actuators,” in *2011 IEEE International Conference on Mechatronics*, pp. 967–972, IEEE, 2011.
- [87] D. Schindele and H. Aschemann, “P-type ILC with phase lead compensation for a pneumatically driven parallel robot,” in *American Control Conference (ACC), 2012*, pp. 5484–5489, IEEE, 2012.
- [88] P. Sachdeva, S. Sueda, S. Bradley, M. Fain, and D. K. Pai, “Biomechanical simulation and control of hands and tendinous systems,” *ACM Transactions on Graphics (TOG)*, vol. 34, no. 4, p. 42, 2015.

- [89] T. Karnjanaparichat and R. Pongvuthithum, “Adaptive tracking control of multi-link robots actuated by pneumatic muscles with additive disturbances,” *Robotica*, vol. 35, no. 11, pp. 2139–2156, 2017.
- [90] C. Wu and W. Zheng, “Position and force control of a twisted and coiled polymeric actuator,” *IEEE Access*, vol. PP, pp. 1–1, 07 2020.
- [91] M. Suzuki and N. Kamamichi, “Control of twisted and coiled polymer actuator with anti-windup compensator,” *Smart Materials and Structures*, vol. 27, 04 2018.
- [92] M. Suzuki and N. Kamamichi, “Displacement control of an antagonistic-type twisted and coiled polymer actuator,” *Smart Materials and Structures*, vol. 27, no. 3, p. 035003, 2018.
- [93] K. Masuya, S. Ono, K. Takagi, and K. Tahara, “Feedforward control of twisted and coiled polymer actuator based on a macroscopic nonlinear model focusing on energy,” *IEEE Robotics and Automation Letters*, vol. 3, no. 3, pp. 1824–1831, 2018.
- [94] M. Suzuki and N. Kamamichi, “Robust control with disturbance observer for twisted and coiled polymer actuator,” *Smart Materials and Structures*, vol. 27, no. 8, p. 085006, 2018.
- [95] T. Li, Y. Wang, K. Liu, H. Liu, J. Zhang, X. Sheng, and D. Guo, “Thermal actuation performance modification of coiled artificial muscle by controlling annealing stress,” *Journal of Polymer Science Part B: Polymer Physics*, vol. 56, no. 5, pp. 383–390, 2018.
- [96] T. Luong, K. Kim, S. Seo, J. H. Park, Y. Kim, S. Y. Yang, K. H. Cho, J. C. Koo, H. R. Choi, and H. Moon, “Modeling and position control of a high performance twisted-coiled polymer actuator,” in *2018 15th International Conference on Ubiquitous Robots (UR)*, pp. 73–79, IEEE, 2018.
- [97] N. Hovakimyan and C. Cao, *L1 Adaptive Control Theory: Guaranteed Robustness with Fast Adaptation*. SIAM, 2010.

- [98] H. Jafarnejadsani, H. Lee, and N. Hovakimyan, “L1 adaptive sampled-data control for uncertain multi-input multi-output systems,” *Automatica*, vol. 103, pp. 346–353, 2019.
- [99] C. S. Haines, M. D. Lima, N. Li, *et al.*, “Artificial muscles from fishing line and sewing thread,” *Science*, vol. 343, no. 6173, pp. 868–872, 2014.
- [100] M. C. Yip and G. Niemeyer, “High-performance robotic muscles from conductive nylon sewing thread,” in *2015 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 2313–2318, IEEE, 2015.
- [101] K. Masuya, S. Ono, K. Takagi, and K. Tahara, “Nonlinear dynamics of twisted and coiled polymer actuator made of conductive nylon based on the energy balance,” in *2017 IEEE International Conference on Advanced Intelligent Mechatronics (AIM)*, IEEE, 2017.
- [102] A. Kato, M. Hirabayashi, Y. Matsumoto, Y. Nakashima, Y. Kobayashi, M. G. Fujie, and S. Sugano, “Continuous wrist joint control using muscle deformation measured on forearm skin,” in *2018 IEEE International Conference on Robotics and Automation (ICRA)*, IEEE, 2018.
- [103] M. Lay, N. L. N. Thajudin, Z. A. A. Hamid, A. Rusli, M. K. Abdullah, and R. K. Shuib, “Comparison of physical and mechanical properties of pla, abs and nylon 6 fabricated using fused deposition modeling and injection molding,” *Composites Part B: Engineering*, vol. 176, p. 107341, 2019.
- [104] S. Bell, A. Bangel, T. Weerakkody, X. Song, and C. Lamuta, “Automated manufacturing system for carbon fiber-based twisted and coiled artificial muscles (tcams),” *Manufacturing Letters*, vol. 33, pp. 19–23, 2022.
- [105] A. M. Taylor, “Science review of internal combustion engines,” *Energy Policy*, vol. 36, no. 12, pp. 4657–4667, 2008.

- [106] D. S. Mintz, B. R. Woodley, T. M. Schuh, Y. Huang, and M. R. Williams, “Medical robotics systems implementing axis constraints during actuation of one or more motorized joints,” June 8 2021. US Patent 11,026,758.
- [107] A. Borboni, M. Mor, and R. Faglia, “Gloreha—hand robotic rehabilitation: design, mechanical model, and experiments,” *Journal of Dynamic Systems, Measurement, and Control*, vol. 138, no. 11, p. 111003, 2016.
- [108] S. Kim, C. Laschi, and B. Trimmer, “Soft robotics: a bioinspired evolution in robotics,” *Trends in biotechnology*, vol. 31, no. 5, pp. 287–294, 2013.
- [109] N. El-Atab, R. B. Mishra, F. Al-Modaf, L. Joharji, A. A. Alsharif, H. Alamoudi, M. Diaz, N. Qaiser, and M. M. Hussain, “Soft actuators for soft robotic applications: A review,” *Advanced Intelligent Systems*, vol. 2, no. 10, p. 2000128, 2020.
- [110] P. Polygerinos, Z. Wang, K. C. Galloway, R. J. Wood, and C. J. Walsh, “Soft robotic glove for combined assistance and at-home rehabilitation,” *Robotics and Autonomous Systems*, vol. 73, pp. 135–143, 2015.
- [111] M. D. Lima, N. Li, M. Jung de Andrade, S. Fang, J. Oh, G. M. Spinks, M. E. Kozlov, C. S. Haines, D. Suh, J. Foroughi, *et al.*, “Electrically, chemically, and photonically powered torsional and tensile actuation of hybrid carbon nanotube yarn muscles,” *science*, vol. 338, no. 6109, pp. 928–932, 2012.
- [112] C. S. Haines, N. Li, G. M. Spinks, A. E. Aliev, J. Di, and R. H. Baughman, “New twist on artificial muscles,” *Proceedings of the National Academy of Sciences*, vol. 113, no. 42, pp. 11709–11716, 2016.
- [113] L. Saharan, A. Sharma, M. J. de Andrade, R. H. Baughman, and Y. Tadesse, “Design of a 3d printed lightweight orthotic device based on twisted and coiled polymer muscle: igrab hand orthosis,” in *Active and Passive Smart Structures and Integrated Systems 2017*, vol. 10164, pp. 504–513, SPIE, 2017.

- [114] B. P. Edmonds and A. L. Trejos, “Stiffness control of a nylon twisted coiled actuator for use in mechatronic rehabilitation devices,” in *2017 International Conference on Rehabilitation Robotics (ICORR)*, pp. 1419–1424, IEEE, 2017.
- [115] N. A. Atikah, L. Y. Weng, A. Anuar, C. C. Fat, I. Z. Abidin, and K. S. M. Sahari, “Development of nylon-based artificial muscles for the usage in robotic prosthetic limb,” in *AIP Conference Proceedings*, vol. 1883, AIP Publishing, 2017.
- [116] K. H. Cho, M. G. Song, H. Jung, J. Park, H. Moon, J. C. Koo, J.-D. Nam, and H. R. Choi, “A robotic finger driven by twisted and coiled polymer actuator,” in *Electroactive Polymer Actuators and Devices (EAPAD) 2016*, vol. 9798, pp. 275–281, SPIE, 2016.
- [117] L. Wu, M. J. de Andrade, T. Brahme, Y. Tadesse, and R. H. Baughman, “A reconfigurable robot with tensegrity structure using nylon artificial muscle,” in *Active and Passive Smart Structures and Integrated Systems 2016*, vol. 9799, pp. 950–960, SPIE, 2016.
- [118] P. Kotak, T. Weerakkody, and C. Lamuta, “Physics-based dynamic model for the electro-thermal actuation of bio-inspired twisted spiral artificial muscles (tsams),” *Polymer*, vol. 222, p. 123642, 2021.
- [119] L. Zhao, H. Cheng, and T. Wang, “Sliding mode control for a two-joint coupling nonlinear system based on extended state observer,” *ISA transactions*, vol. 73, pp. 130–140, 2018.
- [120] K. Xing, J. Huang, Y. Wang, J. Wu, Q. Xu, and J. He, “Tracking control of pneumatic artificial muscle actuators based on sliding mode and non-linear disturbance observer,” *IET control theory & applications*, vol. 4, no. 10, pp. 2058–2070, 2010.
- [121] T.-Y. Choi, B.-S. Choi, and K.-H. Seo, “Position and compliance control of a pneumatic muscle actuated manipulator for enhanced safety,” *IEEE Transactions on Control Systems Technology*, vol. 19, no. 4, pp. 832–842, 2010.

- [122] M.-K. Chang, “An adaptive self-organizing fuzzy sliding mode controller for a 2-dof rehabilitation robot actuated by pneumatic muscle actuators,” *Control Engineering Practice*, vol. 18, no. 1, pp. 13–22, 2010.
- [123] H. Jahanabadi, M. Mailah, M. Md Zain, and H. Hooi, “Active force with fuzzy logic control of a two-link arm driven by pneumatic artificial muscles,” *Journal of Bionic Engineering*, vol. 8, no. 4, pp. 474–484, 2011.
- [124] S. Hussain, S. Q. Xie, and P. K. Jamwal, “Robust nonlinear control of an intrinsically compliant robotic gait training orthosis,” *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 43, no. 3, pp. 655–665, 2012.
- [125] N. N. Son, C. Van Kien, and H. P. H. Anh, “A novel adaptive feed-forward-pid controller of a scara parallel robot using pneumatic artificial muscle actuator based on neural network and modified differential evolution algorithm,” *Robotics and Autonomous Systems*, vol. 96, pp. 65–80, 2017.
- [126] T. Arakawa, K. Takagi, K. Tahara, and K. Asaka, “Position control of fishing line artificial muscles (coiled polymer actuators) from nylon thread,” in *Electroactive Polymer Actuators and Devices (EAPAD) 2016*, vol. 9798, pp. 545–556, SPIE, 2016.
- [127] M. Suzuki and N. Kamamichi, “Simple controller design based on internal model control for twisted and coiled polymer actuator,” in *Actuators*, vol. 7-3, p. 33, MDPI, 2018.
- [128] H. Jafarnejadsani, *Robust adaptive sampled-data control design for MIMO systems: Applications in cyber-physical security*. PhD thesis, University of Illinois Urbana-Champaign, 2018.
- [129] H. Jafarnejadsani, D. Sun, H. Lee, and N. Hovakimyan, “Optimized l_1 adaptive controller for trajectory tracking of an indoor quadrotor,” *Journal of Guidance, Control, and Dynamics*, vol. 40, no. 6, pp. 1415–1427, 2017.

- [130] M. Cianchetti, C. Laschi, A. Menciassi, and P. Dario, “Biomedical applications of soft robotics,” *Nature Reviews Materials*, vol. 3, no. 6, pp. 143–153, 2018.
- [131] F. Rosso, G. Marino, A. Giordano, M. Barbarisi, D. Parmeggiani, and A. Barbarisi, “Smart materials as scaffolds for tissue engineering,” *Journal of cellular physiology*, vol. 203, no. 3, pp. 465–470, 2005.
- [132] D. Morales, E. Palleau, M. D. Dickey, and O. D. Velev, “Electro-actuated hydrogel walkers with dual responsive legs,” *Soft matter*, vol. 10, no. 9, pp. 1337–1348, 2014.
- [133] D. R. Higuera-Ruiz, K. Nishikawa, H. Feigenbaum, and M. Shafer, “What is an artificial muscle? a comparison of soft actuators to biological muscles,” *Bioinspiration & biomimetics*, vol. 17, no. 1, p. 011001, 2021.
- [134] A. Rao, A. R. Srinivasa, and J. N. Reddy, *Design of shape memory alloy (SMA) actuators*, vol. 3. Springer, 2015.
- [135] S. Seelecke and I. Müller, “Shape memory alloy actuators in smart structures: Modeling and simulation,” *Appl. Mech. Rev.*, vol. 57, no. 1, pp. 23–46, 2004.
- [136] Y. Liu, J. Van Humbeeck, R. Stalmans, and L. Delaey, “Some aspects of the properties of niti shape memory alloy,” *Journal of alloys and compounds*, vol. 247, no. 1-2, pp. 115–121, 1997.
- [137] C. Zanotti, P. Giuliani, and A. Chrysanthou, “Martensitic–austenitic phase transformation of ni–ti smas: thermal properties,” *Intermetallics*, vol. 24, pp. 106–114, 2012.
- [138] L. Zhou, L. Ren, Y. Chen, S. Niu, Z. Han, and L. Ren, “Bio-inspired soft grippers based on impactive gripping,” *Advanced Science*, vol. 8, no. 9, p. 2002017, 2021.
- [139] J. Zhu, L. Lyu, Y. Xu, H. Liang, X. Zhang, H. Ding, and Z. Wu, “Intelligent soft surgical robots for next-generation minimally invasive surgery,” *Advanced Intelligent Systems*, vol. 3, no. 5, p. 2100011, 2021.

- [140] Q. Liu, S. Ghodrati, G. Huisman, and K. M. Jansen, “Shape memory alloy actuators for haptic wearables: A review,” *Materials & Design*, vol. 233, p. 112264, 2023.
- [141] M. Cianchetti, C. Laschi, A. Menciassi, and P. Dario, “Biomedical applications of soft robotics,” *Nature Reviews Materials*, vol. 3, pp. 143–153, June 2018.
- [142] J. Jayender, R. V. Patel, S. Nikumb, and M. Ostojic, “Modeling and control of shape memory alloy actuators,” *IEEE transactions on control systems technology*, vol. 16, no. 2, pp. 279–287, 2008.
- [143] N. B. Kha and K. K. Ahn, “Position control of shape memory alloy actuators by using self tuning fuzzy pid controller,” in *2006 1ST IEEE conference on industrial electronics and applications*, pp. 1–5, IEEE, 2006.
- [144] S. Majima, K. Kodama, and T. Hasegawa, “Modeling of shape memory alloy actuator and tracking control system with the model,” *IEEE Transactions on Control Systems Technology*, vol. 9, no. 1, pp. 54–59, 2001.
- [145] N. T. Tai and K. K. Ahn, “Adaptive proportional–integral–derivative tuning sliding mode control for a shape memory alloy actuator,” *Smart Materials and Structures*, vol. 20, no. 5, p. 055010, 2011.
- [146] E. Sgambitterra, C. Greco, S. Rodinò, F. Niccoli, F. Furgiuele, and C. Maletta, “Fully coupled electric-thermo-mechanical model for predicting the response of a sma wire activated by electrical input,” *Sensors and Actuators A: Physical*, vol. 362, p. 114643, 2023.
- [147] M. Hammond, V. Cichella, T. Weerakkody, and C. Lamuta, “Robust and adaptive sampled-data control of twisted and coiled artificial muscles,” *IEEE control systems letters*, vol. 6, pp. 1232–1237, 2021.

- [148] T. H. Weerakkody, M. Hammond, J. H. Neilan, V. Cichella, and C. Lamuta, “Modeling and control of twisted and coiled artificial muscles for soft robotics,” *Meccanica*, vol. 58, no. 4, pp. 643–658, 2023.
- [149] C. Greco, T. H. Weerakkody, V. Cichella, L. Pagnotta, and C. Lamuta, “Lightweight bioinspired exoskeleton for wrist rehabilitation powered by twisted and coiled artificial muscles,” *Robotics*, vol. 12, no. 1, p. 27, 2023.
- [150] P. Kotak, S. Maxson, T. Weerakkody, V. Cichella, and C. Lamuta, “Octopus-inspired muscular hydrostats powered by twisted and coiled artificial muscles,” *Soft robotics*, 2023.
- [151] R. Mamman, P. Kotak, T. Weerakkody, T. Johnson, A. Krebill, J. Buchholz, and C. Lamuta, “Deployable vortex generators for low reynolds numbers applications powered by cephalopods inspired artificial muscles,” *Iscience*, vol. 26, no. 12, 2023.
- [152] P. S. Lobo, J. Almeida, and L. Guerreiro, “Shape memory alloys behaviour: A review,” *Procedia Engineering*, vol. 114, pp. 776–783, 2015.
- [153] Y. Xia, Y. He, F. Zhang, Y. Liu, and J. Leng, “A review of shape memory polymers and composites: mechanisms, materials, and applications,” *Advanced materials*, vol. 33, no. 6, p. 2000713, 2021.
- [154] E. Farber, J.-N. Zhu, A. Popovich, and V. Popovich, “A review of niti shape memory alloy as a smart material produced by additive manufacturing,” *Materials Today: Proceedings*, vol. 30, pp. 761–767, 2020.
- [155] V. Antonucci, G. Faiella, M. Giordano, F. Mennella, and L. Nicolais, “Electrical resistivity study and characterization during niti phase transformations,” *thermochimica acta*, vol. 462, no. 1-2, pp. 64–69, 2007.

- [156] V. Vaijapurkar and Y. Ravinder, “Electrical resistance modeling of shape memory alloy wire using an efficient performance analysis approach,” *Shape Memory and Superelasticity*, vol. 8, p. 168–182, 2022.
- [157] Y. Lu, R. Zhang, Y. Xu, L. Wang, and H. Yue, “Resistance characteristics of sma actuator based on the variable speed phase transformation constitutive model,” *Materials*, vol. 13, no. 6, 2020.
- [158] S. Rodinò, E. M. Curcio, E. Sgambitterra, and C. Maletta, “Sma-polymer composite made by 3d printing: Modelling and experiments,” *Procedia Structural Integrity*, vol. 47, pp. 579–588, 2023. 27th International Conference on Fracture and Structural Integrity (IGF27).
- [159] L. C. Brinson, “One-dimensional constitutive behavior of shape memory alloys: thermomechanical derivation with non-constant material functions and redefined martensite internal variable,” *Journal of intelligent material systems and structures*, vol. 4, no. 2, pp. 229–242, 1993.
- [160] C. Liang and C. A. Rogers, “One-dimensional thermomechanical constitutive relations for shape memory materials,” *Journal of intelligent material systems and structures*, vol. 1, no. 2, pp. 207–234, 1990.
- [161] L. Brinson and M. Huang, “Simplifications and comparisons of shape memory alloy constitutive models,” *Journal of intelligent material systems and structures*, vol. 7, no. 1, pp. 108–114, 1996.
- [162] B. He, X. Dong, R. Nie, Y. Wang, S. Ao, and G. Wang, “Comprehensive shape memory alloys constitutive models for engineering application,” *Materials & Design*, vol. 225, p. 111563, 2023.
- [163] A. Mohammadgholipour and A. M. Billah, “Mechanical properties and constitutive models of shape memory alloy for structural engineering: A review,” *Journal of Intelligent Material Systems and Structures*, vol. 34, no. 20, pp. 2335–2359, 2023.

- [164] H. Sayyaadi, M. Zakerzadeh, and H. Salehi, “A comparative analysis of some one-dimensional shape memory alloy constitutive models based on experimental tests,” *Scientia Iranica*, vol. 19, no. 2, pp. 249–257, 2012.
- [165] M. H. Elahinia and H. Ashrafiuon, “Nonlinear control of a shape memory alloy actuated manipulator,” *J. Vib. Acoust.*, vol. 124, no. 4, pp. 566–575, 2002.
- [166] K. W. Krigger, “Cerebral palsy: an overview,” *American family physician*, vol. 73, no. 1, 2006.
- [167] M. L. Aisen, D. Kerkovich, J. Mast, S. Mulroy, T. A. Wren, R. M. Kay, and S. A. Rethlefsen, “Cerebral palsy: clinical care and neurological rehabilitation,” *The Lancet Neurology*, vol. 10, no. 9, pp. 844–852, 2011.
- [168] S. Østensjø, E. B. Carlberg, and N. K. Vøllestad, “Motor impairments in young children with cerebral palsy: relationship to gross motor function and everyday activities,” *Developmental medicine and child neurology*, vol. 46, no. 9, pp. 580–589, 2004.
- [169] G. J. Snoek, M. J. IJzerman, H. J. Hermens, D. Maxwell, and F. Biering-Sorensen, “Survey of the needs of patients with spinal cord injury: impact and priority for improvement in hand function in tetraplegics,” *Spinal cord*, vol. 42, no. 9, pp. 526–532, 2004.
- [170] R. J. Cox, D. I. Amsters, and K. J. Pershouse, “The need for a multidisciplinary outreach service for people with spinal cord injury living in the community,” *Clinical rehabilitation*, vol. 15, no. 6, pp. 600–606, 2001.
- [171] S. S. Virani, A. Alonso, E. J. Benjamin, M. S. Bittencourt, C. W. Callaway, A. P. Carson, A. M. Chamberlain, A. R. Chang, S. Cheng, F. N. Delling, *et al.*, “Heart disease and stroke statistics—2020 update: a report from the american heart association,” *Circulation*, vol. 141, no. 9, pp. e139–e596, 2020.

- [172] M. A. Delph, S. A. Fischer, P. W. Gauthier, C. H. M. Luna, E. A. Clancy, and G. S. Fischer, "A soft robotic exomusculature glove with integrated semg sensing for hand rehabilitation," in *2013 IEEE 13th International Conference on Rehabilitation Robotics (ICORR)*, pp. 1–7, IEEE, 2013.
- [173] C. G. Canning, L. Ada, R. Adams, and N. J. O'Dwyer, "Loss of strength contributes more to physical disability after stroke than loss of dexterity," *Clinical rehabilitation*, vol. 18, no. 3, pp. 300–308, 2004.
- [174] S. Balasubramanian, "Motor impairments following stroke," *Human Systems Neuroscience*, 2015.
- [175] P. Raghavan, "Upper limb motor impairment after stroke," *Physical Medicine and Rehabilitation Clinics*, vol. 26, no. 4, pp. 599–610, 2015.
- [176] C. D. Wolfe, "The impact of stroke," *British medical bulletin*, vol. 56, no. 2, pp. 275–286, 2000.
- [177] E. J. Benjamin, P. Muntner, A. Alonso, M. S. Bittencourt, C. W. Callaway, A. P. Carson, A. M. Chamberlain, A. R. Chang, S. Cheng, S. R. Das, *et al.*, "Heart disease and stroke statistics—2019 update: a report from the american heart association," *Circulation*, vol. 139, no. 10, pp. e56–e528, 2019.
- [178] M. Nichols, N. Townsend, R. Luengo-Fernandez, J. Leal, A. Gray, P. Scarborough, and M. Rayner, "European cardiovascular disease statistics 2012," tech. rep., University of Bristol, 2012.
- [179] T. S. Olsen, "Arm and leg paresis as outcome predictors in stroke rehabilitation.," *Stroke*, vol. 21, no. 2, pp. 247–251, 1990.

- [180] C. Bütetfisch, H. Hummelsheim, P. Denzler, and K.-H. Mauritz, “Repetitive training of isolated movements improves the outcome of motor rehabilitation of the centrally paretic hand,” *Journal of the neurological sciences*, vol. 130, no. 1, pp. 59–68, 1995.
- [181] J. R. Carey, T. J. Kimberley, S. M. Lewis, E. J. Auerbach, L. Dorsey, P. Rundquist, and K. Ugurbil, “Analysis of fmri and finger tracking training in subjects with chronic stroke,” *Brain*, vol. 125, no. 4, pp. 773–788, 2002.
- [182] J. R. Carey, W. K. Durfee, E. Bhatt, A. Nagpal, S. A. Weinstein, K. M. Anderson, and S. M. Lewis, “Comparison of finger tracking versus simple movement training via telerehabilitation to alter hand function and cortical reorganization after stroke,” *Neurorehabilitation and neural repair*, vol. 21, no. 3, pp. 216–232, 2007.
- [183] P. Lum, D. Reinkensmeyer, R. Mahoney, W. Z. Rymer, and C. Burgar, “Robotic devices for movement therapy after stroke: current status and challenges to clinical acceptance,” *Topics in stroke rehabilitation*, vol. 8, no. 4, pp. 40–53, 2002.
- [184] X. Hu, K. Tong, R. Song, X. Zheng, K. Lui, W. Leung, S. Ng, and S. Au-Yeung, “Quantitative evaluation of motor functional recovery process in chronic stroke patients during robot-assisted wrist training,” *Journal of Electromyography and Kinesiology*, vol. 19, no. 4, pp. 639–650, 2009.
- [185] B. Volpe, H. Krebs, N. Hogan, L. Edelstein, C. Diels, and M. Aisen, “A novel approach to stroke rehabilitation: robot-aided sensorimotor stimulation,” *Neurology*, vol. 54, no. 10, pp. 1938–1944, 2000.
- [186] N. Norouzi-Gheidari, P. S. Archambault, and J. Fung, “Effects of robot-assisted therapy on stroke rehabilitation in upper limbs: systematic review and meta-analysis of the literature,” *Journal of Rehabilitation Research and Development*, vol. 49, no. 4, pp. 479–496, 2012.
- [187] M. L. Dombrov, “Emerging therapies in neurorehabilitation,” *CONTINUUM: Lifelong Learning in Neurology*, vol. 17, no. 3, pp. 530–544, 2011.

- [188] S. E. Fasoli, H. I. Krebs, J. Stein, W. R. Frontera, and N. Hogan, “Effects of robotic therapy on motor impairment and recovery in chronic stroke,” *Archives of physical medicine and rehabilitation*, vol. 84, no. 4, pp. 477–482, 2003.
- [189] G. B. Prange, M. Jannink, C. Groothuis-Oudshoorn, H. J. Hermens, and M. J. IJzerman, *Rehabilitation Robotics: Stimulation restoring of arm function after stroke*. Gildeprint Drukkerijen, Enschede, the Netherlands, 2009.
- [190] B. R. Brewer, S. K. McDowell, and L. C. Worthen-Chaudhari, “Poststroke upper extremity rehabilitation: a review of robotic systems and clinical results,” *Topics in stroke rehabilitation*, vol. 14, no. 6, pp. 22–44, 2007.
- [191] J. Mehrholz and M. Pohl, “Electromechanical-assisted gait training after stroke: a systematic review comparing end-effector and exoskeleton devices,” *Journal of rehabilitation medicine*, vol. 44, no. 3, pp. 193–199, 2012.
- [192] H. S. Lo and S. Q. Xie, “Exoskeleton robots for upper-limb rehabilitation: State of the art and future prospects,” *Medical engineering & physics*, vol. 34, no. 3, pp. 261–268, 2012.
- [193] F. Molteni, G. Gasperini, G. Cannaviello, and E. Guanziroli, “Exoskeleton and end-effector robots for upper and lower limbs rehabilitation: narrative review,” *PM&R*, vol. 10, no. 9, pp. S174–S188, 2018.
- [194] F. Aggogeri, T. Mikolajczyk, and J. O’Kane, “Robotics for rehabilitation of hand movement in stroke survivors,” *Advances in Mechanical Engineering*, vol. 11, no. 4, p. 1687814019841921, 2019.
- [195] S. H. Lee, G. Park, D. Y. Cho, H. Y. Kim, J.-Y. Lee, S. Kim, S.-B. Park, and J.-H. Shin, “Comparisons between end-effector and exoskeleton rehabilitation robots regarding upper extremity function among chronic stroke patients with moderate-to-severe upper limb impairment,” *Scientific reports*, vol. 10, no. 1, p. 1806, 2020.

- [196] L. Piggott, S. Wagner, and M. Ziat, “Haptic neurorehabilitation and virtual reality for upper limb paralysis: a review,” *Critical Reviews™ in Biomedical Engineering*, vol. 44, no. 1-2, 2016.
- [197] R. Gopura, D. Bandara, K. Kiguchi, and G. K. Mann, “Developments in hardware systems of active upper-limb exoskeleton robots: A review,” *Robotics and Autonomous Systems*, vol. 75, pp. 203–220, 2016.
- [198] S. Hussain, P. K. Jamwal, P. Van Vliet, and M. H. Ghayesh, “State-of-the-art robotic devices for wrist rehabilitation: Design and control aspects,” *IEEE Transactions on human-machine systems*, vol. 50, no. 5, pp. 361–372, 2020.
- [199] A. U. Pehlivan, S. Lee, and M. K. O’Malley, “Mechanical design of ricewrist-s: A forearm-wrist exoskeleton for stroke and spinal cord injury rehabilitation,” in *2012 4th IEEE RAS & EMBS International Conference on Biomedical Robotics and Biomechatronics (BioRob)*, pp. 1573–1578, IEEE, 2012.
- [200] J. A. Martinez, P. Ng, S. Lu, M. S. Campagna, and O. Celik, “Design of wrist gimbal: A forearm and wrist exoskeleton for stroke rehabilitation,” in *2013 IEEE 13th international conference on rehabilitation robotics (ICORR)*, pp. 1–6, IEEE, 2013.
- [201] J. A. French, C. G. Rose, and M. K. O’malley, “System characterization of mahi exo-ii: A robotic exoskeleton for upper extremity rehabilitation,” in *Dynamic Systems and Control Conference*, vol. 46209, p. V003T43A006, American Society of Mechanical Engineers, 2014.
- [202] E. Pezent, C. G. Rose, A. D. Deshpande, and M. K. O’Malley, “Design and characterization of the openwrist: A robotic wrist exoskeleton for coordinated hand-wrist rehabilitation,” in *2017 international conference on rehabilitation robotics (ICORR)*, pp. 720–725, IEEE, 2017.

- [203] Y.-Y. Su, Y.-L. Yu, C.-H. Lin, and C.-C. Lan, “A compact wrist rehabilitation robot with accurate force/stiffness control and misalignment adaptation,” *International Journal of Intelligent Robotics and Applications*, vol. 3, pp. 45–58, 2019.
- [204] U. Mayetin and S. Kucuk, “Design and experimental evaluation of a low cost, portable, 3-dof wrist rehabilitation robot with high physical human–robot interaction,” *Journal of Intelligent & Robotic Systems*, vol. 106, no. 3, p. 65, 2022.
- [205] R. S. Gonçalves, L. S. F. Brito, L. P. Moraes, G. Carbone, and M. Ceccarelli, “A fairly simple mechatronic device for training human wrist motion,” *International Journal of Advanced Robotic Systems*, vol. 17, no. 6, p. 1729881420974286, 2020.
- [206] L. Zhang, J. Li, Y. Cui, M. Dong, B. Fang, and P. Zhang, “Design and performance analysis of a parallel wrist rehabilitation robot (pwrr),” *Robotics and Autonomous Systems*, vol. 125, p. 103390, 2020.
- [207] D. Xu, M. Zhang, Y. Sun, X. Zhang, H. Xu, Y. Li, X. Li, and S. Q. Xie, “Development of a reconfigurable wrist rehabilitation device with an adaptive forearm holder,” in *2018 IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM)*, pp. 454–459, IEEE, 2018.
- [208] K. Shi, A. Song, Y. Li, D. Chen, and H. Li, “Cable-driven 3-dof wrist rehabilitation robot with optimized human-robot interaction performance,” in *2020 8th IEEE RAS/EMBS International Conference for Biomedical Robotics and Biomechatronics (BioRob)*, pp. 112–117, IEEE, 2020.
- [209] J. Jeong, I. B. Yasir, J. Han, C. H. Park, S.-K. Bok, and K.-U. Kyung, “Design of shape memory alloy-based soft wearable robot for assisting wrist motion,” *Applied Sciences*, vol. 9, no. 19, p. 4025, 2019.
- [210] Y. Wang and Q. Xu, “Design and testing of a soft parallel robot based on pneumatic artificial muscles for wrist rehabilitation,” *Scientific Reports*, vol. 11, no. 1, p. 1273, 2021.

- [211] J. C. Perry and J. Rosen, “Design of a 7 degree-of-freedom upper-limb powered exoskeleton,” in *The First IEEE/RAS-EMBS International Conference on Biomedical Robotics and Biomechatronics, 2006. BioRob 2006.*, pp. 805–810, IEEE, 2006.
- [212] Y. M. Zhou, C. Hohimer, T. Proietti, C. T. O’Neill, and C. J. Walsh, “Kinematics-based control of an inflatable soft wearable robot for assisting the shoulder of industrial workers,” *IEEE Robotics and Automation Letters*, vol. 6, no. 2, pp. 2155–2162, 2021.
- [213] T. Weerakkody, N. Liyanaarachchi, H. Herath, R. Gopura, and D. Thilina, “Development of an active shoulder prosthesis with low-level control validation,” *Model. Identif. Control*, vol. 848, pp. 194–199, 2017.
- [214] B. A. De la Cruz-Sánchez, M. Arias-Montiel, and E. Lugo-González, “Emg-controlled hand exoskeleton for assisted bilateral rehabilitation,” *Biocybernetics and Biomedical Engineering*, vol. 42, no. 2, pp. 596–614, 2022.
- [215] K. Hub, “Musculoskeletal system anatomy kenhub.” <https://www.kenhub.com/en/library/anatomy/the-musculoskeletal-system>. Ken Hub: Berlin, Germany, 2018.
- [216] M. Bernhard Hirt, M. HarunSeyhan, P. Michael Wagner, and O. Rainer Zumhasch, “Hand and wrist anatomy and biomechanics—a comprehensive guide,” 2017.
- [217] J. G. Betts, P. Desaix, E. Johnson, J. Johnson, O. Korol, D. Kruse, B. Poe, J. Wise, M. Womble, and K. Young, “Anatomy and physiology. openstax,” *Rice University: Houston, TX, USA*, 2013.
- [218] C. A. Boles, S. Kannam, and A. B. Cardwell, “The forearm: anatomy of muscle compartments and nerves,” *American Journal of Roentgenology*, vol. 174, no. 1, pp. 151–159, 2000.
- [219] S. Plagenhoef, F. G. Evans, and T. Abdelnour, “Anatomical data for analyzing human motion,” *Research quarterly for exercise and sport*, vol. 54, no. 2, pp. 169–178, 1983.

- [220] S. Pheasant and C. M. Haslegrave, *Bodyspace: Anthropometry, ergonomics and the design of work*. CRC press, 2018.
- [221] S. W. Churchill and H. H. Chu, “Correlating equations for laminar and turbulent free convection from a vertical plate,” *International Journal of Heat and Mass Transfer*, vol. 18, no. 11, pp. 1323–1329, 1975.
- [222] R. L. Frederick and F. Quiroz, “On the transition from conduction to convection regime in a cubical enclosure with a partially heated wall,” *International journal of heat and mass transfer*, vol. 44, no. 9, pp. 1699–1709, 2001.
- [223] C. Lamuta, “Perspective on highly twisted artificial muscles,” *Applied physics letters*, vol. 122, no. 4, 2023.
- [224] D. J. Slutsky and M. Herman, “Rehabilitation of distal radius fractures: a biomechanical guide,” *Hand clinics*, vol. 21, no. 3, pp. 455–468, 2005.
- [225] A. Opheim, “Top hand therapy exercises,” 2021.
- [226] J. J. Crisco, W. M. Heard, R. R. Rich, D. J. Paller, and S. W. Wolfe, “The mechanical axes of the wrist are oriented obliquely to the anatomical axes,” *The Journal of Bone and Joint Surgery. American volume.*, vol. 93, no. 2, p. 169, 2011.
- [227] K. A. Mann, F. W. Wernere, and A. K. Palmer, “Frequency spectrum analysis of wrist motion for activities of daily living,” *Journal of Orthopaedic research*, vol. 7, no. 2, pp. 304–306, 1989.
- [228] C. Zhou, Z. Yang, K. Li, and X. Ye, “Research and development of ankle–foot orthoses: A review,” *Sensors*, vol. 22, no. 17, p. 6596, 2022.
- [229] J. Harlaar, M. Brehm, J. G. Becher, D. J. Bregman, J. Buurke, F. Holtkamp, V. De Groot, and F. Nollet, “Studies examining the efficacy of ankle foot orthoses should report activity level

- and mechanical evidence,” *Prosthetics and orthotics international*, vol. 34, no. 3, pp. 327–335, 2010.
- [230] G. Chen and C. Patten, “Joint moment work during the stance-to-swing transition in hemiparetic subjects,” *Journal of biomechanics*, vol. 41, no. 4, pp. 877–883, 2008.
- [231] M. N. Eek, R. Tranberg, and E. Beckung, “Muscle strength and kinetic gait pattern in children with bilateral spastic cp,” *Gait & posture*, vol. 33, no. 3, pp. 333–337, 2011.
- [232] T. H. Weerakkody, “Ankle dorsiflexion and plantarflexion motions.” designer.microsoft.com, 2024. Generated using Microsoft Designer.
- [233] B. Balaban and F. Tok, “Gait disturbances in patients with stroke,” *Pm&r*, vol. 6, no. 7, pp. 635–642, 2014.
- [234] M. Jacquelin Perry, “Gait analysis: normal and pathological function,” *New Jersey: SLACK*, 2010.
- [235] F. S. Shahar, M. T. H. Sultan, S. H. Lee, M. Jawaaid, A. U. M. Shah, S. N. A. Safri, and P. N. Sivasankaran, “A review on the orthotics and prosthetics and the potential of kenaf composites as alternative materials for ankle-foot orthosis,” *Journal of the mechanical behavior of biomedical materials*, vol. 99, pp. 169–185, 2019.
- [236] E. Russell Esposito, K. A. Schmidbauer, and J. M. Wilken, “Experimental comparisons of passive and powered ankle-foot orthoses in individuals with limb reconstruction,” *Journal of neuroengineering and rehabilitation*, vol. 15, pp. 1–10, 2018.
- [237] N. Eddison, N. Chockalingam, and S. Osborne, “Ankle foot orthosis–footwear combination tuning: An investigation into common clinical practice in the united kingdom,” *Prosthetics and orthotics international*, vol. 39, no. 2, pp. 126–133, 2015.

- [238] K. Z. Takahashi, T. M. Kepple, and S. J. Stanhope, “A unified deformable (ud) segment model for quantifying total power of anatomical and prosthetic below-knee structures during stance in gait,” *Journal of biomechanics*, vol. 45, no. 15, pp. 2662–2667, 2012.
- [239] S. Hood, S. Creveling, L. Gabert, M. Tran, and T. Lenzi, “Powered knee and ankle prostheses enable natural ambulation on level ground and stairs for individuals with bilateral above-knee amputation: a case study,” *Scientific Reports*, vol. 12, no. 1, p. 15465, 2022.
- [240] L. M. Sullivan, S. Creveling, M. Cowan, L. Gabert, and T. Lenzi, “Powered knee and ankle prosthesis control for adaptive ambulation at variable speeds, inclines, and uneven terrains,” in *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 2128–2133, IEEE, 2023.
- [241] J. Mendez, R. Murray, L. Gabert, N. P. Fey, H. Liu, and T. Lenzi, “Continuous a-mode ultrasound-based prediction of transfemoral amputee prosthesis kinematics across different ambulation tasks,” *IEEE Transactions on Biomedical Engineering*, 2023.
- [242] T. Lenzi, M. Cempini, L. J. Hargrove, and T. A. Kuiken, “Design, development, and validation of a lightweight nonbackdrivable robotic ankle prosthesis,” *IEEE/ASME Transactions on Mechatronics*, vol. 24, no. 2, pp. 471–482, 2019.
- [243] L. Nigro and E. S. Arch, “Design and mechanical testing of a novel dual-stiffness ankle-foot orthosis,” *Journal of Medical Devices*, vol. 17, no. 3, p. 034501, 2023.
- [244] D. Madusanka, L. Wijayasingha, K. Sanjeevan, M. Ahamed, J. Edirisooriya, and R. Gopura, “A 3dof transtibial robotic prosthetic limb,” in *7th International conference on Information and Automation for Sustainability*, pp. 1–6, IEEE, 2014.
- [245] S. H. Collins, M. B. Wiggin, and G. S. Sawicki, “Reducing the energy cost of human walking using an unpowered exoskeleton,” *Nature*, vol. 522, no. 7555, pp. 212–215, 2015.

- [246] M. Moltedo, T. Baček, B. Serrien, K. Langlois, B. Vanderborght, D. Lefeber, and C. Rodriguez-Guerrero, “Walking with a powered ankle-foot orthosis: the effects of actuation timing and stiffness level on healthy users,” *Journal of neuroengineering and rehabilitation*, vol. 17, pp. 1–15, 2020.
- [247] J. Kwon, J.-H. Park, S. Ku, Y. Jeong, N.-J. Paik, and Y.-L. Park, “A soft wearable robotic ankle-foot-orthosis for post-stroke patients,” *IEEE Robotics and Automation Letters*, vol. 4, no. 3, pp. 2547–2552, 2019.
- [248] S. J. Kim, Y. Na, D. Y. Lee, H. Chang, and J. Kim, “Pneumatic afo powered by a miniature custom compressor for drop foot correction,” *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 28, no. 8, pp. 1781–1789, 2020.
- [249] D. Totah, M. Menon, C. Jones-Hershinow, K. Barton, and D. H. Gates, “The impact of ankle-foot orthosis stiffness on gait: a systematic literature review,” *Gait & posture*, vol. 69, pp. 101–111, 2019.
- [250] Y. Zhou, “Recent advances in wearable actuated ankle-foot orthoses: Medical effects, design, and control,” *Proceedings of the Institution of Mechanical Engineers, Part H: Journal of Engineering in Medicine*, vol. 237, no. 2, pp. 163–178, 2023.
- [251] B. Shi, X. Chen, Z. Yue, S. Yin, Q. Weng, X. Zhang, J. Wang, and W. Wen, “Wearable ankle robots in post-stroke rehabilitation of gait: a systematic review,” *Frontiers in neurorobotics*, vol. 13, p. 63, 2019.
- [252] T. Bagneschi, D. Chiaradia, G. Righi, G. D. Popolo, A. Frisoli, and D. Leonardis, “A soft hand exoskeleton with a novel tendon layout to improve stable wearing in grasping assistance,” *IEEE Transactions on Haptics*, vol. 16, no. 2, pp. 311–321, 2023.
- [253] H. K. Dan Yoshikawa and Y. Sankai, “Development of a wearable cybernic glove that enables object grasping and gripping force measurement with open finger pad,” *Advanced Robotics*, vol. 37, no. 14, pp. 924–941, 2023.

- [254] P. Tran, D. Elliott, K. Herrin, S. Bhatia, and J. P. Desai, “Evaluation of the flexotendon glove-iii through a human subject case study,” *Biomedical Engineering Letters*, vol. 13, p. 153–163, Jan 2023.
- [255] Q. Xie, Q. Meng, W. Yu, Z. Wu, R. Xu, Q. Zeng, Z. Zhou, T. Yang, and H. Yu, “Design of a sma-based soft composite structure for wearable rehabilitation gloves,” *Frontiers in Neurorobotics*, vol. 17, Feb 2023.
- [256] N. Kladovasilakis, I. Kostavelis, P. Sideridis, E. Koltzi, K. Piliounis, D. Tzetzis, and D. Tzovaras, “A novel soft robotic exoskeleton system for hand rehabilitation and assistance purposes,” *Applied Sciences*, vol. 13, p. 553, Dec 2022.
- [257] I. Rojek, M. Kaczmarek, P. Kotlarz, M. Kempieński, D. Mikołajewski, Z. Szczepański, J. Kopowski, J. Nowak, M. Macko, A. Szczepańczyk, and et al., “Hand exoskeleton—development of own concept,” *Applied Sciences*, vol. 13, p. 3238, Mar 2023.
- [258] M.-S. Kim, J.-K. Heo, H. Rodrigue, H.-T. Lee, S. Pané, M.-W. Han, and S.-H. Ahn, “Shape memory alloy (sma) actuators: The role of material, form, and scaling effects,” *Advanced Materials*, vol. 35, no. 33, p. 2208517, 2023.
- [259] C. Lee, M. Kim, Y.-J. Kim, *et al.*, “Soft robot review,” *Int J Control Autom Syst*, vol. 15, no. 1, pp. 3–15, 2017.
- [260] H. Lipson, “Challenges and opportunities for design, simulation, and fabrication of soft robots,” *Soft Robot*, vol. 1, no. 1, pp. 21–27, 2013.
- [261] L. Ren, B. Li, G. Wei, *et al.*, “Biology and bioinspiration of soft robotics: Actuation, sensing, and system integration,” *iScience*, vol. 24, no. 9, p. 103075, 2021.
- [262] B. Hochner, “An embodied view of octopus neurobiology,” *Curr Biol*, vol. 22, no. 20, pp. R887–R892, 2012.

- [263] W. M. Kier and M. P. Stella, “The arrangement and function of octopus arm musculature and connective tissue,” *J Morphol*, vol. 268, no. 10, pp. 831–843, 2007.
- [264] M. Cianchetti, M. Follador, B. Mazzolai, *et al.*, “Design and development of a soft robotic octopus arm exploiting embodied intelligence,” in *2012 IEEE International Conference on Robotics and Automation*, pp. 5271–5276, 2012.
- [265] C. Laschi, B. Mazzolai, V. Mattoli, *et al.*, “Design of a biomimetic robotic octopus arm,” *Bioinspir Biomim*, vol. 4, no. 1, p. 015006, 2009.
- [266] L. Margheri, G. Ponte, B. Mazzolai, *et al.*, “Non-invasive study of octopus vulgaris arm morphology using ultrasound,” *J Exp Biol*, vol. 214, no. 22, pp. 3727–3731, 2011.
- [267] G. Levy and B. Hochner, “Embodied organization of octopus vulgaris morphology, vision, and locomotion,” *Front Physiol*, vol. 8, p. 164, 2017.
- [268] W. M. Kier and A. M. Smith, “The morphology and mechanics of octopus suckers,” *Biol Bull*, vol. 178, no. 2, pp. 126–136, 1990.
- [269] R. T. Hanlon and J. B. Messenger, *Cephalopod Behaviour*. Cambridge University Press, 2nd ed., 2018.
- [270] J. A. Mather, “How do octopuses use their arms?,” *J Comp Psychol*, vol. 112, no. 3, pp. 306–316, 1998.
- [271] W. M. Kier and K. Smith, “Tongues, tentacles and trunks: The biomechanics of movement in muscular-hydrostats,” *Zool J Linn Soc*, vol. 83, no. 4, pp. 307–324, 1985.
- [272] M. Cianchetti, M. Calisti, L. Margheri, *et al.*, “Bioinspired locomotion and grasping in water: The soft eight-arm octopus robot,” *Bioinspir Biomim*, vol. 10, no. 3, p. 035003, 2015.
- [273] M. D. Grissom, V. Chitrakaran, D. Dienno, *et al.*, “Design and experimental testing of the octarm soft robot manipulator,” in *ProcSPIE*, vol. 2, pp. 62301F.1–62301F.10, 2006.

- [274] W. McMahan, V. Chitrakaran, M. Csencsits, *et al.*, “Field trials and testing of the octarm continuum manipulator,” in *Proceedings 2006 IEEE International Conference on Robotics and Automation, 2006 ICRA 2006*, pp. 2336–2341, 2006.
- [275] M. Cianchetti, T. Ranzani, G. Gerboni, *et al.*, “Soft robotics technologies to address shortcomings in today’s minimally invasive surgery: The stiff-flop approach,” *Soft Robot*, vol. 1, no. 2, pp. 122–131, 2014.
- [276] W. Li, H. Nabae, G. Endo, and K. Suzumori, “New soft robot hand configuration with combined biotensegrity and thin artificial muscle,” *IEEE Robot Autom Lett*, vol. 5, no. 3, pp. 4345–4351, 2020.
- [277] M. Cianchetti, A. Licofonte, M. Follador, *et al.*, “Bioinspired soft actuation system using shape memory alloys,” *Actuators*, vol. 3, pp. 226–244, 2014.
- [278] L. Zhang, M. Xu, and H. Yang, “Research on soft manipulator actuated by shape memory alloy (sma) springs,” in *2017 IEEE International Conference on Real-time Computing and Robotics (RCAR)*, pp. 74–78, 2017.
- [279] T. Ye, Y. Wang, S. Xu, *et al.*, “Modeling and motion control of an octopus-like flexible manipulator actuated by shape memory alloy wires,” *J Intell Mater Syst Struct*, vol. 33, no. 1, pp. 3–16, 2021.
- [280] T. Zheng, Y. Yang, D. T. Branson, *et al.*, “Control design of shape memory alloy based multi-arm continuum robot inspired by octopus,” in *2014 9th IEEE Conference on Industrial Electronics and Applications*, pp. 1108–1113, 2014.
- [281] C. Laschi, B. Mazzolai, V. Mattoli, *et al.*, “Design and development of a soft actuator for a robot inspired by the octopus arm,” in *Experimental Robotics* (O. Khatib, V. Kumar, and G. J. Pappas, eds.), vol. 54 of *Springer Tracts in Advanced Robotics*, pp. 25–33, Springer, 2008.

- [282] M. Follador, F. Tramacere, and B. Mazzolai, “Dielectric elastomer actuators for octopus inspired suction cups,” *Bioinspiration & biomimetics*, vol. 9, no. 4, p. 046002, 2014.
- [283] I. A. Anderson, T. A. Gisby, T. G. McKay, *et al.*, “Multi-functional dielectric elastomer artificial muscles for soft and smart machines,” *Journal of Applied Physics*, vol. 112, no. 4, p. 041101, 2012.
- [284] T. Doi, S. Wakimoto, K. Suzumori, and K. Mori, “Proposal of flexible robotic arm with thin mckibben actuators mimicking octopus arm structure,” in *2016 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 5503–5508, 2016.
- [285] S. Yamane and S. Wakimoto, “Development of a flexible manipulator with changing stiffness by granular jamming,” in *2017 24th International Conference on Mechatronics and Machine Vision in Practice (M2VIP)*, pp. 1–5, 2017.
- [286] E. Guglielmino, N. Tsagarakis, and D. G. Caldwell, “An octopus anatomy-inspired robotic arm,” in *2010 IEEE/RSJ International Conference on Intelligent Robots and Systems*, pp. 3091–3096, 2010.
- [287] S. Furukawa, S. Wakimoto, T. Kanda, and H. Hagihara, “A soft master-slave robot mimicking octopus arm structure using thin artificial muscles and wire encoders,” *Actuators*, vol. 8, p. 40, 2019.
- [288] J. Walker, T. Zidek, C. Harbel, *et al.*, “Soft robotics: A review of recent developments of pneumatic soft actuators,” *Actuators*, vol. 9, p. 3, 2020.
- [289] J. Mohd Jani, M. Leary, A. Subic, and M. A. Gibson, “A review of shape memory alloy research, applications and opportunities,” *Materials & Design*, vol. 56, pp. 1078–1113, 2014.
- [290] Y. Guo, L. Liu, Y. Liu, and J. Leng, “Review of dielectric elastomer actuators and their applications in soft robots,” *Advances in Intelligent Systems*, vol. 3, no. 10, p. 2000282, 2021.

- [291] Y. Jing, F. Su, X. Yu, *et al.*, “Advances in artificial muscles: A brief literature and patent review,” *Frontiers in Bioengineering and Biotechnology*, vol. 11, p. 1083857, 2023.
- [292] Z. Jiao, C. Zhang, W. Wang, *et al.*, “Advanced artificial muscle for flexible material-based reconfigurable soft robots,” *Advanced Science*, vol. 6, no. 21, p. 1901371, 2019.
- [293] S. Mirvakili, A. Rafie Ravandi, I. Hunter, C. Haines, N. Li, J. Foroughi, S. Naficy, G. Spinks, R. Baughman, and J. D. Madden, *Simple and strong: twisted silver painted nylon artificial muscle actuated by Joule heating*, vol. 9056 of *SPIE Smart Structures and Materials + Nondestructive Evaluation and Health Monitoring*. SPIE, 2014.
- [294] X. Leng, X. Hu, W. Zhao, *et al.*, “Recent advances in twisted-fiber artificial muscles,” *Advances in Intelligent Systems*, vol. 3, no. 5, p. 2000185, 2021.
- [295] S. Maxson, P. Kotak, T. Weerakkody, and C. Lamuta, “Manufacturing and underwater cyclic behavior of different types of twisted and coiled artificial muscles (tcams),” *Manufacturing Letters*, vol. 40, pp. 75–80, 2024.
- [296] J. Martínez, J. Dumas, and S. Lefebvre, “Procedural voronoi foams for additive manufacturing,” *ACM Transactions on Graphics (TOG)*, vol. 35, no. 4, pp. 1–12, 2016.
- [297] D. Goswami, S. Liu, A. Pal, L. G. Silva, and R. V. Martinez, “3d-architected soft machines with topologically encoded motion,” *Advanced Functional Materials*, vol. 29, no. 24, p. 1808713, 2019.
- [298] A. A. Laghari, A. K. Jumani, R. A. Laghari, and H. Nawaz, “Unmanned aerial vehicles: A review,” *Cognitive Robotics*, vol. 3, pp. 8–22, 2023.
- [299] G. P. J. IV, L. G. Pearlstine, and H. F. Percival, “An assessment of small unmanned aerial vehicles for wildlife research,” *Wildlife society bulletin*, vol. 34, no. 3, pp. 750–758, 2006.
- [300] J. A. Elmore, M. F. Curran, K. O. Evans, S. Samiappan, M. Zhou, M. B. Pfeiffer, B. F. Blackwell, and R. B. Igley, “Evidence on the effectiveness of small unmanned aircraft sys-

- tems (suas) as a survey tool for north american terrestrial, vertebrate animals: a systematic map protocol,” *Environmental Evidence*, vol. 10, no. 1, p. 15, 2021.
- [301] P. Lissaman, “Low-reynolds-number airfoils,” *Annual review of fluid mechanics*, vol. 15, no. 1, pp. 223–239, 1983.
- [302] Y. Xie, Y. Rao, Y. Cheng, and W. Tian, “Investigation into the laminar separation control of airfoils at low reynolds numbers by dimple vortex generators,” *Aerospace Science and Technology*, vol. 129, p. 107841, 2022.
- [303] Z. Yang and H. Hu, “Laminar flow separation and transition on a low-reynolds-number airfoil,” *Journal of Aircraft*, vol. 45, no. 3, pp. 1067–1070, 2008.
- [304] J. Wang and L. Feng, *Vortex Generator*, p. 48–64. Cambridge Aerospace Series, Cambridge University Press, 2018.
- [305] M. Bakhtiari and H. Ghassemi, “numerical analysis of electromagnetic control of the boundary layer flow on a ship hull,” *Open Journal of Fluid Dynamics*, vol. 2014, 2014.
- [306] G. Godard and M. Stanislas, “Control of a decelerating boundary layer. part 1: Optimization of passive vortex generators,” *Aerospace science and technology*, vol. 10, no. 3, pp. 181–191, 2006.
- [307] P. Martínez-Filgueira, U. Fernandez-Gamiz, E. Zulueta, I. Errasti, and B. Fernandez-Gauna, “Parametric study of low-profile vortex generators,” *International Journal of Hydrogen Energy*, vol. 42, no. 28, pp. 17700–17712, 2017.
- [308] D. Lengani, D. Simoni, M. Ubaldi, P. Zunino, and F. Bertini, “Turbulent boundary layer separation control and loss evaluation of low profile vortex generators,” *Experimental Thermal and Fluid Science*, vol. 35, no. 8, pp. 1505–1513, 2011.
- [309] L. Zhang, K. Yang, and J. Xu, “Effects on wind turbine airfoils by vortex generators,” *Journal of engineering thermophysics*, vol. 31, no. 5, pp. 749–752, 2010.

- [310] X. Li, K. Yang, and X. Wang, “Experimental and numerical analysis of the effect of vortex generator height on vortex characteristics and airfoil aerodynamic performance,” *Energies*, vol. 12, no. 5, p. 959, 2019.
- [311] L. Zhang, X. Li, K. Yang, and D. Xue, “Effects of vortex generators on aerodynamic performance of thick wind turbine airfoils,” *Journal of Wind Engineering and Industrial Aerodynamics*, vol. 156, pp. 84–92, 2016.
- [312] A. Seshagiri, E. Cooper, and L. W. Traub, “Effects of vortex generators on an airfoil at low reynolds numbers,” *Journal of Aircraft*, vol. 46, no. 1, pp. 116–122, 2009.
- [313] S. Xue, B. Johnson, D. Chao, A. Sareen, and C. Westergaard, “Advanced aerodynamic modeling of vortex generators for wind turbine applications,” in *European Wind Energy Conference (EWEC), Warsaw Poland*, 2010.
- [314] K. J. Forster, T. J. Barber, S. Diasinos, and G. Doig, “Interactions of a counter-rotating vortex pair at multiple offsets,” *Experimental Thermal and Fluid Science*, vol. 86, pp. 63–74, 2017.
- [315] C. M. Velte, M. O. L. Hansen, and D. Cavar, “Flow analysis of vortex generators on wing sections by stereoscopic particle image velocimetry measurements,” *Environmental Research Letters*, vol. 3, no. 1, p. 015006, 2008.
- [316] H. Wang, B. Zhang, Q. Qiu, and X. Xu, “Flow control on the nrel s809 wind turbine airfoil using vortex generators,” *Energy*, vol. 118, pp. 1210–1221, 2017.
- [317] H. M. Lee and O. J. Kwon, “Numerical simulation of horizontal axis wind turbines with vortex generators,” *International Journal of Aeronautical and Space Sciences*, vol. 20, pp. 325–334, 2019.

- [318] C. Zhu, T. Wang, and T. Shao, “Investigation into three-dimensional rotational effect on hawt with numerical simulations,” *Acta Energ. Sol. Sin.*, vol. 40, no. 06, pp. 1747–1755, 2019.
- [319] A. Akbarzadeh and I. Borazjani, “Controlling flow separation on a thick airfoil using backward traveling waves,” *AIAA Journal*, vol. 58, no. 9, pp. 3799–3807, 2020.
- [320] E. P. DeMauro, H. Dell’Orso, S. Zaremski, C. M. Leong, and M. Amitay, “Control of laminar separation bubble on naca 0009 airfoil using electroactive polymers,” *AIAA journal*, vol. 53, no. 8, pp. 2270–2279, 2015.
- [321] M. Sato, K. Asada, T. Nonomura, S. Kawai, and K. Fujii, “Large-eddy simulation of naca 0015 airfoil flow at reynolds number of 1.6×10^6 ,” *AIAA Journal*, vol. 55, no. 2, pp. 673–679, 2017.
- [322] G. Jones, M. Santer, M. Debiasi, and G. Papadakis, “Control of flow separation around an airfoil at low reynolds numbers using periodic surface morphing,” *Journal of Fluids and Structures*, vol. 76, pp. 536–557, 2018.
- [323] T. M. Crittenden and A. Glezer, “A high-speed, compressible synthetic jet,” *Physics of Fluids*, vol. 18, no. 1, 2006.
- [324] I. Moralev, V. Sherbakova, I. Selivonin, V. Bituryn, and M. Ustinov, “Effect of the discharge constriction in dbd plasma actuator on the laminar boundary layer,” *International Journal of Heat and Mass Transfer*, vol. 116, pp. 1326–1340, 2018.
- [325] A. Le Pape, M. Costes, F. Richez, G. Joubert, F. David, and J.-M. Deluc, “Dynamic stall control using deployable leading-edge vortex generators,” *AIAA journal*, vol. 50, no. 10, pp. 2135–2145, 2012.
- [326] B. L. Storms and C. S. Jang, “Lift enhancement of an airfoil using a gurney flap and vortex generators,” *Journal of Aircraft*, vol. 31, no. 3, pp. 542–547, 1994.

- [327] S. Nissle, M. Kaiser, M. Hübler, M. Gurka, and U. Breuer, “Adaptive vortex generators based on active hybrid composites: from idea to flight test,” *CEAS Aeronautical Journal*, vol. 9, pp. 661–670, 2018.
- [328] J. J. Allen, G. R. Bell, A. M. Kuzirian, and R. T. Hanlon, “Cuttlefish skin papilla morphology suggests a muscular hydrostatic function for rapid changeability,” *Journal of morphology*, vol. 274, no. 6, pp. 645–656, 2013.
- [329] C.-C. Chiao, C. Chubb, and R. T. Hanlon, “A review of visual perception mechanisms that regulate rapid adaptive camouflage in cuttlefish,” *Journal of Comparative Physiology A*, vol. 201, pp. 933–945, 2015.
- [330] J. J. Allen, G. R. Bell, A. M. Kuzirian, S. S. Velankar, and R. T. Hanlon, “Comparative morphology of changeable skin papillae in octopus and cuttlefish,” *Journal of morphology*, vol. 275, no. 4, pp. 371–390, 2014.
- [331] R. Hanlon, “Cephalopod dynamic camouflage,” *Current biology*, vol. 17, no. 11, pp. R400–R404, 2007.
- [332] P. Kotak, T. Johnson, and C. Lamuta, “Bioinspired fouling-release smart skin powered by twisted spiral artificial muscles (tsams),” *Advanced materials technologies*, vol. 8, no. 4, p. 2201262, 2023.
- [333] M. Paredes and E. Rodriguez, “Optimal design of conical springs,” *Engineering with computers*, vol. 25, pp. 147–154, 2009.
- [334] A. M. Wahl, “Mechanical springs,” (*No Title*), 1963.
- [335] S. Aziz, S. Naficy, J. Foroughi, H. R. Brown, and G. M. Spinks, “Controlled and scalable torsional actuation of twisted nylon 6 fiber,” *Journal of Polymer Science Part B: Polymer Physics*, vol. 54, no. 13, pp. 1278–1286, 2016.

- [336] J. Hagewood, “Technologies for the manufacture of synthetic polymer fibers,” in *Advances in filament yarn spinning of textiles and polymers*, pp. 48–71, Elsevier, 2014.
- [337] V. Giovinco, P. Kotak, V. Cichella, C. Maletta, and C. Lamuta, “Dynamic model for the tensile actuation of thermally and electro-thermally actuated twisted and coiled artificial muscles (tcams),” *Smart materials and structures*, vol. 29, no. 2, p. 025004, 2019.
- [338] K. Sukel, “Dealing with the shortage of rural physicians,” *Medical Economics Journal*, vol. 28, no. 10, 2019.
- [339] X. Zhang, D. Lin, H. Pforsich, and V. W. Lin, “Physician workforce in the united states of america: forecasting nationwide shortages,” *Human resources for health*, vol. 18, no. 1, pp. 1–9, 2020.
- [340] V. Lin, X. Zhang, and P. Dixon, “Occupational therapy workforce in the united states: Forecasting nationwide shortages,” *PM&R*, vol. 7, no. 9, pp. 946–954, 2015.
- [341] X. Zhang, D. Tai, H. Pforsich, and V. W. Lin, “United states registered nurse workforce report card and shortage forecast: a revisit,” *American Journal of Medical Quality*, vol. 33, no. 3, pp. 229–236, 2018.
- [342] S. P. Juraschek, X. Zhang, V. Ranganathan, and V. W. Lin, “United states registered nurse workforce report card and shortage forecast,” *American Journal of Medical Quality*, vol. 27, no. 3, pp. 241–249, 2012.
- [343] I. Andras, E. Mazzone, F. W. van Leeuwen, G. De Naeyer, M. N. van Oosterom, S. Beato, T. Buckle, S. O’Sullivan, P. J. van Leeuwen, A. Beulens, *et al.*, “Artificial intelligence and robotics: a combination that is changing the operating room,” *World journal of urology*, vol. 38, pp. 2359–2366, 2020.
- [344] G. Ciuti, K. Skonieczna-Żydecka, W. Marlicz, V. Iacovacci, H. Liu, D. Stoyanov, A. Arezzo, M. Chiurazzi, E. Toth, H. Thorlacius, *et al.*, “Frontiers of robotic colonoscopy: a compre-

hensive review of robotic colonoscopes and technologies,” *Journal of clinical medicine*, vol. 9, no. 6, p. 1648, 2020.

- [345] A. Zemmar, A. M. Lozano, and B. J. Nelson, “The rise of robots in surgical environments during covid-19,” *Nature Machine Intelligence*, vol. 2, no. 10, pp. 566–572, 2020.
- [346] P. E. Dupont, B. J. Nelson, M. Goldfarb, B. Hannaford, A. Menciassi, M. K. O’Malley, N. Simaan, P. Valdastri, and G.-Z. Yang, “A decade retrospective of medical robotics research from 2010 to 2020,” *Science robotics*, vol. 6, no. 60, p. eabi8017, 2021.

PEER-REVIEWED JOURNALS

1. Rabiul Mamman, **Thilina H. Weerakkody**, and Caterina Lamuta. Bioinspired Active Vortex Generators for Underwater Flow Control (Under Review).
2. Rabiul Mamman, Tatum Johnson, **Thilina H. Weerakkody**, and Caterina Lamuta. Fouling Release Mechanism of an Octopus-Inspired Smart Skin. *Advanced Functional Materials*, p. 2406405, 2024.
3. Sean Maxson, Parth Kotak, **Thilina H. Weerakkody**, and Caterina Lamuta. Manufacturing and underwater cyclic behavior of different types of twisted and coiled artificial muscles (TCAMs). *Manufacturing Letters, Elsevier*, 2024.
4. Parth Kotak, Sean Maxson, **Thilina H. Weerakkody**, Venanzio Cichella, and Caterina Lamuta. Octopus-Inspired Muscular Hydrostats Powered By Twisted and Coiled Artificial Muscles. *Soft robotics*, 2023.
5. Rabiul Mamman, Parth Kotak, **Thilina H. Weerakkody**, Tatum Johnson, Austin Krebill, James Buchholz, and Caterina Lamuta. Deployable vortex generators for low Reynolds numbers applications powered by cephalopods inspired artificial muscles. *Iscience*, 26(12), 2023.
6. **Thilina H. Weerakkody**, Maxwell Hammond, James H Neilan, Venanzio Cichella, and Caterina Lamuta. Modeling and control of twisted and coiled artificial muscles for soft robotics. *Meccanica*, 58(4):643–658, 2023.
7. Carlo Greco, **Thilina H. Weerakkody**, Venanzio Cichella, Leonardo Pagnotta, and Caterina Lamuta. Lightweight Bioinspired Exoskeleton for Wrist Rehabilitation Powered by Twisted and Coiled Artificial Muscles. *Robotics*, 12(1):27, 2023.
8. Samantha Bell, Arnold Bangel, **Thilina H. Weerakkody**, Xuan Song, and Caterina Lamuta. Automated manufacturing system for carbon fiber-based twisted and coiled artificial muscles (TCAMs). *Manufacturing Letters*, 33:19–23, 2022.
9. Maxwell Hammond, Venanzio Cichella, **Thilina H. Weerakkody**, and Caterina Lamuta. Robust and adaptive sampled-data control of twisted and coiled artificial muscles. *IEEE Control Systems Letters*, 6:1232–1237, 2021.
10. Parth Kotak, **Thilina H. Weerakkody**, and Caterina Lamuta. Physics-based dynamic model for the electro-thermal actuation of bio-inspired twisted spiral artificial muscles (TSAMs). *Polymer*, 222:123642, 2021.